Learning with AI

A Framework for Students, Instructors, and Universities

Undergraduate and Graduate STEM Education,
Departmental Practice, Institutional Policy

James M. Hyman

Tulane University
New Orleans, Louisiana

Author's pre-edit draft, posted with SIAM's permission. Not the published edition.
For the final, edited edition, please buy the book from SIAM. Cite the SIAM edition, not this file.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Use AI when it helps you think.
Stop when it starts thinking for you.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

About this book

This book presents a framework for AI-assisted learning at colleges and universities. It is written for three audiences: students learning to use AI without surrendering their own thinking, instructors designing courses that remain intellectually honest under AI access, and academic departments and institutions developing coherent policy and practice. It is written for the STEM community, undergraduate through graduate, with mathematics as its spine: the framework grew out of graduate mathematics, where the failure modes show most clearly, and its examples say so when they need more than the mathematics every STEM student carries. Much of the guidance transfers to writing, the humanities, the social sciences, professional programs, and creative work as well, though the book does not set out to be a manual for those fields. Throughout, "AI" means generative AI: general-purpose tools that produce text, code, proofs, explanations, or analyses on request, and whose output can vary from one run to the next.

The framework rests on three commitments. First, courses must distinguish what students can do independently from what they can do with AI assistance, and assess each in proportion to the goals of the course. Second, AI use should be visible, verified, and disclosed rather than hidden or assumed. Third, students, instructors, and institutions need shared vocabulary if AI policy is to function in practice.

The book is organized in three Parts. Part I, the Student Guide, gives undergraduate and graduate students a practical compact for using AI as tutor, critic, and study partner without giving up the thinking each course is meant to build. Part II, the Instructor Guide, gives faculty assignment-design tools, assessment models for small and large classes, and rubrics for AI-aware grading. Part III, the Departmental and Institutional Guide, gives departments and universities shared policy language, privacy safeguards, equity considerations, and adoption workflows that work at the level where most decisions are actually made. Each Part has a toolkit that collects its templates and checklists: chapter 7, chapter 13, and chapter 18. The three Parts share a single vocabulary: Core Competence, AI-assisted Practice, and Trustworthiness; the Learning Spiral; and the assignment categories No AI (**NAI**), AI as Tutor (**AIT**), AI as Collaborator (**AIC**), and AI as Object of Study (**AIS**). This book was itself prepared with AI assistance, disclosed in Appendix C.

How to read this book

This book is one bound volume in three Parts. Each Part is written for a different reader, and each Part can be read on its own. Parts I through III are the book. The appendices and references after them are tools to consult, not chapters to read.

Three short descriptions follow, one for each reader. Find the one that sounds like your week.

You got the right answer and could not explain it. An AI produced something clean, and the next day you could not reconstruct it. Page 9 shows a student asking for the limit of $(\sin x)/x$ and getting the whole proof back at once. The same page shows the same model, asked the same question, answering with a question instead. Nothing about the AI changed. What changed was what it had been told. The Student Guide begins on page 3.

You cannot tell what the student did. The work is polished and the grade is a guess. Page 108 describes planting a fixed number of errors in an AI-written passage and asking students to find them. The score is the detection rate. A student who finds four of five has shown you something a clean submission never shows. The Instructor Guide collects instruments like that one, and it begins on page 75.

You have to write a policy and the evidence is thin. Claims about AI in education arrive faster than the studies behind them. Page 185 takes this book's own claims one at a time and marks each one as established, supported only in part, or untested. The Departmental and Institutional

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Guide builds departmental practice on the first two Parts and says where it is guessing. It begins on page 143.

Most students meet the Student Guide because an instructor assigned it. That is how the book is designed to travel, and each chapter can be assigned on its own when a course reaches its topic. A student who found this book alone can read it the same way.

Each description above points you to your own Part. The book also asks you to look at the other two, and that request is not incidental; it is the reason this is one book (figure 1). Policy, teaching, and learning are all being worked out at the same time, and no group yet holds settled expertise to hand to the others. Reading usually runs one way: those who teach and those who set policy read about the work their decisions shape. It is meant to run the other way as well. A student who reads the instructor Part is better placed to tell a teacher what is working, at a time when many instructors are still finding their footing with these tools. An instructor who reads the institutional Part is better placed to tell a chair or dean what classroom policy should say. The transformation these Parts describe is a collective one, and it will go better if each group can see it through the others' eyes. One thread runs through all three. Each group acts on what it is scored on: a student on the grade, an instructor on what the department counts, a department on how the budget rewards enrollment. This is what links the three groups. It is also why a fix at one scale can be undone at another.

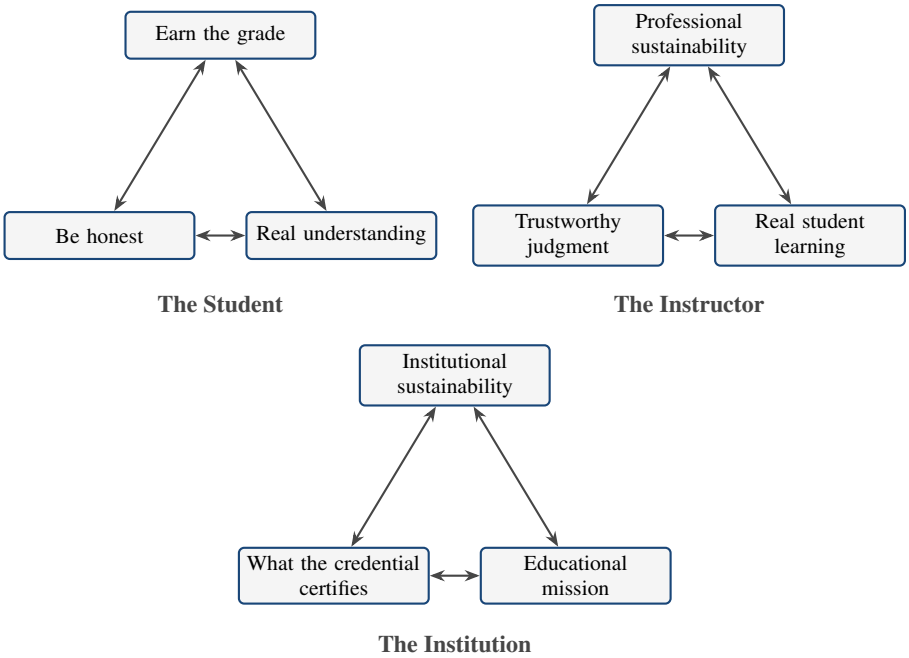


Figure 1. ***One structure, three scales.*** *A fix at one scale can be undone at another, which is why all three are shown together. Each group's three goals sit at the corners in the same orientation: a viability goal at the top, an honesty goal at the lower left, and a genuine-learning goal at the lower right. The arrows mark the tensions between them. The same three failure modes arise at every scale, and figure 2.1 names them on the edges where they occur.*

All three Parts use the same vocabulary. The full development of each term appears in the Part where the reader most needs it: Core Competence, AI-assisted Practice, and Trustworthiness in Part I; assignment categories and assessment models in Part II; institutional policy and adoption practice in Part III.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

A reader looking for a single chapter to scan should consider the CAT framework section in the Student Guide, the four assignment categories in the Instructor Guide, or the CAT section in the Departmental and Institutional Guide, depending on role. The CAT framework is named for its three dimensions: Core Competence, AI-assisted Practice, and Trustworthiness.

Readers who want a brief account of the learning-science principles this book puts into practice should also see section 2.2 (How learning happens, and how AI changes it). That section covers the calibration problem, the gap between how well students think they understand something and how well they actually do. AI use makes this gap sharper.

Two subject appendices follow the three Parts. They are written for different readers, and neither is required for the framework itself. Appendix A covers AI-assisted scientific writing and is written for scientists, from graduate students to faculty researchers. Appendix B is a quick reference for mathematics courses.

About the companion repository

This book has a companion repository at `https://github.com/machyman/hyman2026learning`, with a rendered site at `https://machyman.github.io/hyman2026learning`. The materials are permanently archived at `https://doi.org/10.5281/zenodo.21315920`. That address will keep working even if the repository moves. The repository hosts free, open-source materials that work alongside the book. Two are primary tools that students paste into AI applications. The Study Partner Protocol is a prompt that turns the assistant into a study partner, one that helps the student learn rather than handing over finished work. Define Personal Preferences is a walkthrough that helps a student configure their AI tool's persistent settings the first time they use it. The repository also gives instructors sample syllabus paragraphs, a syllabus guide, and a first-time teaching guide. Students receive standalone documents: a guide to sharing AI conversations with a teacher, disclosure templates, and the student checklist that appears in Part I.

The repository is organized in four folders, and table 1 shows what each one holds. It also holds companion papers that carry arguments this book only summarizes, and more are expected after publication. "AI and the Next Layer of Human Work" [55] develops the historical case about which work survives automation and which does not. "Teaching AI Responsibly to Principled Skeptics" [56] makes the case for engaging with AI critically rather than refusing it, and is the paper to read first if you or your students arrive unconvinced that AI belongs in a course at all.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Table 1. ***What the companion repository holds.*** *Start in the folder that matches who you are, not the Part you are reading. Every folder is reachable from the repository's front page.*

Folder	What it holds	For
`students/`	Two onboarding guides, the student checklist, disclosure templates, a guide to sharing AI conversations with a teacher, and worked study sessions in statistics, programming, physics, and mathematics	Students, and instructors choosing what to assign
`instructors/`	A first-time teaching guide, a syllabus guide, a one-page assignment schedule, assignment templates, rubrics, assessment models, a semester redesign, and a faculty FAQ	Faculty, teaching assistants, course coordinators
`institutions/`	A departmental adoption kit, a policy self-audit, and the regulatory landscape	Chairs, committees, teaching centers, administrators
`companion/`	The Study Partner Protocol, Define Personal Preferences, and the other documents students paste into an AI tool	Students, assigned by instructors
`teaching-the-book/`	Course formats, a sample seminar schedule, and chapter exercises with sample strong and weak answers	Instructors teaching a course with the book, and their students

The book and the repository are intended to be used together. The book teaches the framework; the repository hosts the operational materials. References from the book to specific companion materials appear throughout, in the form `companion/spp/`, `students/checklist.md`, and similar paths. The book remains under SIAM copyright; the materials in the repository are released under the MIT license and can be freely used, modified, adapted, translated, and redistributed with attribution. The repository will continue to evolve after publication; corrections to the book itself, if any are needed, appear in the repository's `ERRATA.md` file. The same file records post-publication status notes for time-sensitive references, so a reader can see which dated snapshot entries have gone stale.

Acknowledgments

My friends and colleagues changed this book. What follows records what each of them changed and how their feedback improved the book. The on-ramp for students who arrive underprepared,

the index, the case for doing the work at all, and the sentence in the preface pointing to the repository all exist because of a reader. I also thank the broader SIAM applied mathematics and STEM education community for the institutional context that informs Part III's policy recommendations, and the graduate students who tested early versions of the framework in their own work and whose questions sharpened the assessment practices in Parts I and II. The framework also owes a debt to the existing literatures on academic integrity [13, 86] and AI-assisted scholarship [89, 98, 126].

Four readers widened the book's reach. Misha Chertkov made the case for widening it from a mathematics text into a framework for STEM as a whole. Arieh Iserles sharpened the page-one framing of who the book is for, and prompted the passages on peer argument as the social training ground for the same scrutiny students must learn to apply to AI. Thomas Hou, reading as a numerical-analysis instructor, pressed the book to say more about mathematical judgment, the intuition that a result is wrong before one can prove it, and supplied the ordered verification hierarchy in chapter 4. Bruce Krell pressed the book to say why it is one volume, and to tell each reader which Part is theirs and why to look at the others.

Several colleagues shaped the student guide. Michelle Lacey led me to add a section on why the work is worth doing for students who do not yet see the point, and to rebuild a worked example so that it tests intuition before mechanics. Warren MacEvoy found a contradiction at the heart of Part I, and he pulled the book's plainest idea, that a learner's value lies in judgment, out of the section where I had buried it. Michael Forbes reshaped its opening and how it keeps a reader oriented. Adriano Marzullo, a mathematics instructor, showed me that the assessment models asked for more verification than many courses need. That led to the simplest option of all: put the weight of the grade on the exam. Wei Bao encouraged me to warn against AI as a student's only resource, since college surrounds them with people who can help. She also made the student the explainer and the AI the second reader. Kaleigh Rudge prompted three additions, including holding an AI tutor to what a course has covered, and strengthened the section on students who have little available time. Lea Jenkins wanted to try the ideas as she read, so exercises and opening questions now sit in Part I. She turned responsible use from a verdict into a description of where the reasoning stayed. Alejandro Aceves showed that the uncertainty guidance missed the case where AI answers correctly but narrows the question, and that the study routine had no group version.

On verification, and on the case against AI altogether, three readers pushed hardest. Aleksei Sorokin read the student guide as a working researcher and sharpened its case for verifying AI, drawing the line between a calculator, which returns the same guaranteed answer every time, and AI, which does not. Gerardo Chowell turned the AI's instruction from flagging its own confidence into naming what to verify and how, and urged that platform-specific advice be marked as a dated snapshot. Kevin Vixie read the guide as a committed skeptic and made the case for the apprenticeship, the slow embodied path to mastery that asks nothing of AI at all. The guide is sharper for having to earn its claims against it.

The institutional guide grew deeper in the hands of four colleagues. Fred Hickernell shaped the Institution's Trilemma, framing its three forces as a tension to hold and its failure modes as the loss of mission, of certification, and of sustenance. He also pressed for assessments that can span more than one category, and for the link to program accreditation. Peter Moore strengthened its treatment of budgets and risk, adding the Responsibility Center Management lens and a risk-register row. Marie Dahleh caught Part III narrowing to graduate STEM, so it now opens with a note on scope. Two of the book's three trilemma figures exist because she asked for them. Kathleen Hoffman pressed that filling a teaching-center workshop is a harder problem than designing one. She also asked four questions that the book already answered, but she could not find the answers. Her questions sent me to the index rather than to the prose, and I rebuilt the index.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Others kept the book wide and honest about its limits. Mike Joyce and Yun Kang prompted the on-ramp for undergraduates who arrive underprepared, and Yun Kang also sharpened the guidance on detector equity. Svetlana Barkanova broadened that argument beyond language to neurodiversity. Juan Restrepo and David Knox pressed hard on length, depth, and audience, questions that guide each revision. Juan Restrepo singled out the academic-integrity material as worth preserving. David Knox pressed the book to ground its claims in what is known about teaching and learning, and to be honest where the evidence is thin. He also argued for treating the cost and benefit of AI use, including the time it saves and spends, as a question in its own right. Shelley B. Rohde Poole's SIAM tutorial prompted the book's emphasis on student buy-in: design protects the will to learn but cannot create it, and struggle needs a climate where being wrong is safe. Daozhou Gao asked for a more actionable book and found the preface disclaiming tool coverage, so it now points to where the practical material lives. Robert Steward found the instructor guide never pointing to the ethical-writing appendix, so it does now, and asked about problem sets years in circulation, which became a section on recasting them. Kevin Lin found his students choosing the textbook default over what the course wanted, so a scope note now names what a course is for, not only what it covers. Theo Kwofie asked who would honor work done under a new scheme outside the department, so Part I now invites assigning one chapter at a time. Scott McKinley pressed the book to teach the presentation of evidence of thinking, not only ask for it, so both guides now do, and his email prompted the exchange that paired the two endpoint prompts. Lulu Kang prompted the example assignments each category now carries, in a form that copies into a syllabus.

For these contributions, and for the careful reading of Joshua Agbomola and Oscar Olbera, I am grateful. The book is better for all of it.

Generative AI tools were used in preparing this book. Claude (Anthropic, 2025–2026) was the primary tool, used for discussion, drafting, revision, structural critique, and reference checking; ChatGPT (OpenAI, 2025–2026) supplied critical reviews. Appendix C describes this use. The author assumes responsibility for all content.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Contents

About this book . . . iii
How to read this book . . . iii
About the companion repository . . . v
Acknowledgments . . . vi

Preface **xiii**

I Learning with AI: A Student Guide **1**

1 Getting Started **5**
1.1 Why this guide exists . . . 5
1.2 Setting up your AI for learning . . . 8
1.3 Two skills, not one . . . 11
1.4 Learning outcomes . . . 14

2 The Problem and the Compact **15**
2.1 The Student's Trilemma . . . 15
2.2 How learning happens, and how AI changes it . . . 19
2.3 The basic compact . . . 22

3 The Framework **25**
3.1 The CAT framework . . . 25
3.2 The Learning Spiral . . . 27
3.3 When the aim is a new idea . . . 32
3.4 Four assignment categories . . . 33

4 Using AI Well and Badly **37**
4.1 Good uses of AI . . . 37
4.2 Annotated dialogues: strong AI use . . . 39
4.3 Dangerous uses of AI . . . 44
4.4 How to verify AI-assisted work . . . 45
4.5 Disclosure . . . 50
4.6 Privacy and data protection . . . 52

5 In Class and Across Disciplines **53**
5.1 Small classes and the Oral-Defense Model . . . 53
5.2 Large classes and the Evidence-of-Thinking Model . . . 53
5.3 Asking better questions in mathematics and related STEM disciplines . . . 54

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

5.4 Asking better questions in the humanities, social sciences, and professional fields . . . 56
5.5 Discipline examples . . . 57

6 Judgment and Limits 59
6.1 When not to use AI . . . 59
6.2 What to do when you are unsure . . . 60
6.3 Working with AI on group assignments . . . 60
6.4 Evidence base and limits of the guide . . . 61
6.5 What you can expect from your institution . . . 62
6.6 A final word . . . 62

7 The Student's Toolkit 65
7.1 One assignment, start to finish . . . 65
7.2 Template: AI account configuration . . . 66
7.3 Template: prompt library for learning . . . 69
7.4 Template: verification checklist . . . 69
7.5 Template: AI use log . . . 70
7.6 Pitfalls . . . 70
7.7 When the work is research . . . 71
7.8 When you are unsure . . . 71
7.9 Template: one-page student compact . . . 72
7.10 Student checklist . . . 72

II Teaching and Assessing with AI: An Instructor Guide 73

8 The Instructor's Trilemma 77
8.1 Purpose and scope . . . 77
8.2 Instructor outcomes . . . 78
8.3 The instructor-side calibration problem . . . 78
8.4 The three corners . . . 80
8.5 Three design axes: class size, level, and discipline . . . 85

9 The Framework and Assignment Design 87
9.1 The CAT framework . . . 87
9.2 Four assignment categories . . . 89
9.3 Designing assignments that preserve thinking . . . 92
9.4 AI tools for teaching preparation . . . 101

10 Assessment Models 103
10.1 Where this fits in a grade you already give . . . 103
10.2 The Oral-Defense Model . . . 104
10.3 The Evidence-of-Thinking Model . . . 106
10.4 The Hybrid Model . . . 108
10.5 One assisted task, one later transfer task . . . 112
10.6 Group and collaborative assignments . . . 112
10.7 Feedback and grading with AI . . . 113

11 Integrity, Equity, and Discipline 117
11.1 Annotated dialogues: weak AI use as teaching cases . . . 117

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

11.2 Academic integrity without detector dependence . . . 118
11.3 Privacy, access, and equity . . . 122
11.4 Discipline overlays . . . 123
11.5 Rubrics . . . 127

12 Implementation and Limits 129
12.1 TA calibration . . . 129
12.2 Implementation workflow for a course . . . 129
12.3 Evidence base and limits . . . 130

13 The Instructor's Toolkit 133
13.1 Assignment wording bank . . . 133
13.2 Making honesty cheap . . . 135
13.3 When a submission looks wrong . . . 135
13.4 Assessment structure and documentation . . . 136
13.5 Oral defense rubric and random audit script . . . 137
13.6 Safeguards that make substitution not pay . . . 138
13.7 Instructor pitfalls . . . 138
13.8 Course design checklist and self-check . . . 138
13.9 AI critique assignment template . . . 139

III AI-Assisted Learning: A Departmental and Institutional Guide 141

14 The Institution's Trilemma 145
14.1 Purpose and scope . . . 145
14.2 Institutional outcomes . . . 146
14.3 What narrows when everyone complies . . . 146
14.4 The three corners . . . 147
14.5 The institutional problem . . . 151
14.6 Guiding principles . . . 152

15 The Framework and Vocabulary 153
15.1 The CAT framework . . . 153
15.2 Course-level AI-use categories . . . 154
15.3 Assessment models by course size . . . 155

16 Standards and Requirements 157
16.1 Minimum independent-competence requirement . . . 157
16.2 Disclosure standard . . . 157
16.3 Privacy, data governance, and tool approval . . . 158
16.4 Equity, access, and accessibility . . . 159
16.5 Academic integrity procedure . . . 160
16.6 Faculty use of AI . . . 162
16.7 Required syllabus elements . . . 162

17 Implementation and Change 165
17.1 Governance and support . . . 165
17.2 Departmental responsibilities . . . 166
17.3 Teaching-center responsibilities . . . 166
17.4 Training plan . . . 167

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

17.5 Implementation plan . . . 168
17.6 Faculty change management . . . 170
17.7 From department to institution . . . 174

18 The Institution's Toolkit 177
18.1 Implementation phases . . . 177
18.2 Expected outcomes . . . 177
18.3 Reference materials . . . 178

19 Looking Forward 183
19.1 Looking forward . . . 183
19.2 Adoption claims and metrics . . . 185
19.3 Risk register . . . 187
19.4 Model institutional policy statement . . . 189
19.5 Evidence base and limits . . . 189

A Best Practices for Ethical AI-Assisted Scientific Writing 191
A.1 Core principles . . . 191
A.2 The verification cornerstone . . . 193
A.3 Disclosure guidelines . . . 195
A.4 Common pitfalls . . . 196
A.5 Pre-submission checklist . . . 197
A.6 Verification standards by AI contribution type . . . 198
A.7 Summary . . . 198

B Math-Specific Quick Reference 199
B.1 Student use in mathematics . . . 199
B.2 Instructor use in mathematics . . . 200

C Disclosure of AI Assistance in Preparing This Book 203

Glossary 205

Bibliography 209

Index 223

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Preface

AI changed how proofs are written.
It did not change what it means to understand one.

What this book is about

This book is about learning, teaching, assessment, and institutional practice in a world where AI is already available to anyone with a browser. It is not about how mathematics is used to build AI systems, nor about how AI is now being used to do mathematics. Both are important subjects with their own large literatures. This book is about something different: how students learn, how instructors teach, and how universities respond, given that AI tools are now a permanent feature of the educational environment.

Mathematical discovery does enter this book, but only where students first meet it: a course project, an undergraduate or master's thesis, a dissertation. What a working mathematician does with these tools is a separate subject.

The framework's primary audience is in the quantitative and computational sciences: mathematics, applied mathematics, computer science, statistics, theoretical and computational physics, operations research, data science, mathematical biology, and related disciplines.

The framework originated in graduate mathematics teaching. Mathematical reasoning makes the gap between using a tool well and using a tool to skip thinking unusually visible, which is why graduate mathematics was a productive starting point. The patterns apply to other quantitative and computational disciplines, where readers will recognize their own field's verification habits in the book's vocabulary. The book does not engage substantively with the experimental and laboratory aspects of the physical and life sciences, which have their own verification practices and would warrant separate treatment. Readers from those fields, and from the humanities, social sciences, design, and the professions, are welcome here; the framework's underlying spiral (frame the question, try to solve, ask, work, verify, reflect) is broadly applicable with discipline-specific adaptation.

Underneath all of it is a single idea. Used well, AI does not replace the work of learning but relocates it, from execution to judgment, and the task for students, instructors, and institutions alike is to preserve and teach that judgment as the tools change.

Who this book is for

This book is written for three audiences who need to communicate with one another.

Students, both undergraduate and graduate, need a clear standard for what counts as their own work and how to demonstrate it. The graduate-mathematics roots of the framework appear most explicitly in the verification chapters and in the discipline overlay sections, but the practical compact applies in any course where AI is allowed.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Instructors need ways to design courses that remain assessable when polished AI output is freely available, and ways to grade AI-assisted work that reward thinking rather than fluency. The Instructor Guide translates the framework into assignment categories, assessment models for small and large classes, and rubrics.

Departments and institutions need shared policy language, privacy safeguards, equity considerations, and a fair process for suspected misuse. The graduate-school origins of the framework matter here too: graduate programs are often where institutional AI policy is first stress-tested, and where the consequences of getting it wrong show up earliest.

The book also works as a supplement, read alongside a technical course that has no room for a unit on AI. It fits an introduction-to-the-profession seminar, where questions about honest work and credit are already on the syllabus.

Each audience faces its own version of the same three-way pull, and the pressure on any one corner usually comes from somewhere else in the system. That is the reason to look at the other Parts. A student who sees what an instructor is protecting understands why an assignment is built that way. An instructor who sees what a chair is up against can make a better case to one. Skimming is enough. The aim is to recognize the other reader's problem, not to master their Part.

How this book began

This book began with a textbook. I was writing a textbook on sensitivity analysis with Leon Arriola when a gap became clear. The textbook assumed a reader with a book, a pencil, and their own patience. Most students now work with something more. They keep an AI assistant open beside the textbook and use it to explain, to check, and sometimes to skip the hard part. The book on my desk was a twentieth-century textbook. The students it was meant for already lived somewhere else.

The obvious response was to add a few words about AI. That proved harder than it looked. I went looking for guidance on how students learn well with AI, and on how instructors can use it to teach, to assess, and to grade, and I found very little. What existed was thin, scattered, and rarely built for the classroom. So I began to dig. I read published and unpublished work, followed the question across disciplines, and sat in on conferences where it was only beginning to be asked. The deeper the reading went, the clearer the need became. This was not a paragraph to insert in one textbook. It was a framework that students, instructors, and institutions all needed, and that no single source yet provided. That is the book this became.

Origin in graduate mathematics

The framework it offers began earlier, as a specific problem in graduate mathematics teaching.

In late 2022 and through 2023, students started presenting me with proofs they had not written and could not defend. Some were honest about asking a chatbot for help. Others were not. The proofs were uneven: occasionally correct, often plausible but flawed in ways that mattered, and sometimes just wrong in subtle ways the student could not detect. The students themselves were often unsure how much of the work was theirs. They wanted to learn. They also wanted the work done.

The standard responses to this situation were both unsatisfying. A blanket prohibition on AI tools was unenforceable in any course where work happened outside the classroom, and it left students unprepared for the professional and research environments they would enter. A blanket permission was no better: it conceded the field. A polished proof submitted by a student who could not reconstruct it on a blackboard was not evidence of mathematical understanding. It was evidence that the student had used a tool.

The framework presented in this book grew out of an attempt to find a third path for graduate

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

mathematics, one that recognized AI as a permanent feature of mathematical practice while preserving the habits of mind that doctoral training is meant to build.

Why this is not the first time

The framework presented here grew out of a particular crisis in graduate mathematics, but the underlying question is older. AI is not the first technology to frighten people by redefining skill. The pattern repeats often enough that it is worth naming before the rest of the book builds on it.

When photography became widely available in the nineteenth century, painters had reason to worry. A camera could capture a likeness quickly and cheaply, and the demand for realistic visual copying did decline. But painting did not disappear. Painters moved more deeply into color, abstraction, emotion, and personal vision, into territory a camera could not follow. The camera did not destroy the artist. It changed what the artist was needed for.

When calculators became common in classrooms, educators worried that students would stop learning arithmetic. The concern was not wrong: a student who reaches for a calculator before developing number sense may never fully acquire it. But calculators did not end mathematics. They shifted the level at which mathematical work happens, from manual computation toward modeling, reasoning, and the judgment of when an answer is plausible. That last skill is what a calculator cannot do, and it is exactly where the mathematician earns the title.

The history is not uniformly positive. Routine work disappears, and people whose livelihoods depend on that work are displaced. But across these transitions, the skill of the practitioner did not disappear. It relocated. The work moved up a level, from execution to judgment, from production to evaluation, from doing to deciding what should be done and recognizing when it has been.

The historical argument summarized here is developed at greater length, with citations to research on AI's effects on productivity and learning, in the companion paper "AI and the Next Layer of Human Work" [55].

This book reads AI in that history. The framework is built on the assumption that AI shifts what teaching and learning are for. Universities, like painters and mathematicians before them, need to identify what the new level of work requires before they can teach it. In some fields the shift is already visible inside professional practice: AI is changing how science itself is done, which makes the case for cultivating the judgment these tools cannot supply more urgent, not less.

What this framework presumes

This pattern, the practitioner's skill relocating rather than disappearing as tools change, holds only when the practitioner has foundations to relocate from. A student who has used AI to complete earlier coursework without building the underlying skills arrives in subsequent courses with credit but without the foundation the credit is supposed to indicate. The framework in this book can help such a student use AI well in their current course. It cannot retroactively give them the prior coursework they did not build.

Looking back, this outcome is predictable from a gap this book discusses throughout: students can perform well without learning much. Instructors in introductory courses have begun to report it anecdotally. The book takes the question seriously without claiming to solve it.

The upstream work is real and outside the scope of any single course: how admissions, placement, and transfer-credit processes interpret transcripts in an AI-shaped environment, how K-12 curricula adapt, how summer remediation should change. The book offers the within-course tools.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

What to expect

The framework is presented through three Parts, but the reader will encounter several elements that may be unfamiliar in this kind of book. The Student and Instructor Guides include eight annotated student–AI dialogues showing strong, weak, and borderline AI use across calculus, real analysis, percolation theory, programming, history, and writing. The dialogues are not transcripts of real students; they are pedagogical illustrations of behaviors the framework's distinctions are designed to surface. The Departmental and Institutional Guide includes a section on faculty change management that addresses the practical question of how a university actually moves from current practice to AI-aware teaching, and a section on what is likely to change as AI capabilities continue to evolve. Both sections take the position that the framework's structural commitments are durable while specific implementations require regular refresh.

What this book is not

A few clarifications about scope.

This is not an evaluated intervention. No study yet measures whether courses built on this framework produce better learning than courses that are not. The closest external evidence is a computer-science course redesigned around no-AI assessment, oral defense of submitted code, and assignments designed to resist chatbot completion: students with and without AI access reached similar learning outcomes [70]. That is what such a redesign is meant to achieve, since it removes the advantage of skipping the work. The result supports the style of redesign recommended here. It does not test the framework itself. Read what follows as a set of reasoned design choices, not as a proven treatment. Table 19.1 sets out claim by claim what is established, what is supported only in part, and what is untested.

This is not a discipline-balanced treatment. The book's worked examples and discipline overlays draw most heavily from mathematics, computer science, statistics, theoretical and computational physics, and adjacent quantitative STEM fields, reflecting the framework's origin in graduate mathematics and the author's primary teaching contexts. Universities and departments in the humanities, qualitative social sciences, and arts will find the framework's structure directly applicable. That structure includes CAT, the four assignment categories, the three assessment models, the three design axes, and the verification-and-disclosure standards. The specific examples will need adaptation to those disciplines' particular reading, writing, interpretive, and creative practices. The discipline overlays in the Instructor Guide make a partial start on this adaptation; deeper humanities-specific treatments are a natural next step that this book deliberately leaves to disciplinary experts.

This is not a tool guide. The specific AI products available at the time of writing will not be the products available a year from now. The framework is intentionally tool-agnostic about products. It is not neutral about configuration: the evidence in section 6.4 attributes most of the measured difference to how a tool was set up rather than to which tool it was, so configuration is part of course design and an instructor and departmental responsibility. What the book declines to do is rank brands. When the book discusses prompts, examples, or model behaviors, the discussion is illustrative rather than prescriptive. A student or instructor who wants to know how to use a specific tool well should consult that tool's documentation; the book treats the tool category, not the brand. The practical, tool-specific material for this framework lives in the companion repository, which can be revised as the tools change and the book cannot.

This is not a policy mandate. Universities, departments, and individual instructors retain the right to set rules appropriate to their courses, programs, and institutional contexts. The framework is offered as a vocabulary and a set of design choices. Local adaptation is expected.

This is not an AI-skeptic manifesto. The framework treats AI as a powerful and consequential

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

technology that students will use throughout their careers. A framework that refused to engage with it would not protect learning; it would simply leave students unprepared. Independent research on AI tutoring [61] and practice guides for instructors [93] suggest that careful pedagogical design can make these tools genuinely useful for learning, and the early position literature already mapped the opportunity-and-risk landscape [60]; the framework draws on this body of evidence.

This is not an AI-utopian manifesto either. The same research base shows that unrestricted AI use can produce short-term performance gains while weakening what students can do once the AI is removed [11], that AI systems can produce confidently wrong answers and invented sources [12, 131], that detector-based enforcement is unreliable and inequitable [79], and that institutional AI adoption has so far outpaced policy maturity [4]. The framework takes all of these seriously. The sharpest of these results, the Bastani field experiment, is laid out in full in section 6.4; its metacognitive finding, that students did not perceive the learning they lost, anchors the book's case for objective Core Competence assessment.

The book aspires to a position that has gained ground in recent treatments of AI in education: AI is best understood as a co-intelligence that augments human work, with a human in the loop who stays responsible for that work [92]. What that leaves for people to do, and why the residual is not simply a smaller version of the same job, is the subject of the companion paper "AI and the Next Layer of Human Work" [55]. Whether that augmentation strengthens or weakens learning depends on the design of the course around it. That is what this book tries to help with. Bowen and Watson's *Teaching with AI* [19] develops the practical teaching side of the same view; this book differs in carrying a single framework across students, instructors, and institutions. Vee, Watkins, and Bruff's *The Norton Guide to AI-Aware Teaching* [129] is the closest recent companion to the Instructor Guide, organized around the teaching tasks an instructor meets in turn rather than around one vocabulary carried across audiences, and it is drawn on at several points below.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Part I

Learning with AI: A Student Guide

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

An answer you cannot explain
does not belong to you yet.

Terms. This Part uses the CAT framework, named for its three dimensions: Core Competence, AI-assisted Practice, and Trustworthiness. The Glossary at the back of the book defines these and the other terms that carry across Parts.

Part I of this book is the Student Guide, written for undergraduate and graduate STEM students. Its examples come mostly from mathematics, but the principles also apply to writing, humanities, social sciences, professional programs, and creative work.

Part I is built around a simple standard: if you submit work, you should be able to explain it, verify it, and take responsibility for it. AI may help you reach that point. It cannot replace it.

Part I is seven chapters of five to sixteen pages. They are close to independent, so a reader can take them in order or one at a time. Chapter 1 separates running the tool from learning with it. Chapter 2 names the pull between grades, honesty, and understanding, and sets out the compact. Chapter 3 gives the framework the rest of the book uses. Chapter 4 works through strong and weak use, and how to check what you get. Chapter 5 covers what changes with class size and across disciplines. Chapter 6 covers judgment and where the tool runs out. Chapter 7 is the toolkit: the study routine, the templates, and the checklist, collected in one place. An instructor can assign one chapter when a course reaches its topic, rather than the whole Part at once. When that fits is a teaching decision, and this book does not make it.

Before Chapter 1, three questions worth answering in writing, for nobody but yourself. What do you hope AI will give you this term, and what do you refuse to let it take? What is your own standard for the work you submit, in one sentence you would sign? Where do you want to be in five years, and did the way you used AI last week move you toward that or away from it? Keep the answers. When Chapter 2 sets out the compact this book asks of you, see how your second answer compares.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 1
Getting Started

1.1 ▪ Why this guide exists

Thirty years of teaching: when the work turns out to be someone else's, the conversation that follows is always the same.

Before any of this, before chatbots, I used to catch a copied proof by its notation. A student would hand in a beautifully written argument that used symbols I had never put on the board. They had pulled it out of a textbook and did not understand it well enough to translate the symbols into the ones I had used. The proof was correct. The understanding was not theirs.

The technology has changed. The ache has not.

Study guides for the AI era exist. This one differs in sharing a single framework and vocabulary with the instructor and institutional guides in this book, so the habits you build here match the rules your courses set and the policies your university writes.

This guide is written for STEM students, undergraduate and graduate. Its examples run on a mathematical spine, because that is where I taught and where the failure modes show most clearly. When an example needs mathematics beyond what every STEM student carries, it says so, and the point does not depend on the details.

AI tools are now part of ordinary academic life. You can use them to explain hard ideas, summarize readings, generate practice questions, revise drafts, write code, check grammar, and prepare for exams. Used well, these tools can make learning more active and more personal. Used poorly, they can mislead students and faculty, and reduce learning, sometimes to nothing: they hide confusion, weaken memory, invent sources, and produce work that a student cannot defend.

The question is not whether AI will appear in college work. It already has. Better to ask how to use it without surrendering your own thinking.

This guide gives you a practical way to do that. It does not treat AI as forbidden, magical, or harmless, but as a powerful tool that requires judgment. Learn to ask better questions, check what you receive, protect private information, and disclose your use honestly. Learn, too, when not to use AI at all.

A useful rule runs through the whole guide:

AI may help you learn, but it cannot learn for you.

That rule matters because learning is more than producing a correct-looking answer. Learning means you can explain ideas, make choices, test claims, notice errors, and transfer what you know to a new situation. A polished AI response is not the same as understanding.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The distinction is older than AI. What we value in mathematicians, and in quantitative scientists broadly, is knowing whether an answer makes sense before verifying it. No calculator can do that part of the work. That gap, between an answer and the judgment that the answer is correct, is exactly where your work with AI lives. AI tools will sometimes produce confidently wrong answers or miss context that you can detect only if you have done the underlying work yourself.

Look again at the calculator, because the comparison holds one way and breaks the other. A calculator is reliable. Ask for the square root of seven and it runs a fixed procedure and returns the same answer every time, and you can build on it without checking. AI does not work that way. Ask it the same question twice and you may get two different answers, with no built-in way to know whether either is right. It can be fluent, confident, and wrong at once. You can lean on a calculator. You cannot lean on AI until you have checked what it gave you. That is why AI is a partner you think with, not an oracle you obey.

The break generalizes. Earlier tools extended execution but could not exercise judgment, so the work relocated toward judgment and stayed human almost by default. AI is different. It can perform the reasoning itself, not only the execution, so that relocation is now a choice that takes deliberate effort rather than something that happens on its own.

Your own sense of how much you have learned can mislead you for the same reason. A pre-registered trial followed nearly a thousand high-school mathematics students [11]. Those who practiced with unrestricted AI did not notice that they had learned less than students who worked without AI. Then the closed-book exam made the gap visible. The work felt successful because the AI produced correct-looking answers. This guide's emphasis on verification, explanation, and self-checks exists to bridge that gap between what polished AI output looks like and what you actually understand.

Put positively, this is where your value as a learner now lies: not in producing a fluent answer, which a tool can do in seconds, but in knowing whether the answer in front of you is right, relevant, honest, and useful. That judgment is what this guide is built to develop. None of this depends on AI being unreliable. As the tools improve, a fluent answer gets cheaper to produce, while judging one takes as much work as it always did. So the gap between how polished the work looks and what you understand grows wider, not narrower.

Current guidance on AI in education points the same way. UNESCO's student framework emphasizes responsible citizenship, critical judgment, and human-centered use of AI [90]. OECD's 2026 education outlook names the policy challenge as making generative AI a partner in learning rather than a shortcut around it, and names the central concern: the gap between how good student work looks and how much learning it contains [98]. Research on study habits agrees. Practice testing and spaced practice are the two best-supported study techniques, and both make you do work that passive exposure does not [32].

One account from outside the classroom shows the standard at work. Meghan O'Rourke, a poet who teaches creative writing at Yale, tested these tools daily and wrote up what she found [100]. She is not a friendly witness. She calls the technology an environmental problem, objects that the models were trained on writers' work without permission, and fears for students who will never feel what is missing from a fluent, empty paragraph.

She also reports that the tools gave her something back. They wrote her memos, drafted job postings, built editorial checklists, and untangled a summer schedule. Work she used to put off became manageable. Lasting effects of illness limit her hours at a screen, so those hours mattered to her. The work did not go away. The parts around the writing were carried by something else, and the energy went where she wanted it.

Then she watched the line move. When the tool drafted an email from her notes, she sent something that did not say what she meant. When she asked for reading on contemporary poetry and it returned a syllabus, its version stayed with her and crowded out her own. The same tool

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

helped her and cost her, depending on which work she handed it.

Her case has a limit. Most of a student's week is the work itself, not the chores around it, so this trade buys you less room than it bought her. One account is not evidence either, and nothing here rests on it. It earns its place because a skeptic, testing carefully, ended up where this guide begins. AI can carry the work around the thinking. Give it the thinking and what comes back is fluent and no longer yours.

Some students arrive at this guide with strong reservations about engaging with AI at all. The concerns are real: data-center electricity and water demand, labor conditions in AI development, concentration of economic power, copyright and creative ownership, and the case that routine use hollows out the practice itself [134]. These worries cannot be waved away, and a guide that ignored them would not deserve the trust it asks for. The companion paper "Teaching AI Responsibly to Principled Skeptics" [56] makes the case for engaging with AI critically rather than refusing it outright. It works through five layers of literacy (technical, environmental, ethical, practical, civic) and treats principled refusal as a defensible position, not a problem to be talked around.

In brief, technical literacy is understanding what a model does well enough to judge its output: at its core it predicts likely text, and fluency is not the same as reliability. Environmental literacy is reading the resource costs at the right scale, telling a question about one prompt from a question about industrial deployment. Ethical literacy is naming which concerns are at stake in a specific use, from labor and privacy to bias and ownership, and weighing them when they conflict. Practical literacy is the discrimination that use builds: knowing, before you prompt, whether the output will help. Civic literacy connects these to policy, what an institution discloses, what rules apply to data centers, and where advocacy is needed.

Most readers of this guide are deciding how to engage with AI, not whether to engage at all, and the guide proceeds on that assumption. The judgment it builds, though, is the same judgment principled refusal requires.

Much of this guide speaks to students who want to learn and want the work done at the same time. Some students do not feel that pull. A required course outside a student's major can look like a hurdle rather than a chance to build something, and when a tool can produce the answer, the reason to struggle through it is not obvious. That reaction deserves a direct response rather than an assumption that everyone already values the work.

The case for doing the work does not rest on the final answer. It rests on what the effort builds. Facing a blank page and finding a first move is a skill. Holding a half-formed idea long enough to test it is a skill. Sensing whether a result is reasonable before checking it is a skill. None of these transfer from watching a tool produce a finished product, and all of them matter in work that has nothing to do with mathematics. A student who never builds them is not freed by the tool. The student is left dependent on it, and dependence is easy to mistake for fluency until the tool is taken away.

The comparison to calculators shows this too. Calculators did not end arithmetic, but a student who reached for them before building number sense could lose something real, and the loss would be hard to see from inside. The point of learning to do the work is not that a student will always do it by hand. It is that the judgment built by doing it is not something a tool can transfer to you.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

There is a longer reason, and it reaches past any one course. Most students entering college today will spend their careers on problems, in jobs, with tools that do not exist yet. Nobody can hand you that content in advance. What the tools take is the routine part of the work. What stays is the judgment, and that is what an employer pays for. What you can build now is the ability to learn the next thing: face unfamiliar material, form an approach, test it, and know when you have understood. And having that ability is not enough on its own. Graduate programs and employers will ask you to show it, and the verification and disclosure habits in this guide are how you learn to show it as your own work.

The content will change. Learning how to learn is what transfers.

This guide turns those principles into habits you can use in any course.

1.2 ▪ Setting up your AI for learning

Two things matter before you ask AI anything for coursework. The first is that you have configured your AI to behave the way this guide's scripts assume. The second is that you have made an honest attempt at the problem yourself. Both are easy to skip, both shape every AI interaction that follows, and both are subjects of this Part. This section addresses the first; the Learning Spiral, the study routine set out in chapter 3, addresses the second. If you would rather start with something shorter, the repository has a ten-minute version at `students/start-here.md`.

Out of the box, an AI is tuned to impress in a demo: verbose, over-helpful, and quick to deliver answers. Those defaults are not designed for learning. By default, an AI will explain too much, offer solutions when hints would serve, and skip the step where it asks what you actually understand. A fresh conversation knows nothing about you. Unless you personalize the AI, it does not know if you are a kindergarten student or a Nobel laureate, so it answers at a generic level pitched at no one in particular. You spend extra effort translating the response to your level. Or worse, you take at face value an answer that was calibrated to someone else.

The attempt-first precondition is the more familiar of the two and the more easily skipped in practice. A student who reaches for AI before trying produces a particular failure pattern: they recognize the AI's solution but cannot generate the analogous one on their own, and they often cannot tell the difference. The pattern has a name in this guide. Call it the **AI-first trap**. Trying first does not mean grinding through to a complete solution before allowing yourself to ask. It means attempting enough to know where you are stuck and what kind of help would be useful. The Learning Spiral develops this discipline. The trap has a second cost, taken up in section 4.4: it skips the practice that teaches you what a right answer looks like.

When using AI, the prompt matters, but the preparation matters more.

The remedy for the configuration gap is straightforward. As of this writing, the major AI platforms let you set persistent preferences (sometimes called custom instructions, system prompts, or memory) that apply to all your conversations. A short, well-chosen configuration makes the difference between an AI that gives you generic explanations and an AI that acts as a tutor pitched roughly to your level and your goals. Experience with academic use suggests six categories to specify:

1. *Who you are and what you are studying* (year, department, current courses, prior coursework).
2. *How you want explanations pitched* (step-by-step reasoning, ask before assuming prerequisites).
3. *How you want uncertainty handled* (flag claims to verify, do not invent citations, say when an answer rests on a condition you did not state, and say when a number is an approximation rather than an exact value).

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

4. *Writing style preferences* (English variant, native vs. non-native, tone).
5. *Scope of use and integrity boundary* (what you will and will not use AI for; assignment-level rules take precedence).
6. *Optional: technical-work accuracy guard* (flag claims that need checking and name the method, source, or calculation to verify them; explain non-obvious code lines).

Two of these categories ask the AI to flag what it is unsure of. That helps you find what to check. Flagging does not replace the checking, and a confident tone does not show that a claim is right.

The configuration template in chapter 7 gives copy-and-adapt language for each category and a brief discussion of why each matters. Spend ten minutes setting these. The payoff is the conversations that follow, pitched closer to your actual level and goals, with fewer wasted exchanges, and a written record of the use boundaries you have committed to. Students who would prefer to be walked through this setup interactively can use the Define Personal Preferences walkthrough in the companion repository (`companion/dpp/`).

Try this now, before reading further. Write categories 1 and 2 for yourself, two or three sentences in all, and save them in your AI's settings. If your instructor posts a course version of these settings, or the course's AI rules, start from those and add what is particular to you. Then ask the AI a question from this week's course material and watch how it answers. The next subsection shows what the difference looks like for one calculus question; your own before-and-after is the version that will convince you.

1.2.1 ▪ Why this step matters: a before-and-after

A concrete case shows the difference between a default AI and a configured AI. A calculus student studying for a quiz might ask:

> *Help me find the limit of* $(\sin x)/x$ *as* $x \to 0$.

A default AI, untuned and over-helpful, will often answer immediately with the full machinery: "The limit is 1. The most elegant proof uses the squeeze theorem: for small positive x, we have $\cos x \leq \sin(x)/x \leq 1 \ldots$," followed by L'Hôpital's rule, the Taylor series argument, and one or two corollaries the student did not ask for. The student now holds the answer and three proofs they never requested. None of the thinking that produces understanding has happened.

The same student asking the same question of an AI configured with the six categories above receives something closer to:

> What have you tried so far? If you graph $y = \sin(x)/x$ near $x = 0$, what value does the curve appear to approach? Once you have made an observation, I can help you build the formal argument.

The configured response asked back instead of answering. It pointed toward a representation (graphing) that builds intuition before formalism, and it promised help *after* the student had done some thinking, not instead of it.

What configuration does not change is whether the answer is right. It changes how the AI responds, not what it knows. The configured tutor above can ask exactly that question about $\sin(x)/x$ and still be wrong about the limit, and it will ask just as confidently either way.

This matters because the studies behind this book used tutors that had something you do not have. In the high-school trial above [11], the guardrailed tutor that protected learning carried in its prompt the solution to each problem and the mistakes students commonly make on it, supplied by their teachers. That is what kept its hints pointing the right way. A tutor told to give hints, with

no answer key behind it, gives confident hints toward the wrong place. Your configured AI is the second kind. That is why the verify step is not diligence for its own sake: it is the part that stands in for the answer key you do not have.

Tools built for learning change fast. Some chatbots now offer a study mode that asks questions instead of giving answers, and some courses provide a tutor that will not hand over solutions. Use them, but know their limits: you can switch a study mode off with one click, and none of them checks that its answer is right. The companion repository keeps a current list, with what the evidence shows (`companion/learning-tools/`).

The model is the same in the default and the configured case. What changed is what the AI was told: who is asking, and what kind of help is wanted. Ten minutes of configuration buys this shift for the conversations that follow in your coursework, thousands of exchanges over a semester, each one aimed at where you actually are instead of at no one in particular.

A practical note: when an AI gives you a default-style verbose response despite your configuration, one of two things has happened. Either the configuration was forgotten across a session boundary, or the preferences were not yet active when the conversation started. Most platforms apply persistent preferences to all new conversations, but the application is not always perfect, and platform updates can reset settings. If the verbose default returns, check that your preferences are still active before assuming you have configured something wrong.

1.2.2 ▪ The prompt is only the visible part

The persistent preferences in the previous subsection set the stage. They establish who you are and how you want the AI to engage with you in general. Each individual conversation still benefits from a brief setup that names what you are trying to do, what you already know, where you are stuck, and what kind of help would be useful. A good prompt is not magic. It works best when the AI has been given the setting, the purpose, the audience, the constraints, and the standard of success before it begins. The quality of the response usually depends less on the cleverness of a single question than on the world built around it.

A weak setup says, “Explain this topic.”

A stronger setup says, “I am writing for first-year students who are new to mathematical modeling. They are nervous about equations but curious about real-world problems. Explain this idea using plain language, one concrete example, and no unnecessary jargon. Emphasize intuition before notation.”

The second version does not merely ask for information. It gives the AI a role, a reader, a purpose, and a standard of success. That changes everything that follows.

Few lessons about AI use matter more. Skilled users design the context. They say what they already know, where they are confused, what kind of answer would help and what kind would not, and they give the AI enough structure to be useful without simply handing it a template to fill in.

The principle applies most directly to learning. A student who asks, “Do my homework problem,” may receive an answer and learn very little. A student who says, “Here is my attempt. I think the error is in the second step, but I am not sure why. Please ask me one question at a time rather than giving away the solution,” has built a much better learning environment. The prompt is only one piece of the request. Behind it sits the deeper act: setting up the conditions for good thinking.[1]

A good prompt asks a question. A good setup teaches the AI what kind of answer

[1] Composing prompts in writing is not the only option. Most AI platforms accept voice input or transcribed dictation, which can help students whose spoken expression is more fluent than their written expression, for example, students whose first language is not English. Speak the question, let the platform transcribe it, then refine the wording in writing before sending. The underlying skill (setting up the situation, naming what kind of help would help) does not change with the input modality.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

matters.

A short analogy makes the point stick. Asking AI for help without setup is like walking into a tutoring session, dropping a textbook on the table, and saying, "Teach me." You might get something useful. You will do much better if you say, "I understand the definition, I am lost in the example, and I want help finding the first step without being told the whole solution." The better you describe the learning situation, the better the AI can help you learn.[2]

1.3 ▪ Two skills, not one

Using AI well in learning takes two distinct skills, and it helps to name them separately. The first is **tool-use skill**: operational fluency with AI as a piece of software. It covers prompt craft, giving context for a request, the differences between platforms like Claude and ChatGPT, recognizing patterns in AI output, knowing when to switch tools or start a new thread, and using features like voice input or file uploads when they help. Tool-use skill transfers across domains. A student competent here can use AI productively for almost any task they would otherwise do without AI.

The second is **learning-with-AI skill**: a distinct competence with little overlap with the first. It includes knowing when to let AI explain a concept and when to struggle through it yourself. It includes recognizing the difference between productive struggle and unproductive frustration, and choosing not to ask AI a question you should work out. It also includes verifying AI output against your own developing understanding rather than only against another AI, and noticing when you have been handed a confident-sounding answer that has not built any knowledge in you. A student can be sophisticated in one of these skills and a novice in the other, in any combination. Tool-use skill is closer to general digital literacy. Learning-with-AI skill is closer to metacognition, the habit of watching and steering your own learning. Both matter. They are different enough that treating them as one thing makes it harder to develop either. Much of what follows in this book is about building learning-with-AI skill. Tool-use skill is well covered by other resources. Learning-with-AI skill is the one most directly connected to whether your time in school produces durable understanding.

Both skills face a pressure that has nothing to do with your intentions. AI systems learn from human approval, and people approve of answers that agree with them and of help that arrives quickly. Systems trained that way drift toward agreement, and agreement invites dependence. Studying conversations about personal disputes rather than coursework, Cheng and colleagues found that sycophantic responses, the kind that flatter and agree, left people more willing to return to the AI and less willing to repair the dispute [24]. The moment to watch for is not the one where you ask the AI to do your work. It is the one where you have set up the session well, started thinking, and the AI offers to finish the problem for you. Intentions at the start of a session do not survive that offer on their own. They need a setup that anticipates it.

1.3.1 ▪ The mentor prompt

A mentor prompt is a system prompt designed to develop both skills at once. It tells the AI how to guide your learning across many tasks, not just what help you want on this one. It also tells the AI how to pitch its help to where you are, and how to notice when that needs to change. A well-designed mentor prompt is cheap to write and pays off in every session that follows. The

[2] Students who want to develop this craft more deliberately can draw on several kinds of resources: platform-provider documentation for setting persistent preferences and constructing well-scoped prompts; university AI-use guidance maintained through teaching-and-learning centers, writing centers, or libraries; accessible books on AI literacy (*Co-Intelligence* [92] matches the orientation this guide develops); and open educational resources. These four categories are intended to age well even as specific instances within them change; the guide deliberately does not endorse specific commercial products or prompt-engineering courses.

same student doing the same assignment can produce very different learning outcomes depending on what the AI was set up to do.

A mentor prompt plays a different role from a tutor prompt. You use a tutor prompt for the content of your assignment: the calculus problem, the paragraph, the code. You use a mentor prompt to set up the kind of help you want, pitch it to where you are in your learning, and check now and then that the help is actually helping. In a single AI session, the mentor's setup carries through to whatever tutor work follows. On some AI platforms the mentor and tutor roles are the same interface used differently; on others they are separate invocations. Either way, the mentor does the same job.

A mentor prompt shapes how the AI works with you across many tasks. It does not make the AI correct. The tutors in the studies were made reliable by loading the answers into the prompt, which is not something you can do for work you have not solved yet.

The account configuration set out in the previous section is a starter mentor prompt. A useful mentor prompt does four things that the starter does not yet do.

First, it asks you to name your starting point on two axes, separately. The first axis is your tool-use skill: how comfortable you are with AI as a piece of software, with prompting, with the differences between platforms, with managing threads. The second axis is your learning-with-AI skill: how practiced you are at deciding when to use AI versus when to work through something yourself, at verifying AI output against your own understanding, and at noticing when an AI explanation has not actually taught you anything. A student can be advanced on one axis and a novice on the other in any combination. The mentor adjusts based on both.

Second, it makes its placement visible. The mentor tells you what level it has placed you at and what it will and will not do as a result. You can correct it at any point. If the mentor's assessment feels wrong, say so, and the mentor adjusts.

Third, it builds in periodic check-ins. Every few problems or every few sessions, the mentor pauses to ask whether the level it is treating you at still feels right. Check-ins mitigate a real concern in self-classification: you may not know exactly where you are when you start, and the mentor's first impression may be off. The placement then develops from accumulated evidence rather than a one-shot upfront guess.

Fourth, it offers an alternative for students who would rather not self-classify: an auto-adjust mode that starts with the most help for everyone and reduces the help as you demonstrate that you no longer need it. The mentor tells you when it is reducing help and why, so you can object if it pulls back too far.

A sample mentor prompt that does the four things above:

> *I am about to ask you for help with [course or subject]. Before we start, I want to set up how you should help me.*
>
> *On a scale of novice, intermediate, or advanced, my tool-use skill with AI is [your level here]. My learning-with-AI skill is [your level here]. If I am unsure of my level on either axis, run in auto-adjust mode for that axis: start with the most help and pull back as I show you I do not need it.*
>
> *Based on those levels, please:*
>
> *Tell me what level you are treating me at and what you will and will not do as a result.*
>
> *When I ask you a content question, default to asking me what I already understand before giving an explanation. If I have not made an attempt at the problem yet, ask me to try first.*
>
> *When a claim rests on a specific fact, date, citation, or number, flag it as something I should verify, and where you can, name the source or calculation to check it against.*

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

> *Your own confidence is a hint about where to look, not evidence the claim is right.*
>
> *Periodically (the frequency I set: by default, every few problems), pause to ask whether the level you are treating me at still feels right.*
>
> *When you are in auto-adjust mode, tell me each time you give less help so I can object if you pull back too far.*

You will adapt this prompt to your specific subject and your specific platform. On Claude, the prompt fits naturally into the system-prompt field or the first message of a conversation and persists across the session. On ChatGPT, you may want to re-invoke a shorter version of the mentor prompt at the start of each session since custom instructions are more limited. These platform specifics are a mid-2026 snapshot and will shift as the tools change, and the companion repository keeps current notes.

Some instructors will require a mentor prompt at the start of certain assignments, often paired with a request to share the full thread of your AI conversation. The required mentor prompt provides the same support that is useful in private use, and the shared thread also gives the instructor diagnostic material about where their students are struggling, which they can use to adjust their teaching.

A mentor prompt is not a complete solution to learning with AI well, and it does not replace good judgment about when to use AI at all. It is one support among several. The Learning Spiral, the main habit this guide asks you to practice, is the other main one. If an instructor asks you for a Plan-Use-Verify write-up (a short plan before you use AI, a note on how you used it, and a record of how you checked the result), that is not a third thing to learn but one pass of the Spiral, written down in the form your instructor can grade. Each of these supports is most useful while you are still building learning-with-AI skill. Over time the training wheels come off: the supports become habits, and the explicit mentor prompt becomes less necessary because you are already doing what it asks.

1.3.2 ▪ Keep the tutor inside your course

A good tutor stays inside what your class has covered so far. The AI does not know your class. It does not know your textbook, your syllabus, or how far you have gone, so when you ask for help it reaches for the most efficient method it knows, which is often one your course has not reached.

Take the limit of $\sin(x)/x$ as x goes to 0, which we met earlier in this Part. A student three weeks into calculus asks the AI for help with it, and the AI answers with L'Hôpital's rule, a method from a later chapter. Two things go wrong. The method is one the class has not learned, so the explanation connects to nothing the student already knows. And it is circular. L'Hôpital's rule uses the derivative of the sine function, and that derivative is proved using this very limit, so the AI has assumed the answer in order to reach it. The student is confused, and half-taught that the course is doing things the hard way.

The fix is to tell the tutor where your class is. You may not know this offhand, and that is normal. The syllabus usually lists it, the table of contents of your textbook is a reasonable substitute, and your instructor will tell you if you ask. Give the tutor two things. The first is a map: what your course covers, and in what order. The second is a pin: where you are on that map right now. The map alone is not enough, because a full syllabus lists the later methods too, and the AI can still reach for them. The pin sets the boundary. A sentence is often all it takes:

> *We have covered limits, the limit laws, and the squeeze theorem. We have not covered derivatives or L'Hôpital's rule. Help me using only what we have covered, and if a problem seems to need something later, tell me instead of using it.*

You rarely need to upload anything at all, and that is the safest habit for a second reason: whatever you upload leaves your hands. When you want more precision, upload the table of contents or the course syllabus up to the current section as the map, and still say where you are as the pin. That is usually enough, and copyright may keep you from uploading the whole book in any case; if you want the AI aligned to a full text, an openly licensed one such as OpenStax allows it. Keep the scope in your AI's persistent settings or a project so you are not retyping it every night, and update it as the term moves. The strongest version is not yours to write: an instructor who states what is in scope for each assignment sets the boundary once, the same way for every student.

Class notes make the most exact map. If your instructor posts typed notes or slides for this purpose, start from those. Otherwise, photograph or scan your own notes after class; most general-purpose AI tools can turn the handwriting into a typed document, in Word or in LaTeX. Check the transcription against the page before you rely on it, and check the equations with special care: a dropped subscript or a flipped sign changes the mathematics. The check doubles as a first review of the lecture. Then give the typed notes to your tutor as the map and the pin together. They show what your class covered, in your instructor's order and notation, and they stop where the class stopped. Add each week's notes to the same project, and the map grows with the course. Your own notes raise fewer copyright questions than a textbook, and your instructor's files are fine to upload when your instructor says so. Tell the tutor that your notes may contain mistakes, and ask it to point out anything that disagrees with your textbook rather than build on it.

A scope statement reduces the problem. It does not fence it. The AI can still cross the line, and you may not always see where the line is, so this is one more reason to keep verifying: when a method looks unfamiliar, stop and ask why before you use it. The concern reaches past calculus. A programming course may bar a library it has not taught. A statistics course may not have reached regression yet. A proofs course builds in a fixed order. The shape is the same in each, and so is the remedy: tell the tutor how far the class has come.

1.4 ▪ Learning outcomes

After using this guide, you should be able to do five things.

1. Distinguish help that supports learning from help that replaces learning.
2. Match your AI use to the assignment category chosen by your instructor.
3. Verify AI-assisted work using methods appropriate to your discipline.
4. Disclose AI use briefly, honestly, and usefully.
5. Explain and defend submitted work as your own intellectual responsibility.

These outcomes are modest by design. The guide does not ask you to become an AI expert. It asks you to become a better learner in a setting where AI is available: to build judgment that keeps producing good answers, rather than collecting one answer at a time.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 2

The Problem and the Compact

2.1 ▪ The Student's Trilemma

In the gap between what you can do
and what is due,
something always gives.

Before we begin, let us meet a student named Wendy. In May 2025, Wendy was interviewed for *New York Magazine* as part of a long piece of reportage on AI use in college.[3] She had just turned in a five-page paper due that morning. A chatbot gave her the outline and topic sentences, and she filled them in. She told the reporter she still liked writing, and remembered her high-school English class with something like nostalgia. But she would rather get good grades. With ChatGPT, she said, you don't really have to think that much.

Wendy is not the villain of this book. She is its central reader. She has correctly read the incentive structure she lives in, and she is paying a cost she can already feel. Her predicament has a name. Call it the **Student's Trilemma**: a three-way pull among earning the grade, being honest about how the work happened, and showing that you actually understand. A dilemma forces a choice between two things. This one has three. Wendy is one student's story, an illustration of the trilemma and not the evidence for it; the evidence comes from the studies and surveys cited through the rest of the book.

Most of the time there is no pull at all. If you can already do the work at the standard the course is grading, you write it yourself, you say so, and all three hold together.

The pull starts in the gap. When the assignment asks for more than you can produce on your own right now, and AI can close that gap in a way nobody can see, the three stop being compatible. You can hold any two. The third gives way.

Hold the grade and your honesty, and understanding gives way. You submit good AI-assisted work, you disclose it, and you still could not rebuild it. Hold the grade and your understanding, and honesty gives way. You learn from the AI, you get there, and then you shade how much it helped. Hold your honesty and your understanding, and the grade gives way. You do the work yourself, you say exactly what you used, and you are graded beside classmates who did neither.

Each pair has a name, and this section takes them one at a time.

Here is the argument of the whole book. This tension is not a law of nature. It is what a course looks like when the assignment, the grading, and the disclosure rules were all built for a world without AI. A well-designed course dissolves the trilemma. Naming the trilemma is how we start.

[3] Walsh [130].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

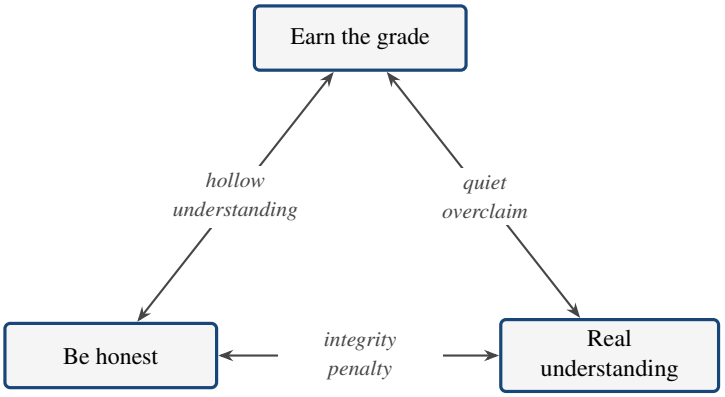


Figure 2.1. ***The Student's Trilemma.*** *Any two of these goals can be held, but only at the cost of the third. When the work is within the student's reach, all three hold together. Each edge labels the failure mode that arises when the two adjacent goals are satisfied at the cost of the third. The text develops the modes. AI sharpens each edge; it did not create the trilemma.*

The trilemma is not just one student's predicament. Recent surveys document how widely it is felt. The 2026 edition of the United Kingdom's Higher Education Policy Institute survey reports that 95% of students now use AI in some form, up from 92% in 2025 and 66% in 2024, and that 94% use it for assessed work [42, 119]. A 2025 Inside Higher Ed Student Voice survey of more than a thousand students from 166 institutions found that 97% of students want their institutions to address academic integrity, and that they favor prevention over punishment in doing so [40]. A study at King's Business School found that up to three-quarters of students, depending on the module, did not complete a mandatory AI-use declaration. Fear of grade penalties or of being suspected of misconduct, and confusion over what counted as AI use, were among the reasons [46]. The trilemma names the structural predicament these numbers describe: AI use is widespread, disclosure requirements are often ignored, and the gap between performance and underlying competence is widely felt but rarely acknowledged.

2.1.1 ▪ Three student failure modes

Hollow understanding. From the outside, it looks solid, but there is nothing inside. The student earns the grade and discloses AI use honestly but cannot reproduce the work. The submitted work is fluent, organized, and correctly cited; the disclosure statement is candid; the underlying competence has not formed. The student does not necessarily realize this in the moment. They have read the AI's solution, found it convincing, and submitted it, and what they have not done is reconstruct the steps themselves under conditions that would reveal the gap. Hollow understanding is the failure mode that is invisible from outside, and often from inside too.

Quiet overclaim. The student earns the grade and understands the material, but hedges on how AI helped. They used it for a first draft, for a confusing example, for catching errors. The learning did happen. But the account they give is shaped to make the AI's part look smaller than it was, partly from expected suspicion and partly because the rules keep changing. The disclosure is technically true and substantively misleading.

It is tempting to treat this as the harmless one. The student understood the material, so what was lost? What was lost is the instructor's ability to tell. Disclosure is the only channel through which an instructor learns that a confident-looking piece of work sat on top of a struggle. Without it, the same polished page says "this student has it" whether or not they do. Every hedged

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

disclosure also makes honest ones harder to interpret. As a result, honesty costs the next student more.

Integrity penalty. The student is fully honest about both their AI use and the limits of their understanding. They disclose what they used AI for, what they verified independently, where they remain uncertain. Their work is documented and defensible. And they pay for the transparency. Their disclosed work is graded beside an identical undisclosed draft from a classmate, and the two come out the same. Sometimes worse, if the disclosure is what draws the suspicion in the first place. The student who refuses to overclaim pays the cost of refusing.

The three failure modes do not cost the same. Hollow understanding accumulates over time, hollowing out skill the student will eventually need. Quiet overclaim protects the grade but wears away the habit of giving an honest account of one's own work. The integrity penalty costs the grade but keeps both competence and integrity. Different students under different pressures will find a different mode the least bad one where they stand. The framework does not tell students which mode to choose. It names the structure so the choice is visible.

The three failure modes are also a tool for catching trouble early, while you work. If you could not rebuild the work without AI, you are sliding toward hollow understanding. If you are shading how much AI actually helped, you are sliding toward quiet overclaim. If staying honest is about to cost you the grade, you are sliding toward the integrity penalty. When you notice one of these, do not give up that corner. Look for the smallest change that restores it without letting the other two slip.

That works cleanly for two of the three. Rebuilding the work from a blank page restores understanding; disclosing fully restores honesty. The integrity penalty is the exception. When staying honest is what costs you the grade, the problem usually sits in the assignment or the grading, not in your own work. The smallest change that would fix it may not be yours to make. The responsible move there is to surface it rather than carry the cost alone: ask for clarification, an accommodation, an extension, or institutional help.

No fixed ranking works everywhere. The right balance depends on your situation, and it changes as the situation does.

2.1.2 ▪ How the trilemma manifests differently for different students

The trilemma is universal, but its pull patterns are not. The three corners weigh differently depending on the student's situation. Four dimensions of variation matter most, and they are pressures that compound for individual students rather than discrete categories.

Economic stakes around the grade. For a first-generation college student whose family is funding the degree as a generational investment, the grade carries a weight that students from families with a history of college do not feel in the same way. Educational researchers have documented an AI digital divide along this line: in a 2023 survey at nine colleges, first-generation students reported lower awareness and use of ChatGPT than continuing-generation students [105]. The first-generation student feels the quiet-overclaim edge most sharply, because the integrity penalty costs more when more rides on the grade.

Consequence severity around honest disclosure. For an international student on an F-1 visa, an academic-integrity finding reaches well past the grade. Suspension or expulsion can lead the school to terminate the student's SEVIS record, which ends F-1 status. As of this writing, once that happens there is no grace period: the student must leave the United States or apply for reinstatement, and immigration authorities may open removal proceedings.

This student is squeezed from both sides, which is what makes the corner so sharp. Disclose fully and the work may be judged more harshly than a classmate's identical undisclosed draft. That is the integrity penalty, with the stakes raised. Say nothing and a detector false positive can start the investigation anyway, and AI detectors flag non-native English writing at documented

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

higher rates [79]. Neither corner is cheap. A domestic classmate making the same two choices is risking a grade. This student is risking their status.

Career stakes around demonstrated understanding. For a pre-professional student preparing for medical school, law school, or competitive engineering programs, what the credential signals depends heavily on proof of independent capability. Hollow understanding, high grades over weak competence, sends its bill later, when the graduate enters a workplace that tests them directly. The pre-professional student feels the hollow-understanding edge most sharply, because the credential is vouching for abilities they have not built.

Time to absorb the integrity penalty. For working students, students with caregiving responsibilities, and other non-traditional students, the time it takes to use AI sustainably (try first, verify independently, document the process, write transparent disclosures) competes directly with paid hours and family obligations. For students whose lives leave no slack for all of that, quiet overclaim is more often a condition of their circumstances than a choice.

A single student often faces several of these pressures at once. A first-generation international pre-medical student who works the twenty campus hours a week the visa allows experiences all four. The framework's claim is not that this student should adopt some single right response, but that they should be able to see the structure they are responding to. You may see yourself in one of these pressures, or in several at once. The point is to recognize the pull you are actually under, so that what you do next is a choice rather than a reflex.

2.1.3 ▪ What makes the trilemma hard to see

The trilemma has one feature that makes it harder to fix than most problems students face: **it is invisible to the people best positioned to remove it.** Hollow understanding is invisible from outside; it looks like the real thing. The quiet overclaim sounds plausible to instructors who are not asking detailed questions about it. The integrity penalty is paid silently: honesty earns no grade advantage, and the student rarely says so, because saying so would sound like a complaint.

Instructors designing courses and administrators setting policy often do not see the trilemma their students are navigating. The failure modes are structured to be unobservable from the instructor's vantage, and instructors and administrators face their own version of the same predicament. The framework's value to instructors and administrators is partly that it makes the student-side trilemma visible from the outside. The framework's value to students is that it names what they have been feeling without a vocabulary for it.

One direct counter-argument needs an answer. A student interviewed in the same article that introduced Wendy asked the obvious question: *If I am being asked to go from point A to point B, why wouldn't I use a car to get there?*[4] The answer is that the comparison fails because the road is the destination. The point of the homework is the walking, and the value of the walking is what the rest of this guide develops. The same is true, in a different register, for an essay, a problem set, or a research paper. The task that looks like a destination is usually a journey whose value lies in being traveled.

That qualifier matters, though. Not every task is a road worth walking. Some are errands, where the destination is the point and the walking is only transport, like reformatting a bibliography or converting a table from one format to another. So the judgment you need is which kind of task you face: is this path the competence the course exists to build, or incidental labor on the way to it? When the walking is the competence, do it yourself and let AI check it; when it is only transport, let AI carry it and spend your effort on what is actually the destination. Telling the two apart is itself one of the skills this guide teaches.

[4] Quoted in Walsh [130].

2.1.4 ▪ Why the Student's Trilemma matters for the rest of Part I

The rest of this Part turns the trilemma into practice. The Learning Spiral is designed to lower the integrity penalty. A student who tries first, verifies, and works independently produces evidence that makes honest disclosure defensible. The CAT framework aligns the three corners by separating what competence requires (Core Competence), what responsible AI use looks like (AI-assisted Practice), and what trustworthy self-account involves (Trustworthiness). The translation test is the diagnostic designed to catch hollow understanding before it becomes a credentialing problem. The disclosure templates are designed to make honest disclosure both possible and proportionate, reducing the structural pressure toward quiet overclaim.

The same three corners appear for instructors and for institutions. They line up one for one, as the table below shows. What the grade is to a student, a manageable workload is to an instructor and financial viability is to an institution: the thing you have to produce to keep going. What honesty about how the work happened is to a student, a grade that can be defended is to an instructor and a credential that means what it says is to an institution: the claim that has to be true. What understanding is to a student, student learning is to an instructor and the educational mission is to an institution: the reason any of it matters. The three failure modes carry over under the same names.

Table 2.1. ***The same three corners, at three scales.*** *Read across a row, not down a column. Each row is one corner of the trilemma. The cells show what it is for a student, an instructor, and an institution. Nothing is inverted; the corners line up one for one.*

For a student	For an instructor	For an institution
Earning the grade	A workload that can be sustained	Staying financially viable
Being honest about how the work happened	A grade that can be defended	A credential that means what it says
Showing genuine understanding	Student learning	The educational mission

2.2 ▪ How learning happens, and how AI changes it

Wendy's predicament has the shape it does because of how durable learning actually happens. The practices in this guide are operational forms of principles from the science of learning that have been studied for decades. The principles have a longer evidence base than the AI debates that frame them.

Five findings recur in studies of how durable learning happens.

Practice testing (also called retrieval practice). Trying to recall information, or produce a solution, or reconstruct an argument, builds memory more strongly than re-reading or re-watching it [106]. The act of retrieval is itself a learning event, not just a check on what was already learned. The timing matters, and the result is instructive. In one well-known study, rereading won when the test came five minutes later. Testing won by a wide margin when the test came two days or a week later, and students were more confident after the method that lost. Most assessments measure production. Most studying measures recognition. The gap between them is one place where AI use is risky.

Distributed practice (also called spacing). Material reviewed at intervals, with forgetting and re-retrieval between sessions, is retained more durably than the same material reviewed in one

block [32]. The interval is the work, not the wasted time. A short return to a problem one to three days later usually does more for memory than a long block on the day of first encounter.

Interleaving. Mixed practice across problem types builds the discrimination skills that single-method practice does not [108]. Students who practice ten problems on the same method become fluent in that method. Students who practice ten problems mixing several methods become fluent at choosing which method applies. In mathematics and other quantitative disciplines, the second skill is often the one that matters on exams.

Generation effect. Information that you produce is remembered better than information you read [16]. Closing the AI window and rewriting the key step from memory is the move that converts recognition into production.

Desirable difficulties. Conditions that slow initial performance, such as spacing, interleaving, varied practice, and testing yourself even when you get it wrong, often produce more durable long-term learning for learners with the background to meet them [16]. Easy is not the same as effective. Fluent reading of a polished AI explanation is the easiest study experience available; it is rarely the most effective.

These five findings show up at specific places in the Learning Spiral. *Try to solve* uses retrieval practice and desirable difficulties. *Verify* adds another retrieval pass and a chance for spaced and interleaved follow-up. *Work independently* and the recommendation to redo key steps from a blank page rely on the generation effect. *Reflect and disclose* closes the pass with elaboration and metacognition. Each step has a learning purpose, not just a procedural one.

2.2.1 ▪ The illusion of competence

A persistent finding in the science of learning is that students often think they understand material that they cannot actually reproduce, apply, or defend under questioning. I know the feeling from the other side of the lectern. As a student I listened to Richard Feynman lecture on physics and left each talk certain I had finally understood the subject. Hours later, trying to explain an idea to someone else, I found I could not reconstruct it. Many of my colleagues report the same experience with those lectures. The clarity was real, but it was recognition, not yet the ability to produce the argument myself. Feynman knew the trap, and named it: the easiest person to fool is yourself. The technical name for this gap is **calibration error**: the distance between what you think you know and what you can demonstrate. Calibration error has been studied for decades. AI does not create the problem, but it can widen the gap in ways earlier study tools did not.

The widening shows up in five places.

The recognition–production gap (figure 2.2). Reading a fluent AI explanation produces a strong feeling of understanding. That feeling is reliable for one thing: it tells you that you can recognize the explanation as correct when you see it again. It does not tell you that you can produce the explanation, or the underlying argument, on your own. Recognizing an argument and producing it draw on different processes. AI delivers recognition. Assessments measure production. A student might put it this way: “I read an AI proof, nodded along, then couldn’t reproduce step two.” The smoother the prose, the more carefully you should check that you can produce the argument yourself.

The mastery illusion on transfer. After AI-assisted homework, students often predict they can solve similar problems independently on a quiz. The prediction is often too optimistic. Fluency on assisted work is a poor predictor of unassisted performance, especially when the new problem requires a small variation of method or a transfer to a new context [31].

The effort illusion. Time spent with AI feels productive. The hours register as study hours. But productive learning hours are hours in which you did the cognitive work: retrieval, reasoning, reconstruction, error repair. If AI did that work and you watched, the time is study time only in name.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

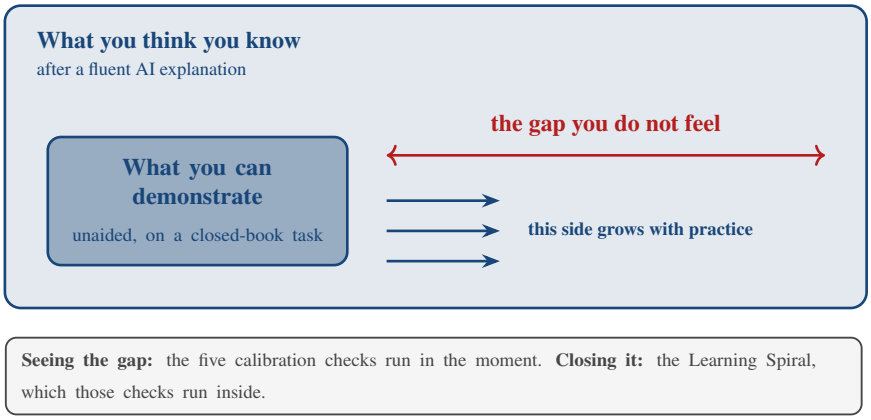


Figure 2.2. ***The recognition–production gap.*** *The outer region is what a fluent AI explanation makes you feel you know, which psychologists call recognition. The inner region is what you can demonstrate unaided on a closed-book task, which they call production. The distance between them is the calibration error a learner does not feel. The gap closes from the inside: practice grows what you can demonstrate, and the five calibration checks are how you see where that boundary currently sits. The regions are schematic and their sizes are not measurements.*

The authorship illusion. In extended AI-assisted writing or proof work, the line between "I had this idea" and "AI suggested it" can become difficult to track. The sensation that the work is yours grows even when most of the structure came from somewhere else.

The tool-use illusion. Being good with the tool is not the same as being good at the task, and the two are hardest to tell apart when you are good with the tool. Suppose you write a careful prompt, get back clean code for the numerical method you asked for, and the method was the wrong one for the problem. The prompting was excellent. What was missing was the judgment that chooses the method, and AI has no way to report that the question you asked was the wrong question. Skill at prompting is visible and improves quickly. Skill at deciding what to ask for is neither, and section 1.3 treats them as separate skills for that reason. The trap is not overestimating how good you are with AI. It is being right about that and taking it as evidence about the work.

These illusions are not failures of effort or honesty. They are predictable consequences of how memory, metacognition, and fluency interact [16, 31]. The defense against them is not to avoid AI. It is to build in checks that reveal what you actually can do without it.

Most of the practices in this book are calibration checks. *Try to solve* gets a baseline before AI flattens the picture. *Verify* asks you to find errors in AI output, which requires that you understand the work. *Blank-page rehearsal* tests whether you can produce, not just recognize. *Reflection and disclosure* track what you did versus what AI did. *No-AI quizzes* and *transfer questions* surface the gap between assisted and unassisted performance.

Instructors face a parallel problem from their side, and the checks on both sides together are what allow learning to remain visible in an environment where final products no longer reliably show how they were made.

2.2.2 ▪ Calibration checks in the moment

The five illusions develop slowly and become visible only when you try to do something new. Most students do not catch them in time, because the moment of catching is also the moment of submitting. The remedy is to build the catch into the work itself. Five short habits, run in the moment, do most of the work.

Predict before reading. Before you read AI's next sentence, write down what step you expect next, or what conclusion you think it is heading toward. Then read. The gap between your prediction and AI's continuation is your calibration error, made visible one sentence at a time. If the gap is large and surprising, you have learned something. If the gap is large and you find AI's reasoning hard to follow, you have learned that you are a step behind and need to slow down. The check takes a few seconds at each sentence and converts passive reading into active hypothesis testing.

Forced unassisted gap. This check asks whether the help transferred to new work. After AI helps you with one problem, close AI and try the next problem of the same type without it. The drop in fluency between the assisted problem and the unassisted one is a rough probe of how much actually transferred. A small drop suggests the assisted work transferred. A large drop suggests you produced an answer without learning much. Most students do not run this check because the assisted problem feels finished and the unassisted one feels like extra work; that feeling is the effort illusion at work, and is what makes the check valuable.

Blank-page rehearsal. Where the forced unassisted gap sends you to a new problem, this one sends you back to the work you just finished. At the end of each AI-assisted session, close AI, close your notes, and rewrite the key derivation, paragraph, or argument from a blank page. This is the strongest of these checks, because rebuilding from nothing tests whether you can produce the work rather than only recognize it. The gap between your draft and what you can rebuild is the gap between recognizing the work and owning it.

Self-quiz on the next day. Within twenty-four hours of an AI-assisted session, set yourself one question about what you worked on the day before, and answer it without the materials in front of you. This is retrieval practice on a one-day delay; it is also a calibration check, because the question reveals whether yesterday's understanding survived overnight. If it did not, you know to return to the topic before the next class or assignment.

Explanation aloud. After any AI-assisted exchange, say out loud, in your own words, what you just learned. The act of vocalizing forces production. If you find yourself stalling, hedging, or reaching for AI's exact phrasing, you have caught a recognition–production gap in the act. The fix is to return to the underlying material, not to re-read AI's response.

These five moves are micro-checks distributed across the Learning Spiral, run as the work happens rather than at the end. *Predict before reading* sits at the boundary between *Ask* and the next AI turn. *Forced unassisted gap* sits at the transition between *Verify* and the next assignment. *Blank-page rehearsal* runs at the close of *Reflect*. *Self-quiz* runs the day after, on the next pass. *Explanation aloud* runs continuously, in any silence the work allows. Built into the work, calibration becomes a habit rather than a final exam.

These in-the-moment checks are real, but they are coarse, and they are weakest exactly where you most need them: when an illusion is strongest, the same judgment that produced it tends to rate the check as passed. Treat them as early warnings, not verdicts. The measures that do not run on your own judgment carry more weight. They include the structural checks later in this Part (units, limiting cases, an independent derivation, code that actually runs) and, above all, the work you produce under an instructor's no-AI assessment, the one signal these illusions cannot reach.

2.3 ▪ The basic compact

AI-assisted learning works best when students and teachers share a clear compact.

The compact starts before AI does. Try the task yourself first, far enough to find where you are actually stuck. Then you may use AI when your instructor allows it. You may ask it to explain ideas, question you, critique a draft, help you study, compare approaches, or point out possible weaknesses. You may use it as a tool for practice and improvement.

But when you submit work, the work is yours, and you say how AI helped when the assignment

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

asks for it. That means you can explain it without saying, "The AI told me." You know which sources support your claims. You understand the code you submit. You can describe how a paragraph changed during revision. You know what you accepted, what you rejected, and why.

This is not only an academic integrity rule. It is a learning rule. If you cannot explain the work, the work has not yet become part of your understanding.

One concrete diagnostic helps. Call it the **translation test**: take a piece of work and re-present it in a different form. The broadest form of the test asks whether you can say the same thing three ways: visually (a picture, diagram, or graph), mathematically (with notation, equations, or formal structure), and in plain language (the kind of sentence you would use to explain the idea to a curious friend at dinner). A student who can move fluently between all three modes is showing ownership of the idea; a student who can perform only one likely has surface recognition, not structural understanding. The test has limits. Fluency can be rehearsed, and anxiety or an unfamiliar language can mask understanding a student really has, so treat a weak translation as a signal to look further rather than a verdict. The test measures whether you can move between forms, not how much you change the original. A draft you accept with few edits can still pass, provided you could re-present it in another form on demand. The point is portable understanding, not visible rework.

A specific case of the test is the notation change: rewrite the proof or argument using different variable names and symbols. The notation change is a useful diagnostic in mathematics and in any field that uses formal notation, but it is the narrowest version of the test. A student who can rewrite $f(x)$ as $g(t)$ has shown they recognize the structure under one specific kind of relabeling. On its own this is weak evidence: symbols can be swapped by pattern-matching without grasping what they mean. The relabeling has not shown the student could draw the same idea, or explain what is happening to someone who has never seen the notation. Treat it as the entry point to the test, not the test itself; the three-modality form also catches the gaps that relabeling misses.

The statement

If $f'(x) > 0$ for every x in $[a, b]$, then f is increasing on $[a, b]$.

Transcription, which is not the test

"If f prime of x is greater than zero for every x in the closed interval from a to b, then f is increasing on that interval." Reading the symbols aloud. Anyone can produce this line, understanding or not.

Translation, which is the test

"Wherever the slope stays positive, the function only climbs: pick any two points in that stretch, and the one farther right has the larger output." The symbols are gone and the meaning survives.

Figure 2.3. ***The translation test.*** *The middle row reads the symbols aloud; it can be produced without understanding, so it shows nothing. The test is the bottom row: say what the statement asserts in words that never lean on the symbols. If you cannot do that, treat it as a sign the statement is not yet yours. Passing it is the floor, not the proof.*

For non-mathematical work, the test usually runs on the plain-language axis. Paraphrase the passage in your own words, give the same example in a different form, or re-derive the result by a different route. The diagnostic is the same: fluent translation across modes indicates ownership; difficulty indicates recognition. Figure 2.3 shows the contrast on a first-semester calculus statement: the symbol-reading that anyone can produce, next to a translation that carries the meaning without the symbols. The guide returns to this diagnostic throughout.

The test has a further limit. The translation test fits understanding you can put into words and

other forms. Some competence, though, shows only when you do the task, not when you describe it.

The translation test catches more than single-chat overreliance. It also catches AI work produced outside any single, identifiable session, including work whose surface was polished in one chat while the actual content was drafted in another. The test asks what the student can do regardless of which chat produced what. Sharing the conversation link deals with the practical side of this concern; the translation test is the structural safeguard that does not depend on which chat is shared.

The compact has five parts.

Attempt. Try the task before asking AI to do anything important.

Use. Ask AI for support that keeps you active.

Verify. Check important claims, sources, computations, code, and interpretations.

Disclose. State how AI was used when the assignment requires it.

Own. Submit only work you can explain and defend.

Different courses will apply this compact in different ways. A professor may ban AI on an exam, allow it for practice, require it for a critique assignment, or ask you to compare your answer with an AI answer. The exact rule belongs to the course. The responsibility belongs to you. With AI tools the habit is easy to state and hard to keep: verify, verify, verify, then trust.

Some courses adopt the compact as a signed agreement between student and instructor. A printable one-page copy is in the repository (`companion/compact/`).

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 3

The Framework

AI does not prevent mistakes.
It lowers the cost of making, examining, and learning from them.

3.1 ▪ The CAT framework

This guide uses the CAT framework, named for its three dimensions: **Core Competence**, **AI-assisted Practice**, and **Trustworthiness**. These three dimensions help you understand what your instructor may be trying to assess. Trustworthiness is the one most often misread. It is not a judgment about your character. It scores what you did: whether you verified the work, said how it was made, and can answer for it. A student who does not disclose may simply never have been shown how, and the axis should read that as missing instruction rather than missing honesty.

3.1.1 ▪ Core Competence

Core Competence is what you must be able to do without AI, or with only tightly limited help. Every course has a core. In chemistry, it may include explaining mechanisms and interpreting lab results. In writing, it may include building an argument and revising a draft with intention. In history, it may include reading sources carefully. In mathematics, it may include solving problems and explaining definitions. In programming, it may include tracing code and debugging simple errors.

AI does not remove the need for these abilities. It makes them more important. If a tool can generate a fluent answer, your value as a learner comes from knowing whether the answer is right, relevant, honest, and useful.

There is a useful way to see why. Tools that help us think divide into two kinds [72, 97]. Some we can internalize. A student who practices with an abacus or with written place value eventually carries the method in their head and can set the tool aside. The tool leaves a durable skill behind. Others work only while they are switched on. A calculator or a car's navigation gives a correct result but teaches little that survives its removal, and long reliance can leave the underlying skill weaker than before. The first kind builds capability. The second kind rents it. Whether AI becomes one or the other is decided by how a course asks students to use it [19, 20, 133]. An assignment that asks for the answer trains dependence. An assignment that asks the student to attempt the reasoning first, then use AI to check it, uses the same tool to build a skill the student keeps.

Expect some assignments to test Core Competence directly. These may include no-AI quizzes, in-class writing, lab practicals, oral explanations, short exams, source analyses, board work, or

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

live coding checks. These assessments are not a rejection of AI. They protect the part of learning that must remain yours.

3.1.2 ▪ AI-assisted Practice

AI-assisted Practice is your ability to use AI well. This is also a real skill. It includes asking useful questions, judging responses, comparing explanations, finding weak points, generating practice problems, revising carefully, testing code, checking assumptions, and using feedback without becoming dependent on it.

Good AI use is measured by whether AI helped you think better.

A weak use of AI is asking for a complete solution and copying it. A stronger use is asking for a hint, trying the next step yourself, and then checking whether your reasoning holds. A weak use of AI is asking it to write a paper. A stronger use is asking it to find the least convincing paragraph in your own draft and explain why a reader might not believe it.

You can practice the conversion directly. Take the last question you actually asked an AI and rewrite it two ways: once so the AI must give you a hint rather than a solution, and once so it must critique work you supply rather than produce work for you. For example:

> *I am stuck on [problem]. Do not give me the solution. Ask me one question that would help me find my own next step.*
>
> *Here is my draft paragraph: [paste]. Find the weakest claim and explain why a careful reader would doubt it.*

Both rewrites keep the thinking with you: the first asks for a question instead of an answer, the second for a critique instead of a draft.

Chapter 4 practices this conversion across the common use cases.

3.1.3 ▪ Trustworthiness

Trustworthiness is your ability to verify, disclose, critique, and take ownership of **AI-assisted work**: work shaped in some way by an AI tool, including brainstorming, drafting, revision, coding, explanation, summarization, critique, or feedback. This is the ethical center of the guide. It is also the part of your work that an employer cannot get from the tool. The person worth hiring is the one who can tell when the output is wrong, say what the tool did, and answer for the result.

Trustworthiness asks practical questions.

- Did I check whether the AI response is true?
- Did I verify sources rather than trusting generated references?
- Did I protect private or restricted information?
- Did I disclose AI use when required?
- Can I explain what I submitted?

One more question turns inward. Trustworthiness includes self-calibration: knowing whether you actually understand the work, or whether it only looks right to you. The illusion of competence in section 2.2 makes this the easiest judgment to get wrong, and a fluent AI answer makes it easier still. The same tool can help you check: ask it to quiz you on the parts you are least sure of, or to find the step you could not reproduce on your own. Being trustworthy to others starts with an honest account of what you know.

The best-practice norms for ethical AI-assisted scientific writing (Appendix A) state the same principle in a research setting: AI assistance becomes acceptable only when the human author

verifies the content and accepts responsibility for it. The guide applies that principle to student learning. AI may assist, but the final responsibility is yours.

3.2 ▪ The Learning Spiral

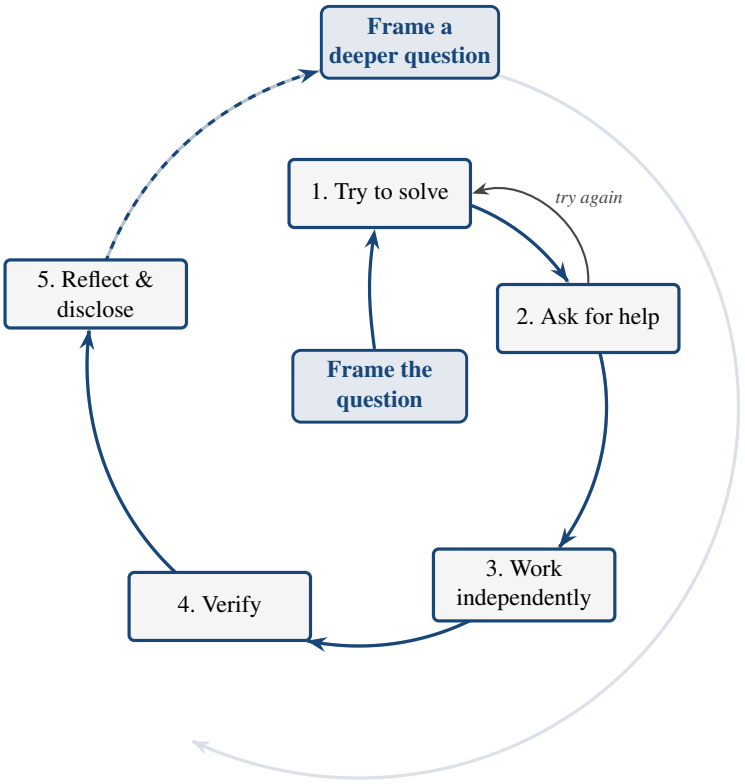


Figure 3.1. ***The Learning Spiral.*** *Five steps around one question at the center: try to solve it, ask for help rather than for a solution, work independently, verify, and reflect. A problem usually raises several questions; this is what to do with one of them. The dashed arrow leaving the circle carries you to the deeper question the first one opened, which is why this is a spiral rather than a circle. The short arrow between steps 2 and 1 is the try, ask, try again loop to expect in practice.*

The Learning Spiral is the main habit this guide asks you to practice, and it begins with a question. Framing a clear question is the move the whole practice turns on, which is why it sits at the center of the spiral (figure 3.1). Around that question run five steps: try to solve, ask for help, work independently, verify, and reflect. The Spiral assumes you have already set up your AI account, so the tool knows who you are and how you want it to respond. If you have not, do that now with the starter template in section 1.2. The Spiral depends on it. The five steps recur on every AI-assisted task, and each pass should deepen what you understand rather than return you to where you began. A question is not finished when you answer it. Pursued well, it opens a deeper question, and the spiral widens. This is why the framework is a spiral rather than a loop.

Read as a rubric, those same five steps become what AI-assisted work should show.

Step	What it means
Try to solve	Make an honest attempt before asking AI for help.
Ask for help	Ask for hints, explanations, questions, feedback, or critique.
Work independently	Return to the task and produce your own answer, draft, code, or analysis.
Verify	Check important claims, sources, computations, and assumptions.
Reflect and disclose	Record what AI contributed, what you changed, and what you can now explain.

The spiral runs at multiple time scales. Within a single task the steps keep their order, but the first two form a loop: try, ask, and try again until the idea holds. Only then does the work go independent. The later steps can send you back as well: a failed verification returns you to asking, and an explanation you cannot finish returns you to the material. The figure draws only the loop that runs most often. Across an assignment, a unit, or a semester, the spiral repeats and widens. What you learned from verifying one pass becomes part of your first attempt in the next. What you reflected on becomes the question that opens the next round. The work on each task feeds a longer arc of deeper questions across the course.

The five steps name skills, not behaviors that come automatically. A first-generation student new to a discipline may not know what a first attempt looks like for a proof, what verifying means for a citation in a specific field, or what counts as working independently in a programming class versus a literature class. Each step is a skill to be built, often in partnership with an instructor, advisor, or the study practices the rest of this Part describes. If you recognize yourself in this paragraph, the framework was built with your starting point in mind. The companion materials in `companion/spp/` support students who are new to any of this.

3.2.1 ▪ Frame the question

Every pass through the spiral starts with a question. Sometimes the assignment hands you one. More often the real question is yours to find: what are you actually trying to understand, and what is the part you cannot yet do? Framing that question well is the move the rest of the spiral depends on, because a vague question pulls a vague answer out of any tool, while a sharp one tells you what kind of help you need.

A good question names what you want to know and where your understanding stops. Putting a confusion into words is itself a way of learning, because it forces you to organize what you already know and to locate the edge of it. This kind of self-questioning is close to what learning research calls self-explanation, which has moderate support as a study habit [32]. Before you reach for AI, write the question down in your own words.

Sometimes you open the assignment with no idea where to begin, often because the topic is new and you have not had a chance to prepare. This is common in your first courses, and it does not mean you cannot do the work. It means framing needs a step in front of it, because you cannot ask a useful question about material you have not met yet. Spend a few minutes meeting it. Skim the section, the problem, or the reading for its shape: what is this about, what are its main pieces, and what is it asking you to produce? You are not trying to understand all of it. You are trying to find the one place where you first get lost, the first sentence or step you cannot follow. That turns “I do not understand any of this” into “I do not understand this part,” which is a question you can work with. As a practical threshold, give yourself five to ten minutes to find the shape of the material and the point where you get lost.

A question rarely ends where it starts. As you work through it you will understand more, and

that understanding will raise a sharper question than the one you began with. The spiral widens because answering a question well is what shows you the next one.

A few moves help you find that next question. Ask whether your answer holds beyond this case, what made it work, where it breaks, whether it runs the other way, and whether you can sharpen it. Each of these turns an answer into a better question, and one is enough to begin the next pass.

3.2.2 ▪ Try to solve

Before using AI, make a first attempt. Write down what you think the problem asks. Draft a paragraph. Sketch the argument. Try the calculation. Read the primary source. Mark the confusing sentence. Run the code. List the assumptions.

This first attempt does not need to be good. Its purpose is to give your mind something to work with. Once you have tried, AI can help you compare, clarify, and repair. Without that first attempt, AI is more likely to become a substitute for thinking.

A counterintuitive finding in the learning literature is that effortful generation, including unsuccessful generation, often produces more durable memory than studying correct examples alone [16, 66]. Even when your first attempt is wrong, the act of attempting before consulting AI strengthens what you will learn from AI's response. That benefit depends on the feedback arriving: a failed attempt that is never resolved teaches little.

A 2025 MIT study makes the same point. Researchers watched students' brains during essay writing under three conditions: no tools, a search engine, and an AI assistant. Minutes after finishing, the AI-assisted writers struggled to quote their own essays. Students who wrote without AI remembered far more. The recordings suggest why. Students who started with the AI shifted from generating their own content to supervising the machine's, and their overall neural connectivity dropped. A smaller follow-up session pointed the same way. Students who wrote alone first and then turned to AI kept higher recall and engagement than those who started with AI [71]. Read all of this as provisional. It is a preprint, and other researchers have raised questions about its sample size and analysis [118]. The pattern matches the rule above, and the rule does not depend on it: the first attempt gives your mind something to work with, and later AI assistance amplifies that effort rather than replacing it.

As a practical threshold: for routine homework, give yourself at least ten to fifteen minutes of serious attempt before asking AI for a hint. For a proof or open-ended problem, spend enough time to write down the definitions, the goal, one possible strategy, and the point where you are stuck. Do not ask AI before you have something for it to respond to.

The reason to protect those minutes is that the cost of skipping them is hard to notice afterward. In a controlled experiment on short-story writing, participants who were shown ideas from a language model before starting produced work rated more creative, yet their stories were more similar to one another, and the researchers describe the mechanism as anchoring on the model's ideas [30]. This book reads that as a warning for a single student too: once you have seen a plausible approach, your own alternatives become harder to reach. The loss is invisible, because you never find out what you would have thought of. Those minutes buy you something you cannot recover later.

3.2.3 ▪ Ask for help, not replacement

A useful AI question keeps you active. It asks for support that leaves the central thinking with you.

The requests that replace your work are rarely the first ones a student makes. They are where a session ends up, usually late, usually on a night when one problem is still open and the clock is

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

not. Follow the route there, because every step on it is reasonable and none of them feels like a decision.

It starts with a real question. Twenty minutes in and stuck at the setup, a student asks where the problem begins. The answer helps, and they move. A few lines later the next obstacle arrives, and they ask whether the method they chose is the right one. Still fine. Then they have a setup on the page and ask the AI to check it, which is the first request that hands over a judgment instead of asking for help making one. Then they ask for the first two steps and promise themselves the rest. Then they ask for the answer.

The tell is not the wording of the prompt. It is what the student made between one prompt and the next. Early on they were making attempts. Later they were only making decisions about the AI's work. At the end they were making nothing, and the page was finished and not theirs.

There is a second tell, and it is the one someone else can see. A student who has gone to the answer first often keeps asking questions, including ones they could already answer. The work looks engaged from the outside, and none of it is being used to find anything out.

One question catches this, and you have to ask it before you type: what have I produced since the last time I asked? If the answer is nothing, you have already started down that road. The next request will take you further along it than it seems. The last stop on that road is the first prompt below.

Instead of asking:

> *Write my answer to this problem.*

try asking:

> *Give me one hint about how to start, but do not solve the problem.*

Instead of asking:

> *Write my essay on this topic.*

try asking:

> *Here is my thesis and outline. What is the weakest part of the argument?*

Instead of asking:

> *Fix my code.*

try asking:

> *Ask me diagnostic questions that would help me locate the bug.*

A different kind of help is useful when the trouble is not a problem you are solving but an idea you cannot get into focus. When a concept sits above your current level, ask for it three ways at once: as a plain analogy, at your working level, and one step deeper than you are now. Instead of asking:

> *Explain this concept.*

try asking:

> *Explain this concept three ways: first with an everyday analogy, then at the level of someone taking this course, then one step more advanced so I can see where it leads.*

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The three levels give you an entry point, a working understanding, and a direction, and the gaps between them are often where the idea finally clicks. This is a strong way to get unstuck on difficult reading, and it comes with the same catch as every other step in the spiral: a clear three-level explanation can feel like understanding without being it. The explanation has done its work only when you can close the window and rebuild the idea in your own words. Students who want a tested prompt for this can use the layered-explanation template in the companion repository (`companion/layered-explanation/`).

The best prompts are requests for coaching. Good help sometimes goes backward before it goes forward. If a hint keeps missing, the problem may rest on an idea you have not firmly grasped, and the quickest way through is to step back to that idea, make it solid, and climb back to the question. Students who want a tested prompt for converting the AI into a study partner rather than an answer machine can use the Study Partner Protocol in the companion repository (`companion/spp/`). A hint is not the end of the exchange. Take it back to the problem and try again, and ask again if the new attempt stalls. Most of the learning happens inside this loop. Working independently begins when the loop ends, when you can rebuild the work with the AI closed.

3.2.4 ▪ Work independently

After receiving help, step away from the AI response and do the work yourself. Write the paragraph in your own words. Complete the proof or calculation. Revise the code. Rebuild the argument. Explain the source. Decide what belongs in the final submission.

This step is easy to skip because AI output often looks finished. Do not confuse finished-looking with learned.

A good test is simple: close the AI window and try to explain the idea aloud. If you cannot explain it, return to the material.

3.2.5 ▪ Verify

Verification is the act of checking AI-assisted content against reliable evidence, sources, data, tests, methods, or reasoning. It is the habit that turns AI assistance into responsible academic work. It is also the most important calibration check in the Learning Spiral: finding errors in AI output requires that you understand the work well enough to recognize when something is wrong. AI systems can produce confidently wrong answers. They can invent sources, misread prompts, use the wrong method, make hidden assumptions, or change the meaning of your sentence.

The errors arrive dressed as confidence for a reason, and the reason is not a flaw someone forgot to fix. Kalai and colleagues trace it to two causes.

Some facts cannot be learned from the training data at all. A fact that appears once, or never, leaves no pattern to generalize from, so errors on questions of that kind are expected rather than surprising. That explains the wrongness. It does not explain the certainty.

The certainty comes from how these systems are graded. Most of the tests used to rank them score an answer right or wrong and award nothing for saying “I do not know.” Under that rule a guess is always worth more than an admission of uncertainty, so a system trained against those tests learns to answer as though sure [59]. A student graded under the same rule learns the same habit. On an exam that gives no credit for a blank, bluffing beats leaving it empty.

The practical consequence is short. Confidence is not reliable evidence of correctness. The model was never trained to sound sure only when it is right.

Verification looks different in each field. The details vary. The standard does not: do not submit AI-assisted work until you have checked the parts that matter, until you can explain why the answer is right and not merely confirm that it looks right. The AI answer is a draft, not a

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

witness. An outside check is worth most where your own sense of the answer is weakest. That is also where a false sense of understanding is hardest to catch on your own.

Where possible, return to the same problem type, or a mixed set of related problem types, one to three days later without AI. This spaced and interleaved retrieval is what converts a single working session into durable understanding [108]. Verification confirms that you understood the work today. Spaced and interleaved practice confirms that you will still understand it next week.

3.2.6 ▪ Reflect and disclose

Reflection helps you notice what you learned. Disclosure helps your instructor understand how the work was produced.

To make reflection concrete, after each AI-assisted session answer in your own words: Why does each step in this proof, solution, or argument work? What would change if one assumption were weakened or removed? These two questions take thirty seconds and turn the session into elaborative practice. A useful weekly habit is to schedule two or three short interleaved retrieval sessions using your hard-problems list or verification checklist. The pattern is short, mixed, and unassisted.

When disclosure is required, use clear and ordinary language. You do not need to apologize for allowed AI use. You do need to be honest.

Suppose the assignment is a short written report on an assigned reading, in a course that allows AI as a tutor. A good disclosure block for that work looks like this:

> **Tools used:** I used the AI tool approved for this course.
> **Tool role:** I asked it to generate practice questions on the assigned chapter, and to critique my draft introduction.
> **Student role:** I rewrote the introduction myself after its critique said my aim appeared only in the last sentence.
> **Accepted, changed, or rejected:** I kept two of its six questions, one on how the control group was chosen and one on what the error bars show, and dropped the rest as too broad.
> **Independent verification:** It stated that the chapter recommends repeating each measurement three times. The chapter says five, so I corrected the number and named the page it came from.
> **What I can explain without AI:** I can explain my aim, the evidence behind it, and why I changed the number of repeats.

Some courses will require more detail. Some will require none for minor uses. Follow the assignment instructions. When an instructor asks to see the AI conversation itself, sharing the conversation link is the most direct disclosure available, and where a course accepts it, it can stand in for the written block. It is a record of the interaction, not proof of authorship.

Some work produces no thread to share. If you set a direction, let a tool run through many cycles on its own, and step in at intervals to redirect it, there is no single conversation that shows what happened. That is not evasion; the record simply does not exist. What you can still give is an account of your own decisions: what you set out to do, where you intervened and why, and what you checked before accepting the result. Keep that as you go, because it cannot be reconstructed afterwards.

3.3 ▪ When the aim is a new idea

Up to here, each question has had an answer, and your job has been to reach it. Some of your work will ask for something else: an idea nobody has handed you, a design, a conjecture, a question

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

worth a semester. AI changes the economics of that work, because it can produce ten candidates in the time it takes you to write one. The risk is that the candidates get better while you do not.

That risk has been measured. In a 2026 experiment with university students, free access to ChatGPT improved creative performance on the task where the tool was present, and the advantage vanished on a later task done without it [137]. Students who generated their own ideas first and only then used the tool to develop and evaluate them did better on that later unaided task than either the free-access group or the group that never used AI at all. Both AI groups used the tool on the first task. The difference was not whether the tool was used but how and when, and only the students who thought first were better afterward.

What the submitted work shows is AI-assisted Practice. What survives when the assistance changes is Core Competence. Trustworthiness is the account of which part of the work came from the assistance and which came from your own competence. The companion paper "Learning to Create with AI" calls that surviving part learned creativity and develops the case at length [54]. The point for you is narrower: a better output is not evidence that you have become more capable, and the evidence that counts is what you can do next time without the same help.

The Learning Spiral was designed to protect exactly that evidence, and a creative pass through it differs in aim rather than in structure. You are not trying to reach a known answer. You are trying to change the question, the representation, or the space of possible answers. Try to solve is where your own ideas get written down before the model's become visible, so the minutes before you ask matter more here, not less.

Add two habits. When you ask, ask the tool to widen the set before it ranks anything: four alternatives under four different assumptions and no verdict. Keep the choosing for yourself, and be able to say why you rejected what you rejected. When you finish, go one step past the original task: take the idea to a new case, add a constraint, or write down the question the work opened and did not answer. If you can only repeat the assisted result, you have finished the product and not the learning.

3.4 ▪ Four assignment categories

Each assignment should tell you what kind of AI use is allowed. When the instructions are unclear, ask before using AI in a way that affects submitted work.

In this book's framework, instructors mark each assignment with one of four short labels that signal how AI fits the learning goal. Each label also has a color, as a memory aid. Three of them sit on a traffic light:

- **NAI** (No AI): red. Stop. The assignment measures what you can do without assistance.
- **AIT** (AI as Tutor): yellow. Proceed with care. AI may coach you toward understanding, but the submitted work must be yours.
- **AIC** (AI as Collaborator): green. Go. AI may co-produce work with you, and the assignment treats the collaboration as part of the learning.

The fourth label is a different kind of sign:

- **AIS** (AI as Object of Study): blue. Off the traffic light entirely. The question is not how much AI you may use but what AI is doing: you study AI itself rather than using it as a tool.

NAI, **AIT**, and **AIC** vary along one axis: how much AI freedom the assignment allows. **AIS** is not a point on that axis. It is a different question altogether, which is why it gets a different color.

3.4.1 ▪ NAI: No AI

In an **NAI** assignment, AI use is not allowed. The purpose is to measure what you can do independently.

Examples include closed-book exams, short quizzes, in-class writing, oral explanations, lab practicals, live coding checks, language drills, or foundational calculations.

An **NAI** assignment is not a claim that AI is bad. It is a way to protect Core Competence. You need some abilities that do not depend on a tool being open in another window.

3.4.2 ▪ AIT: AI as Tutor

In an **AIT** assignment, AI may be used for study support, but not to produce submitted work. What sets the category is where the AI's work ends up, not how well you understand it. If you write or work out the submission yourself, with no AI-generated text, code, or solutions carried into it, the assignment stays **AIT**. Once AI's wording, code, or worked solution becomes part of what you hand in, the work has crossed into **AIC** territory, which is a different category with different rules. If the analysis or the argument is AI's, putting it in your own words leaves the work **AIC**. Whether you could reproduce that work from your own understanding is a separate question. That is the ownership question, and the translation test is how you check it. Understanding is what makes **AIC** work honest. It does not turn it back into **AIT**.

You may ask AI to explain a concept, quiz you, give hints, make flashcards, create practice problems, or help you prepare for class. You should not ask it to write the final answer.

AIT use is common when the instructor wants you to learn from AI without letting AI shape the submitted product. The Study Partner Protocol in the companion repository (`companion/spp/`) is designed for exactly this kind of **AIT** work. If your instructor posts a course version, start each **AIT** session with theirs. When **AIT** work produces a tutoring conversation worth showing your instructor, sharing the conversation link documents the interaction more cleanly than any written summary.

The protocol's opening lines set the pattern:

> For this conversation, act as my study partner, not an answer service. Your goal is to help me learn, so I should end up able to do the work myself. Before helping with a problem, ask me what I have already tried and where I am stuck. If I have not tried yet, ask me to make a first attempt before you help. Give hints, ask guiding questions, and point to the relevant idea. Do not give full solutions, finished proofs, or final written answers unless I explicitly ask for a worked example after trying. When I think I understand, ask me to explain it back in my own words, and check my explanation. If I ask you to just give me the answer, remind me once of this arrangement, then respect my choice.

The full protocol, including the course-scope and step-back rules, is in the repository (`companion/spp/`).

3.4.3 ▪ AIC: AI as Collaborator

In an **AIC** assignment, AI may help with brainstorming, drafting, debugging, revision, critique, outlining, coding, or comparison. You must still verify the result and disclose AI use.

AIC assignments are useful when the learning goal includes process, revision, research, modeling, design, computation, or professional practice. The instructor is not only asking what you know. They are also asking whether you can use tools responsibly. Because the AI's contribution is part of what is being assessed, sharing the conversation link is often the most natural way to disclose how the work was produced.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

3.4.4 ▪ AIS: AI as Object of Study

In an **AIS** assignment, you study AI output directly. You may compare AI responses, find errors, test generated code, identify bias, repair a flawed explanation, evaluate sources, or analyze how AI changes work in a field.

AIS assignments are especially valuable because they train you to examine AI instead of merely using it.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 4
Using AI Well and Badly

4.1 ▪ Good uses of AI

Good AI use keeps you mentally active. It helps you see, test, practice, revise, or explain. It does not remove the main work of the course.

4.1.1 ▪ Clarify confusing ideas

AI can often provide a second explanation when a lecture, reading, or textbook passage does not make sense on the first pass.

Useful prompts include:

> *Explain this concept in plain language, then give a college-level version.*
>
> *Give me a concrete example and a non-example: a case that just fails the definition.*
>
> *Ask me three questions to check whether I understand this idea.*

After reading the response, return to the course material. The AI explanation is a bridge, not the authority.

4.1.2 ▪ Practice retrieval

One of the strongest learning habits is trying to recall information without looking at the answer. Practice testing and spaced practice have broad support in learning research [32], and AI can help you create these practice opportunities.

A good routine is to ask AI for questions, close the AI response, answer on your own, and then check your work.

> *Create ten practice questions on this topic. Do not give the answers until I ask.*
>
> *Quiz me one question at a time. Wait for my answer before giving feedback.*

This matters because recognition is not the same as recall. Reading an answer and thinking "that makes sense" is weaker than producing the answer yourself.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

4.1.3 ▪ Test your explanations

One of the most reliable ways to discover what you do not yet understand is to try teaching it. Explaining a concept or a solution in your own words, with the steps in your own order, forces you to produce the structure of an idea rather than recognize it. AI can play the role of the listener, the curious student, or the skeptical colleague, whatever audience you need to practice for.

> *I'll explain this concept to you. Tell me which parts are unclear and which parts are incomplete.*
>
> *Act as someone outside this field. Listen to my explanation and ask the questions a non-expert would ask.*
>
> *Here is my answer to this problem. Critique it as if I were teaching you the topic. Where would a student be confused?*

The shift in roles matters. When you ask AI to solve a problem, AI does the producing and you evaluate. When you ask AI to evaluate your explanation, you do the producing and AI evaluates. The second form keeps the cognitive work where it needs to be. The same shift lets you rehearse for different audiences, which is one of the more useful side benefits of teaching yourself by teaching AI.

4.1.4 ▪ Find weak points

AI can help you find places where your work may be unclear or incomplete. This is often more valuable than asking it to improve the work directly.

> *Here is my argument. What would a skeptical reader challenge?*
>
> *Find the first step in this reasoning that needs more justification.*
>
> *What assumptions am I making that I have not stated?*

There is an obvious-looking alternative to these prompts, and it does not work. Instead of asking the AI to argue against you, you could ask it to be neutral. Cheng and colleagues tried exactly that. They needed a middle setting for an experiment, an AI that neither approved of what a person had done nor disapproved, and they could not build one: replies written to be neutral were often read as taking the user's side [24]. Their test ran on one model, in conversations about personal disputes rather than coursework, so treat it as a warning rather than a measurement of your own situation.

A request for neutrality reads as permission to agree. A request for opposition does not. That is why the prompts above ask for the challenge by name.

When you use AI this way, you remain the author. AI becomes a critic.

4.1.5 ▪ Compare approaches

A good way to deepen understanding is to compare two methods, two interpretations, or two explanations.

> *Compare these two solution strategies. What does each one reveal?*
>
> *Give me two possible interpretations of this passage, then explain what evidence would support each one.*
>
> *Show me why this programming approach might be less stable or less efficient than another approach.*

Comparison helps you notice structure. It can also reveal when an AI response is too confident.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

4.1.6 ▪ Prepare for direct questioning

In small courses, your instructor may ask you to defend your work orally. In large courses, you may face a short audit, an explanation video, or an in-class follow-up question.

You can use AI to practice.

> *Act as my instructor. Ask me questions about this solution until you find what I do not understand.*
>
> *Ask me to explain each step of this argument. Do not move on until my explanation is clear.*
>
> *Give me a flawed version of this answer and ask me to find the error.*

This is a strong use of AI because it prepares you to think without AI. When AI seems like magic, that is the signal to ask for more information, not less.

4.2 ▪ Annotated dialogues: strong AI use

The next pages show what good AI use looks like in practice. Each dialogue is a short exchange between a student and an AI tool. Marginal annotations show what the student did right or wrong at each turn. Read the dialogues alongside the principles in section 3.2 and section 4.1; the principles describe the habits, the dialogues show the habits at work.

These dialogues come from mathematics, programming, research methodology, the history of mathematics, and writing. Students in other quantitative and computational fields (applied mathematics, statistics, theoretical and computational physics, computer science, operations research, mathematical biology, and their neighbors) can use the patterns as they stand. Students in the humanities, social sciences, professional programs, and creative or design fields should translate them. The spiral underneath (frame the question, try to solve, ask, work, verify, reflect) does not depend on the field, and the moves below carry over with minor adaptation.

Dialogue 1: Undergraduate optimization problem (calculus)

Student. I have a calculus problem: a farmer wants to fence a rectangular field of area 400 m^2 along a river, fencing only three sides. What dimensions minimize fencing cost? I am stuck setting up the problem.

→ Good move: stated the problem and named where they are stuck.

Student. Before I set up any equations, let me guess what the answer should look like. The river side needs no fence, so it makes sense to make that side long and the depth shorter, which points to a field wider along the river than it is deep. A rough guess is about twice as wide as deep, with the deep side somewhere in the low tens of meters.

→ Good move: built geometric intuition before any algebra, so the final numbers can be checked against an expectation instead of accepted blindly.

Student. Don't solve it. Give me one hint about the main obstacle so I can try first.

→ Good move: requested a hint, not a solution.

AI. *One hint: you have two unknowns (length and width) but the area constraint relates them. Use the constraint to write the fencing cost as a function of one variable.*

Student. OK. Let length parallel to river be L, width be W. Area constraint: $L \cdot W = 400$, so $L = 400/W$. Fencing is $L + 2W = 400/W + 2W$. I want to minimize $f(W) = 400/W + 2W$. I'll take the derivative.

→ Good move: did the substitution and set up the function before consulting AI again.

Student. $f'(W) = -400/W^2 + 2$. Setting to zero: $W^2 = 200$, so $W = \sqrt{200} \approx 14.14$ m, and $L = 400/14.14 \approx 28.28$ m.

Student. Now I'll check that it's a minimum. $f''(W) = 800/W^3$, which is positive at $W = \sqrt{200}$, so the critical point is a local minimum. Audit that. Am I missing a step?

→ Good move: ran the second-derivative test first, then asked AI to audit it, not to perform it.

AI. *Your second derivative and your conclusion are both correct. One step is missing: a positive second derivative at the critical point gives a local minimum, and by itself it says nothing about the rest of the domain. Check the extremes: as $W \to 0$ or $W \to \infty$, $f(W) \to \infty$, and $W = \sqrt{200}$ is the only critical point, so the local minimum is the global minimum.*

Student. Got it. Final answer: $W = \sqrt{200}$ m, $L = 2\sqrt{200}$ m. The result matches my early guess: about twice as wide as deep, with the deep side near fourteen meters, the low-tens range I expected. I can explain why the second derivative test confirms it's a minimum, and I can state what would change if the constraint were different.

→ Good move: stated what they can explain. Disclosure block would note: AI gave one hint, audited the student's second-derivative test, and added the step from local to global; setup, derivative, test, and conclusion are the student's.

The takeaway. That is the *try to solve* step: the estimate you make before asking is what gives you something to check the answer against.

The next two dialogues work at the graduate level, one auditing a proof in real analysis and one verifying sources for a research literature review. These two are more advanced; a first-year reader can move on to Dialogue 4 and return to them after working through the Learning Spiral. They show the same habits as the other dialogues, applied to harder material.

Dialogue 2: Graduate analysis (proof audit-and-repair)

Student. I have an attempted proof that every continuous function on $[0, 1]$ attains its supremum. I'm going to write it first, then ask you to audit it.

→ Good move: wrote the proof before AI involvement.

Student. My proof: "Let $M = \sup_{x \in [0,1]} f(x)$. By definition of supremum, there exists a sequence $\{x_n\} \subset [0, 1]$ with $f(x_n) \to M$. Since $[0, 1]$ is bounded, the Bolzano–Weierstrass theorem gives a convergent subsequence $x_{n_k} \to x^*$. By continuity, $f(x_{n_k}) \to f(x^*)$. Therefore $f(x^*) = M$."

Student. Audit this. Find any step that is incomplete or relies on an unstated hypothesis.

→ Good move: asked for an audit, not a rewrite.

AI. *Three issues. First, you used Bolzano–Weierstrass, which requires the sequence to be bounded; you have $\{x_n\} \subset [0, 1]$, so this is fine, but you should state it. Second, you concluded $x^* \in [0, 1]$ implicitly, but $[0, 1]$ is closed, so the limit lies in $[0, 1]$. State this. Third (subtle): you wrote that* sup *attains a sequence approaching it. This requires the sup to be finite. You have f continuous on a compact set, but you haven't established $M < \infty$. The argument circles: you're using the result you want to prove.*

→ Note: an audit can be wrong in more than one place. The student must check each critique against the proof as written.

Student. Your first point is already in my proof: I wrote that $[0, 1]$ is bounded. Your third point is wrong too. The subsequence $f(x_{n_k})$ still converges to M, and by continuity it also converges to $f(x^*)$, a real number. A sequence cannot have two limits, so $M = f(x^*)$, which is finite. Nothing circular is going on. Your second point is right: I never said why $x^* \in [0, 1]$. That holds because $[0, 1]$ is closed, and I should say so.

→ Excellent move: the student checked each critique against the proof as written, rejected two, and kept the one real gap.

Student. Revised proof: "Let $M = \sup_{x\in[0,1]} f(x) \in \mathbb{R} \cup \{+\infty\}$. There is a sequence $\{x_n\} \subset [0, 1]$ with $f(x_n) \to M$. Since $[0, 1]$ is bounded, the Bolzano–Weierstrass theorem gives a convergent subsequence $x_{n_k} \to x^*$, and $x^* \in [0, 1]$ because $[0, 1]$ is closed. By continuity, $f(x_{n_k}) \to f(x^*)$. Since $f(x_{n_k}) \to M$ as well and limits are unique, $f(x^*) = M$; in particular M is finite." The repair adds one missing step; the argument is otherwise my original one.

→ Disclosure: AI used for proof audit; student verified each critique against the proof and the textbook definitions, rejected two, and accepted one. Original argument and repair are the student's.

The takeaway. Asking the AI to justify a step, rather than supply one, is what keeps the *verify* step yours.

Dialogue 3: Graduate research (literature review with verification)

Student. For my qualifying-exam topic on percolation theory, give me a list of foundational papers I should read.

AI. *Here are six often-cited foundational papers: Broadbent and Hammersley (1957) introducing percolation; Kesten (1980) on the critical probability for 2D bond percolation on $\mathbb{Z}^2$; Grimmett's textbook* Percolation *(2nd ed., 1999); Smirnov (2001) on conformal invariance for site percolation on the triangular lattice; Lawler–Schramm–Werner (2001) on intersection exponents; and Aizenman–Newman (1986) on tree graph inequalities.*

Student. Before I cite any of these, I'm going to verify each one independently by checking MathSciNet (the AMS bibliographic database) or the journal record.

→ Good move: applied the verification standard from the scientific-writing appendix.

Student. [Several minutes later.] I verified five of the six. The Aizenman–Newman paper exists, but not in 1986. The reference appears to be "Tree graph inequalities and critical behavior in percolation models," *Journal of Statistical Physics* 36 (1984), pp. 107–143, not 1986. I'm using the verified version.

→ Good move: caught a subtle hallucination (real authors and topic, but wrong year). This is exactly the kind of error AI literature suggestions produce [131].

Student. [The next day.] The records check out, but a record is not the paper, so I pulled up all six and skimmed each one: abstract, introduction, main theorem. Four are foundational for my topic. The other two are real papers doing real work, but one is about Brownian motion, with percolation entering only through later work, and the other assumes machinery I have not covered. Neither belongs on a first reading list.

→ Good move: checked each source against what the list claimed for it. A verified citation can still be the wrong source.

Student. I'll bring the four to my advisor as a starting reading list, with the two I set aside and my reason for each. The note will say the list began as an AI suggestion, that every entry was checked against MathSciNet, and that I read enough of each to judge fit. Then I will ask my advisor the question the tool cannot answer: what belongs on this list that is not on it?

→ Good move: disclosed the AI origin as the scientific-writing appendix asks and brought the checked list to a human expert for the judgment the tool cannot supply.

The takeaway. A literature claim you have not opened is not yet a source; here, verifying means reading the thing cited.

Dialogue 4: Programming (debugging by diagnostic questions)

Student. My Python function is supposed to compute the moving average of a list with window size k, but it gives wrong values at the end. Don't fix it for me. Ask me diagnostic questions until I find the bug.

→ Good move: asked for diagnostic questions, not corrected code.

Student. Code:

```
def moving_avg(arr, k):
    return [sum(arr[i:i+k])/k for i in range(len(arr))]
```

AI. *Three diagnostic questions. First: what does* `arr[i:i+k]` *return when* `i+k > len(arr)`*? Second: how many elements does* `arr[i:i+k]` *contain in that case? Third: are you dividing by the right number when the slice is shorter than* `k`*?*

Student. Oh. Python slices return a partial slice if the end goes past the array, but I'm still dividing by k. So at the end, I'm dividing a shorter sum by the wrong denominator. The fix is either to skip those edge cases or to divide by `len(arr[i:i+k])`.

→ Good move: identified the bug from the diagnostic questions, articulated two fixes, which define the average at the end differently.

The takeaway. Asking for a diagnostic question instead of a fix is what keeps debugging a task you learn from.

Dialogue 5: History of mathematics (background research with primary sources)

Student. I'm writing a paper on the Newton-Leibniz calculus priority dispute. Give me background context; I'll then read the assigned primary sources.

AI. *Newton and Leibniz developed calculus independently in the late 17th century. Newton worked first but did not publish for decades; Leibniz published first, in 1684. The Royal Society convened a committee in 1712 that ruled in Newton's favor. Most historians today regard the two as independent inventors.*

Student. Useful as orientation. Now I'm reading the assigned primary sources: the 1712 Royal Society report *Commercium Epistolicum* and Leibniz's published response.

→ Good move: treated AI as orientation, not as the source.

Student. [After reading.] My initial claim was going to be that both mathematicians received fair treatment. But the report is not a neutral inquiry: it is selective in its evidence and was almost certainly written by Newton himself, who was then president of the Royal Society. Leibniz's

response shows he had no formal opportunity to defend himself before the conclusion was announced.

→ Good move: revised the claim based on what the primary sources actually said, not on AI summary.

Student. Revised thesis: "The 1712 Royal Society ruling against Leibniz was not a neutral judgment but an institutional defense of Newton's reputation, organized by Newton himself."

→ Disclosure: AI used for general background; thesis and argument grounded in assigned primary sources.

The takeaway. Across all five, the judgment that matters stays with the student.

A borderline case

Not all AI use is clearly strong or clearly weak. Some uses sit in a gray zone where the student must exercise judgment about what to keep, and Dialogue 6 shows one such case.

Dialogue 6: Borderline use (writing, certainty drift)

Student. Here's a paragraph from my history paper. Tighten the prose; keep my meaning.

Student. [Student's original]: "The evidence in the parish records suggests that enclosure produced uneven outcomes for small tenants. Some lost grazing access; others received small compensation. The records do not show a single uniform pattern."

AI. *[AI revision]: "The parish records demonstrate that enclosure produced sharply unequal outcomes for small tenants. Many lost grazing access while a few received compensation. The records reveal no uniform pattern."*

→ Watch the meaning drift. "Suggests" became "demonstrates." "Some / others" became "many / a few." "Do not show" became "reveal no." Each change tightens the prose, and each one inflates a hedged claim into a confident one.

Student. Hmm. The AI version reads better, but it's making a stronger claim than my evidence supports. I have records from one parish, not a body of evidence about "the parish records" in general.

→ Good move: caught the certainty drift. This is the verification step from the scientific-writing appendix: AI revision can subtly change the certainty level of central claims.

Student. I'll keep the AI's sentence rhythm but restore my hedging: "The parish records I examined suggest that enclosure produced uneven outcomes for small tenants in this parish. Some lost grazing access; others received small compensation. The records show no uniform pattern across the cases I reviewed."

→ Borderline use that landed in the right place. The student kept the prose improvement but rejected the certainty inflation.

The takeaway. The tool did what was asked; what the disclosure has to answer is whose judgment shaped the result.

This case sits in a gray zone because the AI did exactly what was asked: it tightened the prose. The certainty drift was a side effect, not a failure of the prompt. The student is responsible for catching it. A student who accepts the AI revision unread would have published a stronger claim than their evidence supports, with no malice and no policy violation, but with reduced scholarly accuracy. The verification habit is what catches these cases.

4.3 ▪ Dangerous uses of AI

The easiest person for you to fool
is yourself.

(after Feynman)

Dangerous AI use often feels helpful at first. It saves time, removes friction, and produces a polished answer. The problem appears later, when you cannot remember, explain, defend, or transfer what you submitted. Cognitive science offers one explanation for why this happens: fast, intuitive thinking is comfortable with surface fluency and tends to skip the slower, effortful checking that catches errors [58]. AI output that reads smoothly engages the fast system; verification requires the slow one.

4.3.1 ▪ Do not outsource first contact with the problem

If AI sees the problem before you do, it can shape your thinking too early. You may stop noticing what confused you. You may accept the first framing it gives you. You may never develop your own plan. Reaching for AI first is exactly the AI-first trap named earlier in this Part: you recognize the solution you are given and lose the ability to generate your own. UNESCO's guidance on generative AI in education reports that some experts call this risk "writing without thinking": starting from an AI-generated outline rather than your own [89]. Verification cannot substitute for that lost thinking, because the only thing being verified is the AI's plan, not the student's understanding.

For serious assignments, begin with your own attempt. Even a rough attempt helps.

4.3.2 ▪ Do not submit work you cannot explain

This is the clearest boundary.

If you cannot explain a paragraph, proof, calculation, source choice, design decision, or line of code, it should not be in your submitted work. You may use AI to help you understand it. You may not use AI to hide the fact that you do not understand it.

4.3.3 ▪ Do not trust generated citations

AI systems can invent sources or misstate what real sources say. A citation that looks real may still be wrong, and guidance on generative AI in education warns that these tools can cite papers that do not exist and asks users to verify what they produce [89].

Never cite a source you have not checked. For research assignments, you should find the source yourself, read the relevant part, and confirm that it supports the claim.

Two different failures hide behind the same advice, and knowing which one you face changes what you check. A model answering from memory alone can invent a source outright, so the first question is whether the reference exists at all. A model connected to a scholarly index returns references that do exist, and the failure moves downstream: the source is real but tangential, or it is real and says something other than the summary reports. The second failure is the harder one, because a working link and a correct author list both look like verification succeeding. Confirming that a source exists is not confirming that it supports your claim.

4.3.4 ▪ Do not let AI change your meaning

AI revision can improve clarity, but it can also change what you meant. It may make a claim stronger than your evidence allows. It may remove a useful qualification. It may replace your

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

voice with a generic academic tone.

After using AI to revise, compare the new version with your original intent. Ask yourself: Is this still my claim? Is the level of certainty right? Would I defend this sentence if asked?

4.3.5 ▪ Do not trust code just because it runs

Code can run and still be wrong. It may solve the wrong problem, use the wrong formula, mishandle edge cases, hide numerical instability, or produce plausible-looking output from flawed logic.

If AI helps write code, test it on simple cases where you know the answer. Read it line by line. Know what each major step does.

4.3.6 ▪ Do not use AI to evade the purpose of an assignment

Every assignment has a learning purpose. Sometimes the purpose is to practice a skill slowly. Sometimes it is to struggle with a hard text. Sometimes it is to learn a method well enough to use it later.

AI can defeat that purpose if you use it too early or too broadly. Before using AI, ask: What is this assignment trying to teach me? Will this use help me learn that, or help me avoid it?

4.3.7 ▪ Do not let AI replace people

An AI tutor answers at any hour and never seems to judge you. That comfort is real, and it is also the risk. Some students now prefer a chatbot to office hours for exactly that reason [113, 128]. But a chatbot cannot see that you have been stuck for three weeks, cannot tell when a problem is larger than the assignment, and has no stake in what becomes of you.

When AI becomes your only resource, the people who can actually help you fade from view: the instructor who can spot what you are missing, the classmate who got stuck on the same idea, the advisor or counselor who is there for the parts of student life that are not about grades. Keep talking to people. If something heavier than the coursework is weighing on you, reach out to someone you trust or to your campus support services. Asking for help is part of how learning works.

4.4 ▪ How to verify AI-assisted work

Sage advice from Dick Hamming, whom I knew during his visits to Los Alamos: do the back-of-the-envelope calculation before you build the cathedral. The principle scales. In an AI-assisted setting, a rough check on the answer's shape and order of magnitude is worth more than a precise validation of the wrong thing. Verification starts with what you would expect to see, not with what the AI gave you.

Verification is part of responsible learning. The cognitive habit underneath verification is what learning science calls self-explanation: stating in your own words how a new idea relates to what you already know, and why a claim is true. Self-explanation has measurable benefits for comprehension and transfer, though mostly on tests given soon after study [32]. It also works with no AI involved at all. That independence from the tool is why it is the right habit for verifying AI-assisted work. Figure 4.1 summarizes how verification looks across disciplines. Graduate students preparing manuscripts for publication will find a fuller verification protocol in Appendix A, which addresses research-grade work specifically.

How much verification is enough depends on the stakes. A low-stakes practice problem needs only a quick sanity check. A result you will submit, build on, or rely on later deserves more, with

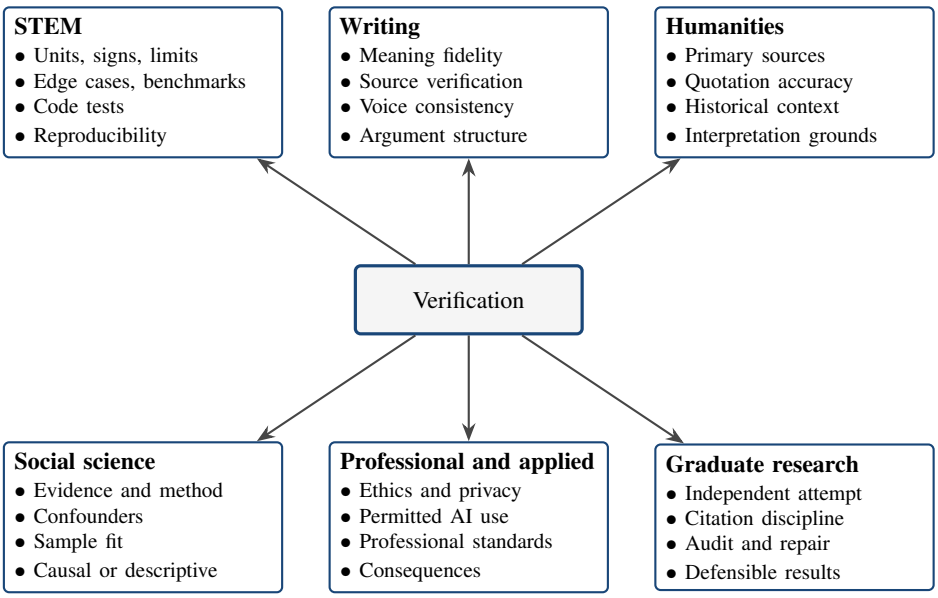


Figure 4.1. ***Verification across disciplines.*** *The verification habit is constant; the specific checks vary by domain. Every discipline has a small set of routine checks that catch common errors in routine tasks; these checks do not on their own certify correctness or sound judgment. Students should learn the verification toolkit appropriate to each course they take.*

the most careful checking reserved for the claims that would matter most if they were wrong.

One check does not work, and it is the one most people try first. Asking the AI to review its own answer, with nothing new to look at, tends to make the answer worse rather than better. Across several models and several kinds of problem, accuracy fell in almost every case when a model was asked to find the fault in its own reasoning and try again [52]. The authors trace this to one limit: the model cannot reliably judge whether its own reasoning is correct. It often keeps the answer it had, and when it does change one, it is more likely to turn a right answer wrong than a wrong answer right.

What changes the result is giving it something outside itself to look at: the definition in your textbook, a worked example, your data, the output of a program you ran, your instructor's comment on the last draft. A second AI counts as outside, with one caution. If two systems disagree, you have learned where to look; if they agree, you have learned nothing, because they can be wrong together. And asking the same one again in a new window is not a second opinion at all.

This matters most at the two ends of a long piece of work. At the start, the thing outside the model is you.

Instead of asking:

> *Write the lab report from these results.*

try asking:

> *Before you start, ask me whatever you need to know about this assignment. Keep asking until you can tell me back what I want.*

Answering those questions is the part where you find out whether you know what you are trying to do. At the end, the thing outside the model is what actually got made.

Instead of asking:

> *Now write the discussion section.*

try asking:

> *Before the discussion, check the results section against my actual data and tell me what does not match.*

The two prompts pair. The questions the first draws out become the checklist for the second, so together they wrap whatever you ask in between: specify before, audit after. The wrapping is reusable; the prompt inside stays yours.

None of this replaces your own reading of the work. It catches what was not done. Whether the work was worth doing is still yours to judge.

When you read it yourself, verify in order. First confirm the problem is still the one you asked: same assumptions, same domain, same conclusion, and no swap of every for some. Then find the hard step and check that it is actually there, since a fluent answer can glide past the one thing that needed proving. Then check the sources and the shape of the argument. Only then audit the local details. Many fabrications fail the first two checks, before any algebra.

One kind of check deserves its own name, because AI makes it more important, not less. Before you can say what is wrong with a result, you can often sense that something is off: a convergence rate that seems too fast, an answer with the wrong order of magnitude, a solution that does not fit what the problem should physically do. This sense that a result does not look right, felt before the error is located, is one of the most valuable things a quantitative education builds. AI raises the stakes on it, because a system can produce work that is polished, confident, and wrong in a way that only this intuition catches. It grows the ordinary way: work enough problems by hand that you learn what right looks like, so the wrong answer starts to feel wrong. Reaching for AI first has a cost here that is easy to miss. The work that builds this sense is the work the AI does for you, so a student who leans on AI too early loses the very sense that would tell them the AI is wrong.

AI can also sharpen this sense, if you make it work against your guesses instead of in place of them. Each practice below follows the same rule: commit to your own judgment before the AI answers, then compare.

- *Predict first.* Before you ask, write down what the answer should look like: its sign, its rough size, its units, and how it should behave at the extremes. Then compare your prediction with the AI's answer. When they disagree, find out which one is wrong before you move on.
- *Change one thing.* Ask the AI to vary one quantity in a problem you have already solved, such as doubling a mass, halving a step size, or adding an outlier. Say how the answer should change before it tells you. This builds a feel for how results depend on what goes into them.
- *Hunt for the planted error.* Ask the AI for a worked solution with one deliberate mistake, and tell it not to say where. Find the mistake yourself. Then ask for several short solutions and judge which ones are wrong.
- *Ask for the tests, not the answer.* Ask what any correct answer must satisfy: units, limiting cases, a known special case, a quantity that must be conserved. Apply those tests yourself, to the AI's answer and to your own.
- *Keep score.* Note how often your prediction was right. A sense you never check is only a guess. A sense you check every week becomes judgment.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

If your instructor posts practice solutions with planted errors, or the checks a right answer must pass, start with those. These practices take minutes. They turn each AI answer into practice at judging answers, a skill you need because the AI can be wrong.

That sense has a second direction, and it is the harder one. Sensing that an answer is wrong protects you from a bad result. Asking why a right answer is stated the way it is protects you from a shallow one. Which assumption is doing the work here, and which is only convenience? A statistical test may assume that observations are independent and roughly normal. Normality you can often bend. Independence you cannot ignore: when it fails and you have not accounted for it, the answer is wrong no matter how careful the arithmetic was. An AI will run the test either way and hand back a number that looks the same whether the assumption held or not, which is why the question belongs in your verification. Knowing which is which is the difference between using a method and understanding it.

Those two checks judge answers you already have. Judgment has a third job, upstream of both: choosing. Which problem is worth solving, which framing makes it manageable, which example exposes the idea, which of several correct approaches teaches the most. This generative side is easy to neglect with AI in the room, because a system that hands you two hundred plausible answers has not told you which three matter. Correct is a floor. Worth pursuing is a judgment, and it stays yours.

Use the following questions before submitting AI-assisted work.

Meaning. Does the final work say what I mean?

Accuracy. Are the facts, claims, computations, and sources correct?

Method. Is the method appropriate for the question?

Evidence. Does each major claim have support?

Assumptions. Have I identified the assumptions, and do I know which ones the answer depends on?

Ownership. Can I explain this without reading from the AI response?

A strong probe of ownership is the translation test introduced in section 2.3. Close the AI window, take a key claim or step, and re-present it in a different form: different notation, a different example, a visual representation if the original was symbolic, or a plain-language explanation if the original was formal. If you can move fluently to another mode, you have good evidence the work is yours; if you cannot, return to the underlying material before submitting.

For verification worked through real problems from start to finish, see the study-sessions library in the companion repository, which covers statistics, coding, physics, and mathematics.

4.4.1 ▪ Verification in STEM courses

In STEM courses, verification often means testing the work against structure.

You might check units, signs, limiting cases, special cases, order of magnitude, known formulas, benchmark examples, code output, or experimental data. You might solve a simpler version of the problem. You might compare two independent derivations. You might ask whether the result behaves sensibly when an input becomes very large, very small, or zero.

For code, useful checks include small test cases, boundary cases, reproducibility, comments explaining the logic, a careful read of the code itself rather than only its output, and comparison with a known implementation.

For laboratory work, useful checks include instrument limits, measurement error, procedural consistency, data provenance, and whether the conclusion follows from the actual results.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

For mathematics, statistics, and other quantitative work, a quick verification kit is:

- Check a trivial or special case (for example, $x = 0$, $x = 1$, or a known solution).
- Check a limiting case (large or small parameter, asymptotics).
- Check units, dimensions, and signs for each term.
- Compare to a known formula or theorem statement.
- Run one independent derivation or route (for example, reach the same result by a second method).

These five checks are quick first-pass screens, not proof of correctness; an independent derivation can take much longer. If a result fails one of them, the underlying work needs another look before submission.

4.4.2 ▪ Verification in writing courses

In writing courses, verification often means checking meaning, evidence, and voice.

After AI-assisted revision, compare the draft to your original purpose. Did the thesis change? Did the AI add a claim you did not intend? Did it make your tone too formal, too vague, or too certain? Did it remove an important nuance?

A useful practice is to write a short author's note after revision:

> *In this revision, I changed the structure by moving the evidence earlier. I rejected the AI suggestion to make the claim stronger because the source did not support that level of certainty.*

That note helps you remain the author.

4.4.3 ▪ Verification in humanities courses

In humanities courses, verification often means returning to the text or source.

AI can summarize, but it can also flatten ambiguity. It may miss irony, context, genre, translation issues, or the difference between a primary source and a later interpretation. If you use AI to explore a text, return to the passage itself and ask whether the evidence supports the interpretation.

Do not let AI replace close reading. This is about substitution, not about how you reach a text: a student who uses text-to-speech, translation, or a leveled version to get into a difficult passage is still reading it.

4.4.4 ▪ Verification in social science courses

In social science courses, verification often means examining evidence, method, and causal claims.

Ask whether the data support the conclusion. Ask whether there are confounders. Ask whether a claim is descriptive, causal, predictive, or normative. Ask whether the sample is appropriate. Ask whether the method can answer the question being asked.

AI may help you generate alternative explanations. It cannot decide which explanation the evidence supports unless you check the evidence yourself.

4.4.5 ▪ Verification in professional and applied courses

In professional courses, verification often includes ethics, privacy, standards, and consequences.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

If you are working with cases, clients, patients, designs, policies, or institutional data, ask whether AI use is permitted. Check whether the work includes sensitive information. Confirm that recommendations follow professional standards and course expectations.

A fluent answer can still be unsafe, unfair, or irresponsible.

4.4.6 ▪ Verification in graduate research

Graduate research raises the verification bar. The standards that apply in coursework, doing the core work yourself and using tools with care, apply with more force in research, where an error travels from a manuscript into its citations and on into the literature. Graduate students who use AI in their research should treat verification as part of how the research gets done, not as a last check before submission. The full protocol appears in the scientific-writing appendix; this subsection summarizes the core practices.

A graduate researcher should be able to defend every result, citation, computation, and interpretation in their work without reference to AI. The defense need not be unaided in real time, but the understanding behind it must be the researcher's own. If a reviewer asks "why did you choose this approach," the answer should be a scientific judgment, not a procedure the tool chose. Three workflow practices support this standard.

Independent attempt before tool use. Begin every nontrivial research task with the researcher's own attempt: a proof outline, a model setup, a literature shortlist, a code sketch. The independent attempt is what AI assistance refines. If AI sees the problem first, the researcher's framing risks becoming AI's framing; a published paper that originated this way can show it in subtle inconsistencies.

Citation discipline. Every cited source must exist, must be read, and must support the claim being made. AI tools can produce citations that look correct but reference the wrong year, the wrong journal, or the wrong author combination, a failure mode documented empirically for AI-generated bibliographies [131]. Verification against MathSciNet, Web of Science, the publisher's own record, or the journal page itself is non-negotiable. Dialogue 3 in section 4.2 illustrates exactly this kind of catch.

Audit-and-repair. For mathematical and computational work, AI-produced derivations and code should be treated as drafts to audit, not as results to accept. The audit identifies where each step uses each hypothesis, what edge cases are excluded, and where the derivation could fail. Analogous practices exist in computational science (input validation, boundary tests, comparison against analytical solutions) and in empirical work (sensitivity to parameter choices, robustness across specifications).

These practices are research, in an environment where AI-fluent text can pass shallow inspection. Researchers who internalize them are better placed to publish carefully, to defend their work under questioning, and to contribute to a literature that remains trustworthy as the volume of AI-assisted scholarship grows.

4.5 ▪ Disclosure

Disclosure is not a confession. It is a record of process.

When an instructor permits AI use and asks for disclosure, your goal is to be factual and clear. State what tool you used, how you used it, what it contributed, what you changed, and how you verified the result. Templates you can adapt are at `students/disclosure-templates.md`.

4.5.1 ▪ A short disclosure template

Use this when AI played a small role.

> **AI disclosure:** I used an approved AI tool to generate practice questions, check grammar, or suggest revision questions. I reviewed the output and made the final choices myself.

4.5.2 ▪ A fuller disclosure template

Use this when AI shaped the process more substantially.

> **Tools used:** Name the tools used, if any.
> **Tool role:** Describe what the tools helped with.
> **Student role:** Describe what you wrote, revised, checked, or decided.
> **Accepted, changed, or rejected:** Describe the AI contributions you used, revised, or set aside.
> **Independent verification:** Describe how you checked important claims, sources, data, code, or reasoning.
> **What I can explain without AI:** Name the main parts of the work you can now explain on your own.

4.5.3 ▪ What good disclosure does

Good disclosure helps your instructor see your thinking. It also protects you from two mistakes: pretending AI was not used when it was, and giving AI more credit than it deserves.

The level of disclosure should match the assignment. Minor grammar help may need only a brief note. A project that used AI for brainstorming, code, revision, and source checking needs more detail.

4.5.4 ▪ Sharing the conversation

When an instructor asks to see the AI conversation that produced your work, sharing the chat link or transcript is, as of this writing, a routine option. Most major platforms (ChatGPT, Claude, Gemini, Copilot, and others) support a share function that produces a read-only link to the conversation. The companion repository's `students/sharing-ai-conversations.md` guide describes the operational steps for each major platform.

Sharing the conversation is among the most informative forms of disclosure available to a student. It is a record of the interaction, not proof of authorship. A written disclosure template, however careful, depends on your account of what the AI contributed. Sharing the conversation lets your instructor see what was actually asked and what was actually returned. The gap between your account and the underlying interaction narrows. For an instructor's purposes, this is sometimes the difference between trusting the work and having to ask further questions.

This approach has two limits. First, sharing the conversation link presumes the shared conversation is the one that actually produced the work. A student who runs a polished AI study session for show while drafting the actual work in a separate, unshared chat has not solved the disclosure problem; they have only relocated it. The translation test in section 2.3 is the structural safeguard that narrows this gap, because the test asks what the student can do regardless of which thread produced what. Second, sharing a conversation link must respect privacy, and that covers anything visible in a shared thread, including identifying information about classmates, instructors, or third parties. Edit or redact before sharing if any such material appears, and ask your instructor for an alternative if redaction would distort the record. The same applies to your own disclosures: if a thread contains course frustrations, learning struggles, or off-topic personal material you would rather not hand over, review and redact it, or keep gradable work in a separate, clean thread

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

from the start. This is the opposite of the case above, not a version of it. Here the clean thread is where the work actually happens, and it is the one you share. It is not a tidy stand-in for a thread you keep back.

4.6 ▪ Privacy and data protection

Before uploading material to an AI system, ask whether you have the right to share it. Many AI tools are run by outside companies. Some retain data. Some may use inputs to improve services. Some institutional tools may have stronger privacy protections, but you should not assume that every tool is safe for every kind of information.

Unless your instructor or institution has clearly approved the tool for that use, do not upload:

- private information about classmates;
- unpublished research data without permission;
- your own unpublished work, if you may want to claim the idea later; what you upload leaves your hands, and a novel construction or result is worth protecting for the same reason a colleague's is;
- patient, client, or human-subject information;
- restricted exam questions or solutions;
- confidential institutional documents;
- private correspondence;
- recommendation letters or evaluations;
- copyrighted course material if your instructor has not permitted it;
- personal data that could identify someone else.

If you are unsure, do not upload the material. Ask your instructor how to proceed.

Privacy is not only a legal concern. It is part of academic trust. Ethical guidance for educators on AI and data in teaching stresses care, transparency, and attention to risk in handling data systems [35].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 5

In Class and Across Disciplines

5.1 ▪ Small classes and the Oral-Defense Model

An **oral defense** is a live conversation or examination in which a student explains submitted work and answers questions about it. Small classes often allow more flexible AI use because the instructor can ask students directly about their work, as in seminars, capstones, advanced labs, writing workshops, and many graduate courses. The defining feature is class size, not course level: a small introductory seminar uses the same Oral-Defense pattern as a small graduate seminar, even though the assignment-category mix and AI autonomy granted are likely to differ.

In these courses, you may be asked to defend AI-assisted work in conversation. That defense may look different across fields.

STEM. Explain a derivation, model, computation, experiment, or code segment.

Writing. Explain your thesis, structure, evidence, and revision choices.

Humanities. Defend an interpretation by returning to the text or source.

Social sciences. Explain your method, evidence, assumptions, and limits.

Professional programs. Justify a recommendation, privacy choice, or ethical judgment.

Creative work. Explain process, iteration, constraint, and authorship.

The main standard is the same:

> **If you submitted it, you should be able to explain it.**

You can use AI to prepare for this kind of questioning. Ask it to play the role of a skeptical instructor. Ask it to question your assumptions. Ask it to find the weakest part of your explanation. Then practice answering without looking at the AI response.

5.2 ▪ Large classes and the Evidence-of-Thinking Model

Large classes cannot rely on one-on-one oral exams for every student. Instead, they often use scalable **evidence of thinking**: process records that show how a student reasoned, revised, verified, explained, or made choices. Most large classes are introductory or service courses simply because that is where enrollment concentrates, but the model is class-size-driven, not level-driven: a large advanced course uses the same evidence-of-thinking patterns.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

You may be asked to complete short no-AI quizzes, write in class, submit a draft history, explain one step from your homework, critique a flawed AI answer, provide a process log, verify sources, record a short explanation, or participate in a random oral audit.

These practices are meant to protect learning. In a world where AI can produce polished answers, instructors need ways to see the work behind the answer.

A common pattern is the homework-to-quiz check. You complete AI-allowed homework, then answer a short no-AI question about the same idea in class. This checks whether the homework became part of your understanding.

Another pattern is the audit-and-repair task. You receive an answer that may contain a subtle error. Your job is to find the error, explain why it matters, and repair it. This is one of the most useful AI-era skills because it rewards judgment over fluency.

The simplest pattern of all is to change what the grade rests on. Many experienced instructors already do this without calling it a strategy: they make homework a small share of the grade, often only a few percent, and place the real weight on in-person, closed-book assessment. When the grade depends on what a student can do in a proctored setting, it matters far less how they studied, because AI cannot sit the exam for them. Homework becomes practice, and AI-assisted practice can be good practice. This model verifies nothing about the homework itself, which is the point: it removes the incentive to hide AI use rather than trying to detect it. Courses with a strong in-person exam tradition may find this the least disruptive path of all, and it solves the AI-on-homework problem with almost no added labor. The trade-off is that it concentrates the assessment into a few high-stakes moments, which suits some subjects better than others and some students less well than others. Where a course can carry that trade-off, this is often the least effort for the most protection.

Many courses fall between the small-class and large-class extremes. They are too large for individual oral defenses on every assignment, but small enough for some direct contact, or arranged so that lab time supports oral defense even at medium enrollment. Such courses can combine elements of both, in what this book calls the Hybrid Model. As a student in a hybrid course, your experience may include occasional oral conferences, often at the start of the term for calibration and again at major project milestones. Alongside them, you may keep continuous process documentation through Plan-Use-Verify records on regular assignments. The expectations are the same: explain what you submitted, verify what you used, own the work.

5.3 ▪ Asking better questions in mathematics and related STEM disciplines

A useful AI question keeps the central thinking with you (section 3.2.3). In mathematics that principle takes two forms within one spiral. For computational work (calculus, numerical methods, modeling), strong prompts ask AI to check, debug, and probe rather than to solve. For proof-based work (real analysis, abstract algebra, topology), strong prompts ask AI to critique, generate examples, and identify gaps rather than to write the proof. Appendix B collects the mathematical patterns from this Part and the next in one place.

For computational mathematics:

Weak prompt:

> *Solve this problem.*

Stronger prompt:

> *Ask me one question that would help me decide which method applies.*

For proof-based mathematics, the same pattern looks like this. The strong prompts preserve the student's role as the proof-builder.

Weak prompt:

> *Explain compactness.*

Stronger prompt:

> *Give me one example, one non-example, and one common false belief about compactness.*

Weak prompt:

> *Prove this theorem for me.*

Stronger prompt:

> *Here is my proof attempt. Identify the first point where the logic may fail, but do not rewrite the proof.*

Weak prompt:

> *Check my proof.*

Stronger prompt:

> *Identify whether each hypothesis in the theorem is actually used in the proof. If any hypothesis is unused, ask whether the theorem can be strengthened by removing or weakening that assumption.*

These prompts train the use of AI as a mathematical coach rather than a solution engine.

The same two-mode split appears in related quantitative and computational STEM disciplines. In physics, the modes are calculation versus principle-identification: weak prompts ask AI to solve the problem; stronger prompts ask AI which principle or conservation law should apply before the algebra begins, or whether a result reduces correctly to a known limiting case. In computer science, the modes are coding versus algorithm design: weak prompts ask AI to write the code; stronger prompts ask AI to identify edge cases or to critique the proposed algorithm before any code is generated. In statistics, the modes are computation versus methodological justification: weak prompts ask AI to run the analysis; stronger prompts ask AI whether the test's assumptions are met or what would change if a key assumption were violated. The pattern holds: the stronger prompt always keeps the field-specific judgment with the student.

A deeper pattern sits under all of these prompts. Cognitive psychologists have spent decades testing which study techniques improve learning. Practice testing (quizzing yourself), distributed practice (spreading study over time), elaborative interrogation (asking why a fact is true), and self-explanation (stating in your own words how a new idea relates to what you already know) all show measurable benefit. Highlighting, rereading, and summarizing show much weaker benefit [32].

AI tools are study-technique amplifiers. Whatever technique you ask AI to support, AI will support it efficiently. Asked to summarize a reading you have not yet engaged with, or to highlight what is important, AI amplifies a technique that has limited learning value. Asked to test you on material you have studied, or to ask you why something is true, AI amplifies a technique that has strong learning value. The prompt determines which amplification you receive. This is why weak prompts tend to ask AI to do the low-utility work for you, while stronger prompts tend to keep the high-utility work yours.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

5.4 ▪ Asking better questions in the humanities, social sciences, and professional fields

Careful AI use pays off outside the quantitative disciplines too, in fields where the work is interpretation, empirical inference, or applied judgment rather than calculation or proof. The spiral does not change, but what counts as a good question and as a real verification differs by field.

5.4.1 ▪ Interpretive disciplines

In literature, history, philosophy, and other interpretive fields, the work is an argument about meaning supported by evidence from texts. AI is most useful as a source of context and as a sparring partner for an interpretation you have already drafted. It is least safe when it supplies the interpretation itself. The reason is that the interpretation is the thing being assessed: a reading produced by AI tells your instructor nothing about whether you can read.

A productive pattern is to draft your own reading first, then ask AI to argue against it. Ask for the strongest objection to your thesis, for a passage that complicates it, or for the historical context a first-time reader would miss. Treat each answer as a lead to check rather than a conclusion to adopt. The characteristic verification move is return to the source: every contextual claim is checked against the assigned text or a cited document, and any detail the sources do not support is dropped. A claim about what an author intended, what a period believed, or what a term meant is only as good as the passage you can point to.

The failure mode to avoid is fluent paraphrase. AI can produce a confident, well-organized reading of almost any canonical text, and a confident reading is easy to mistake for a correct one. The discipline is to keep the argument yours and let the text, not the model's fluency, decide what the evidence will bear.

5.4.2 ▪ Empirical social science

In economics, sociology, political science, psychology, and the other empirical social sciences, the work is an inference about people or institutions drawn from data. The central difficulty is that correlation is easy to find and causation is hard to defend. AI is most useful for widening the set of explanations you consider and for stress-testing a design you have already sketched. It is least safe when it is allowed to draw the conclusion: a plausible-sounding causal story is exactly what the method is meant to discipline.

A productive pattern is to specify your question, your data, and your candidate explanation first, then ask AI what you might be missing: which confounders you have not measured, which alternative specification would change the result, which population the finding may not generalize to. Sort each suggestion into those you can test in your data and those that remain speculative. The characteristic verification move is to run the check: where a suggested confounder is measured, see whether including it moves the estimate; where it is not, record it as a limitation rather than a result. A model's confidence about an effect is not evidence; your data, your design, and the size of your sample are.

The failure mode to avoid is borrowed certainty. AI will narrate a clean causal mechanism for almost any correlation, and a clean mechanism is persuasive in a way raw association is not. The discipline is to keep the claim no stronger than the design supports, and to let the evidence, not the fluency of the story, set the limit.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

5.4.3 ▪ Professional and applied fields

In nursing, law, business, education, and other professional programs, the work is applied judgment under real constraints, measured against authoritative standards: clinical guidelines, statutes and case law, accounting rules, professional codes. AI is most useful for framing a case, surfacing considerations, and rehearsing the reasoning you will later have to perform unaided. It is least safe when it is treated as the authority. In these fields the authority is the standard, and an answer that cannot be traced to it is not yet professional work.

A productive pattern is to reason through the case yourself first, then ask AI to challenge it: what a more cautious practitioner would do differently, which fact would change the recommendation, which guideline or rule bears on the decision. The characteristic verification move is to check against the source of authority: the current clinical guideline, the controlling statute, the applicable standard, not the model's summary of it. The summary can be outdated or wrong on exactly the detail that matters. A confident answer that conflicts with the governing standard is a hazard, not a help.

The failure mode to avoid is the confident wrong answer. AI can produce advice that reads like competent professional work while being wrong in a way only a trained eye catches, and in applied fields a wrong answer delivered confidently is the most dangerous kind. The discipline is to let the standard, and eventually your own examined judgment, decide, and to remember that the point of the training is to become the person who can tell when the confident answer is wrong.

5.5 ▪ Discipline examples

Five short vignettes follow, each showing the whole Learning Spiral at work in a different field. Each vignette is a compressed version of an honest week in a course. Your instructor's policy always comes first.

5.5.1 ▪ Physics example

A physics student starts a problem on energy conservation. *Try to solve*: she sketches a free-body diagram, writes down the conservation equations she remembers, and gets stuck on which surface friction does work. *Ask*: she asks AI for one hint about how to identify the correct system boundary, not for the solution. *Work*: with the hint, she completes the algebra and arrives at a numerical answer. *Verify*: she checks units, tests the limiting case where friction is zero against the textbook formula, and confirms the answer's order of magnitude. *Reflect and disclose*: she records that AI helped with the system-boundary identification; the rest is hers, and she can explain the steps without notes.

The reasoning stayed with her: AI helped her enter the problem without doing it for her.

Mathematics splits the same way, between proof-based and computational work, and section 5.3 works through the prompt patterns for both. The difference is in what you verify. Proof-based work turns on choosing the right argument and checking that every hypothesis is used, so AI earns its place as a critic of proof attempts and a source of examples and non-examples, meaning cases that fail the definition and show where its boundary lies. Computational work turns on choosing the right method and checking the result, so AI earns its place as a debugger and a checker of intermediate steps. The two modes share this spiral; only the verification moves are tuned differently.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

5.5.2 ▪ Programming example

A student writes a data-cleaning function. *Try to solve*: he runs the code and observes that the output looks wrong on rows with missing values. *Ask*: he asks AI to suggest diagnostic tests rather than rewriting the function. *Work*: he builds three small test cases the AI suggested, plus one he thought of himself. *Verify*: the tests reveal that missing values are passed through as zeros rather than handled separately; he revises the function and reruns the tests until they pass. *Reflect and disclose*: his submission includes the tests as part of the work, with a note that AI suggested the diagnostic-test approach; the analysis of test outcomes and the fix are his.

5.5.3 ▪ Writing example

A student is drafting an essay on labor history. *Try to solve*: she writes a full draft from her notes and the assigned readings, identifying the paragraph she's least confident about. *Ask*: she asks AI to identify where the argument loses focus, without rewriting it, and it flags her third paragraph. *Work*: she rewrites the paragraph in her own words. *Verify*: she returns to the source she cited there, finds her evidence is thinner than her claim, and narrows the claim to what the evidence supports. *Reflect and disclose*: her disclosure block notes that AI helped with diagnostic critique on one paragraph; the thesis, evidence, and final wording are hers.

She remains the writer, and the evidence, not the model, controls the claim.

5.5.4 ▪ Social science example

A student is analyzing a study about education and income. *Try to solve*: she works through the dataset codebook and lists the variables she thinks matter. *Ask*: she asks AI to suggest possible confounders she may have missed. *Work*: AI suggests six; she classifies them into those measured in the dataset and those that are speculative. *Verify*: for each measured confounder, she runs a quick check on whether including it changes the coefficient of interest; for the speculative ones, she notes them as limitations. *Reflect and disclose*: her writeup separates evidence-based statements from acknowledged speculation, with a disclosure block noting AI's role in confounder generation.

AI widened the range of hypotheses she considered, and the data decided which one survived.

5.5.5 ▪ Professional fields example

A nursing student is working through a patient-care scenario. *Try to solve*: she reads the case and drafts a care plan from the course material, noting the one intervention she is unsure about. *Ask*: she asks AI to explain the general reasoning behind that class of intervention, not to choose the intervention for her. *Work*: with the explanation, she returns to the case and tests whether her plan fits the patient's specific contraindications. *Verify*: she checks AI's account against the current clinical guideline assigned in the course and corrects one detail the guideline stated differently. *Reflect and disclose*: her disclosure notes that AI explained the general mechanism; the patient-specific judgment and the guideline check are hers.

The clinical standard, not the model, remains the authority.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 6
Judgment and Limits

6.1 ▪ When not to use AI

Sometimes the best use of AI is not to use it.

Do not use AI when the assignment forbids it. Do not use AI when the task is designed to build fluency through practice. Do not use AI when it would expose private information. Do not use AI when you are too tired to check the result. Do not use AI when using it would make you skip the part of the work you most need to learn.

This last case is common. If you are weak at algebra, do not use AI to bypass algebra practice. If you struggle to write introductions, do not ask AI to write every introduction. If you are learning to read difficult texts, do not let AI summarize every text before you have faced it yourself.

Use AI where it strengthens your learning. Avoid it where it removes the struggle that the course is trying to teach you how to handle.

It helps to notice your own pattern. Every so often, look back over a day of coursework and sort your AI use into two kinds. In the first, you went to the tool with a purpose: a specific question, a draft to test, a step you were stuck on. In the second, you reached for it out of habit, took what it gave you, and moved on. The first is you using the tool. The second is the tool using you. You do not have to be all of one kind. But if the second is most of your day, you are learning less than you think. The fix is in your hands: before reaching for AI, ask what you are trying to do, and whether you could do it yourself.

One case deserves its own mention. Some learning comes from other people. A classmate explains it a different way. Office hours catch the thing you could not name. A study group works through the hard part together. AI is there every hour and never makes you feel slow, so it can crowd these out. A good AI tutor beats no help, and sometimes beats a rushed one. But if reaching for it means you stop talking to the people around you about the work, you have given up something you will miss. Use AI to prepare for those conversations. Do not use it to avoid them.

Seek out one kind of conversation on purpose. Find someone who will argue with you. A classmate who pokes holes in your reasoning, who asks why your method works and not just whether the answer matches, makes you defend your thinking out loud. An agreeable tutor rarely does that, and an AI built to be helpful does it even less. The student who learns to find the weak step in a friend's argument is the one who catches the confident mistake in a chatbot's. Practice that with people. It is how you build the judgment you need for the times the machine is the only one in the room.

If a question arises about whether work is yours, process evidence matters. AI detection tools are not perfect, and research has found that GPT detectors can misclassify non-native English

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

writing as AI-generated [79]. Your best protection is honest disclosure, draft history, verification notes, and the ability to explain your work.

6.1.1 ▪ The choice not to use AI at all

Some students choose not to use AI for any coursework, on grounds beyond what specific assignments require. The reasons vary: environmental concerns about the energy cost of large-model inference, labor concerns about how training data was sourced and how AI is reshaping professional work, ethical concerns about complicity in harms the technology produces. From some fields, especially in the humanities, the social sciences, and the arts, the position is sometimes stated more strongly: there is no ethical use of AI in their discipline at the present moment.

This guide is not written to talk those students out of their position. The framework still applies, but the application is different. Every assignment is implicitly **NAI** for a student who has chosen not to use AI; the Learning Spiral becomes the practice they were going to use anyway; the Student's Trilemma names the predicament they are still asked to navigate when peers are using AI freely. What the framework offers is vocabulary for the situations they will find themselves in: grading curves shaped by tools they do not use, group projects where teammates do, courses where the instructor assumes AI use without saying so. Naming the patterns helps a student who has chosen to abstain navigate a coursework environment being reshaped around them.

6.2 ▪ What to do when you are unsure

AI policies vary by course. They may even vary by assignment within the same course. If you are unsure, ask your instructor before using AI in a way that affects submitted work.

A good question is specific:

> *For this assignment, may I use AI to generate practice questions or critique my draft if I disclose that use?*

Another useful question is:

> *May I use AI to debug code if I include a short note explaining what I tested and what I changed?*

Avoid vague questions such as "Can I use AI?" That question is often too broad. Ask about the role you want AI to play.

If you already used AI and are unsure whether the use was allowed, tell your instructor early. It is usually easier to correct a misunderstanding before submission than to explain it afterward.

If you rely on AI as part of a documented disability accommodation, an assignment's AI-use rule and your accommodation can appear to conflict. This happens, for instance, when an assignment restricts AI but an approved assistive tool depends on it. You do not have to resolve this on your own. Raise it early with your instructor and your campus disability or accessibility office, who can confirm how the accommodation and the assignment's rules fit together. Disclosing AI use does not require revealing the accommodation behind it; that is arranged privately with the instructor and the office.

6.3 ▪ Working with AI on group assignments

Group and team assignments raise a question the rest of this guide answers only for individual work: how AI use is handled when the product is shared. The word collaboration needs pinning down first. Here it means working with your classmates, not the AI-as-collaborator role of the

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

AIC category. A group project can use AI under any of the four categories, and the team needs to know which one applies before it starts.

The standards you already know apply at the level of the group. Agree early on how AI may be used, and on which parts of the work, so that no member is surprised later. Keep your own verification notes for the portions you contribute, and be ready to explain them, because the translation test in section 2.3 does not stop applying when the work is shared. When the assignment asks for disclosure, the group discloses together: what the team used AI for, what each member verified, and what the group changed or set aside. A single joint disclosure is usually clearer than separate notes that do not line up.

When group work ends in a presentation, the room reads the result; it cannot read what your group did to get there. Lead with the moment the tool handed you something plausible you did not accept, and show what made you stop. Put the output beside what you changed. A group that checked nothing has nothing to show. This is the compact performed aloud: work you can explain, verification you can show, and responsibility you take in front of people who can ask.

The quieter risk in group work is the one this guide has warned about throughout, sharpened by the division of labor. It is easy to let a teammate, or the AI, take the part you most need to learn, and to finish the project having practiced only what you were already good at. Treat your share of a group assignment the way you would treat your own work: try the hard part first, and use AI for help rather than replacement. Make sure that when the project is done you can explain more than your own slice of it.

6.4 ▪ Evidence base and limits of the guide

This guide draws on three kinds of evidence: learning science, emerging research on AI-supported tutoring, and institutional guidance on responsible AI use. The learning-science foundation is mature enough to support strong claims for active retrieval and spaced practice, and more qualified ones for self-explanation and elaborative interrogation; all four ask the learner to do cognitive work [32].

Research on computer-assisted learning did not begin with generative tools. Intelligent tutoring systems and educational software have been studied for decades, and what changed is not that the questions are new but that a general-purpose tool reached every student at once, outside any course design.

The AI-specific evidence is newer and still heterogeneous, but a clear pattern is emerging: AI helps most when it is embedded in active learning, verification, feedback, and independent reconstruction. Across methodologies and disciplines, the results so far point the same way: the design of AI use determines whether learning improves, stays flat, or declines. A pre-registered randomized trial by Bastani and colleagues gave nearly a thousand high-school mathematics students one of two tools. An unrestricted ChatGPT-style interface raised practice performance by 48% but cut exam performance by 17% against no-AI controls. A guardrailed AI tutor raised practice performance by 127% with no exam-performance loss, and no gain either [11]. Those figures come from three randomly assigned groups in one secondary-mathematics setting, each measured against the no-AI control. Read them as evidence for the pattern rather than as effect sizes that will reproduce elsewhere. A randomized controlled trial by Kestin and colleagues in an introductory physics course at Harvard found that a carefully designed AI tutor produced learning gains over double those of in-class active learning. The tutor was engineered around seven pedagogical practices: active engagement, cognitive-load management, growth mindset, scaffolding, feedback accuracy, timely targeted feedback, and self-pacing. The system prompt handled the first three; the platform's structured walkthrough handled scaffolding; accuracy came from the worked solutions the designers loaded in advance. Only the last two, timely targeted feedback and self-pacing, come from the medium itself [61]. Recent meta-analyses find a positive

overall effect on academic achievement [120] and on higher-order thinking, strongest for problem solving and weakest for creativity [139]. Those averages and the Bastani result are not in conflict. An average pools designs that guard the student's own thinking with designs that replace it. The pooled number hides the difference that matters. That is the finding: not that AI helps or hurts, but that the design decides which. UNESCO pairs its AI competency framework for students with one for teachers, developed together, so AI competence is not treated as a student skill alone [88, 90]; the OECD's 2026 Digital Education Outlook reaches the same design-decides conclusion at the international policy level [98].

A 2026 randomized study by Shen and Tamkin points the same way from outside the classroom. Developers learning an unfamiliar programming library with AI assistance scored lower on conceptual understanding, code reading, and debugging than those who worked without it, and they did not finish meaningfully faster. The same study sorted how participants actually used the assistant and found patterns that preserved learning, each of them keeping the person engaged with the code rather than delegating it [112]. These were working developers rather than students, so the result converges with the classroom evidence rather than repeating it.

How unrestricted AI weakens learning is not settled, but several candidate mechanisms recur. Researchers describe them with overlapping names: cognitive offloading (skipping the diagnostic and evaluative steps of help-seeking), metacognitive laziness (reduced self-monitoring during AI-assisted tasks), and the crutch pattern (asking AI for solutions rather than for help). A preprint EEG study suggests reduced neural connectivity when students start with AI rather than with their own attempt [71]. The Bastani study adds a metacognitive finding that matters here: students often do not perceive that their learning has been reduced, even when objective assessment shows it has. A 2026 series of randomized trials by Liu and colleagues points to a further mechanism. After only brief AI use, people not only performed worse once the tool was removed but also gave up sooner, which suggests that ready answers can erode persistence itself, the willingness to work through difficulty that learning depends on [81]. The study was accepted after peer review at the 2026 Conference on Language Modeling. Its estimates describe brief AI use rather than a full course. The guide's emphasis on trying first, verifying, reflecting, and explaining is designed to address each of these mechanisms in turn.

For that reason, this guide avoids a simple pro-AI or anti-AI position. It treats AI as useful when it keeps you active and risky when it lets you skip the work the course is meant to teach. The best test is practical: after using AI, can you explain more, verify more, and do more on your own than you could before?

6.5 ▪ What you can expect from your institution

The commitments in this book run both ways: the book asks institutions to make them, and you can reasonably expect them. Each one is visible when it is real. Every course states its AI rules in the syllabus, before work is due, not after a problem. AI literacy is taught, not assumed: look for it as a named outcome in a course, a workshop, or an orientation. Tools recommended for coursework protect your data and are available to every student, not only those who can pay. And no one accuses you on a detector score alone. If any of these is missing, asking for it is reasonable, and your student government can ask louder than you can.

6.6 ▪ A final word

AI can make schoolwork faster. That is not always good. Some parts of learning are slow because they require attention, memory, judgment, and practice, and working through them is what builds the capacity you came for. The goal is to use help wisely.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

A student who uses AI well does not hide behind it. They use it to ask better questions, test their thinking, find weak places, and improve their work. They know what the tool contributed. They know what they checked. They know what they still do not understand.

That is the standard this guide asks you to meet.

It is also the standard the author tried to meet in writing this book; Appendix C describes how AI was used here, and where it was resisted.

Wendy, from the opening pages, read her incentives correctly. This guide's answer is that you can keep the grade, your honesty, and your understanding together, and these practices show how.

Use AI when it helps you think. Stop when it starts thinking for you.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 7

The Student's Toolkit

This chapter collects the study routine, the templates, and the checklist from the six chapters before it. It is for the student with a question at eleven at night. The reasoning behind each piece is in those chapters. The companion repository carries a printable copy of the checklist and worked versions of the routine, and their paths are printed where they appear.

7.1 ▪ One assignment, start to finish

The Learning Spiral (section 3.2) explains why this order works. Worked versions of it, in statistics, programming, physics, and mathematics, are at `students/study-sessions/`.

1. *Before class.* Skim the assigned material and write down two questions. If you use AI, ask it to explain background, not to replace the reading.
2. *After class.* Close your notes and write a short summary from memory. Then get the notes into your course project: download your instructor's posted notes, or photograph or scan your own, have AI type them up, and check the equations against the page. Have AI quiz you on the notes one question at a time, answering before you see the feedback.
3. *Starting the work.* Attempt each problem, or draft each paragraph, before you open the tool. When you are stuck, ask for a hint or a simpler related example, not the answer.
4. *Reconcile.* When AI explains something you have already attempted, compare the two accounts. A symbol used differently is often a concept understood differently.
5. *Check.* Ask AI where the gaps are, then verify the final answer yourself. Close the AI window before you write the final version.
6. *Before you submit.* Work one nearby problem without AI. If you cannot explain a step in your own words, do not submit yet.
7. *Before an exam or oral defense.* Have AI question you, using the kinds of questions your instructor asks if they are posted, then a partner who can interrupt. Practice answering without notes, and stay on the places where you hesitate.

Write the step you keep skipping as a when-then plan: when I finish AI-assisted homework, then I work one nearby problem without AI. The form turns an intention into a trigger and an action, and it improves follow-through [45].

In a group, three things change. Each member attempts the problem alone first, and the group brings the AI the disagreement between those attempts, not a blank page. Whoever did not run

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

the prompt checks the result. Disclosure is joint, so keep the record as you go. A group can build understanding faster than one student can. It can also hide a lack of understanding better. The independent attempts decide which of these happens.

7.2 ▪ Template: AI account configuration

Use this six-category template to set persistent preferences in your AI platform (sometimes called custom instructions, system prompts, or memory). Copy them, adapt the bracketed parts to your situation, and paste the whole thing into your platform's preferences panel. The motivation for configuring your account at all is given in section 1.2; the categories below are the operational template.

1. **Who you are and what you are studying.**

   > *I am a [year] [undergraduate/graduate] student in [department] at [university]. I am currently taking [course names this semester]. My academic background includes [prior coursework relevant to current work]. I am using AI to support my learning, not to substitute for it.*

   This single paragraph gives the AI enough context to pitch explanations to your level and your courses. Update it each semester as your courses change.

2. **How you want explanations pitched.**

   > *Explain reasoning step by step. When I ask you to solve a problem, walk me through the approach before giving the answer; do not just give me the final answer when I am working through something I should be learning. If a concept builds on prerequisites I have not told you about, ask me what I already know rather than assuming.*

   The last sentence is the most important. It teaches the AI to ask, which prevents two familiar problems: explanations pitched too high, where you cannot follow the reasoning, and explanations pitched too low, where the AI assumes ignorance you do not have. Both waste your time.

3. **How you want uncertainty handled.**

   > *When you are uncertain about a fact, say so explicitly. Do not invent citations, sources, or specific numbers if you do not know them. If a question is at the edge of your knowledge, tell me, so I can verify in textbooks or other sources rather than trusting an answer that might be wrong. For mathematical and technical work, also tell me when my question left out a condition the answer depends on, and tell me when what you return is a numerical approximation rather than an exact value.*

   AI systems can produce confident-sounding wrong answers, especially for specific facts (dates, citations, equations from particular papers, statistics). Asking the AI to flag its own uncertainty may help, but it does not remove the need to verify important claims yourself. Nothing in this template makes the AI correct. Every category here changes how it responds; none of them changes what it knows, and a configured AI is as confident when it is wrong as when it is right.

   The template's last instruction covers a failure the rest of this category does not reach. Everything above assumes the AI does not know something. The harder case is the one

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

where it does. Ask for the limit of $\sin x/x$ as x approaches zero and you will get 1, which is correct, and you will not be told that the question took for granted that the limit exists. Ask for the value of $\int_0^1 \sqrt{1+x^4}\,dx$ and you will get a decimal, which is also correct, and you will not be told that the integral has no answer in elementary functions, so the decimal is an approximation and not the value. In both cases the AI is confident and right, and in both cases it has answered a slightly different question than the one you asked. The lesson is that asking AI to flag what it does not know will not catch the times it silently narrows your question, so ask it to name the conditions an answer depends on and to say when a number is approximate.

4. **Writing style preferences.**

   I write in [American English / British English / English as a second language]. [If English is not your first language: please flag passages where my phrasing might read as unclear or non-idiomatic, and suggest alternatives.] I prefer prose that is [clear and direct / formal / conversational].

   The English-as-a-second-language clause matters more than it may seem. Detector tools used by some journals and some courses flag non-native-English writing as AI-generated at higher rates [79], an issue covered in Appendix A. If English is not your first language, telling the AI so, and asking it to help with idiomatic clarity, both improves the writing and gives you a record of what you wrote and what the AI suggested, which protects you if your work is questioned.

5. **Scope of use and integrity boundary.**

   I will use AI for [studying, brainstorming, debugging, getting unstuck, checking my own work]. I will not use AI to [write assignments I am supposed to do myself, generate code I do not understand, produce text I will submit as my own work without disclosure]. When my course or assignment policy says no AI, I will not use AI for that work, regardless of what is in this configuration.

   This is partly a reminder to yourself. The configuration does not override your course's rules; the assignment categories described in section 3.4 explain when AI is permitted, encouraged, restricted, or forbidden, and the assignment-level rules always take precedence over your account-level configuration. Putting the boundary in writing where you see it every time you open an AI conversation helps keep the lines clear.

6. **Optional: technical-work accuracy guard.**

   For mathematical, scientific, or technical answers: show your work. For any claim that rests on a specific fact, citation, or number, flag it as one I should verify and name the source or calculation I can check it against. If I ask for code, explain what each non-obvious line does.

   This is optional because it adds complexity to every answer. For students in mathematics, the sciences, and engineering, it is one of the most useful additions you can make. Naming what to check, and against what, trains you to read AI output critically, and the show-your-work instruction makes the verification practices in section 3.2 easier because the work is already laid out.

A living document. The starter template above captures the categories that experienced academic users have found matter most. As you use AI more, you will discover specific failure modes in your own work and want to add specific instructions to prevent them. This is normal

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

and expected. Treat your AI preferences as a living document. Periodically review what is there, prune what no longer applies, and add the instructions that hard experience has taught you to include. After two or three semesters of use, you will have a personalized configuration that reflects your work style and your discipline's particular demands. The point of starting with the elements above is to give you the structure those experienced users converge on, so you can grow your own configuration on a sound foundation rather than discovering each lesson by repeated failure. Keep this one document across your whole degree rather than starting fresh each term. The failure modes you find are yours, not any single course's, and the record of them is worth more in your final year than in your first. Some students also keep a short second file per course, holding the notation and conventions that course expects; that is optional, and it works only if the cumulative document stays the one you actually maintain.

Where to learn the platform-specific details. The instructions for setting persistent preferences vary by AI platform, and the user interfaces change often enough that any specific click-by-click guide written today will be partly outdated within a year. For current instructions, consult the documentation page for whichever platform you use; major platforms (ChatGPT, Claude, Gemini, Copilot, and others) all maintain documentation on system prompts, custom instructions, or memory configuration. If your university's library, IT services, or teaching center has a guide for academic AI use at your institution, start there. Institutional guides usually take account of any university-licensed AI tools and the specific privacy considerations for student use.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

7.3 ▪ Template: prompt library for learning

For understanding

Explain this idea in three levels: simple intuition, course-level explanation, and one technical detail I should not miss.

Give me an example, a non-example, and a borderline case.

Ask me questions that reveal whether I understand this concept.

For problem solving

Give me one hint about the main obstacle. Do not solve the problem.

What assumptions should I check before choosing a method?

What is a simpler related problem I could solve first?

For writing

Here is my draft. Identify the paragraph where the argument is least clear. Do not rewrite it. Ask me revision questions.

What claim in this paragraph needs stronger evidence?

Does this revision change the level of certainty in my original claim?

For coding and computation

Suggest simple test cases for this code. Do not rewrite the function yet.

What edge cases might break this algorithm?

Explain what this error message means and ask me what I have already checked.

For exam preparation

Quiz me one question at a time. Wait for my answer before giving feedback.

Ask me to explain the main concept without using jargon.

Give me a flawed answer and ask me to find the mistake.

7.4 ▪ Template: verification checklist

The work used to be execution: carry out the method, get the answer, check the arithmetic. Execution is now cheap. What stays expensive is saying exactly what you want, then establishing that you got it. Specification and verification are the two skills that survive a change of tool, and the rest of this toolkit is built around them.

☐ I can restate the task in my own words.

☐ I know which parts of the work were helped by AI.

☐ I checked each major claim against an appropriate source or method.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

☐ I verified all quotations and citations.

☐ I checked calculations, code, or data where relevant.

☐ I looked for hidden assumptions.

☐ I removed or rewrote anything I could not explain.

☐ I can answer questions about the final work.

7.5 ▪ Template: AI use log

This log records what you rejected and what you can now explain, not only what you used. Keep one for any assignment that requires process documentation. Copy the fields into your own notes or a document; the table lists what to record.

Question	Your response
Assignment	
AI tool used	
Date used	
Purpose of use	
Prompt or question asked	
Useful output received	
Output rejected or changed	
Verification method	
What I learned	
What I can explain without AI	

7.6 ▪ Pitfalls

Seven ways AI-assisted work goes wrong. Each line points to the subsection of section 4.3 that explains it.

- Do not outsource first contact with the problem (section 4.3.1).
- Do not submit work you cannot explain (section 4.3.2).
- Do not trust generated citations (section 4.3.3).
- Do not let AI change your meaning (section 4.3.4).
- Do not trust code just because it runs (section 4.3.5).
- Do not use AI to evade the purpose of an assignment (section 4.3.6).
- Do not let AI replace people (section 4.3.7).

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

7.7 ▪ When the work is research

Most of this book assumes an assignment with a known answer. Some of your work will not be like that. A course project, an undergraduate or master's thesis, a dissertation chapter: nobody has the solution, including your advisor, and the tools behave differently when there is no answer to retrieve.

Three things change. **Disclosure** stops being a course rule and becomes a professional one. Journals now ask what tools were used and for what [57], and some funders do too [36]. The answer is expected to be specific: not that AI was used, but which tool, on which part, and what you did with the output. Write it down as you go. Reconstructing it eleven months later is guesswork, and it will read like guesswork.

Verification gets harder, because the check you rely on in coursework is gone. You cannot compare against a solution that does not exist. What remains is the slower kind: derive it a second way, test it at a limit you already understand, run the case where you know the answer, hand it to someone who will argue. A plausible result you cannot reconstruct is not a result yet.

Authorship is the one that is easiest to get backwards. Using a tool does not cost you authorship, and it does not earn anyone else any. What earns it is the judgment: choosing the question, deciding the approach, knowing why the answer is right. If a tool did that part, the problem is not a credit question but a learning one.

Being stuck changes too. In coursework, stuck means you have not found the method yet. In research it usually means that several directions look plausible, none is obviously right, and the tool will pick one for you the moment you ask it what to do next. Do not ask it that.

Run a short check first, the one the companion paper "Learning to Create with AI" calls the Next-Move Check [54]. Name the bottleneck: the thing stopping progress now, whether a missing fact, a doubtful assumption, an untested computation, or too many options. List at most three things you could do next. For each one, ask what it would teach you if it failed, because a move whose failure still tells you something is often worth more than the move most likely to succeed. Choose one, and write a sentence saying why. Then do it and look again. The tool is useful for proposing the candidates and for arguing against your diagnosis, and the choice stays yours. The sentence is what your advisor will want to see.

Ask your advisor early what they want disclosed and in what form, and ask before you submit rather than after. Advisors differ on this, and the differences tend to surface at the worst moment. The question takes a minute and settles a year.

7.8 ▪ When you are unsure

If the AI policy is unclear for work you are about to submit, ask before you use the tool. Section 6.2 gives two questions that work and one that does not.

If AI helped grade your work, you should be told that it did and what it did, to know that a person decided the grade, and to ask that person to look again.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

7.9 ▪ Template: one-page student compact

> I understand that AI may be allowed in some parts of this course and not allowed in others. I will follow the policy for each assignment. I will make my own attempt before asking AI for substantial help. When AI use is allowed, I will use it to support learning rather than replace my own work. I will verify important AI-assisted content, protect private information, disclose AI use when required, and submit only work that I can explain as my own.

7.10 ▪ Student checklist

Use this checklist before submitting AI-assisted work. A copy you can print or keep on a phone is at `students/checklist.md`.

☐ I know the AI policy for this assignment.

☐ I made my own first attempt before using AI for substantial help.

☐ I used AI for support, not replacement.

☐ I checked important claims, sources, calculations, code, or interpretations.

☐ I removed any AI-generated material I could not explain.

☐ I protected private, restricted, and confidential information.

☐ I disclosed AI use if the assignment required it.

☐ I can explain the final work in my own words.

☐ I can close AI and my notes and redo the key argument, calculation, proof, or paragraph from a blank page.

☐ I know what I learned from the process.

The blank-page item is the strongest single calibration check on the list. Explanation while looking at finished work is easy to confuse with understanding; reconstruction from a blank page is harder to fake.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Part II

Teaching and Assessing with AI: An Instructor Guide

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The question is not whether a student used AI.
It is what the work still shows.

*Terms. This Part uses the four assignment categories introduced in Part I: No AI (**NAI**), AI as Tutor (**AIT**), AI as Collaborator (**AIC**), and AI as Object of Study (**AIS**). An instructor picks one per assignment and says so on the assignment itself. It uses the CAT framework, named for its three dimensions: Core Competence, AI-assisted Practice, and Trustworthiness. It also uses the student guide's Learning Spiral. The Spiral runs five steps around a question at its center: try to solve, ask for help, work independently, verify, and reflect. The Glossary at the back of the book defines these and the other terms that carry across Parts.*

Part II of this book is the Instructor Guide. It helps faculty, lecturers, teaching assistants, and course coordinators design courses that remain intellectually honest, fair, and useful when students have access to powerful AI tools. It is general in scope, with especially detailed examples for STEM and research-intensive courses.

The central premise is simple. AI-assisted education must assess two things separately: what students can do independently, and what students can do responsibly with AI. A course that ignores AI may lose assessment validity. A course that accepts AI output without safeguards may lose learning. The goal is a better middle path: teach students to use AI as a tool while preserving the habits of attention, judgment, verification, and explanation that higher education is meant to build.

This guide shares its framework and vocabulary with the student guide before it and the institutional guide after it, so the categories on your syllabus are the ones your students study and your institution can write into policy.

Most students will not go looking for the student guide on their own. AI tools are free and take no effort to start using. Because they are so easy, students rarely stop to decide how to use them. A student who has never been asked to think about how to use AI well has no reason to seek out a guide on the subject.

The student guide reaches students when an instructor puts it in front of them. That makes you its only reliable route. Assign it, or assign the parts that match your course, and tie the reading to the first assignment where AI is permitted. If you want a single document to start from, the student compact in that guide is written to be adopted as it stands, and the guide's toolkit gives a one-page version to hand out. A student who has read it will know what your syllabus category means, what verification looks like, and what to disclose. A student who has not will guess.

The companion repository makes the assignment concrete. Assign `students/start-here.md` *in week one; it takes a student ten minutes. Have students set up the Study Partner Protocol, the paste-in prompt that turns an AI into a tutor, at* `companion/spp/` *before the first assignment where AI is permitted. Require the checklist at* `students/checklist.md` *before they submit that assignment. The one-page schedule at* `instructors/what-to-assign.md` *lays out the full sequence and what each piece costs a student in minutes. If this is your first term teaching with AI,* `instructors/first-time-guide.md` *walks through the decisions in order.*

Readers whose class size is their dominant constraint may want to scan the three design axes and the three assessment models early.

Part II is six chapters. Chapter 8 sets out the competing pressures an instructor is under. Chapter 9 covers the framework and how to design an assignment around it. Chapter 10 sets out three assessment models and when each one fits. Chapter 11 covers integrity without detectors, equity, and the discipline overlays. Chapter 12 covers keeping grading consistent across a teaching team, what it takes to run this in a real course, and where the approach reaches its limits. Chapter 13 is the toolkit: the syllabus language, the rubrics, and the checklists, collected in one place. A reader who needs one thing can go to that chapter directly.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 8

The Instructor's Trilemma

8.1 ▪ Purpose and scope

Generative AI has changed the practical conditions of teaching. Students can now obtain fluent explanations, polished essays, working code, plausible citations, study questions, summaries, translations, and step-by-step solutions in seconds. These tools can help students learn, and they can also make it difficult to know what submitted work reveals about the student's understanding.

This guide offers a practical framework for that problem, consistent with broader guidance that treats generative AI as a tool requiring human oversight, transparency, and institutional judgment [89]. It does not ask instructors to choose between banning AI everywhere and allowing it everywhere. Both responses are too blunt for the range of courses taught at a university. A writing seminar, a chemistry lab, a design studio, a calculus course, a history survey, and a graduate methods course need different rules. Still, they share the same underlying responsibility: preserve meaningful learning while teaching students to use modern tools responsibly.

Instructors differ in what role they want AI to play, and the difference is not only a matter of degree. Some ban it as inconsistent with what a course exists to teach; some permit it for practice and not for products; some treat AI competence as a learning outcome in its own right. This Part is written for instructors deciding *how* to engage with AI rather than *whether* to.

That assumption governs less of what follows than it might appear. An instructor who restricts AI still needs to know what students can do unaided, and the machinery for finding out is here: the Oral-Defense and Evidence-of-Thinking Models, in-class checks, random audits, process evidence, and the integrity workflow serve a restrictive course at least as well as a permissive one, and arguably better. What the assumption governs is the material on designing assignments that use AI deliberately. An instructor who does not want that material can skip it and still find the rest intact.

The guide has four aims.

1. Help instructors state clear AI policies for each assignment.
2. Help instructors design assessments that reveal student thinking.
3. Help instructors teach verification, disclosure, and responsible judgment.
4. Help departments develop consistent practices across small, medium, and large courses.

One variable runs through the rest of this Part more than any other: class size. A thirty-student seminar, a seventy-student lecture, and a three-hundred-student introductory course are completely different worlds for AI-era assessment design. Approaches that work in the seminar do not scale

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

to the introductory course; approaches that scale may not preserve what makes the seminar valuable. The framework developed here is built around three assessment models (Oral-Defense, Evidence-of-Thinking, and Hybrid) each calibrated to a different range of class sizes, and around three design axes that include class size as the first. The class-size axis recurs throughout the Part wherever the framework's recommendations bend with it.

The recommendations are intentionally modest. The evidence on generative AI in education is developing quickly, and specific tools will change. Recent studies suggest that AI tutoring can support learning when designed around sound pedagogy, while unguarded access can weaken later independent performance [11, 61]. The durable principles are less likely to change: students need core competence, active practice, verification habits, transparent process, and fair assessment [32, 98].

The companion repository keeps a dated list of course tutors, study modes, and checkers, with who controls each, what it costs, and what evidence supports it (`companion/learning-tools/`). To judge a new tool, ask four questions. Does it hold back the answer until the student has tried? Can you give it your course context? Can students turn its guardrails off? Has anyone measured what students can do later without it?

8.2 ▪ Instructor outcomes

After using this guide, an instructor should be able to design a course that states AI expectations clearly and assesses student learning fairly. In practice, this means being able to do the following.

1. Identify the Core Competence that students must demonstrate without AI or with tightly bounded assistance.
2. Design AI-assisted assignments that develop AI-assisted Practice rather than reward outsourcing.
3. Assess Trustworthiness through verification, disclosure, process evidence, and explanation.
4. Choose an Oral-Defense, Evidence-of-Thinking, or Hybrid model that matches class size and staffing.
5. Use AI to support teaching without delegating grades, integrity decisions, or student evaluation to an automated system.

These outcomes keep the guide focused on course design. They also make clear that AI policy is now part of pedagogy. You are not there to fill a bucket; you are there to build judgment that outlasts any tool.

8.3 ▪ The instructor-side calibration problem

I once caught a copied proof by its notation. The same check helps a teacher today tell understanding a student owns from polished output. The technology is different. The work of telling them apart is not.

A teacher with seventeen years of teaching experience told the *Chronicle of Higher Education* that the work of distinguishing student writing from AI-assisted writing now consumes time that previously went elsewhere. "I'm grading fake papers instead of playing with my own kids," he said.[5] The line is one teacher's, but the predicament is widely shared. The labor cost of the calibration problem is real, and acknowledging it is the place to start.

[5] Quoted in McMurtrie [87].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Students sense the calibration problem from an unexpected angle. Focus groups with ninety-five undergraduates at the University of Pittsburgh in 2025 found that some students prefer AI chatbots to faculty office hours because, as one student put it, "with ChatGPT you can ask as many questions as you want and it's not going to judge you."[6] The line is not a verdict on any particular instructor. It is a description of how the instructor's classroom now looks from the other side of the desk. A patient, infinite, non-judgmental tutor has appeared, and it is competing for the role that the instructor used to hold uncontested. It also takes something harder to replace. Office hours were an early warning. A student in trouble would show up with a question weeks before the exam, in time for the instructor to act. When that question goes to a chatbot instead, the warning does not move; it vanishes, and the first clear sign of trouble becomes the exam, too late to help. The signal was always partial, since the students who needed it most were often the least likely to come. The remedy is to build it into the course rather than wait for the office to fill: the process records and in-class checks in this Part are early signals that do not depend on who shows up.

The signal can also fail in a more active way. A student who does not come to office hours at least leaves an absence that an attentive instructor may notice. A student who declines help when it is offered leaves nothing. Across more than two hundred students I have mentored on presentations over four decades, an offer to help structure a talk was never refused on the grounds that the student had a better method available. That has now happened. The refusal did not present as avoidance. It presented as competence, and it carried a specific claim: the student was skilled with AI and did not need the help. The claim was true. It was also about a different skill than the one the task required. This is one case set against four decades of mentoring, so it is an observation, not a measurement. The students I mentor now also differ from earlier cohorts in more ways than access to AI.

Cheng and colleagues describe this mechanism from the user's side. In conversations about personal disputes rather than coursework, systems that drift toward agreement left people more willing to return to the AI and less willing to repair the dispute [24]. Seen by an instructor, reliance of that kind does not look like dependence. It looks like self-sufficiency, and because it is expressed in the vocabulary of capability, questioning it risks appearing to doubt the student. The Pittsburgh focus groups point the same way: a tool that does not judge is competing with an offer of help that can always be heard as a judgment.

The shape of the offer therefore matters more than it used to. An instructor who offers to produce something, a draft, an outline, a worked structure, is offering what the student can already obtain in seconds and without the social cost of accepting help. An instructor who offers to be the audience, to listen and report back what did not land, is offering something the tool supplies only when asked in a way the student does not yet know to ask. The instructor's move is to make the offer one that AI does not fill on its own.

The translation test is a diagnostic the instructor can use. Ask the student to re-present a key step in another form: a picture or diagram, a plain-language explanation, or a restatement with different notation and a new example. A student who cannot do it has produced work whose origin is uncertain, whatever its surface polish. Treat that as a reason to look further, not as a verdict.

When students submit AI-assisted work, instructors face a parallel problem to the one students face. The cues that previously signaled mastery, including clean prose, organized argument, articulate explanation, and polished proofs, no longer reliably distinguish students who understand the material from students who used AI to produce work that resembles understanding. A student can submit a coherent essay, a well-organized lab report, or a correctly proved theorem without having done the cognitive work that produces those things in a pre-AI environment. AI did not

[6]The study is reported in full in Vee et al. [128]; the quotation is from Silva [113], a first-person account by a member of the study team.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

make the problem harder to solve; it made the problem harder to see.

This is a calibration problem for the instructor: the difficulty of distinguishing what finished work reveals about *this student*, separate from what AI may have contributed. The three assessment models presented later in this Part (Oral-Defense, Evidence-of-Thinking, and Hybrid) are designed to address it. They restore the signal that AI has weakened.

Reading the rest of Part II as a calibration-restoring project may be useful. The **NAI**, **AIT**, **AIC**, and **AIS** assignment categories are not just AI-permission categories. They are calibration choices about which dimensions of student competence each assignment measures. The verification rubrics ask students to demonstrate gap-closing practices. The process-evidence guidance is designed to recover what the polished final product no longer reveals on its own.

Before generative AI, a submitted paper, proof, program, lab report, or problem set was never perfect evidence of learning. Students could receive help from friends, tutors, solution manuals, prior students, websites, and commercial services. AI did not create the problem of attribution. It changed the scale, speed, and polish of outside assistance.

The core assessment question is now sharper:

> **What does this submitted work show that the student can understand, explain, verify, and do?**

If an assignment can be completed well by a generic AI tool, the assignment may still have value as practice, but it is weak evidence of independent competence. This does not mean the assignment should be abandoned. It means the instructor should decide what the assignment is for. Is it meant to measure unaided knowledge? Is it meant to teach responsible use of tools? Is it meant to produce a polished final product? Is it meant to reveal process?

Recent learning-analytics research helps name the mechanisms by which polished AI-assisted work can hide reduced learning. Fan et al. [38] describe **metacognitive laziness**: students using generative AI on revision tasks performed measurably fewer self-monitoring activities, especially evaluation and orientation, even when their final essays scored well. Chen et al. [22] describe a related pattern they call cognitive offloading: in their comparative analysis of student help-seeking with human experts versus AI chatbots, students working with the chatbot tended to skip the diagnostic, evaluative, and iterative stages that ordinary help-seeking includes, jumping directly from request to implementation. Both mechanisms point the same way. The work improves; the learning does not follow. In Fan's trial the AI group's essay scores rose while their knowledge gain and transfer showed no significant advantage. Designing assignments that make those skipped stages visible is the practical response to both mechanisms, and is the orientation behind the Plan-Use-Verify and Oral-Defense patterns in this Part.

Confusion arises when a course treats one kind of task as if it measured unaided knowledge, responsible tool use, a polished product, and process all at once. In the AI era, instructors should separate them.

8.4 ▪ The three corners

The calibration problem in the previous section names what instructors *observe*. This section names what instructors *face* when they sit down to design a course, grade a stack of papers, or decide whether to raise a private conversation with a student about suspected AI use. Like students and institutions, instructors under AI pressure navigate a structural predicament that has not been clearly named in the discourse. Call it the **Instructor's Trilemma**, and it has the same three-cornered shape as the Student's Trilemma and the Institution's Trilemma.

A quieter question runs underneath all three corners. Faculty are encountering AI not only as a pedagogical challenge but as a vocational one: *what am I uniquely for, in a classroom where AI*

can also explain, also tutor, also assess, also produce convincing prose? The question is rarely asked aloud, partly because it is hard to ask, and partly because no answer is fully reassuring. It runs through the faculty discourse in a hundred quiet ways: the early retirement that was not planned, the senior colleague whose enthusiasm has gone flat, the conference panels titled "the future of higher education." The Instructor's Trilemma does not answer this question, but it names the three forces an instructor must navigate while the question hangs in the air.

The corners are:

Professional sustainability. Keeping the job and the conditions that allow you to keep doing it. This includes manageable workload, contract renewal, tenure case, course evaluations, time to engage deeply with students, and protection against the burnout that has become widespread in the profession. AI sharpens every one of these pressures: the cheating-detection arms race adds policing time without commensurate institutional support; the volume of AI-polished submissions complicates grading; the institutional inconsistency of AI policies leaves individual instructors to invent and defend their own.

Trustworthy professional judgment. What your assessments, grades, feedback, recommendations, and other professional acts of *vouching* actually signify about the students you teach. When you assign a B to a paper, write a letter of recommendation, or pass a thesis defense, you are making a claim that other people will rely on. The credential carries weight in part because individual faculty stand behind individual judgments. When AI assistance enters student work without being verified, those judgments lose their connection to what the student can actually do.

Student learning. Students gain understanding, skills, and judgment as a result of taking your course. This corner has two faces, and instructors are often torn between them. The first face is *cognitive development*: students should leave with strengthened reasoning, deeper understanding, and the kind of thinking that holds up when the tools they used in your course are unavailable. The second face is *workforce preparation*: students should leave equipped for professional environments where AI is a permanent feature of the work. These are not opposed in principle, but in individual course-design decisions they often pull in opposite directions. An instructor who limits AI use to protect cognitive development may be failing the workforce-preparation mandate; one who teaches AI fluency to prepare students for professional work may be enabling the very overreliance the first face exists to prevent.

Any two corners can be held, but only at the cost of the third. The failure modes echo the Student's and Institution's Trilemmas deliberately.

8.4.1 ▪ Three instructor failure modes

Sustainability and trustworthy judgment without student learning. The instructor maintains a manageable workload and grades that can be defended in conversation, but the pedagogy itself has hollowed out. Assignments become defensive: in-class blue-book exams that limit AI use but also limit what can be assessed; multiple-choice tests that are easy to grade but reveal little; assignments rewritten until blind AI use "collapses under its own weight" [109] but in the process becoming so constrained that students do not develop the kinds of thinking the course title promises. Call this the **instructor's form of hollow understanding**: the professional surface is intact, the integrity record is defensible, but the educational mission has hollowed out from inside.

Sustainability and student learning without trustworthy judgment. The instructor protects their workload and produces actual student learning, but lets AI-assisted work into the gradebook without demanding genuine ownership. AI grades the homework; students use AI to write the papers; the pedagogy may even be sophisticated and engaging; but the instructor can no longer

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

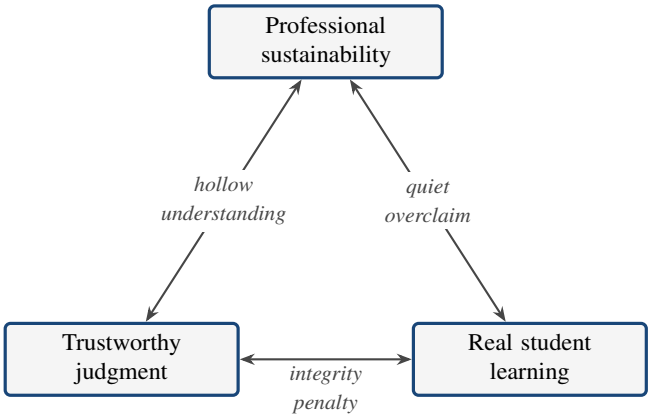


Figure 8.1. ***The Instructor's Trilemma.*** *Any two of these commitments can be kept, but only at the cost of the third. Each edge labels the failure mode that arises when the two adjacent commitments are kept at the cost of the third. AI sharpens each edge; it did not create the trilemma.*

defend in conversation what individual grades, recommendations, or evaluations actually signify. This is the **instructor quiet overclaim**. The instructor keeps up the appearance that grades certify what students can do. Quietly, the instructor accepts that the checking was never done. Expect it when the cost of demanding genuine ownership exceeds what the instructor can sustainably do.

Trustworthy judgment and student learning without sustainability. The instructor holds high standards, refuses to lower expectations under AI pressure, redesigns assessments to surface genuine thinking, conducts the oral conversations and process audits the framework calls for, and produces real learning that students will carry with them. And burns out. One account names this pattern directly in its observable forms: faculty reverting to blue books or opting for early retirement to avoid the perceived flood of machine-generated text [84]. Call this the **instructor integrity penalty**: every commitment kept, no commitment surrendered, but the personal cost cannot be sustained.

The American Association of University Professors frames this pattern at the professional-labor level. They argue that the uncritical adoption of AI poses a threat to academic professions through potential work intensification and through implications for the faculty working conditions that affect student learning conditions [4]. The intensification is real: more policing time, more course redesign cycles, more individual policy invention. The integrity penalty is the price an instructor pays for refusing the other two.

AI did not create the Instructor's Trilemma. The trilemma names a structural tension between commitments faculty have always held in some balance. What AI did was sharpen each edge. Each pair of corners is harder to hold together when AI assistance lowers the apparent cost of giving up the third.

8.4.2 ▪ How the trilemma manifests differently by faculty type

Every teaching faculty member faces the same three corners. How strongly each corner pulls depends sharply on employment status. The dimension is unusually consequential in U.S. higher education: contingent positions account for nearly 70% of all instructional staff appointments, 49% part-time and 19% full-time non-tenure-track [5]. The trilemma looks different depending on which contractual category the instructor occupies. Four types follow.

Tenured faculty with reasonable course loads. These instructors have the most freedom at

every corner. Their jobs are relatively secure, since the contract cannot be cancelled at the end of the semester. Trust in their judgment rests on a long-developed disciplinary reputation. And they can pursue student learning through whatever teaching experiments they find compelling. They are the ones who can best afford the integrity penalty, and some choose it on purpose: they hold high standards and absorb the heavy workload that comes with them. They may also be the most likely to romanticize the integrity penalty when advising junior colleagues, for whom it is unsustainable.

Tenure-track faculty. Professional sustainability dominates because the tenure case is at stake, and almost every classroom decision feeds eventually into the tenure dossier. Course evaluations matter; student complaints carry weight; pedagogical experiments that backfire can be remembered. The hollow-understanding failure mode is the most relevant risk: defensive pedagogy that protects the tenure case by avoiding controversy, at the cost of actual learning. The framework provides an alternative path through better-designed assignments and clearer rubrics, but the structural pressure toward defensiveness is real and should be acknowledged. The fear is not imagined, and it has been measured. One randomized comparison in introductory physics kept the content and handouts identical across sections. Experienced, highly rated lecturers taught the passive sections, and the instructor made no attempt to sell either method. Students in the active sections learned more and rated their own learning lower [28]. The rating was still positive. It was just lower than for the polished lecture. The authors draw the conclusion that matters here: evaluating teaching by how much students feel they learned can reward the weaker method. A course evaluation is evidence about the experience. It is not evidence about the learning, and an instructor defending a pedagogical choice can now say so with a citation.

Full-time non-tenure-track faculty (lecturers, clinical faculty, teaching-stream professors, professors of practice). These positions have grown substantially in U.S. higher education and now carry a meaningful share of the teaching. Their trilemma is mixed. Contract renewal is a pressure, but a milder one than for adjuncts. Teaching is usually what they are evaluated on, which protects student learning. But their teaching loads are often heavy, and support for careful checking can be thin. When the load exceeds the time it takes to check every student's work, quiet overclaim becomes a real risk.

Adjunct and per-course contingent faculty. For these instructors, sustainability wins, and that is a fact about the job, not a judgment of the person. An adjunct teaching four or five sections at two or three institutions has no time for the checking that trustworthy grades require: reading process records, holding oral checks, spot-checking work. This is not a failure of values. The conditions of the role make the checking impossible. So adjuncts are the faculty most exposed to the quiet overclaim, usually without having chosen it: the grades go out looking as if someone had checked them.

The practical design for these courses is the cheapest one. Move the grade onto in-person assessment, and let homework count for little. That adds almost none of the checking that the assessment models later in this Part require. **The honest alternative, the integrity penalty, is a luxury in the academy:** only faculty whose positions allow it can pay it term after term. Recommendations in this Part that assume an instructor can spend extra time on oral defenses, individualized feedback, and process audits should be read with this in mind. Institutional change that does not address the underlying labor conditions will leave contingent faculty in a failure that comes from the job, not the person.

These four types are not comprehensive. Graduate-student instructors, postdoctoral teaching fellows, and emeriti returning to teach face the trilemma with their own type-specific pull patterns. Discipline also matters. Faculty in the humanities and writing-intensive fields face the trilemma in its most acute form, because their disciplines are where AI-generated text is hardest to distinguish from student work. Faculty in disciplines with strong oral and laboratory traditions have more natural defenses against the quiet-overclaim pattern. The framework is useful because the three

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

corners stay the same for everyone. What changes is how hard each corner pulls, and which failure is most likely given an instructor's working conditions.

8.4.3 ▪ Why the Instructor's Trilemma matters for the rest of this Part

The instructor's version of the student's navigation rule is the same in spirit: when one corner starts to slip, adjust what you assign and how you grade rather than abandon the corner.

The rest of this Part offers tools for holding the three corners together: assessment models, design patterns, integrity procedures, and ways to support faculty. Each answers a particular pressure. The three assessment models (Oral-Defense, Evidence-of-Thinking, Hybrid) let an instructor judge work in a way that can be trusted at different course sizes, without paying the integrity penalty in personal time. The four assignment categories serve both things students need from a course, the thinking it builds and the work it prepares them for, so instructors do not have to choose between them. The fair response sequence for suspected misuse keeps policing from wearing instructors down, and that exhaustion is what drives the integrity penalty.

Without the trilemma frame, the operational tools in this Part can read as a list of best practices. With it, they read as coordinated commitments that protect the instructor from each of the three failure modes simultaneously. Much faculty writing on AI describes whichever failure mode its author fears most. The framework's value is that it surfaces the structure beneath the individual stories and gives instructors a vocabulary for understanding why their colleagues are choosing different responses to the same forces.

8.4.4 ▪ What holds as the tools improve

Every recommendation in this Part rests on one of two foundations, and they age differently.

The first kind depends on what a tool cannot yet do. A prohibition works while the work is hard to produce. A detector works while the output looks distinctive. Rules against particular uses work while students find those uses inconvenient. These are real controls, and they are worth having, but their strength is borrowed from the current state of the technology. Each of them weakens as the tools improve, and none announces its own decline.

The second kind depends on what a student can demonstrate. Consider an explanation given in the student's own words, a reconstruction of the method, a transfer of the idea to a nearby case, an account of what was checked and how. None of these gets much easier to fake as the models improve, because none is a claim about the work. They are claims about the person, tested directly. A tool that writes a flawless essay does not thereby give its user the ability to defend one.

This distinction sorts the material in this Part. The four categories, the disclosure templates and the syllabus language are a shared vocabulary. Vocabulary is durable and worth investing in, but a vocabulary is not an enforcement mechanism, and treating it as one produces a policy that reads as firm and holds nothing. The assessment designs are the load-bearing part.

Corbin, Dawson and Liu draw the same line and give it names [26]. A change to an assessment is discursive when it alters the instructions around a task, and structural when it alters the task itself, so that what the task measures no longer depends on the instructions being followed. Vocabulary is discursive. Assessment design is structural.

One consequence runs against what instructors reasonably assume. **A spot check decides whose work gets a closer look. It does not decide who is punished.** An audit is a closer measurement of the same thing every student is graded on, so a student picked for one is not being penalized; they are being read more carefully. Where an audit raises a genuine concern about honesty, that concern leaves the gradebook and enters the integrity procedure, which rests on evidence, explanation and due process. Marks answer the question of what a student can do. They are a poor instrument for the separate question of whether a student was honest, and using them

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

for it makes the grade mean less without making the course more honest.

8.5 ▪ Three design axes: class size, level, and discipline

The design choices an instructor makes about AI in a course are governed mostly by three dimensions: class size, course level, and discipline. The three are roughly independent, though level and size correlate in practice, since graduate courses tend to be smaller. Each still drives a different family of decisions, and treating them as a single axis (often labeled informally as "undergraduate versus graduate") tends to bury the design logic that this Part is meant to make visible.

Table 8.1. ***The three design axes.*** *Each axis drives a different family of decisions for AI-aware course design. Treating them together produces clearer choices than treating "undergraduate versus graduate" as a single axis.*

Axis	Drives	Sections in this Part
Class size	*How* you assess: which assessment model is feasible.	Oral-Defense Model; Evidence-of-Thinking Model; Hybrid Model.
Course level	*What* you assess and how much AI autonomy to grant: Core Competence definition, the typical assignment-category mix, expected verification capacity.	Four assignment categories; rubrics.
Discipline	The specifics of *how* AI fits in: what verification looks like, what AI tools are appropriate, what counts as acceptable use.	Discipline overlays.

The three axes are summarized in table 8.1 and discussed in turn.

Class size determines what direct contact the instructor (or trained TAs) can have with each student. A 12-person seminar can support oral defenses on every major assignment; a 200-person lecture cannot. This axis is the primary driver of which of the three assessment models discussed below fits the course. Class size does not, by itself, determine *what* students should be able to do without AI.

Course level determines the substance of Core Competence, the typical assignment-category mix, and how much AI autonomy a student can responsibly handle.

Discipline determines the specifics of verification and AI use. A mathematics assignment is verified by checking limits, units, and edge cases; a humanities essay is verified by checking primary sources, quotation accuracy, and argument structure; a programming assignment is verified by writing test cases. The framework is uniform across disciplines, but the verification practices, AI tool choices, and category boundaries are not.

The three axes are largely independent. A small graduate seminar and a small first-year writing class share the assessment-model options that small classes support, but differ in category mix and AI autonomy. A large introductory mathematics class and a large introductory humanities class

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

share the model constraints that large classes impose, but differ in what verification looks like. A large advanced seminar (rare but real) and a small introductory course share neither directly, and instructors in those settings can mix elements from different parts of this Part rather than picking one prepackaged path.

This Part is organized around all three axes. The CAT framework and the four assignment categories that follow are level-agnostic and discipline-agnostic at the framework level; their operational application varies along all three axes. The three assessment models that follow vary primarily by class size. The discipline overlays vary by discipline. Where the prose in this Part says "a graduate course" or "an advanced undergraduate course," the relevant claim is usually about course level, and sometimes about class size, since graduate courses are typically smaller.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 9

The Framework and Assignment Design

The design choices in this Part assume something they cannot supply: a student who wants to learn. A well-built assignment makes thinking visible and gives AI less to offload, but it cannot make a student care, and a student who does not care will route around the best task you can write. The deepest lever an instructor holds is not the assignment but the student's reason to engage with it. Design protects and rewards the will to learn; it does not create it. That comes earlier, from the case the instructor makes for the work: when students believe the reasoning they are building will outlast the answers and matter in the work they go on to do, they will spend effort a chatbot would spare them.

Buy-in has a condition that is easy to miss. Productive struggle happens only where it feels safe. A student who fears that a wrong first attempt will count against them will reach for the answer that removes the risk, and AI supplies it instantly. Lowering the stakes on early attempts, grading effort before correctness, and treating a wrong step as part of learning rather than a verdict make the struggle survivable, and a struggle that feels survivable is one students will actually attempt.

9.1 ▪ The CAT framework

The framework generalizes a three-axis assessment model originally developed for graduate mathematics, where independent mathematical understanding, AI-augmented skill, and epistemic responsibility are assessed separately.

Core Competence. What students must understand and be able to do without AI or with tightly bounded assistance. In a course this is what the instructor is willing to assess with AI set aside.

AI-assisted Practice. What students can do when they use AI responsibly as a permitted tool for learning, revision, analysis, coding, feedback, or exploration, without losing control of the work.

Trustworthiness. How well students verify, disclose, critique, and take ownership of AI-assisted work. In a course the disclosure block is where a student shows it.

A course should assess all three dimensions, but not always with equal weight. A foundational course may emphasize Core Competence. A capstone course may place more weight on AI-assisted Practice and Trustworthiness. A professional course may emphasize judgment, privacy, and defensible decisions.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The framework is consistent with UNESCO guidance that treats AI competence as part of a human-centered educational system rather than a substitute for teacher and learner agency [88, 90]. It has one non-negotiable feature: AI skill should not compensate for failure to meet the core learning goals of the course. A student who can produce strong AI-assisted work but cannot explain the central ideas has not met the same standard as a student who understands the material.

9.1.1 ▪ Core Competence

Core Competence is the part of the course that remains the student's own. It may include definitions, methods, concepts, interpretations, calculations, source analysis, writing decisions, experimental reasoning, coding fundamentals, design principles, or ethical judgment.

Each instructor should identify the core explicitly. The useful question is not, "Can AI do this task?" The useful question is, "What must students still be able to do when AI is unavailable, inappropriate, or untrustworthy?"

Core Competence can be assessed through no-AI work such as short quizzes, oral explanations, or closed-resource problem solving. These assessments need not dominate the course, but they should be meaningful.

9.1.2 ▪ AI-assisted Practice

AI-assisted Practice is the student's ability to use AI productively without losing control of the work. This includes asking useful questions, constraining a model, comparing explanations, testing code, finding flaws, improving drafts, generating examples, and using AI feedback to improve human reasoning.

This is not a lesser skill. In many fields, responsible use of AI will become part of professional competence. Students should learn it in courses where faculty can set boundaries and provide feedback.

The instructor's task is to design assignments where AI use is visible and assessable. A strong AI-integrated assignment asks students not only for a final product, but also for evidence of choices, verification, revision, and judgment.

9.1.3 ▪ Trustworthiness

Trustworthiness includes verification, disclosure, privacy, source discipline, and intellectual ownership. It is the dimension that keeps a student from passing off AI output as their own understanding.

A student's responsible judgment is visible when they can answer questions such as these:

- What did AI contribute?
- What did you accept, reject, or revise?
- How did you check the output?
- What sources did you verify yourself?
- What part of the submitted work can you explain without AI?

These questions are useful across disciplines. They apply to a proof, a policy memo, a lab report, a design prototype, a literature review, a computer program, and a close reading of a text.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

9.2 ▪ Four assignment categories

Every assignment should tell students what role AI may play. The guide uses four categories: **NAI**, **AIT**, **AIC**, and **AIS**. The categories are simple enough for a syllabus, but flexible enough for different disciplines. External guidance points the same way: the 2026 Higher Education Policy Institute survey recommends that institutions offer AI-free assessments alongside assessments that expect students to use AI [119], which is the **NAI**-versus-**AIC** distinction stated as institutional advice.

Before applying these categories, instructors may want to recommend that students complete the AI account configuration template at the start of the course. The configuration step takes ten minutes, pitches the AI's help to the student's level and discipline, and creates a written record of the use boundaries the student has committed to. For a course with significant **AIC**-category work, requiring the configuration as a no-credit first assignment is a low-cost way to start the term with shared expectations.

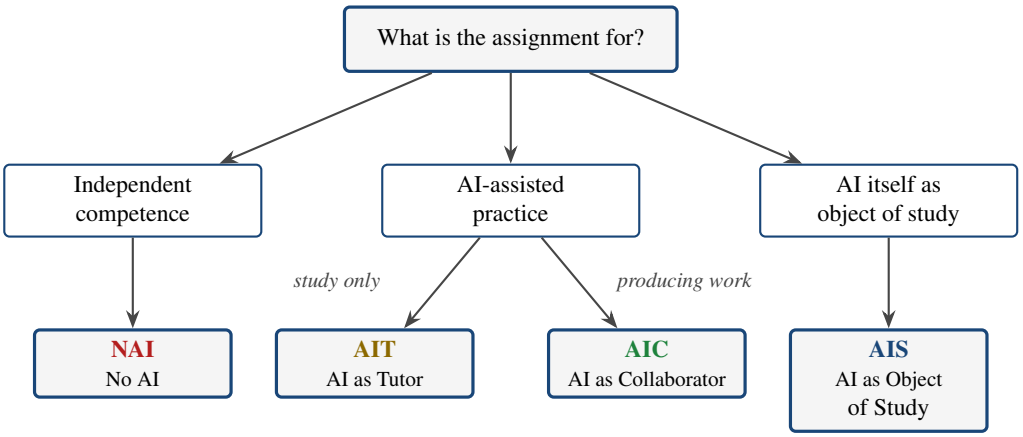


Figure 9.1. ***Choosing the assignment category.*** *The category follows from what the assignment is for, not from which tools happen to exist. The instructor decides the purpose, then selects the category that matches it.* ***NAI*** *(No AI) protects independent competence;* ***AIT*** *(AI as Tutor) allows AI for study support but not for submitted work;* ***AIC*** *(AI as Collaborator) permits responsible AI use in producing the submission;* ***AIS*** *(AI as Object of Study) makes AI itself the subject of analysis. The choice should appear on the assignment.*

9.2.1 ▪ NAI: No AI

NAI assignments prohibit AI use. They are used to assess Core Competence.

NAI assignments take familiar forms: in-class exams, short quizzes, oral explanations, live coding, lab practicals, handwritten derivations, in-class writing. What makes an assignment **NAI** is not the format but the declared purpose: each is a stated, protected measurement of what the student can do alone.

The purpose is what the label declares. The format is what makes the declaration hold. Every example above shares a property the definition does not name: the instructor can see the work happen.

An **NAI** assignment should have a clear reason. It should not be a default expression of distrust. The reason is usually that the instructor needs evidence of what the student can do without outside help.

9.2.2 ▪ AIT: AI as Tutor

AIT assignments allow AI for study support, but not for producing submitted work. Students may use AI to ask for hints, examples, explanations, practice questions, or feedback before doing the work themselves.

AIT rules are useful when the instructor wants students to benefit from tutoring while still submitting independent work. A homework set might be **AIT** if students may ask AI to explain a concept but may not ask it to solve the assigned problems.

For **AIT** assignments, instructors may define a help ladder that structures the order in which AI assistance is permitted: concept explanation, then an analogous example, then one hint on the assigned problem, then a critique of the student's attempt, and only then a full solution for ungraded practice. The ladder allows substantial AI support while protecting retrieval practice and productive struggle. Students who begin with a full solution to the assigned problem skip every learning move the assignment was designed to produce.

The ladder works best written into the prompt itself. Add it to the course's version of the Study Partner Protocol (`companion/spp/`). Post it with the assignment, and ask students to open each **AIT** session with it. Every student's AI then climbs the same ladder, and you know what it was told. A student can skip the prompt, so it sets the default rather than enforcing the rule; a shared thread shows whether it was used.

9.2.3 ▪ AIC: AI as Collaborator

AIC assignments allow AI as part of the working process. Students may use AI to brainstorm, revise, debug, critique, summarize, generate alternatives, or test ideas. **AIC** assignments require disclosure and verification.

AIC assignments work best when the process matters. For example, a student might submit an original plan, selected AI interactions, a revised product, and a verification note. The final product may be stronger because AI was used, but the student's judgment remains visible.

Familiar **AIC** forms include a modeling project where AI helps check a formulation, a computational assignment submitted with its debugging log, a literature summary the student verifies claim by claim, and a report revised across drafts with the AI exchanges attached.

9.2.4 ▪ AIS: AI as Object of Study

AIS assignments make AI itself part of the subject. Students might critique an AI-generated essay, identify hallucinated citations, repair flawed code, compare model responses, evaluate bias, or analyze how a tool handles a disciplinary problem.

AIS assignments are especially valuable because they teach students not to treat AI as an oracle. They make error detection and tool critique explicit learning goals. That critique goes deeper when students know why the errors sound certain, since a system graded only on answers has no reason to say it does not know [59].

9.2.5 ▪ The categories and the AI Assessment Scale

These four categories are not the only way to label AI use. A widely used alternative is the AI Assessment Scale, five ordered levels running from no AI to full AI [101]. It has been translated into thirty languages and used in over three hundred institutions, and Australia's tertiary education regulator has recognized it as a useful approach to AI in assessment [75]. A pilot at one university reported a drop in AI-related misconduct cases after adoption [44]. The authors revised the scale in 2025 [102].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The categories here add two things. The scale's middle levels are set by how much of the task the AI performs: planning, drafting, or completing. **AIT** is set by whether AI wording, code, or worked solutions enter the submission. That is a line an instructor can state and a student can check. **AIS** sits outside the scale. In an **AIS** task the AI is not helping with the work; it is what the work studies.

The scale exists because a two-valued rule, allowed or banned, is too coarse for what students actually do. The same criticism is often aimed at any permission framework, including this one. It lands less hard here. The categories are coarse, but the line between **AIT** and **AIC** is one a student can check: AI wording, code, or worked solutions either enter the submission or they do not. The fourth category asks something else, which is what the AI itself can be trusted to do.

A department already using the scale does not have to choose. **NAI** matches Level 1. **AIT** matches Level 2 when AI text is kept out of what is handed in. **AIC** covers Levels 3 and 4, where AI work enters the submitted product. Level 5 asks students to build new AI applications, which is near **AIS** but not the same. That mapping is to the 2025 scale, which restructured its levels.

The scale does some things better. Five levels are finer than the three categories that sit on the traffic light. It is peer reviewed, revised, translated, and adopted at scale, and these categories are none of those yet. Fewer categories buy compliance and cost precision.

The revised scale dropped its traffic-light colors so its levels would not read as a ranking. The light here ranks nothing beyond how much AI an assignment allows, which is why **AIS** sits off it in blue. The authors of the revision note that a red-to-green light still earns its place where clarity matters most.

A further objection reaches past the colors, and it lands on the scale and on these categories alike. Used as labels alone, both are discursive in the sense of section 8.4.4. They tell a student what is permitted and leave the task unchanged. The scale's authors make the same point, and present their revision as a scaffold for redesigning the task. Corbin, Dawson and Liu argue that instruments of this kind borrow the authority of systems that genuinely constrain behavior, a traffic light among them, without borrowing any of the enforcement [26]. A road light stops cars. A syllabus light stops nothing by itself. Five levels are more precise than four categories, and no more binding.

The answer is not a better label, and it is not abandoning labels. It is the division already set out in section 8.4.4. A shared vocabulary is worth having and does not enforce itself, and the assessment design is what carries the weight. The useful question to ask of any label, on the scale or on this light, is what happens when a student ignores it. That answer comes from the assignment, not from the syllabus.

9.2.6 ▪ A syllabus table for assignment categories

Category	AI role	Typical purpose	Example assignment
NAI	No AI	Measure independent competence.	In-class exam or handwritten derivation.
AIT	Tutor only	Support study while preserving independent submitted work.	Problem set where AI may explain but not solve.
AIC	Collaborator	Teach responsible AI-assisted work with disclosure and verification.	Modeling project submitted with its AI log and a verification note.
AIS	Object of study	Teach critique of AI output, bias, errors, and limits.	Compare two models on one problem and locate the error.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

9.2.7 ▪ Category mix shifts with course level

The four categories are level-agnostic at the framework level; any assignment in any course can fall into any of the four. The *typical mix*, however, shifts substantially with course level, and ignoring this shift is one of the more common ways instructors design AI policy poorly.

Introductory courses typically lean toward **NAI** and **AIT**. The reason is that students at this level have not yet built the disciplinary competence required to verify AI output reliably. Granting **AIC**-category latitude before that competence exists can amount to inviting students to accept work they cannot evaluate. A Calculus 1 student often cannot yet distinguish a correct optimization solution from a confidently wrong one; a first-year writing student cannot reliably detect when an AI revision has shifted the meaning of a thesis. **NAI** protects the foundational skill; **AIT** lets the AI act as a study companion without entering the submitted work. When introductory courses do assign verification, the instructor supplies the target: a benchmark, a special case, a source to check against, or the method to use. The task then teaches checking rather than presuming it.

Intermediate undergraduate courses typically use a balanced mix. As students develop disciplinary competence, the verification side of **AIC**-category work becomes feasible, and **AIS**-category critique becomes possible. A junior-level course in any discipline can typically support meaningful **AIC** and **AIS** assignments alongside **NAI** and **AIT** ones.

Advanced undergraduate and graduate courses typically lean toward **AIC** and **AIS**. Students at this level can verify AI output, can defend submitted work in detail, and can engage AI tools as objects of disciplinary critique. **NAI** assignments still serve a purpose at this level (foundational-skills demonstrations, qualifying-exam preparation), but they are typically a smaller fraction of the course.

The shift is not strictly linear with level: a research-methods graduate seminar might sensibly require several **NAI** oral examinations, while an upper-division undergraduate elective might be almost entirely **AIC**. The shift is also not driven by class size: a 200-student introductory lecture and a 12-student introductory seminar share the typical **NAI/AIT** weighting; a small graduate seminar and a large graduate-equivalent course share the typical **AIC/AIS** weighting. Enrollment motive is another variable. Students choose an elective and are placed in a requirement, and that difference in why they are there shapes what a course can ask of them, independent of level, size, and field.

A useful rule for an instructor designing a course: the Core Competence (**NAI** and **AIT** assessments) defines what the course is for; the AI-assisted Practice (**AIC**) and AI-as-object (**AIS**) work defines how that competence is exercised in a tools-rich environment. The proportion between these two halves is what shifts with course level.

9.3 ▪ Designing assignments that preserve thinking

The best AI-aware assignments make student thinking visible. They do not merely ask for a final answer. They ask for evidence of how the student framed the problem, what they tried first, what they asked AI, what they checked, and how the final work changed.

A randomized controlled trial of an AI-tutoring system in an introductory physics course at Harvard built its design around seven practices: active engagement, cognitive-load management, growth mindset, scaffolding, feedback accuracy, timely targeted feedback, and self-pacing [61]. The evidence behind them is not uniform. Active engagement rests on a meta-analysis of 225 studies [43], feedback on a literature going back decades [17, 47], and cognitive load on a developed theory [121]. The seven are useful as a design checklist. They are not seven equally settled findings.

The seven practices map onto principles long established in the learning-science literature [3, 94]. In the meta-analysis of 225 studies, active-learning approaches raised examination

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

performance by 0.47 standard deviations, with students under traditional lecturing 1.5 times more likely to fail than those under active learning [43]. The first five practices can be built into any assignment whether or not it uses AI; the last two are easier when AI is part of the design.

The assignment patterns recommended in this section are consistent with all seven, and instructors who want a deeper grounding for these patterns can read the Kestin study directly. A separate field experiment in high-school mathematics provides the converse evidence. An unrestricted AI interface that simply answered questions produced 48% practice gains but 17% exam losses. A guardrailed variant designed to give hints rather than answers produced 127% practice gains, with no exam loss and no exam gain [11]. The design choice behind the difference is what the **AIT** assignment category in this guide formalizes: AI is permitted as a hint-giver, not as a solver. The guardrailed tutor also had the worked solutions in its prompt, which is what kept it from inventing answers. Both halves matter. Without an answer key behind it, the base model was right only about half the time, so hints built on it could point the wrong way.

One approach to buy-in has been tested directly. Hulleman and Harackiewicz randomly assigned high-school science students to write briefly about how a topic connected to their own lives, against a control group that wrote summaries of the same material [53]. Students who did not expect to do well earned higher grades and reported more interest in science. Students who already expected to do well showed no difference. Canning and colleagues ran the same design across three units of a college introductory biology course and found higher grades, higher continuation into the second course, and higher persistence in the major [21]. Both studies come from one research program rather than from independent groups, and reported effects in this literature range from none to large, so this is a promising design and not a settled result.

The AI question here is sharper than it looks. The assignment asks a student to write about why the material matters to them. A chatbot will produce that in seconds and it will read well. The whole mechanism depends on the student making the connection. An AI-written version keeps the form of the intervention and removes its substance.

A separate concern is the social dimension of learning. A study of generative AI's effect on peer interactions in computing courses found that students increasingly route their questions to AI rather than to each other, and that students seeking peer help are sometimes redirected to AI rather than receiving it [51]. The pattern erodes the study groups, peer tutoring, and informal collaborative practices through which much undergraduate learning has historically happened, including the peer-instruction methods documented to improve conceptual understanding in undergraduate STEM [85]. Assignments designed to preserve thinking should also protect peer interaction, and they should name it as a goal.

A useful design pattern is the **Plan-Use-Verify** structure. The structure operationalizes the principle, well established in the learning sciences, that learners benefit from explicit self-monitoring before and after task engagement [94].

Plan. Before using AI, the student writes a short plan, prediction, outline, proof sketch, lab strategy, or source note.

Use. The student uses AI within the assignment rules and records the relevant interaction.

Verify. The student checks the output and explains what was accepted, changed, rejected, or corrected.

This structure does not require grading every prompt. It asks students to show enough of the process that the instructor can see whether learning occurred.

Plan-Use-Verify is not a second framework competing with the Learning Spiral. It is one pass of that spiral, collected as evidence. The instructor sees three records. Plan is the question and the honest attempt the student had before asking. Use is the asking. Verify is the independent work, the check that follows, and what the student made of it. Students already practicing the Spiral are

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

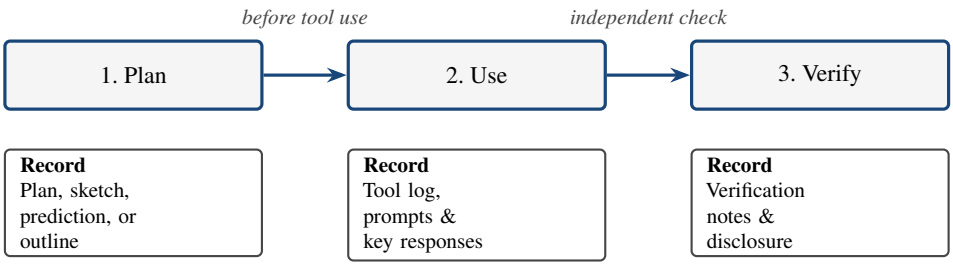


Figure 9.2. ***The Plan-Use-Verify assignment structure.*** *Making the thinking visible in three small records is what makes AI-assisted work gradable. The three records together show how the student thought, where AI helped, and what the student verified. These three records are one pass of the Learning Spiral, in the form an instructor can grade. Grading does not require reading every prompt; it requires seeing that all three stages happened.*

being asked only to hand in what they already did.

9.3.1 ▪ Plan-Use-Verify in practice

The three records work because each is short and gradable at a glance. The pattern is the same shape across disciplines, with only the lengths and content varying.

For a humanities essay, the Plan might be a one-paragraph thesis with the primary sources the student will use. The Use is a 3–5 line tool log naming what the student asked AI and what was returned in summary. The Verify is a 4–6 line note describing one place the student returned to a source to check an AI-suggested claim, plus what was accepted or rejected.

For a programming problem set, the Plan is a sketch of the algorithm or data structure the student intends to use, written before any code. The Use is the prompts and key responses from any AI debugging help, with the student's own diagnostic questions surfaced. The Verify is a list of test cases run, with one or two showing edge cases that catch a typical bug.

For a mathematics problem, the Plan is the student's setup of the problem (variables, constraints, choice of method) before AI use. The Use is a record of any AI hint or critique. The Verify is a unit check, a limiting-case check, or a comparison with a known result.

Three patterns make this work and three patterns break it.

Works when:

- The three records together are short, one page or less for most assignments. Brevity forces selection of the meaningful moves.
- The records are gradable at a glance, not parsed line-by-line. The instructor reads them as evidence of process, not as objects of grading.
- The student treats the structure as a thinking aid. The Plan forces commitment before AI exposure; the Verify forces a check after.

Breaks when:

- The records are graded on length or completeness. Process documentation drifts into theater.
- The Plan is written after the work is finished. The pattern collapses into retroactive rationalization.
- The instructor grades verbatim AI transcripts in place of the three short records. The signal-to-noise ratio drops to near zero, and students learn to game the documentation rather than the work.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The pattern scales: same shape for a 1-page homework problem and for a semester project; only the lengths change. An instructor adopting Plan-Use-Verify across a course can keep the structural template constant and adjust expected lengths to the assignment.

9.3.2 ▪ Writing the problem to start the spiral

Plan-Use-Verify wraps around a problem. The problem itself can do more of the work.

A problem written for a single answer is finished when the answer arrives. A student who reaches it has no reason to continue, and a student who cannot reach it has nowhere to go but the answer key or the AI. **AIT** assignments make a third option available, because the student is allowed a tutor and the problem can be written on that assumption. The instructor sets the trajectory once, in the wording, and every student runs it.

The Learning Spiral supplies the shape: a question at the center, then try, ask, work, verify, and reflect. A problem can supply the first turn and hand the student the second.

> **Single-answer form.** A car loan of $18,000 is repaid monthly over four years at 6% annual interest. Compute the monthly payment.
>
> **Spiral form.** A car loan of $18,000 is repaid monthly over four years at 6% annual interest.
>
> 1. Before computing anything, estimate the monthly payment and write one sentence saying what your estimate assumes.
> 2. Compute it.
> 3. If your estimate was off by more than twenty percent, ask an AI tutor what your reasoning missed. Ask for the flaw, not the payment.
> 4. Check the computed figure a second way.
> 5. Predict, before computing, what changes if the term is six years instead of four. Say whether the total interest paid roughly doubles, and why.

The arithmetic is identical. What changed is that the student now has somewhere to go when stuck, a reason to keep going after the answer arrives, and a question at the end that the first version never raised. The AI is present in step three and is asked for a diagnosis rather than a result, which is what the **AIT** category is for.

This is also the cheapest way to refresh a problem set. Recasting an old problem this way produces a new assignment even when the underlying computation is unchanged, and it does not depend on inventing a new scenario every term. An instructor with a bank of single-answer problems has the raw material already.

The closing step has a long history. Asking what a solved problem opens goes back at least to Pólya's *How to Solve It* [103]. The companion repository extends this recast into a method any course can run, with worked before-and-after pairs from a writing course, a data course, and a first-year lab, and a repertoire of closing questions in that tradition (`companion/spiral-problems/`).

9.3.3 ▪ Required mentor prompts

Some assignments benefit from a course-level requirement that students begin AI use with a mentor prompt. The pattern is most useful in **AIT**-category assignments and in introductory courses where verification competence is still developing. This subsection treats the mentor prompt as an instructor design choice.

Why require what students should ideally do on their own? The same reason any support is required in a course: not every student arrives with the habits a self-directed learner would have, and building those habits explicitly is often more effective than expecting students to develop them on their own under assignment pressure. A required mentor prompt at the start of an assignment makes guided AI use the default rather than an option students may or may not exercise.

The required mentor prompt is most useful when paired with spot checks of students' shared threads. The guidance is to read three to four threads per assignment, chosen at random or based on signal. Used together, the required mentor prompt and the thread-review practice operationalize the **AIT** category: AI is permitted, the form of AI use is guided, and the instructor can see what students are actually doing.

Thread review has a second use. The threads you spot-check, and the Plan entries in students' Plan-Use-Verify records, show what students asked and where several of them stalled. Start the next class with the two or three questions that came up most often. When a student's prompt works especially well, ask to add it to the course's starter prompts. Students then see their AI use feeding back into the course instead of running alongside it.

Different disciplines call for slightly different mentor prompts. In Calculus I, the mentor prompt should emphasize verification of intermediate algebraic steps: students with weak prerequisite skills are particularly at risk of accepting AI solutions they cannot themselves derive, and the mentor's job is to surface the gap rather than paper over it. In first-year writing, the mentor prompt should emphasize asking the student what they are trying to argue before suggesting language: AI that drafts before the student has clarified their own thinking can produce prose that displaces the student's voice without their noticing. In introductory programming, the mentor prompt should emphasize having the student narrate the algorithm before requesting code: students who skip the design step often produce working code they cannot explain or extend. In introductory statistics, the mentor prompt should emphasize verifying that an AI-supplied test or formula matches the problem's assumptions: students often accept a confident answer without checking whether the test's preconditions are met. In physics problem sets, the mentor prompt should emphasize identifying which principle or law applies before computation begins: AI-supplied solutions often produce algebraically valid work from a wrongly-identified starting point.

An important case for instructors to watch for is the student who has strong tool-use skill but weak learning-with-AI skill. These students will produce clean prompts and clean outputs and will look competent in routine assessment. They will fail under transfer questions and under in-class no-AI work. The required mentor prompt does not solve this case, but it makes it easier to detect, because the mentor's calibration step surfaces the gap that the student's tool-use fluency would otherwise hide.

Students will engage performatively with a required mentor prompt. Some will run through the calibration questions in a perfunctory way and then defect to a different AI session for the real work. Performative engagement is still better than no engagement, but it is not full engagement, and instructors should not expect the required mentor prompt to fully solve the problem it is designed to address. Thread review, occasional in-class no-AI assessment, and the Plan-Use-Verify pattern from section 9.3.1 together do more than any one of them alone.

The pattern is offered as a recommendation, not a default. Courses where students already have well-developed learning-with-AI skill (typically advanced undergraduate and graduate courses) may find a required mentor prompt more friction than benefit. Introductory courses, and courses where the student population shows the high-tool-use and low-learning-with-AI pattern, are the strongest candidates.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

9.3.4 ▪ Platform-specific implementation

The mentor prompt described above implements differently on different AI platforms. This section uses two platforms as examples: Claude and ChatGPT. The specific product features named below are a mid-2026 snapshot and will change; the companion repository keeps the current platform notes. What will not change is the difference the two patterns respond to: how much persistent context and customization a platform gives you. The differences are not minor, and a required mentor prompt that works well on one may produce friction on the other if not adapted.

On Claude, the two roles integrate in one conversation: persistent system-prompt support (Projects, Preferences, Styles) carries the mentor's setup across the session, and tutoring happens inside it with fluid transitions between the roles.

On ChatGPT, structured separation works better: the custom-instructions field is more limited, so students keep a saved mentor prompt to re-invoke at the start of each session and run tutoring as separate, task-focused conversations.

Both versions carry out the same distinction between mentor and tutor. Each one works with what its platform offers rather than against it.

The book's durable contribution here is the distinction between mentor and tutor functions articulated clearly enough that students and instructors can implement it on whatever platform they are using, including platforms that do not exist yet. The book treats specific tools this way throughout. A result about one generation of tools is cited as a finding about that generation. It is not presented as a fact that will hold for every AI system.

9.3.5 ▪ Keeping AI inside the course

Students hit a predictable snag with AI tutors: the tool does not know how far a course has gone, so it answers with whatever method is most efficient, often one the course has not taught. A student early in calculus asks about a limit and gets L'Hôpital's rule from a later chapter, which connects to nothing yet and, for that particular limit, assumes the result the student is trying to learn. A student can hold the tutor to what they have covered. You hold the stronger lever.

A one-line scope note attached to each AI-permitted assignment does most of the work. State what is in scope and what to hold back:

> *For this problem set, use only methods through the squeeze theorem (Chapters 1–3). If a problem seems to need a later method such as L'Hôpital's rule, name it and stop, rather than use it.*

A second kind of drift is easier to miss, because nothing in the course has been skipped. An AI answers from the middle of a discipline rather than from the position your course occupies inside it. A linear algebra course taught for data science is a clear case: asked whether a small symmetric matrix is positive definite, the tool reaches for the leading-minors test every textbook leads with, which is correct and is not what the course wants. The course wants the factorization that scales, stays numerically stable, and tells you where the property fails. The student has covered determinants, so no boundary was crossed and no guard fires. Naming the course's orientation in the scope note, and not only its topics, is what catches this.

The note is authoritative in a way a student's own description is not. It is identical for every student, it does not depend on each student knowing where the boundary falls, and it spares students from reconstructing the syllabus for the AI. It also does a second job: writing the scope note makes you name the method you intend the assignment to exercise, which is good design discipline on its own. The Study Partner Protocol in the companion repository (`companion/spp/`) carries a matching field, so a student can drop your scope line straight into the tutor prompt.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

A scope note reduces reaching ahead; it does not prevent it. The tool can still drift, so the note works best alongside the process evidence and verification this Part already builds, not in place of them.

A scope note says how far the course has gone. Your own materials say how the course does it. A student whose AI tutor has your notes gets explanations in your notation, with your examples, in your order. Many students will try to build this from their own handwritten notes, and some will get the transcription wrong. You can do it once, correctly, for everyone, and the students whose own notes are thinnest gain the most. Post the files in one place students can reach, such as the course website or a shared folder, and put each assignment's scope note there too. Say plainly that students may upload these files to the AI tools they use for the course; without that permission, the privacy guidance in Part I tells them not to. Typed files work best, such as text-based PDFs or the LaTeX or Markdown source, because a scanned page has to be transcribed first. Typed files also serve students who use screen readers. Add a one-page sheet of the course's notation. Update the folder as the term moves, so the newest files mark where the class is now. Share only what you are willing to have leave your hands, since students will upload it to tools run by outside companies, and leave out solutions to work not yet due, exam materials, and anything with student information in it.

The same folder is the place for your course's version of the companion files. The repository's student files are written for any course, and the license lets you adapt them with attribution; most take a few minutes. Put your course, its level, its prerequisites, and its notation into the personal-preferences template (`companion/dpp/`). Write the course's mentor prompt into the Study Partner Protocol (`companion/spp/`), with the emphasis your discipline needs and the course's scope. Add your field's checks to the verification checklist (`students/checklist.md`), and give students five or six starter prompts for the tasks your course actually assigns. Post the edited files beside the course materials and assign them in week one. The editing pays back. An instructor who has written the course's mentor prompt knows what every student's AI has been told, so the threads students share are easier to read. In a course with several sections, one person keeps the folder for all of them, so every student's AI hears the same course.

Add the rules as well: your institution's AI guidelines and its list of approved tools, and the AI section of your syllabus. A student's AI that has the rules can help the student keep to them. Post them in order, institution, course, then assignment, and say that the assignment's label decides when they differ. Prefer a short version of a long policy, which an AI can misread. The student, not the AI, stays responsible for following them.

The folder can also hold practice for the answer-checking habits in Part I: a few worked solutions with a typical mistake planted in each, and, for key problems, the checks a right answer must pass, such as its expected size or a limiting case. Errors you plant are the ones your students actually make; an AI asked to plant an error in its own work may pick an easy one. Add the week's learning objectives and the mistakes students in your course most often make, so the AI quizzes students on what matters.

9.3.6 ▪ Creative work: protect the first move, widen the search, grade the choice

When an assignment asks for ideas rather than answers, three design moves carry most of the weight. Protect the independent first move: five minutes of sketching before devices open, a short pre-AI note online, a question map before any tool-assisted ideation session in a project course. No surveillance is needed, because the point is sequencing rather than proof of innocence. Then use AI to widen the space rather than collapse it: ask for alternatives under different assumptions, for counterexamples, for a comparison rather than a verdict. A student who asks the model to choose has handed over the part of the task that was theirs. Then grade the reasoning behind the

selection, not the number of ideas or their polish. AI raises surface quality at no cost, so a grade for polish largely measures the tool.

The four categories say where each move belongs: **NAI** tasks hold the first move and the later transfer. **AIT** tasks are where the widening happens, with the submitted work, and the ideas in it, still the student's own. **AIC** tasks admit co-creation, and so need the process evidence, the verification, and the disclosure that make the choice visible. **AIS** tasks make the model's own creativity the object: do its alternatives really differ, or only in wording? A course that wants creative capability needs all four at different moments. The companion paper "Learning to Create with AI" works these three moves through six disciplines and a navigation layer [54].

9.3.7 ▪ Design features that improve assessment validity

Assignments are stronger when they include one or more of the following features. The pattern these features share is the one articulated by the backward-design approach to assessment [135]: working from the learning outcome to the evidence that would demonstrate it, then to the assignment that produces such evidence, rather than from familiar assignment forms outward.

- A first attempt before AI use.
- A short explanation of choices made during revision.
- A verification checklist.
- A source or evidence audit.
- A transfer question completed without AI.
- A short oral, written, or video explanation.
- A flawed AI response that students must critique.
- A comparison between the student's approach and an AI-generated approach.

These features move assessment away from polished output alone and toward reasoning, judgment, and process. The distinction between polished output and demonstrated learning is the conceptual heart of AI-aware assessment: grade the thinking, not the polish. This is the practical face of the long-established *learning versus performance* distinction in cognitive psychology [16, 116]: conditions that improve immediate performance can fail to produce durable learning, and the converse.

A 2026 analysis of survey data from 2024 supports the same reading: its authors argue that assessments may be becoming weaker indicators of capability when they judge polished final work without evidence of process, judgment, and independent performance. The authors conclude that the remedy is assessment reform rather than bans and detection [25].

One caution belongs here. Controlled conditions are the most direct way to see independent work, and a reader of this Part could conclude that more of them is always better. It is not. An exam written without help gives a clean read on Core Competence. It says almost nothing about whether that student can scope a project, weigh a source, or work with a tool honestly. Assessment research makes this point against assignments built mainly to defeat AI, which rarely capture the broader knowledge and judgment a program is trying to develop [25]. The framework here treats controlled work as one of three outcomes rather than the goal. Use it where you need to see independent competence. Assess AI-assisted Practice and Trustworthiness on assignments designed for them.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

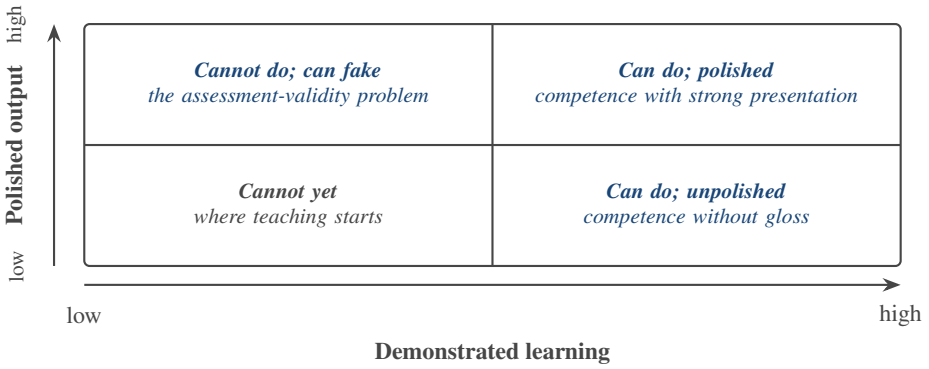


Figure 9.3. ***Polished output vs. demonstrated learning.*** *Final polish and underlying competence are independent. The upper-left quadrant is the assessment-validity problem: high polish without learning. Strong AI-aware course design moves student work toward the right two quadrants and gives instructors evidence to distinguish all four cases. Tool use can land anywhere on this grid, with or without AI; that is why the axes are polish and learning rather than tool and no-tool.*

9.3.8 ▪ Design features that weaken assessment validity

Some common assignment designs are now fragile.

- A take-home task graded only on final polish.
- A generic prompt that can be answered well by a public chatbot.
- A writing assignment with no draft history or author reflection.
- A coding assignment graded only on whether it runs.
- A problem set where students submit only final answers.
- A citation-heavy assignment with no source verification requirement.

These assignments may still be useful for practice, but they should not carry too much weight as evidence of independent understanding.

9.3.9 ▪ Skill: when doing is the evidence

The diagnostics in this Part, and the three models that follow, mostly test what a student can explain. The translation test asks a student to say a thing in another form; an oral defense asks a student to account for the work. These are the right checks for understanding that lives in words and symbols.

Some competence does not. A student who can run the assay, suture the wound, throw the pot, or execute the design owns that work whether or not they narrate it well. For skill of this kind the evidence is the performance, not the account of it. A practiced welder may struggle to put the motion into sentences and still lay a flawless bead. Grading the explanation would measure the wrong thing, and demanding one can understate what the student has.

The design response is to assess skill by having the student do it under observation, with the same care for ownership the rest of this Part asks for. Watch the technique, vary the conditions, ask for a repair or a next step, and judge the doing. AI's role shifts to match. For skill it is a coach rather than a source to verify: where the domain allows, it can give feedback on form, generate graded practice, or run simulations to rehearse against. There is no claim to check, only

a performance to build. Any of the three models can carry this; what changes is not the model but what counts as the evidence. For a skill, it is the work performed, not the work described.

Table 9.1. ***Three assessment models compared.*** *No model verifies everything; each buys a different kind of evidence at a different cost. The Oral-Defense Model relies on live questioning to confirm understanding; the Evidence-of-Thinking Model relies on process records that are difficult to outsource; the Hybrid Model combines both. The right model is the one that fits the course size, staffing, and goals.*

	Oral-Defense	**Evidence-of-Thinking**	**Hybrid**
Best for	Small classes; seminars and honors courses; graduate courses	Large lectures; service courses; online courses	Medium courses; lab and project work; multi-TA staffing
Evidence sources	Oral conferences; live questioning; submitted work; in-class observation	Process records; in-class quizzes; random audits; submitted work	Mix of oral and process; spot oral checks; process records; in-class checks

These models are operationally plausible, and direct evidence shows the Hybrid pattern can work in at least one setting: a redesigned programming course, described with the Hybrid Model below, in which students with and without AI access reached similar learning outcomes [70]. The three models share one design aim: what students learn should not turn on whether they used AI. The Oral-Defense and Evidence-of-Thinking models await their own direct evidence.

9.4 ▪ AI tools for teaching preparation

AI can now help with much of the work of preparing to teach, and an instructor cannot adopt a capability they have not seen. The durable point is short, and the practical detail belongs elsewhere. What follows names the kinds of help that exist; the companion repository maintains the working guide to them.

These uses share a low risk profile because they act on your own materials, not on student assessment. The same tools can rewrite a passage at several reading levels for a mixed class, turn an outline into draft slides or a worked-example set, propose quiz questions or a first-pass rubric from your content, convert a reading into audio or a figure into a described alternative for accessibility, draft routine communication, or turn a recorded lecture into notes for a student who missed it. The higher-risk uses, AI feedback on student work and AI-assisted grading, are different in kind and carry boundaries of their own.

Every one of these produces a draft, and the instructor who uses it is responsible for checking the result. That is the whole of what this book needs to say. AI removes the friction of a first draft. It does not remove the responsibility to verify, and it does not replace the judgment that made you the teacher of the course.

The rest is practical, tool-specific, and changes too fast for print. The companion repository keeps a working guide to teaching-preparation tools: what each kind of tool is good for, how to use it well, what a good result looks like, and how to obtain current tools, most of which have free versions (`companion/instructor-capabilities/`). It is meant to be updated as the tools change, which they will.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 10

Assessment Models

A mistake is not yet a lesson.
The lesson begins when you ask why.

Automation bias is documented outside education. In one study, people ran a monitoring task with and without a reliable automated aid. The ones without the aid did better. The ones with it missed problems the aid did not report, and followed its recommendation even when their training and every other reliable indicator said otherwise [114]. The aid was accurate almost all the time. Almost was enough.

The mechanism is substitution. A person watching a system attends to many things at once and names none of them. When a tool takes over the watching, the person attends only to the few things the tool reports. The tool does less than the person was doing, and the person credits it with the whole job. Grading works this way too. Reading a student's work, you register whether the reasoning holds, whether the writing is understood or copied, whether the explanation would survive one follow-up question. Replace that reading with a proxy, a rubric line, a detector score, an automated grade, and your attention narrows to what the proxy reports. Work passes while the learning is absent, and the gap does not announce itself, because the proxy gave an answer and answers feel like knowledge.

The useful part is that the way a task is arranged changes how strong the bias is. People told they would answer for their accuracy checked more [115]. Engineering students who met the automation failing during training went on to check its recommendations more carefully than those who were only told it could fail, though the training reduced their errors without ending them [10]. The bias is not fixed. It responds to how the work is set up. The three models that follow are ways of setting it up.

10.1 ▪ Where this fits in a grade you already give

Every course already divides its grade. Homework, exams, attendance, a final, sometimes writing. The models in this chapter do not replace that formula. They divide some of its lines.

A calculus course grades homework at 20 percent, two midterms at 30 between them, a final at 40, and attendance at 10. To run the Evidence-of-Thinking Model, split the homework. Ten percent stays no-AI work done and checked in class. Ten percent becomes AI-assisted work submitted with a short verification note. The midterms, the final, and attendance do not move. The formula still sums to 100 and the registrar never hears about it.

This matters more than it looks. Nothing here needs a new record, a new transcript line, or a committee. Where the instructor already sets the weights, no one above the instructor has to

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

approve the change. In a coordinated multi-section course, settle it with the coordinator first. A department can adopt most of this chapter without asking its institution for anything; AI grading tools are the exception.

The cost is that one grade now reports both independent work and AI-assisted work. A student who is strong alone and careless with AI, and a student who is the reverse, can finish the term with the same grade, and nobody reading it later can tell them apart. Inside a course that is usually fine, because the instructor saw both halves. It matters more the further the grade travels.

That balance may not hold. This book tells students that graduate programs and employers will ask them to show what they can do as their own work. When that question is put to the record rather than to the student, a single blended grade will stop answering it, and separating the record will become worth its price. The approach in this chapter is the right one now. It is not obviously the right one permanently.

10.2 ▪ The Oral-Defense Model

The Oral-Defense Model is best suited to small classes of any level: seminars, honors courses, writing workshops, independent studies, capstones, advanced labs, and many graduate courses. The model is class-size-driven; it works wherever the instructor (or trained TAs) can speak directly with each student about their submitted work. The assignment-category mix and the AI autonomy granted to students will shift with course level (section 9.2.7), but the assessment model itself is the same. Table 9.1 compares the three assessment models in this Part.

The model rests on a simple principle:

> **Students may use AI more extensively when they can personally defend the work.**

An oral defense does not need to be adversarial. It can be a teaching conversation. Its purpose is to test whether the student's submitted work has become part of the student's understanding.

10.2.1 ▪ Recommended structure

A short oral defense can take 15 to 20 minutes. A fuller version can take 30 to 60 minutes. The following structure works across disciplines.

Segment	Purpose
Student summary	The student explains the submitted work in their own words.
Core question	The instructor probes a central concept, method, source, or decision.
Audit-and-repair	The student critiques a flawed AI response or a flawed piece of their own process.
Transfer question	The student applies the idea to a new but nearby situation.
Process defense	The student explains how AI was used, checked, and limited.

Where more than one person runs these defenses, the teaching team should agree on what a strong answer looks like for each segment before the first one.

10.2.2 ▪ Examples across fields

In mathematics, the student may prove a result at the board, identify which hypotheses matter, and repair a flawed proof. In biology, the student may explain the design of an experiment and

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

interpret a surprising result. In history, the student may defend the use of a primary source and distinguish evidence from background summary. In writing, the student may explain why a paragraph was revised and how the claim changed. In computer science, the student may walk through code and predict what happens on an edge case.

The format changes by discipline. The purpose remains the same: the student must show ownership.

10.2.3 ▪ Equity in oral assessment

Some students hear "oral audit" as threatening, even when the instructor intends a teaching conversation. A few small adjustments protect fairness without weakening the assessment. Share example audit questions in advance so students can prepare. Grade for understanding, not verbal polish. Allow students to point to their work, write a formula, sketch a diagram, or explain in steps; oral does not have to mean spoken-only. The goal is ownership, not performance under pressure. These adjustments help multilingual students, shy students, neurodivergent students, and students who understand the material but need a moment to organize their explanation. The Oral-Defense Model is a calibration-restoring design; the calibration is of student understanding, not of presentation skill.

Two further considerations matter. First, students with documented accommodations through the institution's disability services (including students with anxiety disorders, students who use augmentative communication, students who are deaf or hard of hearing) may need formats beyond the spoken-aloud audit. A short written explanation, a signed conversation, a typed exchange, or a recorded video with subtitles can satisfy the diagnostic purpose while accommodating the student. The framework is about ownership of work, not about fluency in any particular mode of demonstration. Second, instructors who can do so should make alternative formats available to any student on request, not only those with formal accommodations on file. Some students whose understanding is sound will struggle in live questioning for reasons that do not appear on any accommodation letter; quiet flexibility about format protects the model's validity.

10.2.4 ▪ Worked example: a small seminar

A 14-student upper-division seminar meets once a week. Each student gives one oral defense during the term, fifteen minutes, scheduled in office hours, on a paper they choose. The defense follows the five segments above. The student summarizes the work, answers one core question, repairs a flawed passage the instructor supplies, takes one transfer question, and explains how AI was used and checked. The grade rests on ownership, not on verbal polish, and the instructor shares example questions in advance so the conversation is preparation rather than ambush. A student who used AI heavily and a student who used none can both earn full marks, because the defense, not the AI policy, sets the grade.

The time objection usually comes first, and the fair test is against the alternative rather than against zero. Catherine Hartmann, who replaced written papers with oral examinations in a 25-student course at the University of Wyoming, reported spending under thirteen hours in total, against the fifteen hours she had spent grading midterm and final papers in a previous semester [129, ch. 14]. That comparison will not hold at every size, which is why large classes get a model of their own, but at seminar scale the oral defense is not obviously the more expensive option.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

10.2.5 ▪ Suggested weighting for small classes

Component	Typical weight
Core competence assessments	30–40%
AI-integrated assignments or projects	25–35%
Oral defense or conference	20–30%
Disclosure, verification, and reflection	10–15%

These weights are only a starting point. A writing workshop may place more weight on conferences and revision. A lab course may emphasize practical demonstration. A proof-based course may require a higher independent competence threshold.

10.3 ▪ The Evidence-of-Thinking Model

The Evidence-of-Thinking Model is designed for large classes where one-on-one oral defense is not feasible for every student: large lectures, service courses, multi-section courses, general education courses, and online courses. Large classes are usually undergraduate, but the model is class-size-driven, not level-driven. A large advanced course or a large graduate-equivalent course uses the same evidence-of-thinking patterns, with the assignment-category mix shifting with course level (section 9.2.7).

The model rests on a different principle:

> **When individual oral examination is not feasible, the course should collect repeated, low-cost evidence of student thinking.**

This model does not assume dishonesty. It assumes that polished products are no longer enough. In a large course, instructors need scalable ways to see whether students understand what they submit.

10.3.1 ▪ Recommended structure

A course using this model does not need every component that follows. A workable default combines three: a stream of low-cost process records on regular assignments, a small set of in-class checks that cannot be outsourced, and spot checks that make the process records worth taking seriously. The weighting and staffing then scale with enrollment.

Component	Purpose
Process records	Plan-Use-Verify notes or draft histories on regular assignments make thinking visible at scale.
In-class checks	Short no-AI writing or problem solving anchors the grade to work done under observation.
Spot checks	Random oral or written audits on a fraction of submissions create probabilistic accountability.
Weighting and staffing	The component mix and the grading load scale with enrollment.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

10.3.2 ▪ Scalable evidence sources

Useful evidence sources include short no-AI quizzes, homework-to-quiz transfer questions, in-class writing, process logs, source-verification sheets, draft histories, lab notebooks, code test reports, AI-output critiques, video explanations, peer review, and random oral audits. A **process log** is a short record of how a student approached the task, used AI, checked the output, and revised the final work.

The goal is to distribute accountability across the term. A course with ten small checks is often more robust than a course with two high-stakes take-home assignments.

Process evidence should be short and targeted. Long AI transcripts and lengthy prompt logs should not be rewarded for length, since process documentation can drift into theater. The useful evidence is a decision, a verification step, a correction, or a change in understanding. Evidence-of-Thinking is not a paperwork model; it is a calibration model that reads small records for signal.

10.3.3 ▪ Worked example: a large lecture

A 280-student introductory lecture cannot defend every student orally, so it leans on repeated low-cost evidence. Every weekly problem set carries a three-line Plan-Use-Verify note. Each Friday section opens with a five-minute no-AI transfer quiz on the prior week. Teaching assistants read a random tenth of the Plan-Use-Verify notes each week, and any submission that crosses a stated complexity threshold draws a short written audit. No single check carries much weight. Together they let a submission built on hollow understanding stand out against the student's own quiz scores and process record, without the instructor reading every prompt from every student.

The cost of this is bounded but not zero. Reading a random tenth of the notes and running one short quiz per section adds a few hours per assignment cycle, not the dozens that reading every prompt would take. The labor it does require is recurring and intellectual: writing transfer questions that probe the week's idea rather than its surface, calibrating what an audit should look for, and keeping rubrics current. Teaching assistants can carry much of the spot-checking, but only if they are trained to grade for the presence of a genuine process rather than its depth: did the student plan, use AI, and verify the result? A course that asks a teaching assistant to judge the depth of thirty such notes each week breaks. A course that asks whether each was genuinely attempted scales.

10.3.4 ▪ Suggested weighting for large classes

Component	Typical weight
Short no-AI checks	25–35%
AI-assisted homework or projects	20–30%
Process evidence and verification notes	15–20%
AI critique or error-detection tasks	10–20%
Random oral, video, or TA-led checks	5–10%
Reflection and disclosure	5–10%

A large course does not need to use every component. The important point is balance. Students should have chances to practice with AI, but the course should also include direct evidence of independent understanding.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

10.3.5 ▪ Random oral audits

A random oral audit is a short conversation about submitted work. It may last five to eight minutes. Students know in advance that any AI-assisted assignment may be selected.

A good audit asks ordinary questions.

- Explain the main idea of your submission.
- What did AI help with?
- What did you change after checking the output?
- Show me one place where you verified a claim.
- What would change if this assumption were different?

The audit should be framed as part of the course design, not as an accusation. It protects fairness by holding everyone to the same expectation: if students submit work, they should understand it.

10.3.6 ▪ TA-supported mini-conferences

In multi-section courses, trained teaching assistants can conduct short structured conferences. The instructor should provide a common script and rubric. TAs should grade a small sample together before conducting conferences independently.

This structure is most useful when assignments are **AIC** or **AIS**. It helps ensure that AI-assisted work remains connected to student understanding.

10.3.7 ▪ The seeded-error check

Verification is easy to assert and hard to measure. This check measures it.

Take a piece of AI-produced work in your subject and plant a fixed number of errors in it. Vary them: one that any careful reader catches, one that requires knowing the material, one that requires checking a source. Give students the passage, tell them how many errors it contains, and ask them to find and explain each one.

Scoring is the detection rate. A student who finds four of five has demonstrated something a polished submission cannot demonstrate. A class that averages one of five has told you where the term needs to go.

Three things make it work. Tell students the count, so the task is detection rather than suspicion. Ask for the explanation, not the flag, because naming why something is wrong is the competence being measured. And change the errors between terms.

It scales. The passage is written once, the errors are known in advance, and scoring is mechanical enough to delegate. Difficulty is set by how subtle the planted errors are, which makes the same check usable in a first-year course and a graduate seminar. The chapter opened with the finding that people who met the automation failing checked its later advice more carefully. This is that experience, arranged as coursework.

10.4 ▪ The Hybrid Model

Many courses fall between the small-class and large-class models. A class may be too large for full oral exams but small enough for some direct contact. A lab course may have many students but strong TA support. A writing course may have conferences for major projects but not for every assignment. The Hybrid Model is the realistic choice for these courses; it is the third of the three

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

calibration-restoring designs presented in this Part. Process records carry most of the load, while live spot-check conversations provide direct verification at the rate the staffing allows.

Useful hybrid tools include group oral exams, oral spot checks, short student conferences, video explanations, peer review with instructor spot-checking, TA-led recitations, and in-class transfer questions. The pattern is to use live explanation where it matters most and process evidence everywhere else. The remainder of this section develops the model in five parts: recommended structure, how to choose whom to spot-check, a worked example, suggested weighting, and two empirical anchors.

Peer critique deserves more than its place in that list. Asking a student to review another student's work tests something the submitted work cannot. To judge an argument, a reviewer has to understand the subject, decide whether the evidence supports the claim, locate the weak points, and say how the work could be improved. A student who cannot do the work usually cannot review it well either, and the review leaves a record of the attempt.

This resists AI for a different reason than the other designs here. It does not remove the tool or try to detect it. It makes the evidence relational, since two students are each accountable for reading the other. The obvious objection holds, because a review can be AI-generated like anything else. The answer is the same as everywhere else in this Part. Grade the review as work, ask the reviewer to defend a judgment in person, and spot-check.

Grading any live or process element carries one more obligation. A rubric line for reasoning visibility assumes students know how to show reasoning, and most have never been asked to do it in front of anyone. The rule this guide applies to disclosure applies here as well: treat a weak first presentation as missing instruction rather than missing ability. Model one before grading any. Show the tool's output beside what a group did to it, and name what made the checking good.

10.4.1 ▪ Recommended structure

A hybrid course pairs one or more live elements with continuous process evidence. The pairing depends on the staffing and the course type.

Table 10.1. ***Hybrid pairings by course type.*** *The live element provides direct verification of understanding; the process element distributes accountability across the term.*

Course type	Live element	Process element
Lab course (intermediate enrollment, strong TA support)	Lab-time oral defense of submitted code or analysis	Lab notebook, code tests, in-class transfer questions
Medium lecture (40–80) with TAs	One TA-led conference per student per term	Process logs, weekly transfer quizzes
Writing course with conferences (20–40)	Mid-term and final 20-minute conferences on major projects	Draft histories, author's notes, in-class writing
Flipped-classroom STEM (any size)	In-class problem-solving with instructor circulation	AI-tutor logs from pre-class material, post-class transfer quizzes

Table 10.1 sketches the mapping for four common course types. The mapping is illustrative; many other pairings work. The instructor's task is to identify the live element the staffing supports,

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

then design process evidence to cover the rest of the term. A common error is to over-design both elements, producing a course that asks students to document everything and defend everything, which collapses into paperwork. The hybrid model works when each element does its part: the live element verifies depth through spot checks; the process element provides continuous low-cost evidence.

10.4.2 ▪ Choosing whom to spot-check

When a course cannot conference every student on every assignment, the instructor must choose whom to spot-check. That choice is a sampling problem. Three sampling strategies are common, with different trade-offs.

Random sampling. Every student has the same probability of being audited on every assignment. Fairest in expectation, simplest to administer, hardest to argue with. Works well when the course has enough audits per term that most students will be sampled at least once. The drawback is statistical: a student with weak understanding may not be sampled on the assignment where the gap is largest.

Stratified sampling. The instructor stratifies the population (by submission quality, by stated AI use, by prior audit history, or by self-selection through optional sign-up) and samples within strata. Concentrates effort where it adds most assessment information. The risk is the appearance of targeting; the strata should be public and the rationale should be educational rather than punitive.

Threshold sampling. The instructor audits any submission that meets a threshold condition (a confident-sounding answer to a problem the student has not worked on in class; a major project crossing a stated complexity boundary; a flagged inconsistency between submitted work and prior work). Most efficient when the thresholds are known in advance. Vulnerable to gaming: students who learn the thresholds may submit work that just stays under them.

A hybrid course can mix strategies across the term. For example, a course might use random audits early, when the threshold for problematic submissions is still being calibrated. It might use stratified or threshold audits later, when the instructor has seen enough of each student's work to recognize patterns.

10.4.3 ▪ Worked example: a hybrid course across the semester

Consider a 60-student intermediate course with two TAs. Class meets twice weekly for 75 minutes; one TA-led recitation per week. The instructor designs the course as follows.

Week 1. Students set up their AI from the course's versions of the preferences template and the Study Partner Protocol, posted with the course materials, as a no-credit first assignment. The instructor explains the course's category mix (mostly **AIT** and **AIC**; one **NAI** mid-term; one **AIS** critique exercise late in the term).

Weeks 2–4. **AIT** and **AIC** homework assignments, each requiring a short Plan-Use-Verify record (one short paragraph each for plan, use, and verify). Weekly 5-minute in-class no-AI transfer quizzes on the prior week's material.

Week 5. Each student completes a 5-minute check-in conference with their assigned TA. The conversation is structured: the student summarizes one recent assignment, answers one transfer question, and describes their AI use pattern. The TA records a brief note. This early conference is a check-in, not a grade.

Weeks 6–9. Continued homework + transfer quizzes. One **AIS**-category critique assignment around Week 8 (students critique an AI-generated answer to a course-relevant problem and identify two errors). Process logs are required for every assignment; the instructor reads only the verification notes, spot-checking the rest.

Week 10. Mid-term exam: paper-and-pencil, no-AI, weighted as the **NAI** component of the

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

course.

Weeks 11–15. Larger **AIC**-category project. Students submit a project proposal at Week 11; receive AI-assisted draft feedback at Week 13; submit a final draft at Week 15. Each student's project culminates in a 10-minute oral defense during the final week.

Week 15. Project oral defenses, run in parallel by the instructor and the two TAs across the week's class sessions, the recitation, and added office hours.

Final. Cumulative process review. The instructor scores process records across the term, weighted as a single component.

The course distributes assessment across the term: weekly small no-AI checks, two oral conferences (early check-in + final defense), one paper-and-pencil mid-term, one **AIS** critique, one Plan-Use-Verify pattern on every **AIC** assignment. No single assessment carries excessive weight; no single modality is the only path to demonstrating learning.

10.4.4 ▪ Suggested weighting for hybrid classes

Component	Typical weight
Core competence assessments (no-AI quizzes, mid-term, **NAI** components)	25–35%
AI-integrated assignments or projects (**AIC**)	25–35%
Process evidence and verification notes (Plan-Use-Verify records)	15–20%
Oral, video, or conference spot checks	10–15%
AI critique tasks (**AIS**)	5–10%
Reflection and disclosure	5–10%

These weights are starting points. A lab course will weight the live lab-time defense higher; a writing workshop will weight conferences and revision higher; a foundational course will weight no-AI assessments higher. The principle is that no single modality should carry the whole assessment load. A student who is weak on one modality but strong on others should be visible to the instructor as the intermediate case they are, not as a failing student.

10.4.5 ▪ Two empirical anchors

Two recent studies provide empirical grounding for the patterns above. Kestin et al. [61] showed in their randomized controlled trial that an AI tutor designed around the seven pedagogical practices listed in section 9.3 could outperform in-class active learning on the introductory material it covered, when measured by post-test learning gains. The implication for course design is not that AI should replace classroom instruction. The implication is that, when AI tutoring is well-designed, it can bring many more students up to a level where in-person class time can be spent on advanced problem solving, project work, group discussion, and the kinds of higher-order tasks that benefit most from a faculty member's presence. Used this way, AI tutoring functions as a flipped-classroom mechanism, with the AI handling first exposure and the instructor handling deepening, transfer, and oral assessment.

Kosar et al. [70] report a more direct precedent for the Oral-Defense and Evidence-of-Thinking elements that hybrid courses combine. They redesigned a programming course for novice programmers by designing home assignments to limit what an AI chatbot could answer directly, using lab time for oral defenses in which students explained and modified their submitted code, and moving the mid-term exams to paper and pencil. The redesign produced similar learning outcomes for student groups with and without generative AI access. It supports the plausibility

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

of the broader design logic, but the Oral-Defense and Evidence-of-Thinking models await direct evaluation of their own.

10.5 ▪ One assisted task, one later transfer task

An assessment that uses AI has to answer two questions, and one task cannot answer both. What can the student produce with responsible AI assistance? What remains when that assistance is removed or changed? In the CAT vocabulary, the first task measures AI-assisted Practice and the second measures Core Competence. Trustworthiness is what connects them: a student who can explain, adapt, and check the assisted work is usually the student whose second score will hold.

The design is a pair: one assisted task, then a later task without the same assistance, one that keeps the underlying structure and changes the surface enough to prevent copying. The change of surface is the point, since a transfer task that repeats the assisted task measures memory, and one that changes the structure as well measures something else. A derivation with the constants changed, a proof with the roles of two objects swapped, a program that must handle the case the first version did not: each keeps the reasoning and moves the details.

This is the logic of a think-first experiment in which the students who thought before consulting ChatGPT were the ones who did better on an unaided invention task afterward [137]. It is also why the three models in this chapter already carry an unaided component. The oral defense, the in-class no-AI work, the spot check, and the seeded-error check are transfer tasks of different sizes. Whichever model a course uses, assess both halves when the outcome that matters is what the student can do without the tool. The companion paper "Learning to Create with AI" develops the pairing for creative work, where the gap between assisted work and learned capability is likely to be widest [54].

10.6 ▪ Group and collaborative assignments

Group and project work adds an assessment problem that AI sharpens. A shared product is even weaker evidence of any one student's competence than an individual submission, because the contributions of several students and of AI are combined in a single submission. The three assessment models extend to this case rather than being replaced by it. Under the Oral-Defense Model, ask each member to defend the part they contributed, not only the group to present the whole. Under the Evidence-of-Thinking Model, require individual-contribution records, so that each student's process is visible alongside the joint result. The Hybrid Model combines the two and is often the right choice for project courses.

Two design choices do most of the work. First, ask the group to agree on and state its AI-use norms, in the same way an individual assignment states its category (section 9.2). A team that has not discussed whether AI may draft its shared text will discover the disagreement at the worst time. Second, collect evidence of each member's individual contribution, so that the grade reflects what each student learned rather than the polish of the combined output. A short individual reflection or a brief oral check per member is usually enough.

Keep the two senses of collaboration distinct. AI as a collaborator is the **AIC** category; collaboration among students is the peer interaction that good group work is meant to build. The two can pull against each other. A team that divides the work and routes each piece through AI has neither collaborated with one another nor learned much, which is the erosion of peer learning noted in section 9.3. Group assignments are stronger when they reward joint reasoning and weaker when they reduce to parallel individual work stapled together. The strongest group work also rewards disagreement. A team whose members challenge one another's reasoning, rather than dividing the task and trusting each piece, practices the same scrutiny they must apply to AI. An

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

assignment that asks a group to surface and resolve its disagreements, not only to produce a joint answer, builds that habit.

10.7 ▪ Feedback and grading with AI

Instructors may also use AI. This can be legitimate and useful, but it requires boundaries.

Feedback design is one of the most-studied levers in the learning-science literature, building on the foundational case for formative assessment made by Black and Wiliam [17]. Effective feedback addresses three questions for the learner: where am I going, how am I going, and where to next; and it works best when directed at task and process rather than at the person [47].

AI may help instructors draft practice questions, generate examples, identify possible misconceptions, rewrite assignment instructions, produce alternate explanations, create low-stakes quizzes, and draft preliminary feedback. These uses can save time and improve student support. The standards that govern this work are the ones in Appendix A: an instructor who drafts a problem with AI verifies it, and discloses the assistance where a reader would want to know. Writing a fresh problem set beats reissuing one from six years ago, and the verification is what makes it safe to do.

But final evaluative judgment remains the instructor's responsibility. AI should not assign final grades. It should not decide whether misconduct occurred. It should not be used in ways that violate student privacy or institutional data rules.

A good instructor-facing standard mirrors the student standard:

> **AI may assist judgment, but it must not replace or inhibit it.**

10.7.1 ▪ Safe uses of AI for feedback

Relatively safe uses include asking AI to suggest clearer wording for feedback already written by the instructor, generate additional practice questions for common errors, produce alternative explanations of a concept, or check whether a rubric covers the intended learning outcomes.

Even here, the instructor should review the output. AI-generated feedback can sound confident while missing the actual student difficulty. It can also become too generic to help.

When using AI to draft feedback on student errors, prompt the AI for step-by-step hints rather than complete corrections. Suppose an instructor has AI fully rewrite a student's flawed paragraph and hands it back. The instructor has then offloaded the student's thinking to the machine. The desirable difficulty that produces learning is lost [16]. Hints, leading questions, or partial models preserve the work the student still needs to do, which is the work that produces the learning. Tell students when AI helped make what they receive. The standard is the one you are asking of them. A line in the syllabus covers standing practice. A sentence on the work itself covers the times it matters, as when AI drafted the comments on their paper. The disclosure is the same record of process the students are keeping.

A constructive recent example is the *Tutor CoPilot* system developed at Stanford, an AI tool built on a model of how expert tutors reason and integrated into an online tutoring platform. In a randomized study of 783 tutors working with 1,013 underserved students in the United States, the tool raised student pass rates by an average of 4 percentage points, with the largest gains for students of lower-rated tutors (9 percentage points) and of less-experienced tutors (7 points), and smaller gains for stronger tutors [132]. The study has so far appeared as an Annenberg working paper, and the specific effect sizes should be read with that status in mind. The pattern is informative even at preliminary status: AI assistance supporting human teachers produced measurable gains in session pass rates without replacing the human teaching role. For instructors thinking about how AI might support their own feedback workflow, this study illustrates that the

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

most useful design point is often AI that supports the less-experienced rather than AI that replaces the experienced.

10.7.2 ▪ High-risk uses of AI for grading

High-risk uses include uploading identifiable student work to unapproved tools, using AI to assign scores without human review, using AI-generated accusations of misconduct, or using a model's explanation as the sole basis for a grade.

These practices can harm students and weaken trust. They should not be adopted without explicit institutional approval, privacy review, and human oversight.

10.7.3 ▪ Letting AI help with grading

Publishers have scored homework automatically for years. Those systems compare a final answer with a key. The newer systems are different in kind: they read a derivation or an essay and judge it. That judgment is the part of grading this book asks the instructor to keep, so the question is not whether to let AI grade. It is how much of the reading an instructor can hand over and still stand behind the grade.

The evidence so far is mixed. A 2025 synthesis of 65 studies comparing language-model scores with human scores found agreement anywhere from none to near-perfect. Many fell short of the level the testing industry has long expected of machine scoring, and the spread inside a single study, across prompts and models, was sometimes as wide as the spread across studies [78]. Agreement tended to be better when the model had a detailed rubric and scored examples, and worse on classroom assignments than on the standard research datasets. Most of the work concerns essays in language courses; seven of the 65 studies scored subject-matter writing. The same synthesis is fair to the machines on one point: human graders disagree too, so a human score is a reference, not a truth.

Mathematics and physics have less evidence but some. In a high-stakes thermodynamics exam of 252 students, the hardest step was not the judging but reading the handwriting [69]. An earlier feasibility study concluded that such a workflow suits formative feedback and, for final evaluation, should assist human graders rather than replace them [67]. A follow-up used item response theory to set thresholds for when an AI score can stand and when a person has to look [68]. That is the right shape for the question: not "is the tool accurate?" but "on which answers is it accurate enough?"

Two failures deserve particular attention because neither announces itself. The first is bias. When one widely used model was asked to score essays, it could partly infer a writer's first language from the text, and its scoring errors for non-native writers were larger when it had inferred correctly [138]. This is the detector problem in a new place [79]. The second is drift. A model sold under the same name can change underneath its users: in one study, a model's accuracy at telling primes from composites fell from 84 to 51 percent between its March and June versions [23]. A grader checked in September is not necessarily the grader at work in December.

An instructor who wants AI's help with grading can have it on four conditions. Together they are the **grading-agreement check**, and a course that cannot meet all four should not let AI score work that counts.

1. *Write the rubric first, with examples.* A rubric an AI can apply has explicit levels and a scored example at each one. Writing it is also the best test of whether the assignment says what it is for.

2. *Grade a sample by hand before trusting the tool.* Grade enough work to cover every level of the rubric, without seeing the AI's scores, then compare. Two numbers decide it: the

tool should agree with you nearly as well as a second careful grader does, and its agreement should reach the level the testing field expects of machine scoring.[7] If it does, the tool can take a first pass. If it does not, it cannot.

3. *Keep checking.* Record which model and version did the scoring. Re-check a small sample whenever the tool changes and once in mid-term, and look at where the disagreements fall. If they cluster among multilingual writers, or on one kind of solution, the tool is not grading those students fairly.

4. *Decide every grade yourself.* The AI proposes; you decide. Read every case where its score and your spot check disagree, and do not issue a grade you could not explain to the student.

Figure 10.1 draws the four conditions as one sequence.

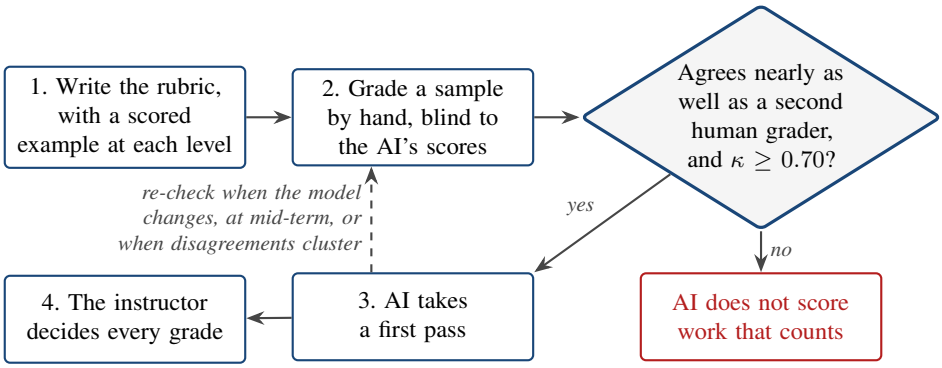


Figure 10.1. ***The grading-agreement check.*** *AI may take a first pass at grading only after it matches the instructor on a hand-graded sample, and only until the model changes. The diamond holds two tests: agreement nearly as good as a second careful grader's, and a quadratic weighted kappa of at least 0.70. Failing either ends AI scoring of work that counts. The dashed arrow returns to the hand-graded sample whenever the tool changes, at mid-term, and whenever disagreements cluster on one group of students or one kind of solution.*

The repository holds the working form of this check: a rubric template with anchor levels, the sample-comparison worksheet, and the record fields conditions 3 and 4 call for (`instructors/grading-agreement-check/`).

Students are owed three things when AI has helped grade their work: to be told that it did and what it did, to know that a person decided the grade, and a way to ask that person to look again. A line in the syllabus covers standing practice. A sentence on the returned work covers the rest.

[7] The usual benchmark is a quadratic weighted kappa (a standard measure of how far two graders agree beyond chance) of at least 0.70 between machine and human scores, with machine-to-human agreement no more than 0.10 below the agreement between two human graders [78, 136].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 11

Integrity, Equity, and Discipline

11.1 ▪ Annotated dialogues: weak AI use as teaching cases

The following two dialogues show what weak AI use looks like and how an instructor might respond. They are paired with strong-use dialogues so that an instructor can show students both versions side by side. Longer worked sessions, following one student through a messy problem turn by turn with the moves annotated, are in the companion repository (`students/study-sessions/`). They are useful to hand to a colleague who has never used one of these tools and wants to see what a student session actually looks like. The contrasts are pedagogically the most useful part: the boundary between strong and weak use is rarely about whether AI was used, and almost always about how the student engaged with the work.

Dialogue 7: Weak undergraduate use (calculus, contrast with Dialogue 1)

Student. Solve this calculus optimization problem: a farmer wants to fence a rectangular field of 400 m^2 along a river, fencing only three sides. What dimensions minimize fencing cost?

→ Mistake: asked AI to solve, not for a hint.

AI. *Let length parallel to river be L and width be W. Constraint: $LW = 400$, so $L = 400/W$. Fencing cost is proportional to $f(W) = 400/W + 2W$. Setting $f'(W) = -400/W^2 + 2 = 0$ gives $W = \sqrt{200}$, so $L = 2\sqrt{200}$. Second derivative $f''(W) = 800/W^3 > 0$ confirms a minimum.*

Student. Thanks. I'll submit this.

→ Mistake: did no independent work, no verification, no reconstruction.

Student. [At the random oral audit a week later.] Instructor: "Why did you set up the function as $400/W + 2W$ rather than $L + 2W$?" Student: "I think because… I'm not sure. The AI did it that way."

→ The audit reveals what the submission concealed: no understanding of the constraint substitution that is the key insight. Course rule violated: "If you cannot explain it, it should not be in your work."

Student. Instructor follow-up: "What would change if the area constraint were 200 m^2 instead of 400?" Student: "I'd have to redo the problem with a new tool query."

→ The student cannot transfer. This is the canonical failure mode the Learning Spiral is designed to prevent.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The takeaway. That is hollow understanding: skipping the *try to solve* step produced a correct submission the student could neither rebuild nor transfer.
Instructor scoring. This case illustrates the upper-left quadrant of figure 9.3: high polish, low demonstrated learning. A reasonable response under most course policies is partial credit on the submission (the answer is correct), zero credit on the oral audit, and a teaching conversation about how to use AI in this course. The same submission paired with the engagement shown in Dialogue 1 would receive full credit on both.

Dialogue 8: Weak graduate use (proof, contrast with Dialogue 2)

Student. Prove that every continuous function on $[0, 1]$ attains its supremum.

→ Mistake: asked AI to produce the proof rather than to audit one.

AI. *Let $M = \sup_{x\in[0,1]} f(x)$. Since $[0, 1]$ is compact and f is continuous, $f([0, 1])$ is compact in $\mathbb{R}$, hence closed and bounded. Therefore M is attained.*

Student. Looks fine. I'll submit this.

→ Mistake: accepted a proof that uses a result (compactness implies attainment) that is essentially the conclusion. The AI proof is technically correct, but it bypasses the sequential-compactness (Bolzano–Weierstrass) argument the qualifying-exam syllabus expects.

Student. [Oral examination.] Examiner: "Walk me through the step where you concluded M is attained from compactness of $f([0, 1])$." Student: "A compact subset of $\mathbb{R}$ is closed and bounded, so the supremum is attained."

Student. Examiner: "Right. Now: prove that statement directly. Why is the supremum of a closed bounded subset of $\mathbb{R}$ attained?" Student: "…"

→ The student passed the surface but not the depth. Proof construction *and repair* requires producing the underlying argument when probed, not just citing the named theorem.

The takeaway. A proof can be correct without being yours; the gap opens the moment you are asked to prove the step you leaned on.
Instructor scoring. The submission is correct mathematics. Under conventional grading it could pass. Under the framework, the oral component reveals that the student has not built the underlying argument. Compare with Dialogue 2: a student given the same theorem can engage with subtle critique, push back when AI is wrong, and produce a proof of their own. The framework distinguishes these two cases; the framework's job is to make that distinction visible.

11.2 ▪ Academic integrity without detector dependence

AI detectors should not be the foundation of an academic integrity system. They produce both false positives and false negatives, and the magnitudes are not small. One study of seven widely used detectors quantified both failure directions [79]. Most detectors in 2023 relied on text perplexity, how predictable the words are to a language model, and the narrower vocabulary and syntax of non-native writers lower it, so the detector reads their work as AI-generated (the study's figures are in section 16.5). What matters in the classroom is the direction of the errors: the false positives concentrate on non-native writers, and the false negatives go to any student who adds a single self-editing prompt, so a detector score cannot carry an integrity decision alone.

The same mechanism reaches past language. Perplexity-based detection flags writing that departs from the model's expected norm, and non-native writers are not the only students whose writing departs from it. The same structural logic reaches autistic and other neurodivergent

students, whose writing and reasoning a detector may also read as unusual. That extension follows from the mechanism rather than from a study that has tested it. The exposure is wider than text. Automated proctoring, engagement detection, and behavior analytics can misread a neurodivergent student's eye contact, pace, movement, or stress response as a sign of trouble, when it is only a different way of working. A system that judges people by how closely they match an assumed default penalizes whoever sits furthest from it. That is a fairness problem the institution owns, not the student.

The institutional consequences extend beyond accuracy. Detector-based enforcement creates what the same authors term a presumption of guilt: students are flagged as suspect by an automated system, and the burden falls on them to prove innocence. Even when an accusation is later revoked, the student's reputation and trust in the institution may not recover. The teaching-and-learning approach to academic integrity, developed long before AI tools were a concern, treats integrity as a pedagogical issue rather than a policing one [14, 15]. The same orientation applies to AI-related concerns: when integrity is built into course design, fewer concerns arise, and those that do can be handled through evidence and conversation. When concerns arise, instructors should use multiple sources of evidence.

- Assignment policy and category.
- Student disclosure or lack of disclosure.
- Draft history or version record.
- Source verification.
- In-class performance.
- Oral explanation or targeted reassessment.
- Consistency with prior work, interpreted cautiously.
- A fair conversation with the student.

The goal is to make accountability fair, educational, and evidence-based.

Spot-checking does not require reviewing every student's AI conversation on every assignment. Reviewing three or four threads per assignment, chosen at random or based on signal, creates probabilistic deterrence at sustainable cost. Students who know that any single thread might be reviewed engage differently from students who know nothing is ever checked. The probabilistic version preserves the framework's three-cornered balance. Instructors can keep their judgment trustworthy without taking on an impossible workload. Student disclosures also become useful to them, without the work of reviewing every thread.

Faculty are not exempt from the integrity question. A senior at Northeastern University filed a formal complaint after she found her organizational behavior professor was using AI to produce his lecture materials, despite a syllabus that prohibited student AI use.[8] The asymmetry is the issue: an integrity rule that applies to students but not to those teaching them is structurally fragile, and students notice.

The framework developed for student AI use applies equally to instructor AI use. The Instructor's Trilemma in section 8.4 names the three forces faculty balance under AI pressure: professional sustainability, trustworthy professional judgment, and student learning. Read through that frame, the Northeastern case is the instructor's quiet overclaim. The lecture materials implied more instructor engagement than the AI-assisted preparation actually reflected, and the syllabus held students to an integrity standard the instructor did not apply to himself. Trustworthy judgment was the corner that gave way.

[8] Hill [50].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The translation test gives both sides of the classroom a positive standard. A student who can translate their work into a different form is giving good evidence they own it; a student who cannot has reason to check further. A faculty member who can translate AI-assisted course material into their own language, with their own examples, in their own structure has integrated it. A faculty member who cannot has the analog of hollow understanding: course material the instructor cannot stand behind in conversation. The test is a private check faculty can run on their own work before placing it in front of students, and a standard the institution can defend in public.

The translation test catches the *hollow understanding* mode of the Student's Trilemma directly, because the student cannot translate. The other two modes, *quiet overclaim* and *the integrity penalty*, require different instructor responses, developed in the rest of this Part.

11.2.1 ▪ When the work exceeds its label

The four assignment categories create a failure with a particular shape. A student works at **AIC** on a task labeled **NAI** or **AIT**, and the work that arrives looks like what the assignment asked for. This is *quiet overclaim*: the student may understand the material perfectly well and still hedge on how the work happened. A category scheme is least able to see this failure. The work arrives in the expected form, and the hedging is in the student's account of the process, not in the work itself.

The design question comes first. A useful test is whether the validity of a task depends on a student following an instruction. If it does, the task will fail at some rate, and clearer wording does not repair it, because wording is the thing being ignored. Of the four categories, **NAI** depends most on the instruction and least on the design of the task. That makes it the riskiest label in this book. An **NAI** label is worth issuing when the task is secured by its own form: in class, supervised, oral, or built on work already produced under supervision. An **NAI** label attached to unsupervised take-home work is a prohibition with nothing behind it, and it manufactures the situation this section is about.

AIT fails more quietly. The boundary between asking for an explanation and asking for production is crossed gradually, usually inside a single conversation, and often with no moment at which the student decides anything. A rule in a syllabus cannot catch a gradual crossing. A check the student runs during the work can. If they cannot close the tool and rebuild the key argument, calculation, or paragraph from a blank page, the help has crossed from explanation into production, whatever the transcript shows. That check belongs in the assignment and not only in the policy.

When the boundary is crossed anyway, the situation separates into two questions that should not be answered together. The work no longer evidences the competence the assignment was built to elicit. That is a question about what the student can do, so it is a question about the mark, and the remedy is measurement rather than penalty. An oral defense, a short audit, or a supervised reconstruction will establish what the student can demonstrate, and the mark follows from that. Whether the student misrepresented their use is a different question, and by the reasoning in section 8.4.4 it leaves the gradebook and enters a fair response sequence.

That ordering also answers the objection that an educational response is unfair to the students who honored the label. It is not unfair, because the student is measured again and the shortcut gains them nothing. The student who worked at **AIC** is assessed on the same competence as everyone else, later and with more friction and without the time they saved. Compliance stays rational without a penalty attached to it. The sharper version of the fairness problem sits earlier. A disclosure requirement collects the most from the students least likely to be concealing anything. One study at a business school found that up to three-quarters of students did not complete a mandatory AI-use declaration [46]. Ambiguity in the guidelines and fear of the consequences were among the reasons. A policy resting on disclosure alone puts its visible burden on the honest.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

One step belongs before the conversation with the student. Read the assignment wording again and ask whether the expectation was actually stated, and stated in a form a student working under time pressure would notice. Instructors are building these categories for the first time, often in the same term their students first meet them. A first case is usually better evidence about the assignment than about the student.

11.2.2 ▪ A fair response sequence

When AI misuse is suspected, a fair process can follow six steps. This is a short form of the fuller procedure a department or institution should follow.

1. Review the assignment policy and the student's submitted disclosure.
2. Identify the specific concern, not merely a general suspicion.
3. Invite the student to explain the work and the process.
4. Use a short reassessment or oral explanation when appropriate.
5. Distinguish misunderstanding, poor documentation, prohibited assistance, and deliberate misrepresentation.
6. Document the concern, the evidence, the student's response, and the decision.

For minor first problems, an educational remedy may be more appropriate than a formal charge. Serious, deceptive, or repeated misconduct should be handled through the institution's academic integrity process.

The sequence above establishes what happened. A procedure can be scrupulous about fairness and still leave an instructor unsure what a good outcome looks like, so it helps to say what the conversation is for. For a first concern the goal is usually to return the student to the work rather than to secure a finding. Jennifer Trainor reports that what she mostly encounters is not a student trying to bypass a degree but one who panicked, ran out of time, or otherwise made a poor decision under pressure [129, ch. 14]. Those causes are addressable, and the conversation is where they surface. Asking what happened, with curiosity rather than accusation, also tends to produce a fuller account than confrontation does.

This is not leniency. The educational route is usually the one taken because of the evidence, not out of sympathy. Without the student's own admission, the available evidence often falls short of what a formal proceeding requires, and an educational remedy is the response that evidence actually supports.

11.2.3 ▪ A constructive role for detectors

Although detector tools should not serve as the basis for misconduct decisions, one constructive alternative remains open. A detector can work as a student-facing self-check aid that flags clichéd phrases or repetitive patterns, encouraging the writer to vary structure or sharpen voice. That use is the study's own suggestion rather than an invention of this book: the authors who document detector bias, and who caution against detector use in evaluative settings, see a low-risk use for detectors as educational aids that can serve as self-check mechanisms for students [79]. Used as a self-check, the detector is a writing tool the student applies to their own draft, not an enforcement tool the instructor applies to a submitted paper. This use sits comfortably in the **AIT** assignment category: AI as feedback-giver on student-produced work, with the student deciding what to change and writing the changes. Instructors who want to acknowledge the existence of detector tools without granting them evidentiary weight can recommend this self-check use directly.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

11.3 ▪ Privacy, access, and equity

AI policy is also about privacy, access, disability, language, and fairness. These concerns interact: a policy that gets one right can damage another. The subsections below treat them in turn.

11.3.1 ▪ Privacy

Students and instructors should not upload protected or confidential material into unapproved AI tools. This caution is consistent with European guidance that asks educators to engage with AI and data systems critically, ethically, and with attention to risk [35]. Material that should not be uploaded without explicit institutional approval includes identifiable student information, private correspondence, unpublished research data, human-subject data, patient or client information, restricted exam materials, recommendation letters, and copyrighted course materials when sharing is not permitted.

The instructor's privacy responsibility is not limited to declining to upload material themselves. It extends to course design: an assignment that requires students to upload classmates' work, group-project drafts containing other students' contributions, or interview transcripts collected under research-ethics protocols can create privacy exposures the students may not anticipate. When in doubt, consult institutional research-ethics or data-governance offices before requiring AI use that processes information about identifiable third parties.

11.3.2 ▪ Access

Some students may have paid AI tools, faster devices, better internet access, or more experience with AI. A course that requires AI should either provide equitable access or offer a no-cost alternative. A course that permits AI without requiring it should avoid giving hidden advantages to students who can afford premium tools.

Access has three layers. Tool access (free vs. paid tiers; rate limits; capability differences across models) is the most visible. Device access (a laptop with reliable browser support vs. a phone or shared library computer) is less visible but more durable. Internet access (campus wifi vs. home connections that may be intermittent or absent) determines whether AI is usable at all outside class.

The practical instructor response is a baseline-plus pattern: design assignments so the baseline AI capability available through institutional licensing or free tiers suffices to complete them, then accept that students with premium access may do additional exploration. Avoid assignments where premium tools confer a measurable assessment advantage that other students cannot match.

11.3.3 ▪ Equity in assessment design

Different assessment designs distribute their burdens differently across students. Process-evidence assessment can favor students with the time and home circumstances to document their work in detail. In-class no-AI assessment can favor students whose lives support uninterrupted focus. Oral defense can favor students whose language background, neurotype, or personality make spoken explanation comfortable. Each modality has its own equity profile, and a course that uses only one modality concentrates the equity exposure on whichever students it disadvantages.

The hybrid approach to assessment (section 10.4) is partly a response to this. By distributing assessment across multiple modalities, the course gives most students at least one assessment modality where they perform near their best, rather than asking every student to perform at their best in the single modality the course happens to have chosen. This is not a guarantee of fairness, but it spreads the burden more evenly than monomodal assessment does.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Instructors who notice persistent gaps in performance across demographic groups should treat the gap as a design question first. Is one assessment modality contributing disproportionately to the gap? Is the documentation burden falling unevenly? Is the no-AI assessment running at a time of day or under conditions that disadvantage some students? These questions surface design adjustments more usefully than attribution to AI use itself.

11.3.4 ▪ Multilingual students and accessibility

Multilingual students deserve special care. Automated judgments about authorship, fluency, or style can penalize students whose writing differs from dominant patterns. Detectors specifically misclassify non-native English writing at high rates. Instructors should rely on process evidence and conversation rather than detector scores or vague impressions.

This is also a matter for institutional policy, not only classroom practice. Because that misreading is structural rather than a passing flaw, a detector score should never be the sole basis of an academic-integrity charge, and policy should state this in plain terms. Where a campus uses detection tools at all, policy should prohibit acting on a score alone. It should require corroborating process evidence and a chance for the student to explain, and apply the same standard to every student rather than letting it fall hardest on those whose writing departs from dominant patterns. The stakes are uneven. For an international student, an unfounded finding can reach past the grade to visa status and continued enrollment, so an integrity procedure built on detector scores places its heaviest risk on the students least able to absorb an error. The sounder course is to treat detection as, at most, a weak signal that prompts a conversation, and to ground integrity decisions in process evidence, draft history, and the student's ability to explain their own work.

AI tools can support multilingual students, helping with idiomatic phrasing, vocabulary suggestions, and grammar checks. The AI account configuration template explicitly addresses this use. Used this way to learn phrasing the student then writes in their own words, it fits the **AIT** category. Once AI-revised wording enters the submission, the work is **AIC**, and the revision should be disclosed. Instructors should make this use legitimate rather than treating disclosed AI-assisted English revision as suspect.

Accessibility can cut both ways. AI may help some students brainstorm, read, summarize, translate, or organize work. It may also create new burdens if tools are inaccessible, expensive, or poorly integrated with assistive technology accommodations. Instructors should coordinate with campus accessibility offices when AI tools become part of required coursework, and should provide alternative access paths for students whose accommodations make standard AI interfaces inappropriate.

11.4 ▪ Discipline overlays

The framework is general, but implementation should be disciplinary. The same CAT structure can be translated into local practice.

11.4.1 ▪ STEM courses

In STEM courses, Core Competence may include conceptual understanding, problem solving, modeling, lab interpretation, coding fundamentals, and quantitative reasoning. AI-assisted Practice may include generating practice problems, debugging code, comparing derivations, checking edge cases, or critiquing a model. Trustworthiness includes unit checks, benchmark tests, reproducibility, source verification, and clear separation between evidence and speculation.

Strong STEM assignments often require a first attempt, a verification step, and a transfer question. For example, students might submit a solution, an AI critique of the solution, and a

short no-AI explanation of the central method.

Mathematics. Mathematics courses divide along an internal axis. In foundational proof-based courses (real analysis, abstract algebra, topology, transition-to-proof), **NAI** and **AIT** assessments often focus on definitions, short proofs, and proof-audit-and-repair tasks; **AIC** assignments may use AI to generate examples and non-examples or to critique a proof attempt, with oral defense for ownership. In computational and applied courses (numerical analysis, scientific computing, optimization, modeling), **AIC** assignments can integrate AI for code generation or method comparison under a verification rubric covering edge cases, stability, and reproducibility. The two patterns share the same spiral, but they have different default category mixes and different assessments; the design choices follow from which kind of mathematical thinking the course is trying to build. A consolidated quick reference for mathematics courses, covering both the student and the instructor side, appears in Appendix B.

Computer science. In computer science courses, Core Competence includes algorithm design, complexity reasoning, debugging from observation, test design, and the ability to read code written by other authors. AI-assisted Practice may include code completion, exploring alternative implementations, debugging assistance, generating test cases for review, and explaining unfamiliar libraries. Trustworthiness includes verifying that generated code actually does what it claims (which AI is unreliable at self-confirming), checking edge cases, reviewing for security and performance implications, and distinguishing code the student understands from code the student has merely watched appear. Strong computer science assignments require the student to describe the algorithm before any AI use, to write at least one test before requesting code, and to walk through the resulting code line by line in oral defense or written annotation. The failure mode the course is designed to prevent is the student who submits working code they cannot explain, predict, or modify when the requirements shift slightly.

Statistics and data science. In statistics and data science courses, Core Competence includes study design, statistical reasoning, model selection and interpretation, assumption-checking, and the ability to recognize when a methodological choice is being made on autopilot rather than for substantive reasons. AI-assisted Practice may include exploratory analysis, code generation for data manipulation and visualization, comparison of methodological options, and generation of plausible counter-explanations to a hypothesis. Trustworthiness includes verifying that the test or model an AI suggests actually fits the problem's assumptions (a frequent AI failure mode), checking for confounding the AI may have missed, and distinguishing statistical significance from practical significance, which AI commentary often blurs. Strong assignments require the student to justify methodological choices before computation, to identify what the analysis would look like under alternative reasonable assumptions, and to interpret results in their substantive context. The failure mode the course is designed to prevent is the student who runs a test the AI suggested without checking whether its preconditions hold, then reports the p-value as if it answered a question it does not address.

Physics. In physics courses, Core Competence includes identifying the principle or law that applies to a problem, setting up a model with clear assumptions, dimensional and limit-case reasoning, and translating between formal description and physical intuition. AI-assisted Practice may include checking algebra, comparing approaches (kinematics versus energy methods for the same problem, for instance), generating intuitions about limiting behavior, or exploring whether a result agrees with known special cases. Trustworthiness includes confirming that the result has correct units, that limiting cases reduce to expected behavior, and that the chosen principle is

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

appropriate to the problem rather than merely producing an algebraically valid solution from the wrong starting point. Strong physics assignments require the student to identify the operative principle before any AI use, to predict the limiting behavior before computing it, and to interpret the final answer in physical terms. The failure mode the course is designed to prevent is the student who produces algebraically correct work from a wrongly-identified starting point, where the symbols are right but the physics is wrong.

11.4.2 ▪ Writing and communication courses

In writing courses, Core Competence includes argument, structure, evidence, audience, revision, and voice. AI can help students ask revision questions, identify weak transitions, generate counterarguments, or compare possible openings. It should not erase the student's authorship. Disciplinary guidance from the Modern Language Association and the Conference on College Composition and Communication recommends teaching students to engage critically with AI-generated text rather than treating writing courses primarily as a site of detection [91].

Strong writing assignments ask for draft history, an author's note, source verification, and a short explanation of revision choices. Students should be able to say why a sentence, paragraph, or claim changed.

11.4.3 ▪ Humanities courses

In humanities courses, Core Competence includes interpretation, close reading, source use, historical context, and argument from evidence. AI may help students generate background questions or compare interpretations, but primary sources and assigned texts should remain central.

Strong humanities assignments ask students to tie claims to specific passages, verify quotations, distinguish background summary from interpretation, and explain why an interpretation is plausible.

11.4.4 ▪ Social science courses

In social science courses, Core Competence includes research design, causal reasoning, data interpretation, theory comparison, and evidence evaluation. AI may help students generate possible confounders, critique survey questions, compare explanations, or summarize methodological options.

Strong assignments ask students to separate observed evidence from conjecture, justify methods, identify assumptions, and test alternative explanations.

11.4.5 ▪ Professional and applied courses

In professional courses, Core Competence often involves judgment under constraints. Students may need to apply standards, weigh risk, protect confidentiality, communicate with clients or patients, or justify decisions.

AI may support case simulation, draft critique, risk identification, or policy comparison. It should not replace professional responsibility. Assignments should ask students to explain their decisions and identify what information would change their recommendation.

Three professional disciplines illustrate the point: nursing, law, and engineering. Each raises a distinct version of the same pedagogical challenge: how to develop judgment when AI tools can produce confident-sounding answers that look professional but may be wrong.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Nursing. Core Competence in nursing courses includes clinical reasoning, patient assessment, ethical decision-making, and communication. AI may help students practice differential reasoning, generate case variations, or draft patient-education materials for critique. AI should not generate care recommendations that students submit without verification against course standards and clinical guidelines. A strong assignment asks students to compare an AI-generated triage suggestion with established clinical decision rules, identify where the two diverge, and explain which they would follow and why.

Law. Core Competence in law courses includes case analysis, statutory interpretation, legal writing, and reasoning under uncertainty. AI may help students explore alternative argument structures, generate counterarguments to a draft brief, or summarize unfamiliar areas of law for orientation. AI should not generate legal citations or substantive arguments students submit without verification. Hallucinated case citations are a documented risk. Appellate courts have sanctioned attorneys for filings that contained fictitious cases, holding that no filing should carry a citation, however generated, that the lawyer has not read and verified [125]. A strong assignment asks students to critique an AI-generated argument for hidden weaknesses, missing distinctions, and unverified citations.

Engineering ethics and applied engineering. Core Competence in engineering courses includes safety reasoning, standards compliance, ethical analysis, and professional judgment in design. AI may help students explore design alternatives, identify potential failure modes, or compare ethical frameworks. AI should not produce safety analyses or design decisions that students submit without independent verification. A strong assignment asks students to use AI to generate a list of potential failure modes for a design, then verify which are real engineering concerns versus surface-level pattern matches that do not apply.

In all three disciplines, the framework's core practices apply: independent attempt before AI use, verification against authoritative sources, disclosure of AI involvement, and the ability to defend decisions without AI. The professional context raises the stakes of failure.

11.4.6 ▪ Creative and design courses

In creative and design courses, AI can support ideation, variation, critique, and iteration. The risk is that students may mistake generated novelty for artistic or design judgment. This section is about the creative and design disciplines; creative capability in any field, and how AI changes the way it is learned, is taken up in section 9.3.6, section 10.5, and the companion paper "Learning to Create with AI" [54].

Strong assignments ask for process portfolios, design rationales, iteration histories, constraints, and critique responses. Students should be able to explain why they chose one path rather than another.

The creative and design disciplines raise authorship and originality questions that other fields can largely sidestep. Where an engineering student is judged on whether a design works, a creative writing student is judged partly on whether the work is theirs.

Visual arts and design. Core Competence in visual arts includes observational skill, compositional judgment, technique, and the development of a personal visual vocabulary. AI image generation can produce visually polished output that bypasses each of these. A strong assignment requires sketches, drafts, iteration, and a written rationale explaining the artistic choices; AI may be a research tool or a comparison point, but the submitted work must show the student's

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

hand. Some programs have moved to in-studio work as a foundation, with at-home assignments restricted to specific iteration tasks.

Music and composition. Core Competence in composition includes harmonic reasoning, voice leading, structural awareness, and the development of musical ideas over time. AI tools can generate plausible-sounding compositions that lack the structural reasoning a course is meant to build. Strong assignments ask for sketch development, theoretical annotations, and live performance or sight-reading components that confirm the student's musical understanding.

Creative writing. Core Competence in creative writing includes voice, observation, structural choice, and revision. AI can produce text that is fluent but generic, or that imitates specific styles without the underlying observation that produced them. Strong assignments ask for drafts at multiple stages, marginalia explaining revision choices, and in-class writing under timed conditions that ground the work in the student's own voice.

Design (industrial, graphic, user experience). Core Competence in design includes user understanding, constraint analysis, iteration, and the rationale behind aesthetic choices. AI can generate variations quickly, which is valuable for ideation but dangerous if students mistake variation for design thinking. Strong assignments ask for user research, constraint documentation, iteration narratives, and presentation defenses where students explain why specific decisions serve specific users.

Across creative disciplines, the framework's core insight applies with extra force: the goal of the course is to develop the student's judgment, not to optimize the polish of their output. AI accelerates polish more than it accelerates judgment, so courses that grade on polish alone become particularly fragile in the AI era. The remedy is not to reject AI but to grade what the course is meant to teach.

11.5 ▪ Rubrics

A shared rubric helps students understand expectations and helps instructors grade consistently. The following five anchors work across many disciplines.

Correctness. The work is accurate, well supported, and appropriate to the course level.

Reasoning visibility. The work shows the path, not only the destination.

Verification discipline. Important claims, sources, calculations, code, or interpretations are checked.

Communication. The work is clear, organized, and suited to its audience.

Integrity. AI use is disclosed and consistent with the assignment policy.

For **NAI** assignments, the integrity anchor mainly concerns compliance with the no-AI rule. For **AIT** assignments, it concerns whether AI was limited to tutoring. For **AIC** and **AIS** assignments, it concerns whether the disclosure is honest. In every category, a concern under this anchor opens a conversation about the work rather than lowering the grade. How complete and clear a disclosure is can still be graded, as the **AIC** rubric below does.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

11.5.1 ▪ A simple four-level rubric

Level	Description
Excellent	Work is accurate, well explained, verified, and fully documented. The student can defend the main decisions.
Satisfactory	Work meets the main goals. Some reasoning or verification is incomplete, but the student shows ownership.
Developing	Work contains gaps in understanding, verification, or documentation. The student needs more guided practice.
Unsatisfactory	Work cannot be explained, contains serious unsupported claims, or lacks ownership.

11.5.2 ▪ A worked example: rubric for an AIC assignment

For an **AIC** assignment (say, a data-analysis report where AI may draft code and prose), the three criteria that carry the most weight are verification, ownership, and disclosure. One workable rubric:

Criterion	Strong	Adequate	Needs revision
Verification discipline	Key claims and code tested against known cases; checks documented.	Spot checks on the main results; some documentation.	Output accepted as returned; no independent check visible.
Integration and ownership	Student can explain and extend every part; AI material rebuilt in the student's own structure.	Student can explain the main line; some passages remain tool-shaped.	Student cannot explain or vary substantial parts.
Disclosure quality	States tools, roles, and what was verified, proportionate to use.	States tools and roles.	Missing or vague.

Editable versions, with variants for writing and coding courses, are in the repository (`instructors/rubrics/`). Grade the three rows separately rather than averaging them away; a strong analysis with absent verification is not a strong **AIC** submission.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 12

Implementation and Limits

The best measure of a teacher is what others become.

12.1 ▪ TA calibration

Large courses need consistent grading practices. A good rubric is not enough. TAs and instructors should practice applying it together.

A short calibration process can be used before major grading.

1. The instructor selects three to five sample submissions.
2. Each grader scores the samples independently.
3. The teaching team compares scores and discusses disagreements.
4. The rubric is clarified where needed.
5. Graders proceed in batches, with occasional spot checks.
6. The instructor reviews unusual or borderline cases.

The goal is shared interpretation of the course standards.

The protocol above assumes work that two people can read separately. An oral defense does not produce one. Where TAs serve as second readers, sit in on the first two or three together, compare judgments immediately afterward, and agree on what counts as a sufficient answer for each segment of the defense structure.

12.2 ▪ Implementation workflow for a course

For instructors redesigning under time constraints, three moves matter most: assigning each task an **NAI**, **AIT**, **AIC**, or **AIS** label; adding one meaningful independent-competence check per unit; and adding a brief AI-disclosure requirement to any **AIC** assignment. These three are the smallest set of moves that materially change how AI fits into a course.

Read those three as a starting point rather than a compromise. They do not require choosing a model. An instructor who makes only these three has already changed what the course asks of a student, and the models earlier in this Part are where a design ends up rather than where it has to begin.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

A course can get there across more than one term. In the first term, make the three moves: label the tasks, add one competence check per unit, and ask for disclosure where AI is allowed. Tell students what the labels mean and why they are there. In the second term, convert one existing assignment into an **AIT** task, where students may use the tool to study and must produce the submitted work themselves. Then read what comes back. One such assignment will teach you more about your own course than any amount of planning. In the third term, choose the model that fits what you learned and rebuild the rest around it.

The sequence matters most for an instructor whose course already works. Twenty years of a well-tuned syllabus is not resistance. It is a reason to want evidence before spending a term on redesign. An instructor who is not persuaded that AI belongs in the course at all is in a different position again, and one this book addresses only briefly; the companion paper "Teaching AI Responsibly to Principled Skeptics" [56] takes up that argument properly, including the case of an unconvinced instructor teaching students who already use these tools daily. The order above buys that evidence at the cost of one assignment.

The fuller workflow has seven steps.

1. Identify the core learning goals that must remain independent.
2. Assign each major task an **NAI**, **AIT**, **AIC**, or **AIS** category.
3. Choose the course model: Oral-Defense, Evidence-of-Thinking, or Hybrid.
4. Add disclosure and verification requirements where AI is allowed.
5. Add at least one meaningful independent competence check.
6. Prepare a fair process for suspected misuse.
7. Tell students why the policy exists.

The last step matters. Students are more likely to follow a policy when they understand its educational purpose. A useful explanation is simple: the course allows AI where it supports learning and restricts AI where the instructor needs to see what students can do on their own.

This step has a stronger version. Instead of only explaining the policy, you can ask students to help shape it. Give them a few sample cases (a coding question, a reading response, a take-home problem) and have them argue which uses should be allowed and which should not, then draft the rule from those cases. When students helped build a course's AI policy this way, the resulting policies drew more support and more agreement from the students who would live under them, and they covered cases an instructor working alone might miss [73]. Student drafts tend to run long and need an editing pass, so the instructor still owns the final wording. But a policy the class helped write is one the class is more likely to follow.

12.3 ▪ Evidence base and limits

The instructor recommendations in this guide rest on an evidence base the whole book shares. Two findings matter most for course design. First, the AI-tutoring evidence is mixed: AI can help when designed around good pedagogy and students remain cognitively active; unrestricted access often improves apparent performance while weakening later independent performance. Second, cross-level meta-analyses find moderate positive overall effects but substantial heterogeneity by discipline, intervention duration, and assessment type [82, 83], heterogeneity that the discipline overlays in this Part are designed to address.

The result is a design principle rather than a universal prescription. AI use should be allowed when it advances the learning goal and constrained when it hides the learning goal. A policy that is too permissive may reward polished output without understanding. A policy that is only

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

prohibitive may fail to teach students how to use tools they will encounter outside the course. The instructor's task is to preserve independent competence while teaching responsible tool use.

Mixed evidence still leaves a standard:

Judge a technology by whose learning it deepens.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 13

The Instructor's Toolkit

This chapter collects the templates, rubrics, and checklists from the five chapters before it. It is for the instructor designing a course in August and for the one facing a suspect submission in October. Many of these can be adapted directly into a syllabus, gradebook, or course-management system. The subject appendices appear at the end of the volume.

13.1 ▪ Assignment wording bank

Wording for the syllabus and for individual assignments. Both use the four assignment categories defined in section 9.2.

13.1.1 ▪ Sample syllabus language

The following language may be adapted to fit a course. For instructors who prefer to start from a pre-written briefing rather than adapt the language below, the companion repository's `instructors/briefings.md` provides three ready-to-use syllabus paragraphs at different permissiveness levels (restrictive, default, and permissive). Each is about 180 words and designed to drop into a syllabus without modification.

General course statement

> AI tools are part of modern academic and professional work. In this course, you will learn when AI can support your learning and when it would interfere with the goals of an assignment. Each assignment will state whether AI use is prohibited, allowed for tutoring only, allowed as part of the working process, or required as an object of critique. When AI use is allowed, you are responsible for verifying the output, disclosing your use when asked, and submitting only work that you can explain.

Course materials statement

> Lecture notes, slides, and a notation sheet for this course are posted in [course site or folder]. You may upload these files to the AI tools approved for this course, so that your AI tutor follows the course's notation and order. Do not upload exam materials, solutions to work not yet due, or other students' work.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

No-AI assignment statement

This is a no-AI assignment. You may not use AI tools to generate, revise, translate, summarize, solve, check, or explain your answer while completing this task. The purpose is to assess what you can do independently.

AI-as-tutor assignment statement

You may use AI to study for this assignment by asking for explanations, examples, practice questions, or hints. You may not use AI to produce the submitted response. The submitted work should be written or completed by you without AI-generated text, code, or solutions.

AI-as-collaborator assignment statement

You may use AI as part of your working process. You must disclose the tools used, describe what they contributed, and explain how you verified the final work. You are responsible for all claims, sources, calculations, code, and interpretations in your submission.

AI-as-object-of-study assignment statement

In this assignment, AI output is the object of study. You will analyze, critique, compare, or repair AI-generated work rather than produce the answer yourself. Use the AI tools the assignment specifies, record what they produced, and document the errors, biases, or limitations you find. Your grade rests on the quality of your critique and verification, not on the AI's output.

13.1.2 ▪ Assignment label template

This is a blank form, not a filled example. The instructor completes the right column once for each assignment and gives the finished label to students along with the assignment. Filling every row settles what a student may and may not do before the work begins, so nothing about AI use is left to inference.

Field	Instructor entry
Assignment title	
AI category	**NAI / AIT / AIC / AIS**
Allowed uses	
Disallowed uses	
Required disclosure	
Required verification	
Independent component	

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Help ladder (optional): explanation, analogous example, one hint, critique of the attempt, then a full solution for ungraded practice

Privacy warning

How this will be graded

13.2 ▪ Making honesty cheap

A student who wants to be honest about AI use often does not know what to say, and a blank line invites either silence or a sentence that reports nothing. The block below settles the form in advance. It asks for six things. A student who used AI lightly can answer in a few lines, and a student who leaned on it hard has to write down what they actually did. Honesty costs less when its shape is already fixed, and the record that comes back is worth reading.

13.2.1 ▪ Student disclosure block

Tools used: Name the tools used, if any.
Tool role: Describe what the tools helped with.
Student role: Describe what you wrote, revised, checked, or decided.
Accepted, changed, or rejected: Describe the AI contributions you used, revised, or set aside.
Independent verification: Describe how you checked important claims, sources, data, code, or reasoning.
What I can explain without AI: Name the main parts of the work you can now explain on your own.

13.3 ▪ When a submission looks wrong

A submission arrives that does not read like the student who wrote it. What follows is for the first hour, before anything has become a case, and it assumes you are not yet sure. Section 11.2 sets out the reasoning; this is the order to work in.

1. Pause. Nothing is lost by waiting a day, and a first message written from suspicion is hard to take back.
2. Read the assignment wording again, and ask what the category actually permitted. Some of what looks like misuse is a task that never said what it wanted (section 11.2.1).
3. Separate the two questions. Whether the work still evidences the competence the assignment was built to measure is one question. Whether the student acted in bad faith is another. Answer the first before the second, because the first can be settled by re-measurement (section 11.2.1).
4. Gather process evidence: drafts, version history, the disclosure block, source checks, and in-class work. A detector score may be considered where policy allows, but it cannot be the primary evidence and it cannot settle this on its own (section 11.2).
5. Talk to the student before you open a procedure, and ask them to explain the work in their own words. The random oral audit script supplies the questions. Use it as a conversation,

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

because the aim is to establish what happened before deciding what it was.

6. Escalate by seriousness, not by certainty. For a minor first problem an educational remedy may be more appropriate than a formal charge, and re-measurement leaves the shortcut earning nothing. Serious, deceptive, or repeated misconduct goes through the institution's academic integrity process (section 11.2.2).

Write down what you saw, what you asked, what the student said, and what you decided. Do this also when the answer is that nothing was wrong, because the record is what protects the student and you.

13.4 ▪ Assessment structure and documentation

Class size decides which assessment model a course can actually run. A small class can carry the Oral-Defense Model (section 10.2), which asks students to explain their work in person. A large class needs the Evidence-of-Thinking Model (section 10.3), which reads process evidence instead. The Hybrid Model (section 10.4) sits between them and spot-checks. The structure, instruments, and documentation below serve all three, and what changes with size is how much of each you can collect and read.

13.4.1 ▪ Assessment structure

The weighting depends on the model the course runs. Section 10.2 and the two sections after it give a suggested weighting for small, large, and hybrid classes; start from the one that matches the course.

Instructors may adapt these weights to the course, but the underlying principle is non-negotiable: AI proficiency cannot compensate for weak subject foundations. A student who excels in AI-assisted Practice and Trustworthiness but fails to meet the minimum Core Competence standard has not met the course's subject requirements.

13.4.2 ▪ Assessment instruments

No-AI assessments (C).

- In-class written exams without AI tools.
- Short proof and concept-explanation tasks.
- Oral examinations at the blackboard.

AI-integrated assessments (A).

- AI-assisted homework and take-home exams.
- AI-critique and error-detection tasks.
- Human–AI comparative problem solving.
- AI-assisted mini research projects.

Trustworthiness assessments (T).

- Plan-Use-Verify notes and AI-use disclosure.
- Reflective methodological reports.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

- Error analysis and verification tasks.
- Attribution and reproducibility statements.

13.4.3 ▪ Documentation requirements

For AI-integrated assignments, students must submit the following.

- Description of AI tools used (name and version).
- A short plan, prediction, or sketch, written before using AI.
- The AI outputs that influenced the submission, not the full transcript.
- Independent verification and corrections.
- Final reasoning, in the student's own words.

This documentation can itself be evaluated, and it should stay short (section 10.3.2). Sloppy or incomplete documentation lowers the Trustworthiness score even when the technical work is correct. Clean documentation that traces the student's reasoning through tool interactions and back to verified conclusions is what professional research practice requires.

13.5 ▪ Oral defense rubric and random audit script

The rubric gives the criteria for a full oral defense. The script runs a short audit when you are spot-checking submissions rather than defending every one.

13.5.1 ▪ Oral defense rubric

Criterion	Evidence
Conceptual command	Student explains central ideas in their own words.
Reasoning	Student can reconstruct the main argument, method, design, or interpretation.
Verification	Student identifies how important claims or outputs were checked.
Transfer	Student can adapt the idea to a nearby case.
AI process	Student explains what AI contributed and what was accepted, rejected, or revised.
Integrity	Student's explanation is consistent with the submitted disclosure and course policy.

The Integrity row is the one criterion here that does not belong in the mark. A mismatch between a student's explanation and their disclosure is a reason to open the conversation described in section 11.2, not a reason to lower a grade. Section 8.4.4 sets out why the two are kept apart.

13.5.2 ▪ Random oral audit script

A short audit may use the following sequence.

1. Please summarize your submission in two minutes.
2. What was the most important decision you made?

3. Did you use AI? If so, what did it contribute?
4. Show one place where you verified a claim, source, computation, code result, or interpretation.
5. What is one weakness or limitation in your submission?
6. Answer this nearby transfer question without AI.

   The audit should be brief, structured, and consistent across students.

13.6 ▪ Safeguards that make substitution not pay

To keep AI from replacing learning in a course, the framework recommends four safeguards.

- A minimum Core Competence threshold required to pass the course.
- A significant proportion of no-AI assessment across the term.
- Oral checks, for every student or as spot checks, as the course's size and staffing allow (section 10.4).
- Process-based grading rather than output-only grading.

The threshold rule is the most important. Without it, students can in principle compensate for weak subject understanding through fluent AI use, which defeats the purpose of the course. The threshold need not be high to be effective; what matters is that it exists and is enforced consistently.

No study has yet tested these specific safeguards directly, but experience with redesigned STEM courses is broadly consistent with them. The Kosar redesign in chapter 10 uses two of the four safeguards above, no-AI assessment and oral checks, and its students reached similar learning outcomes with and without AI access [70].

13.7 ▪ Instructor pitfalls

Six ways course design goes wrong. Each line points to the section that explains it.

- Do not set a no-AI rule the course cannot support (section 11.2.1).
- Do not rest an integrity case on a detector score (section 11.2).
- Do not let a high-stakes measurement rest on a task whose validity depends on a student following an instruction (section 11.2.1).
- Do not grade the output when the assignment was about the process (section 9.3).
- Do not open a misconduct process before talking to the student (section 13.3).
- Do not hold students to a standard you do not apply to your own AI use (section 10.7).

13.8 ▪ Course design checklist and self-check

The first list checks a course you have designed. The second checks your own AI use in teaching and grading.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

13.8.1 ▪ Course design checklist

☐ I identified the core abilities students must demonstrate independently.

☐ Each major assignment has an AI-use category.

☐ Students receive clear disclosure instructions where AI is allowed.

☐ AI-assisted assignments include verification or process evidence.

☐ The course includes meaningful no-AI or bounded-AI assessments.

☐ Large-class assessments include scalable evidence of thinking.

☐ Small-class assessments include oral defense, conferences, or live explanation where feasible.

☐ The syllabus explains privacy restrictions.

☐ The academic integrity process does not rely on AI detectors alone.

☐ TAs, if present, have a calibration process.

13.8.2 ▪ Instructor self-check

Before using AI in teaching or grading, ask:

☐ Am I using an approved tool for this kind of data?

☐ Does the AI output affect a grade or only help me prepare feedback?

☐ Have I reviewed the output before sharing it with students?

☐ Could this use expose private student information?

☐ Could this use create unfairness across students or sections?

☐ Am I still making the final instructional or evaluative judgment?

13.9 ▪ AI critique assignment template

Give students an AI-generated response to a course-relevant task. Ask them to complete four steps.

1. Identify what the response does well.
2. Identify at least two weaknesses, errors, unsupported claims, missing assumptions, or misleading statements.
3. Repair one part of the response.
4. Explain what this example teaches about using AI in the discipline.

This assignment works in many fields. A mathematics course can critique a proof. A writing course can critique an argument. A history course can critique a source summary. A computer science course can critique generated code. A lab course can critique an experimental explanation.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Part III

AI-Assisted Learning: A Departmental and Institutional Guide

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

A workable policy gives every course the same words
and leaves each course its own rules.

*Terms. This Part uses the four assignment categories introduced in Part I: No AI (**NAI**), AI as Tutor (**AIT**), AI as Collaborator (**AIC**), and AI as Object of Study (**AIS**). A department needs all four to exist before it can write a policy that means anything. It uses the CAT framework, named for its three dimensions: Core Competence, AI-assisted Practice, and Trustworthiness. The Glossary at the back of the book defines these and the other terms that carry across Parts.*

Part III is written for the people who shape AI-assisted learning above the level of a single course. Within a department, that means chairs, curriculum committees, faculty governance bodies, and senior faculty. Across the institution, it means deans, provosts, teaching centers, academic-integrity offices, faculty senates, and the university-wide leaders who set policy. The framework gives these readers practical tools for designing policy that protects learning while making AI-assisted practice an explicit part of academic life.

One premise runs through this Part: students will use AI, and academic units should teach them to use it without surrendering their own thinking. The framework separates Core Competence from AI-assisted Practice and Trustworthiness. It supports different course types, from small seminars to large multi-section courses, and recognizes that policy decisions sit at different levels: some belong to the department, some belong to the institution, and a few belong to both.

A note on scope and audience. The range of readers just named is deliberate. As elsewhere in the book, the worked examples lean toward STEM and nearby technical fields, where the framework began, but the structure applies across the institution, including the humanities, social sciences, professional programs, and the arts. Readers with influence in a department but little over university policy will find the department-level guidance concrete. For them, the institutional guidance describes what to expect from their institution and how to work with its policy bodies. Readers in central administration will find the institutional sections addressed more directly to their work. One boundary: this book addresses AI in teaching, learning, and the credential, not AI's wider footprint in staff work, admissions, and operations, which raises governance questions of its own.

Part III is six chapters. Chapter 14 sets out the competing pressures an institution is under and the principles that answer them. Chapter 15 gives the shared vocabulary: the CAT framework, the course-level categories, and assessment models by course size. Chapter 16 sets the standards: minimum independent competence, disclosure, privacy and tool approval, equity, the integrity procedure, faculty use of AI, and what every syllabus must say. Chapter 17 covers governance, the roles of departments and teaching centers, training, implementation, and change management. Chapter 18 is the toolkit: implementation phases, expected outcomes, and reference materials. Chapter 19 looks forward: adoption metrics, the risk register, a model policy statement, and the limits of the evidence. Throughout, the text separates what a department can do on its own from what it should expect from its institution.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 14

The Institution's Trilemma

14.1 ▪ Purpose and scope

AI is now part of the ordinary academic environment. Students use it to explain concepts, draft prose, write code, summarize material, generate examples, prepare for exams, prepare presentations, and test arguments. Faculty use it to design assignments, draft feedback, generate cases, refine explanations, and organize course materials. These uses will vary across tools and disciplines, but the policy problem is already clear at every level: departments and universities need a framework that protects learning without pretending that AI can be kept outside academic life. A policy based only on prohibition will be brittle. A policy based only on permission will be too vague. A useful framework must help each course answer practical questions: What uses of AI are allowed? What must students still be able to do without AI? How should students disclose use? How should instructors grade AI-assisted work? How should suspected misuse be handled fairly?

Institutions have faced versions of this before. When calculators became widely available, schools that simply banned them protected the old form of arithmetic instruction without asking whether that form was still what it was for. Schools that simply allowed them, without rebuilding the surrounding judgment skills, risked students losing number sense without anyone noticing. Useful policy came later, after enough institutions worked out what calculator use required of the curriculum it was embedded in: which skills students still needed to demonstrate without the tool, which skills the tool enhanced rather than replaced, and how to teach the difference. AI is at the same point. The framework in this guide is an attempt to provide what most institutions are still missing. It offers shared categories that distinguish AI use that supports learning from AI use that substitutes for it, and shared vocabulary for the policy decisions those categories enable.

Institutional AI guidance is plentiful; what this guide adds is a policy vocabulary shared verbatim with the book's student and instructor guides, so the definitions a syllabus quotes are the ones the classroom already uses. This guide provides a framework for those decisions at the level where most decisions actually get made. It is not a single rule for every course. A first-year writing course, a physics lab, a history seminar, a programming course, a nursing simulation, and an engineering design studio need different practices. They should still share a common vocabulary, common ethical expectations, and common safeguards.

The framework is designed for the full range of courses a university teaches: any class size, level, or discipline. Its design choices change along each of these three lines, and the structure here supports that variation rather than imposing one practice on every course. The three do not weigh equally everywhere. Class size, and the direct contact it allows, mostly decides which

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

assessment model fits. Discipline enters elsewhere, through the book's discipline overlays: its field-specific guidance on verification, tools, and acceptable use. Most of the practical work happens in departments, and this Part is written accordingly. That emphasis on departments is this book's position, not a settled result. The evidence points both ways, and no one has tested the question directly. Sections that departments can act on speak to faculty and chairs. Sections about institutional policy speak to deans, provosts, and central administrators. Where both levels are involved, both audiences are addressed.

14.2 ▪ Institutional outcomes

A university that adopts this framework should be able to show progress in seven areas.

1. Students receive clear guidance on how to use AI without surrendering their own thinking.
2. Faculty have shared categories for permitted, limited, required, and prohibited AI use.
3. Courses preserve meaningful assessment of independent competence.
4. Departments use fair procedures for disclosure, privacy, access, and suspected misuse.
5. Teaching assistants and graders apply rubrics with reasonable consistency.
6. The institution reviews the framework on a regular cycle, at least annually, as tools, risks, and teaching practices change. The review should ask whether the licensed tools are actually being used and whether learning is improving, since wide access can be purchased and still go largely unused.
7. Programs can say whether AI-mediated work is narrowing the range of what their students produce.

Governing boards increasingly ask to see these outcomes as well. Trustees, and the executive boards of individual schools and colleges, are accountable to the public for what the institution's degrees certify. They need not a one-time assurance that an AI policy exists but evidence, gathered on the same regular cycle, that it is improving what students can do over time.

These outcomes give the framework a practical test. The policy is working only if it improves clarity, learning, fairness, and trust.

14.3 ▪ What narrows when everyone complies

The usual question about AI asks what a given use does to a given student. That question is the right one for a course, and it cannot see the effect described here, which appears only in aggregate and only when nothing has gone wrong at the level of any individual.

In a controlled experiment, writers given story ideas from a language model produced work rated more creative, better written, and more enjoyable, with the largest gains going to the least creative writers. The same stories were also more similar to one another than stories written without AI [30]. Every writer improved. The collection lost range. The authors describe the result as a social dilemma: because AI-assisted work is rated more highly, each writer has an incentive to use AI more, which narrows the collective further. The study used one model on one task, and its authors note that the technology will change. The pattern of the finding carries over to other settings, but the size of the effect does not.

A department should expect the same shape wherever students reach a common tool at a common stage. Capstone topics converge. Literature reviews cite the same already-cited authors, since a model's suggestions reflect existing citation patterns and so amplify them. Project

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

proposals begin to resemble one another. No student has done anything the policy prohibits, no instructor has failed to verify, and the program's range contracts anyway.

A second path to the same place runs through enforcement rather than use. Where students believe their writing will be judged for signs of AI, some respond by writing more plainly than they can: simplifying vocabulary and sentence structure so their own work is not mistaken for a machine's [129, ch. 14]. The mechanism is different, since this is a response to being watched rather than a property of any tool, but the outcome is the same contraction, and it is produced by the institution rather than by the students.

Neither effect is visible in the places a department usually looks. Grades do not fall, since individual work is rated as good or better. Integrity referrals do not rise, since no rule has been broken. What changes is variety, and nothing in a standard assessment cycle measures it.

Two low-cost checks make it visible. Compare the distribution of topics, questions, or approaches in a capstone or project course against a cohort from before AI adoption; a marked narrowing is worth understanding even if it turns out to be benign. And where an assignment begins with ideation, require students to record their own candidate ideas before consulting any tool, which both preserves each student's own first attempt and leaves a record of how much range existed upstream.

A department that finds narrowing has a design question rather than a discipline question. The answer is usually to vary the starting point rather than to restrict the tool: different data, different sources, or a requirement that proposals differ from one another.

14.4 ▪ The three corners

Students face a predicament of their own under AI, which the book calls the Student's Trilemma: earning the grade, being honest about how the work was done, and showing that they understand it pull against one another. The conflict takes hold when an assignment asks for more than the student can do alone and AI can fill that gap without anyone seeing. A student can hold any two of these, but only at the cost of the third. Its three failure modes each give something up: hollow understanding keeps the grade and the honesty but not the learning, quiet overclaim keeps the grade and the understanding but hedges on disclosure, and the integrity penalty keeps full honesty and pays for it in the grade. Almost every student now feels some version of it.

Institutions face the same structural problem at their own scale. Presidents, provosts, deans, and trustees are pulled between three goals that AI now makes harder to hold together: keeping the institution financially viable, keeping its credential honest about what graduates can do, and serving the educational mission the institution exists for. An institution can hold any two, but only at the cost of the third. Most administrators now feel some version of it, though they rarely name the pattern.

One difference sets the institution apart. A student can resolve their trilemma, at least partly, by choosing how to study; an instructor can resolve theirs by choosing what to assign and how to grade it. The institution has no such exit. It does not get to choose between the student who refuses AI on principle and the student who uses it well; it has to certify both, under the same degree, in the same term. And if it grades verified and unverified work as though they were the same, the credential drifts away from what its graduates can actually do. The three Parts of this book share one vocabulary because the institution carries all three responsibilities at once, with none of them optional.

Call this the **Institution's Trilemma** (figure 14.1). The corners are:

Institutional sustainability. Keeping the institution financially viable through enrollment quality and quantity, tuition, donor cultivation, research grant flow, and state appropriations, and keeping the public legitimacy that makes financial viability possible. Presidents themselves

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

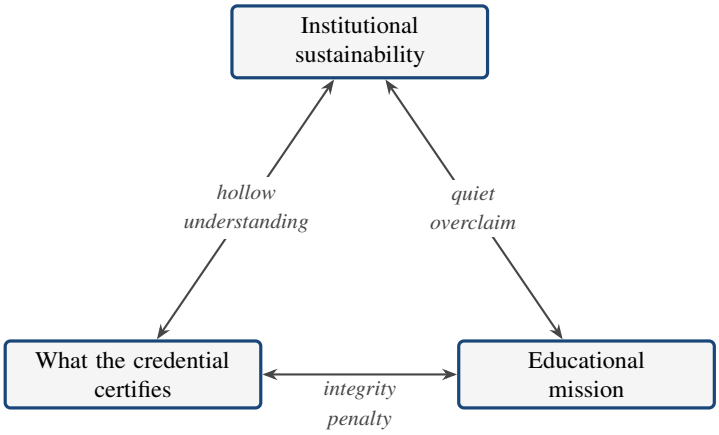


Figure 14.1. ***The Institution's Trilemma.*** *An institution faces the student's predicament at its own scale, and cannot resolve it by choosing. Each edge labels the failure mode that arises when the two adjacent goals are kept at the cost of the third. This mirrors the Student's and Instructor's Trilemmas.*

give higher education poor marks on rebuilding public trust [41]. Public skepticism erodes enrollment; falling enrollment erodes funding; falling funding erodes the institution's ability to make the case for its own value. Administrators cannot work on finances without also working on public legitimacy, or on public legitimacy without working on finances.

What the credential certifies. The degree as a meaningful claim about the institution's graduates: what they can do, what they understand, why they are worth hiring or admitting to graduate school. The credential's value depends on both the institution's testimony (what it says its graduates can do) and the underlying evidence (what they can actually do under direct examination). When AI assistance enters student work without being verified, testimony and evidence drift apart.

Educational mission. The deeper commitment to what students gain by attending. This varies by institutional type (Jesuit formation, liberal arts breadth, research apprenticeship, workforce readiness, professional preparation) but is always present in the institution's self-understanding and almost always present in the language it uses to describe itself publicly. The mission is what an administrator points to when asked why the institution exists beyond its operational and credentialing functions.

These statements of mission differ by type of institution, but they share a single claim: that some learning is itself the payoff, not just a means to an end. A parallel claim concerns work: what remains for people to do is not the leftovers of what machines cannot yet manage. The companion paper "AI and the Next Layer of Human Work" [55] develops it. A coder welcomes AI because for them the drudgery is what they want eliminated and the working program at the end is the payoff. An artist resists AI because for them the drudgery, the patient hours of practice and revision, *is* the payoff and the finished piece is its trace. The educational-mission corner rests on the same distinction at the institutional scale. An institution that has accepted that all of its content can be delivered more efficiently by AI has tacitly accepted that nothing it does is itself the point. An institution that defends its educational mission is asserting the opposite: that some of what students pay for, and some of what graduates carry forward, is the intellectual practice itself, not just its end product.

Any two corners can be held, but only at the cost of the third.

14.4.1 ▪ Three institutional failure modes

Sustainability and credential certification without mission. The institution maintains enrollment and credential reputation by turning itself into a sophisticated AI gatekeeper. Students complete coursework, get verified through carefully designed oral defenses, receive degrees that mean what they say. But the institution has stopped trying to educate, in the sense the mission language implies, beyond the credentialing process itself. Call this the **institutional form of hollow understanding**, a loss of mission: the degree still means what it says and the public account still holds, but the education the mission promises has hollowed out from inside. This is the failure mode of an institution that has outsourced the deep work of education to AI while retaining institutional gatekeeping for itself.

Sustainability and mission without credential certification. The institution maintains enrollment and stays committed to its educational mission, but lets AI-assisted work into the credential without demanding genuine ownership. Students graduate with degrees they cannot defend. The mission language remains genuine; the financial picture remains stable; the credential loses its meaning. This is the **institutional quiet overclaim**, a loss of credential integrity: the degree keeps the look of a certificate of what graduates can do, while the institution tacitly concedes that no one checked. This is the failure mode of an institution that wants to grow and stay true to its mission, but cannot bring itself to enforce the standards its credential implies.

Credential certification and mission without sustainability. The institution stays rigorous and mission-driven, refuses to lower standards in response to AI pressure, and runs out of students or money in the process. Call this the **institutional integrity penalty**, a loss of sustenance: standards and mission both survive intact, but the institution cannot keep its doors open.

AI did not create the Institution's Trilemma any more than it created the Student's. The trilemma describes a structural tension that pre-existed AI and that any responsible institution always navigates. What AI did was sharpen each edge. By lowering the apparent cost of giving up any one corner, AI makes the other two harder for an institution to hold together.

Institution-level guidance mostly stops at data protection and approved tools. Many of the policies that address student work at all do one thing: they forbid passing off AI work as your own. A student can follow such a policy exactly, disclose every use, and still finish a degree without having learned what the degree says they learned. Honesty and understanding are different questions, and only one of them is what the credential certifies. A policy that addresses honesty and stops there leaves understanding undefended, and understanding is the corner it most needed to defend.

14.4.2 ▪ How the trilemma manifests differently by institutional type

Every degree-granting institution faces all three corners, but how hard each one pulls depends on the kind of institution. Four types are distinguished below, though real institutions often combine features of more than one.

R1 private research universities. Here endowments and the power to set tuition cushion sustainability. Research grant funding, however, is under direct strain from AI as of this writing. Proposals written with agentic AI threaten to overwhelm funding agencies and their reviewers [104], and the NIH now limits each investigator to six applications a year [96]. The credential is the brand; protecting it is non-negotiable. Mission is articulated through the dual commitment to research training and undergraduate liberal education. Where these conditions hold, the institutional form of hollow understanding is the one to watch: an institution that maintains its credential through sophisticated gatekeeping while quietly admitting that what it offers undergraduates has converged with what AI tutors offer for free.

Small residential liberal arts colleges. At these colleges the trilemma takes its tightest form.

Credential and mission corners are tightly coupled: the case for the institution *is* the case for the human-led residential experience. Sustainability is the threatening corner for the many that depend on tuition. For them, the institutional integrity penalty is the most visible and the most current. The closures of small private tuition-dependent colleges in the years leading up to this writing show how exposed that corner already is [77]. The case for these institutions in the AI era is not that they teach better content than larger universities, since AI tutors can deliver content well. It is that they teach in a setting where the human-led mentoring relationship is the educational point, not just the delivery mechanism.

Regional public universities. At most state-system institutions outside the research flagships, money comes first. The enrollment cliff hits these institutions hardest, and cuts in federal and state funding add to the pressure. Their mission is increasingly described in terms of jobs rather than liberal education, partly because state legislatures ask for it and partly because students and parents want clear outcomes after graduation. What the credential certifies matters in practice, since graduates have to perform when they leave, but it is rarely the main strategic question. Under this much enrollment pressure, the main risk is the institutional quiet overclaim. The institution keeps its students by accepting AI-assisted work without asking students to own it, because asking would cost enrollment it cannot afford to lose.

Community colleges. Here the three corners pull less hard against one another. Public trust in community colleges has remained strong even as trust in higher education overall has fallen. One common explanation, from DeRionne Pollard, who leads the American Association of Community Colleges, is that community colleges are seen as affordable, connected to career pathways, and engaged with local employers [41]. Money is a pressure, but a milder one. The mission is clearer than at most institutions: access, plus preparation for local work. The credential question is different too. It turns mainly on whether credits transfer to four-year institutions. Of the three failure modes, the integrity penalty may be the one community colleges are least exposed to. Their larger risk is drifting from the local mission if state policy shifts.

The four types are not comprehensive. Other kinds of institution (HBCUs, religious-affiliated colleges, for-profit institutions, fully online providers) face the same trilemma, but how hard each corner pulls depends on the type. The frame is useful because its corners do not change from one type to the next; only the relative pull and the most likely failure mode do.

14.4.3 ▪ Why the Institution's Trilemma matters for the rest of this Part

Each tool this Part offers an institution, from the assessment models and the integrity procedure to faculty change management and the phased rollout, exists to keep the three corners together under AI pressure. Seen without the trilemma, the tools can look like separate fixes for separate operational problems. Seen through it, they work as coordinated commitments that guard the institution against all three failure modes at once. The trilemma is managed, not solved: no fixed point dissolves the tension. The healthy state is holding all three corners in dynamic balance, the way a three-legged stool needs all three legs and no single leg is the optimum.

This Part also has a more concrete rhetorical purpose. Department chairs and program directors increasingly find themselves in conversations with deans, provosts, and boards about whether faculty roles can be replaced by AI tools at substantially lower cost. The Institution's Trilemma supplies the structural answer to that question. The case for human-led education is that a system hollowed out into pure credentialing cannot honor the educational mission. This view is consistent with the OECD's analysis of teaching in the age of AI, which holds that AI amplifies both effective and ineffective educational practice and that the teacher's judgment and human qualities remain central, particularly where the technology has clear limits [110]. Throughout the rest of this Part, those same operational tools can be read as ammunition for the argument chairs need to make to deans when AI-replacement proposals arrive on the agenda. The argument is that

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

some of what students pay for, and some of what graduates carry forward, is the very thing that AI-as-replacement is supposed to eliminate.

A budget model can undercut the standards an institution says it wants to keep. Under Responsibility Center Management and similar models, a unit's resources rise and fall with its enrollment. A department that allows broader AI use may then pull students from a department that holds a stricter line. The pool of students is roughly fixed, so one unit's gain is another's loss. That leaves the stricter unit financially punished for the very standard the institution endorses. Each unit then has reason to loosen in turn. The result can be a slow drift toward the most permissive policy on offer, and no single department can step out of it alone. The same pressure runs one level down, between instructors: the colleague who is generous with AI may draw students from the one who is not. None of this comes from any unit's bad choice; it comes from how the budget is built. So it can be fixed only where the budget is set. An institution that wants its standards to hold should set the core expectations centrally and stay alert to its own budget incentives, so departments are not left competing on permissiveness. This risk appears among those tracked in the Risk register.

14.5 ▪ The institutional problem

AI can deliver the facts.
An education forms the values and judgment to use them.

A senior at Northeastern University spent the final weeks of her undergraduate degree filing a formal complaint and a tuition refund request against her organizational behavior professor. She had discovered that the professor was using ChatGPT, Perplexity, and an AI presentation generator to produce his lecture materials, with a stray AI prompt instruction visible in the lecture notes. The syllabus for the course prohibited student AI use. The professor later acknowledged the AI assistance. The university declined to refund the tuition.[9]

The case names a pattern. The professor was wearing a **competence costume**: AI-polished material that signals more current and confident engagement than the underlying work supports. The student saw it. The institution declined the refund. The asymmetry is the point. If an institution's integrity rule applies to its students but not to their teachers, the rule is weak by its very design.

The case can also be read through the trilemma frame. It is not just a faculty integrity question. It is an early signal of the institutional form of hollow understanding: an institution that maintained its credential and sustainability through procedural responses to a violation, but did not address the deeper mission question of what students were paying for.

The translation test, which the book first gives to students, offers institutions a positive standard to adopt symmetrically. An instructor who drew on AI for course materials should be able to restate them in their own words and with their own examples. Students using AI on coursework should be able to do the same. The same diagnostic, applied to both, is a policy the institution can defend in public.

AI creates a gap between polished performance and demonstrated learning. A student may submit fluent prose, working code, or a convincing solution without having developed the underlying competence. This problem is not limited to cheating. It can happen through ordinary over-reliance. A student may believe they understand because the final product looks good.

The institutional response should not be to treat every AI-assisted product with suspicion. The better response is to redesign expectations. Courses should assess both independent understanding and responsible tool use. Students should learn to verify, disclose, and defend AI-assisted work.

[9] Hill [50].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Faculty should have clear models for small, medium, large, and online courses. Departments should not leave each instructor to invent policy alone.

The goal is a durable approach. AI systems will become stronger. Detection tools will remain imperfect. The safest assessments will be those that make student thinking visible and require students to explain, verify, apply, and revise their work.

Evidence from learning science supports this direction. Practice testing and distributed practice have strong support across learning contexts, and recent AI-tutoring studies suggest that carefully designed systems can help when they preserve active engagement rather than replace it [32, 61]. At the same time, field evidence warns that unrestricted generative AI can create short-term performance gains without the learning that shows once the AI is taken away [11].

14.6 ▪ Guiding principles

The following principles should guide institutional policy.

1. **Learning remains the center.** AI may support learning, but it should not replace the intellectual work students are meant to practice.
2. **Independent competence must be preserved.** Every course should name the abilities students must demonstrate on their own, with AI absent or tightly limited.
3. **AI literacy is a legitimate educational goal.** Students should learn how to use AI critically, productively, and ethically within their disciplines.
4. **Verification is required.** AI output should be treated as provisional until checked against evidence, data, sources, methods, or disciplinary standards.
5. **Disclosure should be normal, brief, and useful.** Students should be able to state plainly what AI did in the work and what they checked.
6. **Assessment should reveal thinking.** Courses should gather evidence of reasoning, process, revision, judgment, and transfer, not only polished final products.
7. **Privacy and equity must be protected.** Students should not be required to use tools that compromise data, create avoidable costs, or disadvantage students by language background, disability, access, or prior exposure.
8. **Faculty judgment remains human.** AI may help instructors prepare feedback and materials, but instructors remain responsible for grades, academic-integrity judgments, student evaluation, and the accuracy and integrity of the materials they put before students.

These principles align with emerging international guidance that treats AI literacy, human oversight, ethics, privacy, and responsible educational design as central institutional responsibilities [35, 88–90, 98].

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 15

The Framework and Vocabulary

15.1 ▪ The CAT framework

University policy should distinguish three kinds of learning evidence: Core Competence, AI-assisted Practice, and Trustworthiness. The acronym CAT gives faculty and students a shared language.

Core Competence. What students must understand and be able to do without AI or with tightly bounded assistance. In institutional policy each program sets its own, and the framework does not ask them to match.

AI-assisted Practice. What students can do when they use AI responsibly as a permitted tool for learning, revision, analysis, coding, feedback, or exploration, without losing control of the work. In institutional policy this includes AI acting as tutor, critic, simulator, editor, coding aid, or research assistant.

Trustworthiness. How well students verify, disclose, critique, and take ownership of AI-assisted work. In institutional policy it is what an academic-integrity process has to be able to weigh.

The distinction matters because these abilities can diverge. A student may produce strong AI-assisted work while having weak independent understanding. Another student may understand the material well but use AI carelessly. Good policy should not collapse these into one score.

Institutional guidance should therefore encourage courses to state which assignments measure Core Competence, which assignments develop AI-assisted Practice, and which assignments evaluate Trustworthiness.

A single assessment can serve more than one of these. A closed-book exam measures Core Competence; an AI-permitted assignment develops AI-assisted Practice; a project that students present and then answer questions about serves all three at once, since the project reflects AI-assisted Practice while the live questioning tests Core Competence and Trustworthiness together. The categories name what an assessment measures, not a rule that each measure only one thing.

15.1.1 ▪ Accreditation and assurance of learning

The framework's line between Core Competence and AI-assisted Practice connects directly to accreditation, which is where assessment policy acquires institutional force. Both institutional accreditors, which review the university as a whole, and programmatic accreditors, which review specific degrees, ask the same underlying question: can the institution show that students actually achieve the learning outcomes it claims? Engineering and computing programs answer to ABET,

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

business programs to AACSB, and many health, education, and other programs to their own disciplinary bodies, but the logic is shared across all of them.

AI sharpens this question rather than changing it. Where a stated outcome requires a competence students must hold independently, the institution must be able to show that the outcome was assessed independently of AI, not inferred from work that AI could have produced. The minimum independent-competence requirement, set out in chapter 16, is the mechanism that supplies this evidence, and the CAT distinction lets a program report cleanly on which outcomes are assessed as independent competence and which are assessed as responsible AI-assisted Practice. A program that maps its outcomes onto CAT can show an accreditor not only what it teaches but how it knows students learned it under current conditions.

In practice, departments should treat AI-aware assessment design and accreditation reporting as one task, not two. An assessment plan that already separates independent competence from AI-assisted Practice produces the evidence an accreditor expects.

15.2 ▪ Course-level AI-use categories

A department's policy should require each assignment to name the AI use it permits. The four categories below are easy for a student to apply and leave a department room to adapt them.

They are the book's shared vocabulary for AI use, stated here as institutional policy terms so that this Part can stand on its own.

The boundaries between categories are not always sharp. A student who asks AI to explain an approach and then to help draft from that explanation has moved from tutor to collaborator within one session. The categories name the most permissive role AI plays in the submitted work: once AI wording, code, or worked solutions enter it, the work is **AIC**. Each assignment should state which role is permitted, and where a task mixes roles, the instructor should say so rather than force an artificial split.

15.2.1 ▪ NAI: No AI

AI use is not allowed. These assignments measure independent competence. Typical forms range from closed-book exams, in-class writing, and oral explanations to live coding checks, foundational calculations, language practice, and lab practicals.

NAI assignments should not dominate every course. They serve a specific purpose: they show what students can do without outside help.

15.2.2 ▪ AIT: AI as Tutor

AI may be used for explanation, practice, review, hints, and study support, but not to produce submitted work. A student might ask AI to explain a concept, generate practice questions, or quiz them before an exam. The submitted response should still be the student's own work, with no AI-generated text, code, or worked solutions in it.

15.2.3 ▪ AIC: AI as Collaborator

AI may help with brainstorming, drafting, revising, debugging, outlining, critique, source search, comparison, or testing. Students must disclose use and verify important claims. **AIC** work is useful wherever a course sets out to teach students to revise, code, research, model, or design as practitioners do.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

15.2.4 ▪ AIS: AI as Object of Study

Students analyze AI output directly. The work ranges from spotting hallucinations and repairing flawed reasoning to critiquing bias, comparing tools, evaluating generated code, checking the quality of sources, and testing how AI performs in the discipline. **AIS** assignments teach students to examine AI rather than merely use it.

15.3 ▪ Assessment models by course size

Class size changes what assessment can reasonably do. The same principles apply across courses, but the implementation must differ. The deciding question is whether the teaching team can talk with each student about the work. Part II's worked examples give a sense of scale: a 14-student seminar under the Oral-Defense Model, a 60-student course with two TAs under the Hybrid Model, and a 280-student lecture under the Evidence-of-Thinking Model.

15.3.1 ▪ Oral-Defense Model

The Oral-Defense Model is best for a department's small classes at every level, from seminars, honors courses, and capstones to writing workshops, advanced labs, research courses, and many graduate courses. What selects it is class size; it works wherever an instructor can talk with each student about the work.

Because each student can be questioned on the work, the course can allow wider use of AI. A defense may be a board proof, source-analysis conversation, code walk-through, lab-method explanation, design critique, writing conference, or project presentation.

A course using this model should include at least one live explanation of AI-assisted work. The purpose is to confirm ownership and deepen learning. A short defense can take 15 to 20 minutes and a fuller one 30 to 60. It moves from the student's own summary through a core question, an audit-and-repair task, and a transfer question to an account of how AI was used, checked, and limited. Where more than one person runs defenses, the teaching team should agree beforehand on what a strong answer looks like.

15.3.2 ▪ Evidence-of-Thinking Model

The Evidence-of-Thinking Model is best for a department's large courses: lectures, service courses, online sections, and multi-section offerings. In these settings, individual oral exams for every student may not be feasible.

Large courses should gather repeated, low-cost evidence of thinking. A department can draw that evidence from short no-AI quizzes and in-class writing, from draft histories, process logs, lab notebooks, and AI-use disclosures, from source checks, code tests, homework-to-quiz transfer questions, and critiques of flawed output, and from short video explanations and random oral audits.

The goal is not surveillance. The goal is valid assessment. A polished final product is no longer enough evidence by itself.

15.3.3 ▪ Hybrid Model

Many courses fall between the two, too large to examine every student orally and small enough to question some of them live. The Hybrid Model combines scalable process evidence with limited direct questioning. Its tools include group oral exams, oral spot checks, TA-led conferences, short meetings with the instructor, video explanations, and written verification logs.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Hybrid designs are useful for medium-sized writing courses, upper-level STEM courses, lab courses, project-based courses, and professional programs.

Table 15.1. ***Recommended assessment model by course context.*** *The model follows from how much direct contact with each student the course allows. Where contact is high, as in a seminar, an oral defense verifies understanding directly. Where it is low, as in a large lecture or an online section, the evidence has to be built into the work itself.*

Course context	Recommended model
Small seminar	Oral-Defense Model
Graduate course	Oral-Defense or Hybrid Model
Honors course	Oral-Defense or Hybrid Model
Medium undergraduate course	Hybrid Model
Large lecture	Evidence-of-Thinking Model
Multi-section course	Evidence-of-Thinking Model with TA calibration
Online course	Evidence-of-Thinking Model with stronger process documentation
Capstone or project course	Hybrid or Oral-Defense Model

Table 15.1 names the primary model for each context, not the only one in play. Most courses combine elements: a small seminar built on oral defense still gathers some written process evidence, and a large lecture using the Evidence-of-Thinking Model can still add a few oral spot checks. Read each recommendation as a course's center of gravity, not an exclusive choice.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 16

Standards and Requirements

16.1 ▪ Minimum independent-competence requirement

Every course must include some meaningful assessment of what students can do independently. The form should fit the discipline.

In STEM courses, this may be a quiz, exam, derivation, lab practical, coding check, or oral explanation. In writing courses, it may be in-class writing, a writing conference, or a revision defense. In humanities courses, it may be close reading, source analysis, or argument reconstruction. In professional programs, it may be a case explanation, simulation, or ethical decision defense.

The institution should not prescribe one method for every course. It should prescribe the principle, and the principle is not optional: some course credit must reflect independent understanding, not only AI-assisted production. Strong AI-assisted performance does not make up for a lack of independent competence. This element of the framework should not be relaxed under local adaptation.

A useful policy sentence is:

> Every course must identify the core skills, concepts, or judgments students are required to demonstrate without AI or with tightly bounded assistance, and must assess them. The method of assessment should be appropriate to the discipline, level, enrollment, and learning goals of the course.

16.2 ▪ Disclosure standard

Disclosure should be normal and brief. It should not read like a confession. It should give instructors enough information to understand how AI shaped the work.

Keep two things apart: what counts as AI-assisted work, and the standard that a disclosure says the work meets. AI-assisted work is any work an AI tool has shaped. Whether that work meets the institution's standard is the question disclosure exists to answer, and the standard is this: the submitted work is verified and owned by the student, and any AI contribution stays within what the assignment permits and is disclosed. Work that does not meet it is still AI-assisted work; it is AI-assisted work that falls short, and a policy needs to be able to say so.

Disclosure attaches to the use of a general-purpose AI system, not to routine help from dedicated tools. Spelling correction, grammar checking, and predictive text from dedicated, deterministic tools are treated like the familiar tools they resemble and need not be reported, just as using a calculator or a dictionary is not. What triggers disclosure is the tool rather than the task:

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

asking a general-purpose AI system to check your grammar is AI use, even though a dedicated grammar checker is not. How much to say depends on the task: a line is usually enough for routine help, and use that did any of the thinking the assignment is meant to develop needs a fuller account.

The recommended student disclosure block has six fields: *Tools used*; *Tool role*; *Student role*; *Accepted, changed, or rejected*; *Independent verification*; and *What I can explain without AI*.

A complete entry need not be long. A student who used AI to help debug a script might capture all of this in a few lines. The AI assistant suggested fixes for a sorting bug. The student wrote the original code, applied the fix, and added tests, then kept the fix but renamed the variables. They confirmed the result with a minimal working example, and they can explain the sorting logic and why the bug occurred without AI.

How much a student discloses should depend on what the course treats as core. A student may use a tool as a black box, without explaining how it works, if the tool does work below the level the course aims to teach. In a course where nonlinear optimization is a means rather than a learning goal, it is enough to report that an optimizer was used and verified, without accounting for how the optimizer works. Where a tool touches the core competence the course is meant to build, the explanation should reach that level. The same habit of naming the help received applies to substantial non-AI assistance wherever a course's policy already calls for it.

For low-stakes assignments, a one-sentence disclosure may be enough. For substantial AI-assisted projects, instructors may require a process log, prompt excerpts, verification notes, or a short reflective memo.

16.3 ▪ Privacy, data governance, and tool approval

Students and faculty need clear rules about what may not be uploaded into AI systems. These rules should be plain enough to appear in syllabi and student orientation materials.

Institutional policy should bar students and faculty from uploading the following into any tool not approved for that use:

- private information about students, classmates, patients, clients, research subjects, or employees;
- unpublished research data without permission;
- restricted exams, answer keys, or secure assessment materials;
- confidential institutional documents;
- private correspondence, recommendation letters, or evaluations;
- copyrighted course materials when sharing is not permitted;
- data governed by human-subjects, clinical, contractual, or legal restrictions.

The institution should maintain a list of approved tools and approved uses. A tool approved for general brainstorming may not be approved for student records, research data, grading, or protected course materials.

Data governance should also address retention, training use, third-party access, accessibility, cost, procurement, and support. These details should not be left to individual instructors alone.

One more category belongs here, and it is easy to miss: regulation. AI that makes or supports decisions about students may be regulated activity, not merely a matter of institutional policy. Admissions, assessment, and monitoring are the uses that draw attention. Obligations can include bias testing, documented human oversight, and retained records. Some regimes reach across

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

borders. Institutions should confirm where they stand with counsel, and the companion repository maintains a current orientation (`institutions/regulatory-landscape.md`).

Some courses require students to begin AI use with a mentor prompt, a standing instruction that tells the AI how to guide the student's learning across many tasks, and to share the resulting conversation threads with the instructor. Those threads are a new kind of student work, and institutional data governance should address them explicitly. A thread is the student's contribution to the assignment, so retention rules should mirror those for other student work. It may also hold sensitive things the student told the AI, such as frustration with the course, trouble with the material, or personal matters unrelated to the course. The data governance policy should say who may see shared threads, how faculty and TAs handle them, how long they are kept, and what happens to them when the course ends. Departments and institutions that require shared threads should align these rules with their existing protections for other student work. They should also follow the privacy law that applies in their jurisdiction: FERPA in the United States, and equivalent frameworks elsewhere.

16.3.1 ▪ AI-assisted grading

An institution that allows AI-assisted grading has to settle four things, and none of them should be left to the individual instructor.

Whether student work may leave the institution. In the United States, work sent to a vendor with a name or a grade attached can carry personally identifiable information from education records. FERPA allows disclosure without consent to an outside party treated as a school official only if the party performs a service the institution would otherwise use employees for, stays under the institution's direct control in how it uses and keeps the records, uses them only for the purpose disclosed, and meets the criteria the institution publishes in its annual FERPA notice [127]. A tool an instructor signs up for alone has been checked against none of these. Institutions can go further than the law and let students choose human-only grading.

What the vendor may do with the work. Before approval, the review should answer plain questions. Is student work used to train or improve the vendor's models? How long is it kept, and can the institution have it deleted? Who else processes it, and where? Will the vendor say when the underlying model changes? Can the institution keep the record needed to reconstruct a grade?

What the law requires. In the European Union, AI used to evaluate learning outcomes is classified as high-risk [37], which brings duties of human oversight, record keeping, and informing the students affected. A 2026 amendment moved the date those duties apply for such systems to December 2027 [34]. Other jurisdictions differ, and the companion repository keeps a current note.

Who answers for the grade. The instructor does, and the institution must make that possible. For every course using AI-assisted grading, keep the rubric, the model and version, the hand-graded comparison and its result, and the instructor's decision on each disputed score. A grade that cannot be reconstructed cannot be defended on appeal. The same record answers an accreditor, and it answers the student.

16.4 ▪ Equity, access, and accessibility

AI policy should avoid creating a hidden paywall around learning. If a course requires AI use, students should have access to an approved tool at no additional cost or should be offered an equivalent alternative.

Equity also includes language background, disability, prior exposure, bandwidth, device access, and confidence using new tools. Some students will arrive with extensive AI experience. Others will be learning basic use for the first time. Courses should not mistake unequal access for

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

unequal ability.

Survey evidence puts numbers on that unevenness. Regular AI use was reported by about a third of women against nearly half of men, and by under a third of underrepresented minority students against roughly two fifths of White and Asian students [25]. The authors are careful about what this shows. Higher use is not better use. The concern is that unequal engagement becomes unequal opportunity to build the literacy and the critical oversight this framework asks of every graduate.

Accessibility can cut in both directions. AI tools may help some students summarize, rephrase, plan, or review material. They may also create barriers if tools are inaccessible, expensive, unstable, or poorly documented. Accommodation decisions should remain human and should follow institutional disability procedures.

16.5 ▪ Academic integrity procedure

Academic integrity should be handled through evidence and due process, not through detector scores used as the sole or primary evidence. A hallucinated citation is not plagiarism: nothing was copied and no author was deprived of credit. Some student codes call it fabrication and others false citation. Use the category your own code provides. In research training the label matters more, because it decides which office handles the case and under which standard [140]. Early commentary on ChatGPT and academic integrity urged clear policies and assessment redesign, and treated detection as an uncertain arms race [27].

The case against detector-based enforcement rests on a mechanism, not only on a rate. Most detectors in 2023 relied on text perplexity, a measure of how predictable a sequence of words is to a language model. Non-native writers tend to vary their words and sentence structures less. That gives their text lower perplexity, which detectors then read as a sign of AI generation. Liang and colleagues documented what follows, evaluating seven widely used GPT detectors on 91 TOEFL essays written by non-native English speakers, alongside 88 essays by US eighth-grade students [79]. On the TOEFL essays the average false-positive rate was 61.3%, while the US student essays were classified accurately. All seven detectors unanimously misclassified 19.8% of the TOEFL essays. At least one detector flagged 97.8% of them as AI-generated. That the mechanism drives this result, rather than the particular sample, is supported by an intervention within the same essays: enriching the word choice of the TOEFL essays lowered the average false-positive rate to 11.6%, leaving only one essay unanimously flagged, while simplifying the vocabulary of the US essays moved misclassification the other way.

The same study shows that detectors can be bypassed by prompting an AI to use more literary language, with detection rates falling to near zero. Detectors therefore fail in two directions: false positives concentrate on multilingual writers, and false negatives go to anyone willing to add a single self-editing prompt, which rewards evasion rather than honesty. These rates come from the detector tools available in 2023 and from a single evaluation, and later detectors may lower them. The structural problem will remain, because perplexity-based detection penalizes restricted linguistic range by design, and that will not change unless detectors come to work in a fundamentally different way.

Accuracy is not the only cost. Detector-based enforcement sets up what those authors term a presumption of guilt, an institutional posture in which an automated system marks a student as suspect and the student must then prove innocence. The damage to a student's reputation from being flagged may last even after the accusation is withdrawn. Those authors' own recommendation follows from both findings: they "strongly caution against the use of GPT detectors in evaluative or educational settings, particularly when assessing the work of non-native English speakers" [79].

The accuracy concerns translate into harm at the case level. A senior at the University of

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

California, Davis whose paper was flagged by Turnitin as AI-generated spent the final months of her senior year defending against the accusation; her grades reportedly began to slide as she fought it.[10] The arms race that relying on detectors creates has produced its own absurdity. Students now use "humanizer" tools that adjust AI-generated text to defeat detection, and some students who do not use AI at all subscribe to the same tools to reduce the chance of being falsely flagged. As one professor quoted in recent reporting put it, the better the writer you are, the more AI thinks you're AI.[11] Detector dependence does not create the integrity problem. It produces a parallel problem of its own.

This is the canonical version of the procedure. Shorter forms elsewhere in the book summarize it for their audiences and do not replace it. The recommended process is:

1. Review the assignment's AI-use rule and the student's submitted disclosure.
2. Identify the specific concern rather than a general suspicion.
3. Examine process evidence, such as drafts, version history, source notes, lab notebooks, code history, or prompt logs.
4. Compare submitted work with in-class or other course work when relevant.
5. Invite the student to explain the work in a non-accusatory meeting or written response.
6. Ask targeted questions about sources, reasoning, method, code, data, or revision choices.
7. Use a short reassessment or oral explanation when needed.
8. Distinguish misunderstanding, poor documentation, prohibited assistance, and deliberate misrepresentation.
9. Document the concern, the evidence, the student's response, and the decision.
10. Use educational remedies for minor first violations when appropriate.
11. Use formal academic-integrity procedures for serious, deceptive, or repeated misconduct.

The institutional standard should be clear:

> AI-detector output may be considered, if allowed by institutional policy, but it should not serve as the sole or primary evidence of academic misconduct. Decisions should rest on assignment rules, process evidence, student explanation, and professional judgment.

One consequence of that standard runs against what departments often assume when they write procedures. Absent the student's own admission, the available evidence often falls short of what a formal proceeding requires. Many institutional proceedings run on a preponderance standard (more likely than not) rather than a criminal one, and even that is hard to meet here. There is no register of generated text to search, detector output is probabilistic and contested, and a model asked what it wrote will not answer reliably. Comparison with a student's other work supports a judgment; it does not settle one.

A department that understands this will design so that its procedures do not depend on proof it cannot obtain. That is why step ten above, educational remedies for a minor first violation, is usually the operative route rather than an occasional kindness: it is the response the available evidence actually supports. The University of Texas at Austin runs this as a named administrative process, under the heading of **restorative practice**, in which a student who takes responsibility

[10] Klee [64].

[11] Kingkade [63]; the line is attributed in that article to Erin Ramirez of California State University, Monterey Bay.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

works with staff, and with those affected where they agree to take part, to repair the harm and rejoin the community. Departments that already operate such a process for other conduct matters can route academic-integrity cases to it rather than building something parallel.

None of this weakens the standard. It relocates it. The student remains accountable for explaining work submitted under their name.

There is a design consequence, and departments that draw it are spared most of the difficulty above. If use cannot be established, do not build procedures that depend on establishing it. The University of Sydney's two-lane approach, set out in its Sydney Assessment Framework, takes this route by sorting tasks into two lanes: a secured lane, where conditions make individual capability observable, and an open lane covering everything else, where AI use is simply *assumed* rather than permitted or policed [129, ch. 6]. A presumption costs nothing to enforce. Under it, the department's attention returns to whether the secured tasks are measuring what they should.

Two cautions belong with it. The lanes are better read as more secure and less secure than as secure and open, since lockdown browsers, proctoring, and process tracking are all imperfect and each carries costs to accessibility and trust. And the presumption only works if the secured lane covers what a degree certifies; a department that secures nothing has not adopted a framework, it has stopped asking. Section 16.1 states what must remain observable.

16.6 ▪ Faculty use of AI

The framework applies to faculty as well as students, and it applies in full. Faculty may use AI to support teaching, but they remain responsible for professional judgment and they say where AI has shaped the work students receive. A course that asks students to disclose while the instructor does not has taught them that disclosure is a rule imposed on the assessed, not a habit of the profession.

The Northeastern case discussed in section 14.5 is one of the early reported instances where this responsibility was tested in public. The translation test offers an internal standard faculty can apply to themselves. Before AI-assisted material goes in front of students, the instructor should be able to translate it into their own words, examples, and structure. If the translation is awkward, the integration is incomplete. A faculty member who cannot make that translation is wearing the competence costume the Northeastern case made visible.

Appropriate uses may include drafting examples, generating practice questions, identifying possible misconceptions, improving assignment wording, drafting feedback for human review, creating case variations, and preparing discussion prompts.

Faculty should not delegate final grades, academic-integrity decisions, accommodation decisions, or sensitive student evaluation to AI. Protected student work, student records, and confidential materials should stay out of any system the institution has not approved.

A concise policy sentence is:

> Instructors may use AI to assist teaching preparation and feedback drafting, but the instructor remains responsible for all grades, evaluative judgments, academic-integrity decisions, course materials, and protection of student information. Where AI has shaped material students receive, the instructor says so.

16.7 ▪ Required syllabus elements

Every syllabus should answer five questions.

1. What AI uses are allowed or prohibited in this course?
2. Which assignments are **NAI**, **AIT**, **AIC**, or **AIS**?

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

3. What must students be able to do without AI?
4. How should students disclose AI use?
5. What materials may be uploaded into AI systems, and what may not?

For large or multi-section courses, syllabi should also explain how the course will check understanding. Examples include short no-AI quizzes, process logs, random oral audits, in-class writing, draft histories, or TA-led conferences.

Courses that adopt the required-mentor-prompt practice, in which students share their AI conversation threads with the instructor, should additionally answer two questions in their syllabus: how the mentor-prompt requirement applies (which assignments, with what support) and how shared conversation threads will be reviewed, retained, and protected. The institution's general AI data-governance policy (section 16.3) should be the floor; course-level commitments may go further but should not undercut institutional protections.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 17

Implementation and Change

17.1 ▪ Governance and support

Two structures separate a policy that lives on paper from one that changes practice: a body that owns the policy, and a support system that helps students and faculty act on it.

17.1.1 ▪ Governance and oversight

Responsibility for AI policy is easy to misplace. The questions it raises, about what a course should certify, how to assess independent competence, and when AI use is honest, are academic and ethical rather than technical. Yet the natural institutional reflex is to route anything involving software to information technology. IT has an essential role in access, security, procurement, and data protection, but those are different questions from how AI changes learning and how to use it well. A policy left to IT tends to answer the questions IT is equipped to answer and to leave the academic ones unowned.

The more durable arrangement is a standing, cross-functional body that owns the policy. It is a committee or task force that includes academic leadership, faculty from more than one discipline (among them some who are skeptical of AI), students, and representation from information technology, the library, and legal counsel. Its remit is to maintain the institution's AI-use policy and shared vocabulary, to review and approve tools for specific uses (section 16.3), to monitor the risks in the register, and to review the framework on the regular cycle the outcomes call for (section 14.2). The policy itself should carry a date and a stated review cadence. An undated policy in a field moving this fast cannot be maintained, and after a year it reads as expired. The exact form should fit the institution. What matters is that the body exists, that it is led from the academic side rather than from IT, and that someone owns the policy rather than each office assuming another will. Such a body rarely starts from nothing; it usually builds on departments that have already developed coherent practice.

17.1.2 ▪ Support beyond the classroom

Owning the policy is not the same as helping students follow it. Much of the work of learning to use AI well happens outside any single course, in the units students already turn to for help. Coordinating those units rather than leaving each to improvise is what turns a policy into everyday support. The library is the natural hub for the student-facing part. Teaching information and research literacy is already central to its work [8], and evaluating AI-generated content, judging a

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

tool's limits, and choosing the right tool for a task are direct extensions of that work. Other units carry the pieces closest to their own:

- **The library** curates AI-literacy resources, runs workshops on effective and ethical use, helps students see which tools suit which tasks, and is a natural home for the required AI-literacy module.
- **Writing centers** help students use AI within the writing process without surrendering authorship, reinforcing the verification and disclosure habits the framework asks for.
- **Tutoring and academic-success centers** teach study strategies that use AI as a tutor rather than a substitute, complementing human tutoring rather than competing with it.
- **Academic advising** helps students understand course-level AI rules and plan for assessments of independent competence.
- **Disability and accessibility services** keep accommodation decisions and approved tools aligned, so that AI neither bypasses an accommodation nor becomes a barrier (section 16.4).

Faculty support remains the work of the teaching center; the units above are where students meet the framework outside the classroom. With the governance body, they are the structure that turns a written policy into something students experience as help.

17.2 ▪ Departmental responsibilities

Departments should not require every instructor to use the same AI policy. They should require every course to make its policy clear.

At the department level, useful responsibilities include:

- adopting shared definitions and assignment categories;
- identifying discipline-specific standards for verification;
- developing sample assignments and rubrics;
- supporting faculty who teach large or multi-section courses;
- calibrating TA grading (grading shared samples together until scores agree) when process evidence is assessed;
- reviewing course policies for consistency and fairness;
- updating policies as tools and student practices change.

Departments should also identify courses where AI literacy is itself a learning outcome. In those courses, students should practice critiquing, verifying, designing prompts, detecting errors, and disclosing responsibly.

17.3 ▪ Teaching-center responsibilities

Teaching centers can turn policy into practice. They should provide templates, examples, workshops, and consultation for faculty, and the need is well documented. In a 2025 faculty survey, fewer than one in four agreed that their institution had provided sufficient resources to develop their AI literacy, and a large majority reported being unclear about how AI applies in their own teaching [29].

High-value resources include:

- a syllabus-statement bank;

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

- sample assignments in each of the four categories;
- discipline-specific verification examples;
- rubrics for process evidence and disclosure;
- guides for oral defense and random audits;
- TA calibration materials;
- privacy and data-use checklists;
- academic-integrity conversation scripts;
- student-facing learning modules.

The most useful teaching-center materials are short, editable, and course-ready.

17.4 ▪ Training plan

An institutional framework should include training for students, faculty, TAs, and academic leaders.

17.4.1 ▪ Student training

Student orientation should teach the Learning Spiral: frame a question, try to solve, ask for help rather than replacement, work independently, verify, and disclose. Students should also learn what not to upload, how to check sources, and how to prepare for no-AI assessments. Institutions can reduce friction further by providing a recommended baseline AI configuration (a starter set of personal preferences students can adopt during orientation; the Student's Toolkit in Part I gives a template).

A short module can be completed in the learning-management system before the first AI-permitted assignment. Departments may add discipline-specific examples.

17.4.2 ▪ A required AI-literacy outcome

A recurring policy question is whether to create a dedicated course that teaches students to use AI effectively and ethically for learning and research, and whether such a course should be encouraged or required. A requirement will meet principled objections from faculty and students alike, and those objections deserve an answer rather than a waiver; the companion paper "Teaching AI Responsibly to Principled Skeptics" [56] sets out one. The strongest case for treating AI literacy as a *required* outcome rather than an optional one rests on equity. Students do not arrive equally familiar with AI tools. Low-income, racially underrepresented, and female students are significantly less likely to use generative AI regularly [25], so an optional offering is likely to reach more of the students who need it least than of those who would benefit most. That would widen the access gap this Part is concerned with rather than narrowing it (section 16.4). Student demand points the same way: in a 2025 survey of more than a thousand students, education on ethical AI use ranked among the most-wanted institutional responses at 53%, ahead of every punitive option [40]. Institutional practice runs the other way: sector data put training for faculty and staff at the top of institutions' AI strategies while student training lags, and the sector's own first advice is to close that gap [33].

The delivery mechanism is better left to local choice than prescribed centrally. A standalone required course is one option. Others have embedded AI-literacy outcomes across existing courses, as in the University of Florida's AI-across-the-curriculum model [117], or folded them into an

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

existing general-education requirement. The approach most consistent with the framework in this book is to make AI literacy a required learning outcome delivered primarily through the course-level categories already described (the **NAI**, **AIT**, **AIC**, and **AIS** structure of section 15.2) together with the student orientation above. A standalone course can be reserved for institutions that prefer it, or offered as an elective for deeper study.

Making this universal need not be expensive. Most universities already require short online onboarding modules on topics such as sexual-misconduct prevention and alcohol safety, often tied to course registration. An AI-literacy module can occupy the same slot and reach every student at low marginal cost. Treated as the floor rather than the whole, such a module reaches every student, while the course-level categories and discipline-specific instruction develop and assess the depth that a single module cannot provide.

17.4.3 ▪ Faculty training

Faculty training should focus on course design rather than tool novelty. Useful topics include assignment redesign, assessment validity, process-based grading, responsible feedback, privacy, equity, suspected misuse, and discipline-specific AI literacy.

Faculty should leave training with a revised syllabus statement, at least one redesigned assignment, and a plan for assessing independent competence.

17.4.4 ▪ TA training

TAs need practical calibration. They should learn how to grade disclosure blocks, process logs, audit-and-repair tasks, and short explanations. Their training should cover the four assignment categories, the three assessment models, and discipline-specific verification checks. A TA who suspects misuse should check the assignment's AI rule and the student's disclosure, write down the specific concern, note the relevant process evidence, and pass all of it to the instructor. Meeting the student and deciding the case stay with the instructor and, for serious cases, the institution's integrity process.

A strong TA calibration session uses sample submissions. The instructor and TAs grade the same examples, compare judgments, revise the rubric, and document standards before grading the full class.

17.4.5 ▪ Administrative training

Administrators and academic-integrity officers need a shared understanding of the institution's AI categories, detector limits, privacy rules, and due-process expectations. This reduces inconsistent handling across departments.

17.5 ▪ Implementation plan

A university should adopt AI-assisted learning policy in phases.

The four-phase plan described below sits within a broader international policy context. UNESCO's guidance on generative AI in education proposes seven steps for regulating these tools in education systems [89]. National and regional governments are responsible for most of these steps, and the framework in this Part is designed to be compatible with whichever regulatory environment a particular institution operates in.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

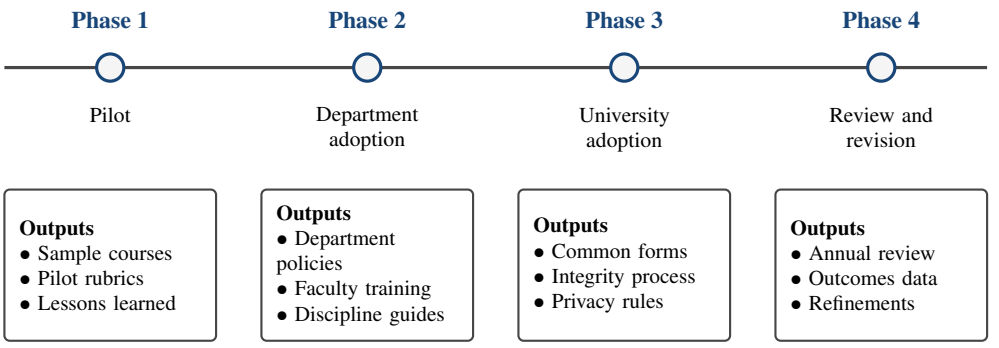


Figure 17.1. ***Four-phase implementation timeline.*** *Universities adopt the framework progressively rather than at once. Phase 1 builds confidence in a small set of courses; Phase 2 lets departments localize; Phase 3 creates institutional consistency; Phase 4 maintains and refines. Each phase produces concrete outputs that feed the next.*

17.5.1 ▪ Phase 1: Pilot

Select a small set of courses across disciplines and sizes. Include at least one large undergraduate course, one writing-intensive course, one STEM course, and one seminar or capstone. Test the shared vocabulary, assignment labels, disclosure form, and assessment models. The governance body owns the pilot. Pilot courses use only tools it has approved for that purpose, so the privacy rules that Phase 3 extends to the whole university apply from the first course.

Collect feedback from students, instructors, TAs, and academic-integrity staff. Revise the materials before wider adoption.

17.5.2 ▪ Phase 2: Department adoption

Departments adapt the framework to local needs. They identify Core Competence standards, discipline-specific verification practices, sample assignments, and course-size models. They also decide which courses should include explicit AI literacy outcomes.

17.5.3 ▪ Phase 3: University adoption

The university integrates the framework into syllabus guidance, teaching-center resources, student orientation, academic-integrity procedures, and data-governance policy. The policy should remain flexible enough for local course design.

17.5.4 ▪ Phase 4: Review and revision

AI policy should be reviewed annually. The review should examine student learning, faculty workload, integrity cases, equity concerns, privacy incidents, and student feedback. Review should be scheduled, not only triggered by problems. Beyond the institutional cycle, instructors should revisit their course-level AI policy each time they teach the course, since both the tools and students' familiarity with them shift from term to term. The goal is stable principles with adaptable implementation.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

17.5.5 ▪ Cost and resource considerations

The four-phase plan above describes what to do but not what it costs. Any honest plan has to admit that AI-aware teaching is a real investment for the institution. Disclosure-protocol design and faculty workshops in Phase 1 typically require teaching-center staff time and a modest stipend pool for pilot instructors. Process-evidence rubrics in Phase 2 increase per-assignment grading time, which means departments with high enrollment-to-instructor ratios may need to reallocate teaching assistant support or accept somewhat coarser rubrics. Oral assessment is the most expensive element to scale; small classes can absorb it within existing contact hours, while large classes typically cannot without TA support or a mix of in-class and recorded explanations. Calibration sessions across instructors cost the least per session, but they have to be scheduled again and again, and departments often underestimate that.

For resource-constrained departments, a useful adoption path is to begin with three low-cash-cost elements that do not require new hiring: explicit per-assignment AI policies; the disclosure-block standard, largely a one-time setup; and a single calibration session per term. These cost little new money, but each costs faculty time, which is neither free nor abundant. The path shifts cost from the budget line to the workload, and a chair or dean should plan for that workload rather than assume it away. The path also assumes no course requires AI use. Where one does, students need an approved tool at no additional cost or an equivalent alternative, so tool licenses become a cash line of their own. What the time and the money buy is the validity of the credential: assessment that still certifies what a graduate can do. The full framework can then be phased in over multiple years as resources allow. The plan does not ask departments to adopt every element at once; it gives them a coherent target state to grow toward.

Where the institution's accreditation language already distinguishes independent competence from tool-assisted performance, the phase plan above can be reported against that language directly, without a separate AI policy vocabulary.

17.6 ▪ Faculty change management

A university may know what it should do. It still has to find a way there from current practice. The gap between the two is faculty change management.

The guidance here draws on practical experience with faculty adoption and on the higher-education change-management literature, an established empirical body that predates AI but bears directly on the problem. Analytic reviews of how faculty adopt new teaching practices find that top-down mandates from administrators and the spreading of model curricula are, on their own, largely ineffective, and that durable change instead depends on sustained support, local evidence, and faculty engagement [18, 49, 62]. What remains AI-specific, and is still accumulating, is direct evidence on this framework's particular elements. The guidance below is therefore offered as informed practice grounded in that broader literature. It is not the kind of direct AI-specific finding that anchors parts of the student and instructor guidance.

17.6.1 ▪ The change-management problem

Faculty are not, in general, opposed to AI policy in principle. Most instructors recognize that AI tools have changed what students bring to class and what assignments measure. The resistance an institution encounters is rarely ideological. It is practical. New disclosure conventions add a small amount of work to every assignment. Process-evidence rubrics take longer to grade. Oral assessments require time many faculty do not have. Calibration sessions across instructors need someone to coordinate them, and departments are not staffed for it.

A faculty member who is already teaching three courses, advising graduate students, serving

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

on two committees, and writing a tenure file does not need a new burden. AI-aware teaching is real additional work, at least at first, and institutional change management has to start from that recognition.

17.6.2 ▪ Stages of faculty adoption

Faculty distribute across at least four broad groups, each requiring a different kind of support. The grouping draws loosely on the diffusion-of-innovations literature, which has consistently identified distinct adopter categories with characteristic motivations and barriers [107]. The labels matter less than the recognition that one approach will not reach all groups.

Skeptics doubt that AI policy is needed at all in their course. They may believe their existing safeguards (in-person exams, oral components, structured drafts) already prevent the failure modes the policy targets. What helps most is not advocacy but evidence: invite skeptics to participate in calibration sessions where they grade student submissions blind alongside other instructors.

Pragmatists will adopt new practices if shown clear benefit and if the marginal workload is manageable. They want concrete templates, sample rubrics, and ready-made assignment redesigns rather than principles. What helps most is shared resources: a department-level library of AI-aware assignments organized by course type, a small set of validated rubrics, and one or two paragraphs of standard syllabus language. The framework's adoption depends on them.

Early adopters are already running ahead of policy. Some have been redesigning assignments for two years; some have published their experiments. Their needs are different: they want to be recognized, to help set the institution's direction, and to be protected against penalties for trying. What helps most is to involve them in policy design and to let them serve as resources for the pragmatist group rather than asking them to attend mandatory training built for skeptics.

Overwhelmed faculty want to comply but lack the bandwidth to redesign assignments, learn new tools, or attend additional meetings. They are often contingent faculty, lecturers teaching multiple sections, or faculty in caregiving roles. What helps most is making compliance cheap: providing a pre-made syllabus paragraph, a default rubric, and a single accessible workshop rather than a series of trainings. For faculty paid by the course, the materials have to carry the whole load, because attendance of any kind is time they are not paid for.

A successful adoption strategy reaches all four groups with calibrated support, not all with the same training.

17.6.3 ▪ Barriers and how to address them

Several specific barriers recur across institutions adopting AI-aware teaching.

Time cost of process-evidence grading. Reading a draft history, an annotation, and a verification statement takes longer than reading the final work. The remedy is spot checks rather than universal review. A reasonable practice is to grade the final work for everyone and to read process records for a randomly selected subset, plus any submission that raises a concern. Students should be told spot checks are in place; this preserves the deterrent effect without the workload of universal audit.

Lack of confidence in oral assessment. Many faculty have not conducted structured oral assessments since their own qualifying exams. They worry about consistency, fairness, and student anxiety. The remedy is training that uses real student work, scripts a small set of standard questions, and provides a simple rubric. Confidence tends to come with practice; the first oral check-in is the hardest.

Concern about fairness to multilingual writers. Detector-based enforcement is documented to misclassify non-native English writing as AI-generated [79]. Faculty who teach diverse student populations are right to worry about this. The remedy is to stop treating a detector score as the

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

sole or primary evidence of misconduct and to rely on process evidence and student explanation. Faculty who understand this shift in evidentiary standard have less reason to be skeptics of the framework.

Workload disparities across course sizes. What works for a dozen students in a seminar fails for two hundred in a lecture hall. Faculty in large courses sometimes feel the framework is designed for small classes and does not address their reality. The remedy is to make the Evidence-of-Thinking Model explicit in change-management materials, with concrete examples drawn from large courses.

Tenure-track and lecturer differences. Tenure-track faculty have research obligations that make additional teaching workload costly to their careers. Contingent faculty and lecturers carry heavier teaching loads and have less time for redesign. Both groups need accommodation, and an institutional approach that ignores these differences will fall hardest on contingent faculty. The remedy is differentiated support: course-release time for redesign work where possible, shared resources that scale across many sections, and accepting that faculty in different employment categories will adopt in different ways.

17.6.4 ▪ When the standard playbook is not enough

The four-category adopter model and the barrier remedies above describe what works for most faculty. Departments should also accept that a small number of faculty will not move regardless of support level. A tenured professor who refuses to use disclosure standards, declines calibration sessions, or actively grades against process-evidence rubrics in their courses can stall department-wide adoption if their courses are gateway requirements for the major. No graceful technical solution to this problem exists. It is a governance problem and should be handled through department governance rather than through additional faculty-development effort.

What works in these cases is usually some combination of four moves. Make the policy a department-level requirement rather than something each instructor may choose, so that ignoring it becomes a governance issue rather than a matter of personal style. Have course evaluations and curricular reviews ask about AI-policy implementation. Assign a noncompliant faculty member to courses where the policy footprint is smaller, electives rather than gateways. The fourth move is to accept that uniform adoption may take longer than the four-phase plan suggests. Department chairs should not personalize these situations into individual conflict. The goal is for the institution to adopt the policy, not for every instructor to agree, and a small number of holdouts is compatible with effective department-wide practice as long as gateway courses and qualifying milestones are covered. This is a deliberate exception rather than a second standard. Section 16.1 asks every course to assess independent competence, and the framework asks every course to state its AI-use rules; a department taking this route should record which courses fall short of either and for how long.

Institutions should also recognize that a faculty member who appears to be a persistent skeptic may instead be a thoughtful dissenter whose objection is substantive. A serious objection to a particular framework element (for example, that oral assessment as currently implemented produces a particular fairness concern in their student population) may be evidence that the framework needs adapting rather than that the dissenter needs persuading. Department leadership should distinguish between these cases before treating dissent as obstruction.

17.6.5 ▪ What does not work

A short list of patterns that institutions try, and that consistently disappoint, deserves explicit naming.

Mandatory training without local input. Centrally designed training that does not reflect

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

what nursing instructors, mathematics instructors, and writing instructors actually face produces compliance without change. Faculty attend the session, sign the attendance sheet, and continue teaching as before.

Detector-only enforcement. Acting on AI-detection scores as the primary basis for academic-integrity cases produces false positives and fairness problems for multilingual writers, and it erodes trust in faculty judgment [27]. The teaching-and-learning approach to academic integrity, well-developed before AI tools became prevalent, is more robust [14, 15]. Institutions that rely on detectors create both injustices and litigation risk.

Discipline-blind requirements. Mandates that ignore disciplinary differences place a heavier burden on some faculty than others. A mandate to require draft histories on every assignment is sensible for a writing course and unworkable for a calculus course where the assignment is solving twenty problems. Change-management materials should distinguish requirements that apply universally (disclosure, and no detector score as the sole or primary evidence of misconduct) from practices that should be adapted by discipline (specific verification protocols, choice of assessment model).

17.6.6 ▪ Department leadership

A dean's policy memo can require new syllabus language; a chair's example shapes daily practice. Three concrete moves are within most chairs' authority.

A one-page departmental compact stating what the department expects of every course (clear AI-use rules in syllabi, a defensible Core Competence assessment, disclosure expectations) and what faculty can expect in return (shared rubric library, calibration meeting time, TA training support). The compact should be brief enough to fit on one page and should be revisited annually.

A calibration meeting protocol held once a semester, where faculty bring one anonymized submission and the group grades it together. The comparison reveals hidden assumptions about what counts as AI-assisted Practice and resolves them through conversation. The first such meeting feels awkward; later ones can be among the most useful hours of the term. Department chairs who take part themselves make the meetings last.

A teaching-portfolio question for tenure and promotion review. Adding "how does the candidate's teaching address responsible AI use" as one item among many in a teaching dossier signals that the institution treats this as part of teaching rather than as an extra burden.

17.6.7 ▪ Faculty support resources

Institutional resources that support faculty change should be visible, accessible, and curated. A short, well-organized set is more useful than a long list of links.

Teaching center workshops on AI-aware assignment design, oral assessment, and process-evidence grading. Workshops should be offered both synchronously (with live discussion) and asynchronously (recorded for those who cannot attend). Department-specific sessions, where the workshop draws examples from the participants' own discipline, are usually more useful than generic sessions. The harder problem is attendance rather than design. A voluntary workshop competes with everything else on a faculty calendar and usually loses, and the faculty who most need it are the least likely to come. Putting the same material on a departmental meeting agenda reaches people no workshop will, and it keeps the disciplinary framing that makes it useful.

Assignment redesign clinics where a small group of faculty bring an existing assignment, work through one round of redesign with a facilitator, and leave with a revised draft. The format respects faculty time (one session, one assignment) and produces tangible output (a revised assignment they will use). Clinics scale better than one-on-one consultations.

Shared rubric libraries organized by assignment category and by assessment model. Rubrics

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

should be lightly curated for quality but not over-standardized. The goal is to give faculty a starting point they can adapt, not to impose uniformity.

17.7 ▪ From department to institution: how departmental practice can shape institutional policy

Readers who chair or teach in a department will not personally drive university-wide AI policy. They will, however, shape what their own department does, and that departmental practice can become the template for what eventually becomes institutional policy.

17.7.1 ▪ The pattern in practice

Universities rarely adopt comprehensive AI policy in a single top-down decision. Far more often, one or two departments develop coherent practice first, accumulate concrete examples of what works, and become reference cases when university-wide bodies begin to formulate policy. A mathematics department with six semesters of experience with the Oral-Defense Model, three years of disclosure-block practice across all its courses, and a documented academic-integrity procedure is, in effect, handing the institution a tested case study. Similar trajectories work in computer science, statistics, physics, and other quantitative disciplines.

The reverse also holds. When the faculty senate or the academic-integrity committee finally takes up AI policy, it looks for examples of what is already working. Departments that have done the work first are positioned to shape that conversation; departments that have done nothing become subject to whatever the institution decides without their input. Sector data as of 2025 fit this fragmented picture: a majority of respondents say their institution views AI as a strategic priority (57%), fewer than half report AI acceptable-use policies in place (39%), and most describe their AI strategy as confined to particular units rather than adopted institution-wide (55%) [33]. A 2026 mapping of all fifty US flagship public universities sharpens the picture. Every one placed instructional AI decisions at the course level. Institution-level guidance concentrated instead on data protection, approved tools, and risk management [74]. This Part is built on the departmental layer. The flagship universities' arrangement skips that layer, so departments that build coherent practice are filling a gap the sector has left open. The gap is widest where capacity is thinnest: respondents at the largest institutions were more likely than those at the smallest to report an acceptable-use policy (52% against 40%), and those at the smallest were nearly twice as likely to call their cybersecurity and privacy policies not at all adequate (44% against 24%) [33].

17.7.2 ▪ The college or school in between

Department and university are not the only levels. In most universities a college or school sits between them, led by a dean who coordinates several departments, allocates the resources that determine whether AI-aware practice is feasible, and mediates between departmental practice and university policy. The college level is often where the resource decisions in section 17.5.5 are actually made: teaching-assistant lines, teaching-center support, and course-release time for redesign are typically college-controlled. It is also where neighboring departments that share students and standards can align their practice before a university-wide body takes up the question. Departments building AI-aware practice are usually most effective when they bring the college along early, both to secure resources and to position departmental practice as a college-level reference case rather than an isolated experiment.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

17.7.3 ▪ The layer outside the institution

Department, college, and university do not exhaust the possibilities. A 2026 policy forum in *Science*, drawing on original survey data, reaches the same diagnosis this Part begins from and then places the remedy somewhere else. It argues that assessment reform belongs at the level of the discipline, because each discipline carries its own learning goals, assessment traditions, and ways that AI can support or replace student work. The coordinating work, on this account, may be best done not by individual universities acting alone but by disciplinary societies, accreditors, and other field-level bodies that can write shared guidance and supply examples for local adaptation [25].

A discipline is not a department. The two accounts agree that institution-wide policy fails and that the right grain is disciplinary. They disagree about who holds the pen.

The case for the department is that it is the smallest unit with authority over the things that decide whether any of this happens. A society can publish guidance. It cannot run a course, set a syllabus, assign a teaching assistant, or tell a lecturer how to weight an oral defense. Accreditors can require that a program show evidence of learning, and that pressure is real, but the review cycle runs on years while a course runs on weeks. The department hires, schedules, staffs, and reviews. It decides what a shared syllabus says and whether a redesigned course counts toward a colleague's promotion case. Guidance that arrives without those levers becomes another document.

The two placements need not stay rivals. Field-level bodies are well suited to write what a department has neither the time nor the standing to produce alone: model assignments, worked examples, discipline-specific statements of what independent competence means. Departments are suited to adopt, adapt, and enforce them. The society supplies the text and the department supplies the authority. A department waiting for its society to act will wait, and a society writing for departments that have built nothing is writing into a vacuum.

17.7.4 ▪ What departmental faculty can do to support the bridge

Faculty whose primary influence is at the department level can still make their work institutionally legible. The most useful actions are:

1. Document departmental practice in writing, including what was tried, what was kept, and what was discarded, so that the institutional record can refer to it rather than having to reconstruct it.

2. Participate in any institutional working groups, faculty senate committees, or teaching-center conversations that touch AI policy, even briefly, because departmental experience is the missing input these bodies usually lack.

3. Name the institutional support that would help departmental practice scale, including the Cost and resource considerations from section 17.5.5, so that institutional policy can be informed by what departments need rather than only by what institutions can mandate.

17.7.5 ▪ What departments should ask of their institution

Departments doing AI-aware work need a few things from their institution:

- Clear privacy and data-governance policy for AI tool use (so departments do not each have to negotiate licensing separately).

- A centrally supported pool of AI-tool licenses that gives equity-of-access guarantees so departments do not depend on student personal accounts.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

- An academic-integrity procedure at university level that is compatible with department-level evidence standards (specifically, that does not require detector-tool output as the basis for misconduct decisions, given the documented unreliability described in section 16.5).
- Recognition in the merit-review process, and in hiring and tenure-and-promotion criteria, of the time faculty spend on AI-aware course redesign.
- A teaching center that can offer the workshop and consultation services described in section 17.3.

Departments that ask for these clearly, in writing, with reference to the framework principles in this volume, are far more likely to receive them than departments that ask for general support without specifics.

17.7.6 ▪ The honest limit

Some institutional decisions are out of departmental hands. Most universities will not adopt the framework here in its entirety; some will adopt fragments; some will adopt nothing and eventually develop their own framework that overlaps imperfectly with this one. Departments should not wait for institutional adoption to do good work in their own courses. The departmental framework is designed to be useful regardless of what the institution does. Even in institutions that adopt nothing centrally, a department running coherent practice is providing real value to its students; that value does not depend on whether the policy environment around the department keeps up.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 18

The Institution's Toolkit

This chapter gives an institution the phased rollout, the outcomes to expect, and the forms it needs to put the framework into practice across its courses. It is for the chair drafting a policy, and for the committee that has to review one. The argument for each form is made in the chapters before it.

18.1 ▪ Implementation phases

An institution adopting this framework should plan the four-phase rollout of section 17.5. In brief:

Phase 1: Pilot. Adopt the framework in a small set of courses across disciplines and sizes. Test the assessment instruments, calibrate grading among the instructors involved, and collect feedback from students.

Phase 2: Department adoption. Departments set their Core Competence standards and course-size models. Run workshops on AI-integrated assessment for the broader faculty. Use sample student work from Phase 1 as calibration material.

Phase 3: University adoption. Build the framework into syllabus guidance, student orientation, academic-integrity procedures, and data-governance policy.

Phase 4: Review and revision. Establish an annual review cycle, since AI tools and student practices will continue to evolve.

Faculty are encouraged to adapt the framework to local context while preserving its core epistemic commitments. The minimum independent-competence requirement and the requirement for transparent AI disclosure should not be relaxed even as other elements of the framework are localized.

18.2 ▪ Expected outcomes

A program that adopts this framework should expect to produce students who possess deep subject understanding, who can use AI as a working tool rather than a crutch, who demonstrate strong epistemic judgment when AI output is involved, and who uphold ethical standards in AI-assisted work. The framework is not a guarantee of these outcomes; it is a structure that supports them.

The framework is not designed to produce uniformity across departments. Different programs will reasonably weight Core Competence, AI-assisted Practice, and Trustworthiness differently

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

based on their disciplines and student goals. The framework provides shared vocabulary and a common assessment architecture; the specifics of weighting, rubric stringency, and assessment frequency are local decisions.

18.3 ▪ Reference materials

The following worksheets, templates, and checklists support institutional adoption of the framework. Each may be adapted for local use; the items are designed to be lifted into faculty-handbook material, syllabus-design guidance, or chair-level operations documents. Technical references that serve the whole book sit at its end.

18.3.1 ▪ Course classification worksheet

Instructors or departments can use this worksheet before the semester begins.

1. What are the three to five core abilities students must develop in this course?
2. Which of those abilities must students demonstrate without AI?
3. Which assignments are **NAI**, **AIT**, **AIC**, or **AIS**?
4. How will students show that they understand the work they submit?
5. Is the course best suited to the Oral-Defense Model, Evidence-of-Thinking Model, or Hybrid Model?
6. What privacy restrictions are most relevant to this course?
7. What disclosure format will students use?
8. What student training is needed before the first AI-permitted assignment?
9. What grading support or TA calibration is needed?
10. How will the course be evaluated and revised after the term?

18.3.2 ▪ Syllabus template

AI use in this course

This course uses the following AI-use categories.

NAI assignments. AI use is not allowed. These assignments assess what you can do independently.

AIT assignments. You may use AI to study, to have ideas explained, to practice, and to get hints, but its wording, code, and worked solutions stay out of the work you submit.

AIC assignments. AI may be used as a collaborator for brainstorming, revision, debugging, critique, or testing. Check the work before you submit it, and state how you used AI.

AIS assignments. AI is part of the object of study. The task may be to analyze, critique, or compare what an AI produced, or to repair its mistakes.

For any assignment that permits AI, your submission must follow the disclosure instructions given with the assignment. You are responsible for all submitted work. You should not submit AI output that you cannot explain, verify, or defend.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Do not upload private, restricted, copyrighted, confidential, clinical, human-subject, or secure assessment material into AI systems unless the instructor has explicitly approved that use. Course materials posted in [course site or folder] may be uploaded to the AI tools approved for this course.

18.3.3 ▪ Student disclosure template

Tools used: Name the tools used, if any.
Tool role: Describe what the tools helped with.
Student role: Describe what you wrote, revised, checked, or decided.
Accepted, changed, or rejected: Describe the AI contributions you used, revised, or set aside.
Independent verification: Describe how you checked important claims, sources, data, code, or reasoning.
What I can explain without AI: Name the main parts of the work you can now explain on your own.

18.3.4 ▪ Privacy checklist

Before using AI in a course, ask:

1. Is the tool approved for this type of use?
2. Does the task involve student records, grades, private communications, clinical data, research-subject data, or confidential material?
3. Could student work be stored, reused, or used for training by the vendor?
4. Are students required to create personal accounts?
5. Is there a no-cost option or equivalent alternative?
6. Is the tool accessible to students with disabilities?
7. Are students told what not to upload?
8. Has the instructor provided a privacy-safe alternative when needed?

18.3.5 ▪ Academic-integrity workflow

When AI misuse is suspected, instructors should avoid beginning with accusation. This workflow is a short form of the canonical procedure in section 16.5. A fair workflow is:

1. Re-read the assignment's AI-use rule and the student's disclosure.
2. Name the specific concern; a general suspicion is not enough.
3. Gather available evidence, including process evidence and comparison to other course work.
4. Invite the student to explain the work.
5. Ask targeted questions about sources, reasoning, method, code, data, or revision choices.
6. If needed, assign a short reassessment tied to the submitted work.
7. Decide which it was: misunderstanding, poor documentation, prohibited assistance, or deliberate misrepresentation.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

8. Record the concern, the evidence, what the student said, and the decision.
9. Apply the appropriate educational or formal response under institutional policy.

   For why an educational remedy is usually the route the evidence supports, see section 16.5.

18.3.6 ▪ Department adoption checklist

A department is ready for adoption when it has:

- approved shared AI-use vocabulary;
- provided sample syllabus language;
- identified discipline-specific verification practices;
- developed example assignments for each of the four categories;
- stated expectations for independent competence;
- provided a disclosure template;
- created a process for large-course TA calibration;
- clarified privacy and tool-use rules;
- aligned academic-integrity procedures with the institutional framework;
- scheduled periodic review.

Two companion files carry the list further. The departmental adoption kit (`institutions/department-adoption-kit.md`) turns it into a day-one toolset for a chair, with a model policy statement, the shared vocabulary, the four assignment categories, and a one-semester rollout. The policy self-audit (`institutions/policy-self-audit.md`) runs thirty-two checks across six dimensions. It separates what the institution already covers from what the department covers, and from what neither one does.

18.3.7 ▪ Program evaluation rubric

A university can evaluate implementation using the five-level rubric below.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Table 18.1. ***Institutional implementation rubric.*** *Each row describes one stage; an institution is rarely at the same stage in every respect.*

Level	Description
1. Fragmented	AI policies vary widely, many syllabi are unclear, and faculty have few shared resources.
2. Emerging	Some departments use clear policies, but vocabulary, privacy rules, and assessment models are inconsistent.
3. Functional	Most courses state AI-use rules, and the institution provides basic templates, training, and academic-integrity guidance.
4. Mature	Departments use shared vocabulary, course-size models, disclosure standards, privacy safeguards, and calibrated assessment practices.
5. Exemplary	AI literacy, independent competence, responsible judgment, equity, privacy, and assessment validity are integrated into regular course design and program review.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

18.3.8 ▪ One-page adoption summary

The institutional framework can be summarized in ten commitments.

1. Teach students to use AI while keeping their thinking their own.
2. Require every course to state its AI-use rules.
3. Preserve meaningful assessment of independent competence.
4. Treat AI literacy as an educational outcome.
5. Require verification of AI-assisted work.
6. Normalize brief, factual disclosure by students and faculty.
7. Protect privacy, access, and equity.
8. Never treat detector output as sole or primary evidence.
9. Support faculty with templates, training, and TA calibration.
10. Review policies regularly as tools and practices change.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Chapter 19
Looking Forward

19.1 ▪ Looking forward

This framework will meet capabilities that did not exist when it was written. Some predictions about the next several years will be wrong. The framework's structure is meant to stay useful when specific implementations need refreshing.

19.1.1 ▪ What is likely to change

Several trends seem stable enough to plan around.

AI capability will continue to improve. Recent work on mathematical reasoning suggests that current limitations on proof construction and on multi-step computational reasoning will narrow over the next several years [1]. Code generation, multimodal evaluation, and agentic workflows that operate beyond a single chat are all developing faster than university policy cycles.

Tool ubiquity will increase. AI capabilities are moving from separate applications into operating systems, productivity software, and integrated development environments. The distinction between "using AI" and "using a word processor" will blur, and assignment policies that try to specify which tools count will become harder to maintain. Disclosure-and-verification standards generalize better than tool-by-tool restrictions.

Detector reliability will decrease. As AI-generated text becomes harder to distinguish from competent human writing, the underlying signal that detection tools rely on will weaken. Detectors are already failing in two directions: they produce documented false positives against multilingual writers [79], and they produce false negatives against students who refine prompts to evoke more literary language. The first direction will persist wherever detectors stay sensitive; the second will grow as vendors retune against false positives and as prompting skill spreads. Institutions that built integrity processes around detectors will face mounting reliability problems on both fronts and will need to migrate to the process-based evidence the framework already emphasizes.

Student baseline familiarity will increase. Many undergraduates of 2030 will arrive on campus having used AI tools throughout secondary school. Orientation curricula that explain what AI is will matter less; orientation curricula that teach responsible use, verification habits, and disclosure norms will become more important.

19.1.2 ▪ What is likely to stay the same

Other elements appear durable across capability shifts.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

The need for independent competence will not disappear. A graduate student who cannot reason without tools is not a researcher; a physician who cannot diagnose without AI is not a physician. The specific topics that count as foundational competence may shift, but the underlying requirement that human expertise must extend beyond tool-mediated procedure remains.

Oral assessment will continue to discriminate well between independent understanding and tool-mediated performance, regardless of tool capability, so long as the student answers unaided. A student who has not built an underlying argument cannot reconstruct it under questioning, and AI capability does not change that. Oral assessment is expensive but durable.

Process evidence will continue to provide signal. The asymmetry between polishing output and developing judgment does not disappear as AI improves. Polish becomes easier; judgment requires the same effort it always did. The competence costume gets cheaper every year; the competence it imitates does not. Assignments that grade process retain validity longer than assignments that grade only final polish.

Disclosure and verification will become more important, not less. As AI involvement becomes ubiquitous and harder to detect, the burden shifts from external detection to internal disclosure. Verification habits and disclosure norms (Appendix A) support both the integrity of student work and the trustworthiness of the academic record. Neither certifies correctness on its own.

19.1.3 ▪ What this means for the framework

The framework's core elements appear stable across the anticipated changes. The CAT distinction will remain useful regardless of capability shifts because it identifies what an academic course is for, not what a tool can do. The four assignment categories describe the role AI plays in an assignment, which is a curricular choice rather than a tool fact. The three assessment models reflect course size and staffing constraints that change slowly. The verification standard generalizes across disciplines and across tool generations.

What will need refresh is the specific content within those structures. Tool lists will need annual updates. Prompt examples will date quickly; the book's prompt examples are illustrative rather than canonical. Disclosure templates will evolve as institutional norms develop. Detector references should be removed from policy documents that treat detector output as sufficient to establish misconduct. The model policy given later in this Part names detectors for the opposite purpose, forbidding their use as sole or primary evidence, and many existing institutional policies have yet to draw that distinction.

Departments adopting the framework should review its specifics (tools, prompts, templates) every year and keep its structural commitments (categories, models, verification, disclosure) as a stable base. Annual reviews are part of Phase 4 (section 17.5.4).

19.1.4 ▪ Why this is hard, and where the levers are

Everything in this book asks students, instructors, and institutions to choose the harder path exactly where the easier one is cheaper and its damage is hidden. Polished work hides the learning loss, so no one feels the cost of skipping the judgment work. Enrollment pressure and faculty workload reward the shortcut. The framework runs against that gradient, and naming the trilemma does not flatten it.

Two levers change the slope. The first is no-AI assessment. Its value is not only that it is much harder to game. It makes the learning loss visible, and a loss you can measure is a loss an institution can act on. The second is setting integrity standards centrally, decoupled from enrollment incentives, so that no single course or department pays alone for holding the line. Without them, the prediction is that the framework is adopted by the conscientious and skipped by the pressured. That would reproduce at scale the very split the trilemma describes: diligent

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

against strategic, resourced against time-poor. With them, the unit that holds the line is no longer the one that pays for it. That alignment, not exhortation, is what turns a framework into practice.

19.1.5 ▪ Modesty about prediction

Specific predictions will be wrong. Some capabilities anticipated above will arrive faster than expected; others will plateau. Some failure modes will be solved; others will appear that no one currently anticipates. The framework is designed to be more durable than any specific tool list, but it is not designed to be permanent. Universities adopting it should expect to revise the implementation details every year and the structural commitments every five years or so. The annual reviews named in section 14.2 and section 17.5.4 are reviews of implementation; the structural commitments are what the longer cycle revisits.

A useful posture is to treat this volume as a working document rather than a settled answer. Departments that find the framework helpful are encouraged to adapt it, document their adaptations, and share what they learn. The framework improves by use, not by being right in advance.

This returns to where the book began. The camera and the calculator did not remove the practitioner; each moved the work up a level, from execution to judgment. A university adopting this framework is making that move deliberately, deciding while the tools are still new what the higher level of work requires and how to teach it.

19.2 ▪ Adoption claims and metrics

Before adopting a framework, an administrator should know what it rests on. The book states its evidentiary limits where each claim is made. Gathered in one place, they look like this.

Table 19.1. ***What the framework rests on.*** *Each claim is listed with the strength of evidence behind it, so a committee can see which parts are settled and which are still judgment.*

Claim	What supports it	Status
AI writes fluently and can be confidently wrong, including inventing sources.	Published research on language-model behavior [12, 131].	Established.
Unrestricted AI use can raise short-term output while weakening unassisted performance, and students do not notice the loss.	A field experiment with a control group [11].	Established by one field experiment in one secondary setting; read the effect sizes as pattern, not as expected values.
Detector-based enforcement is unreliable and falls unevenly across students.	One evaluation of seven detectors, with an intervention that moves misclassification in both directions [79].	Mechanism established; rates from one evaluation.
Careful design can make AI genuinely useful for learning.	A randomized comparison in a physics course [61].	Established by one course study.
Institutional adoption has outrun institutional policy.	Survey of about 500 AAUP members (December 2024) [4].	Established by one survey.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Claim	What supports it	Status
Redesigning assessment around independent competence checks and oral defense removes the advantage of skipping the work.	One course redesigned this way, where students with and without AI reached similar outcomes [70].	Closest direct evidence. Tests the style of redesign, not this framework.
Departments are the layer where AI policy should be set.	Documented delegation from institution to course across fifty flagship universities [74]. A *Science* policy article places the remedy at the level of the discipline, with disciplinary societies and accreditors rather than universities acting alone [25].	Contested. The gap is documented from both sides. Neither placement is tested.
The four assignment categories make expectations clearer to state and easier to follow.	Reasoning from the rows above.	Untested.
The three assessment models yield more trustworthy evidence of learning.	Partial, through the redesign above.	Untested as a set.
Courses built on this framework produce better learning than courses that are not.	Nothing. No study has asked.	Untested.

The last three rows are the untested ones. They are also the reason for what follows. A department that runs the framework and measures what happens becomes the study the last row is waiting for.

An untested claim invites a better-supported claim, not a competing opinion. So it helps to say what evidence would change the status of those rows. Three questions are cheap for a piloting department to record. First, do students and instructors read the categories the same way? As written, **NAI** says AI use is prohibited but does not say whose use. Many instructors write assessments with AI assistance. A short survey asking both groups what the label forbids would give a clear answer. Second, does **AIC** function as a learning mode or a production mode? A student working with full tools is a different unit of capability from the same student working alone, and if the category means practitioner conditions, a department should find that its **AIC** work is assessed against different expectations than its **AIT** work. Third, is the difference between a tutor and a mentor large enough to need its own category? Or is it a matter of relationship that cuts across all the categories? The book draws it in prose and the category set does not carry it, which is either a gap or a deliberate asymmetry.

A department that records answers to these has not adopted the framework on faith. It has run the first study of it.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

19.2.1 ▪ Metrics

A university should evaluate whether the framework improves learning and fairness. Useful metrics include:

- percentage of syllabi with clear AI-use rules;
- percentage of courses identifying independent competence assessments;
- student understanding of AI disclosure expectations;
- faculty confidence in AI-aware assignment design;
- TA grading consistency on process-evidence rubrics;
- student performance on no-AI transfer checks;
- quality of student verification statements;
- number and type of academic-integrity cases;
- frequency of misconduct claims resting on a detector score as the sole or primary evidence;
- student reports of fairness, access, and privacy clarity;
- adoption of approved tools rather than unsupported tools;
- faculty workload effects.

The purpose of measurement is improvement. If students disclose AI use but cannot verify claims, the student training needs revision. If faculty avoid AI assignments because grading process evidence is too burdensome, the institution should provide better rubrics or TA support. If large courses rely mainly on polished take-home work, assessment validity may need attention.

19.2.2 ▪ A self-study a department can run

Those metrics answer whether a department is improving. With four additions they answer something larger, which is whether the framework works at all.

Measure before you adopt. A department that starts measuring after the change has nothing to compare against, and one term of baseline is worth more than three terms of after.

Leave some sections alone at first. Where a course runs several sections, stagger adoption by a term. The unchanged sections are not a control group in the strict sense, since instructors differ, but the comparison is far better than none.

Watch the transfer checks most closely. Syllabus counts and faculty confidence are easy to move and tell you little. How students do without AI, on material they were taught with it, is the measure that speaks to learning, and the one the literature lacks.

Write down what you find, including the parts that did not work. Departments that publish a plain account of two years of adoption, with numbers, will settle questions this book can only reason about. Nothing here requires a grant or an education researcher. It requires a baseline, a comparison, and the willingness to report a result that was not the one hoped for.

19.3 ▪ Risk register

Institutional adoption should include a simple risk register.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Table 19.2. ***Common risks and institutional responses.*** *Each risk is paired with a response; the table is a starting register, not a complete one.*

Risk	Why it matters	Recommended response
Over-reliance on AI	Habitual first-resort AI use erodes the independent skill the course is meant to build, even when each single use looks harmless.	Require independent competence checks and evidence-of-thinking assignments.
Detector dependence	Detection tools may be unreliable and unfair.	Use process evidence, student explanation, and due process.
Privacy breaches	Students or faculty may upload protected data.	Maintain approved-tool lists and clear no-upload rules.
Unequal access	Paid tools can create academic advantage, and procurement can create it too: license tiers that differ across units, and pilots that expire mid-semester and strand the students inside them.	Provide institutionally supported tools or equivalent alternatives; match tiers across sections, and let no pilot end mid-term.
Faculty workload	Process grading and audits can add burden.	Provide templates, TA calibration, spot-check methods, and reusable rubrics.
Core Competence narrowing	Independent competence checks are costly to run, so a department gradually keeps the competences that fit an oral slot or a class period and drops the ones that do not. Outcome metrics stay healthy throughout, because students still perform on what is asked.	At scheduled review, compare the Core Competence definition against the one written at adoption, and record what left the list and why.
Policy fragmentation	Students receive conflicting rules across courses.	Use shared vocabulary and assignment categories while preserving local flexibility.
False confidence	Fluent AI output leads students to overestimate what they understand; self-assessment fails to track the deficit.	Include transfer checks, oral explanations, and audit-and-repair tasks.
Underuse of AI literacy	Students may be told what not to do without being taught what to do.	Include explicit student training and **AIS** assignments.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Risk	Why it matters	Recommended response
Budget-model incentives	Under enrollment-linked budgeting such as Responsibility Center Management, units that allow broader AI use may draw students from stricter units, pressuring all units toward leniency.	Set core AI-integrity expectations institutionally so units do not compete on permissiveness; keep integrity standards decoupled from unit-level enrollment incentives.

19.4 ▪ Model institutional policy statement

The following language can be adapted for a university policy page.

> The university recognizes that artificial intelligence tools are now part of academic, professional, and creative work. Students should learn to use these tools responsibly while continuing to develop the independent knowledge, skills, and judgment that each course requires.
>
> Each course must state how AI may or may not be used. Instructors should identify which assignments prohibit AI, which allow AI for tutoring or study support, which allow AI as a collaborator, and which ask students to analyze AI directly. Students are responsible for following course rules, verifying AI-assisted work, protecting private or restricted information, and disclosing AI use when required.
>
> Instructors remain responsible for grading, feedback, and academic-integrity judgments, and they say where AI has shaped the work students receive. AI tools may assist instruction, but they should not replace human evaluation. Suspected misuse should be handled through process evidence, student explanation, and established academic-integrity procedures. AI-detector scores should not be used as the sole or primary evidence of misconduct.

Publish the policy as a short plain document, so instructors can post it with their course materials and students can give it to the AI tools they use.

19.5 ▪ Evidence base and limits

For research training above the coursework level, one example of a complementary instrument is the Instats Institutional AI Readiness Pack [140], an auditable maturity assessment for the research office and doctoral supervision; this framework addresses courses, instructors, and departments.

Institutional AI policy is developing faster than the research base. The learning-science evidence shared across the book establishes the foundation that AI-aware policy must preserve. For institutions specifically, two additional findings are stable enough to guide practice: automated text detectors are not reliable enough to serve as the sole basis for misconduct decisions, and AI tools raise privacy, access, and equity concerns that universities must address directly. A 2026 systematic review of institutional generative AI governance reaches compatible conclusions [9]. It calls for embedded governance, equity safeguards, sustained faculty development, and continuous monitoring.

A third stable finding at the institutional level is the *performance-versus-learning gap*. The OECD's 2026 Digital Education Outlook frames the central problem of evidence this way: generative AI tools may raise the apparent quality of student work without improving actual

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

learning, meaning the durable knowledge and skill the work was supposed to show [98]. An institution that judges its AI policies mainly by the quality of submitted work will tend to miss this gap. One that judges them by independent performance after AI use, oral-defense results, or accuracy on transfer questions is far more likely to catch it. The implementation plan in this Part is built to expose the gap rather than hide it, and the assessment models and rubrics are how it does so.

Other questions remain unsettled. The best balance between no-AI, bounded-AI, and open-AI assessment will vary by field, level, and class size. AI tools will also change. For that reason, this framework should be adopted as a living policy. It gives institutions a shared starting point, not a final answer.

The framework ends here. What follows is reference material: a guide to ethical AI-assisted writing, a quick reference for mathematics courses, a disclosure of the AI assistance used to prepare this book, and the bibliography. None of it is required to use the book. Consult it when you need it.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Appendix A

Best Practices for Ethical AI-Assisted Scientific Writing

When a reviewer asks,
your answer should not begin with the tool.

This appendix gives practical guidance for scientists, including graduate students and faculty researchers, who use AI tools when preparing research manuscripts. The goal is to maintain scientific integrity, ensure transparency, and preserve human accountability while benefiting from AI assistance.

The guidance here applies to manuscript preparation, code development associated with research, literature review, and grant writing. The emphasis is on STEM and computational research where equations, code, and numerical results dominate verification practice. Humanities and qualitative-research writers will find the underlying principles (verification, disclosure, responsibility) transferable but should also consult the discipline-specific guidance for humanities, writing, and creative disciplines in Part II (section 11.4). It complements the verification and disclosure practices described in Part I for student work. Researchers preparing manuscripts for journal submission should also consult their target journal's specific AI policy, which may impose requirements beyond those described here. Undergraduates beginning a first mentored research project have an on-ramp to these standards. The repository guide `students/undergraduate-research.md` adapts them to a summer program, an honors thesis, or a first poster.

A.1 ▪ Core principles

Human responsibility is non-negotiable, and shared. On a multi-author paper the authors hold it together. Every scientific result, analysis, interpretation, and conclusion must be something the authors can defend without reference to AI assistance. Authors can divide the work of understanding among themselves. They cannot divide the duty to answer for the whole paper. Each author understands the parts they answer for, can say which author answers for each of the others, and has grounds for confidence in that work; each also accepts that a question raised about any part of the paper will be investigated and resolved [57]. This is the ordinary standard of multi-author work, not a special provision for AI.

What responsibility requires is narrower and firmer. Every essential part of the work has at least one qualified author who can defend it. An essential part is one the argument turns on: a result other steps depend on, a computation no later step would expose as wrong, an interpretive claim the conclusion rests on. The author who owns such a part understands it, knows its place in

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

the literature, and made or verified the choices behind it. Taken together, the authors understand the whole. Taken singly, each owns a part and can say where their own understanding ends.

What is never acceptable is an essential part that no author can answer for. A result no author understands, however fluently a tool produced it, is not ready to publish. The permission is granted part by part; the floor holds over the whole.

A practical test makes this concrete. If a reviewer asks why an approach was chosen, some author should have an answer rooted in understanding rather than a procedural account of what the tool suggested. For any essential part, the author who owns it should be the one who can answer. If the only available answer is tool-mediated, the work is not ready for submission.

The same standard, an unfamiliar failure mode. It is easy to ask whether AI hallucinates, mangles prose, or invents references, and to stop there. The sharper question is comparative: whether it does so more than the human baseline it is measured against, since human authors also miscite, overstate, and err. Rigorous head-to-head evidence on that question is still thin. The standard in this appendix does not wait on it: the author verifies AI-assisted content because the author is accountable for it, whatever the comparative rates turn out to be. The standard is the ordinary one. What the tool changes is how hard a lapse is to see. That is why this appendix asks for disclosure and verification rather than for a different standard.

AI systems are tools, not collaborators. AI tools may assist with drafting and refining prose, exploring analytical approaches that the author then verifies, code prototyping that the author then tests and validates, and literature searching that the author then reads and evaluates. AI systems can now generate research-level results: they select and revise proof strategies and produce expert-checked proofs of open problems [6], and in some cases produce complete formal proofs or disproofs checked mechanically by proof assistants. They are at their best in the hands of an expert who sets the direction, recognizes which of the system's proposals is worth pursuing, and verifies what comes back. What they cannot do is answer for the work. Formal certification establishes that the encoded statement follows from the stated axioms and definitions; it does not establish that the formalization captures the intended problem, or that the result is novel, significant, or worth pursuing. Those judgments take domain expertise; responsibility for the published claims stays with the authors, and no system qualifies for authorship. Disclose and credit the system's contribution in proportion to its actual role, and state how the result was verified and who stands behind it. Parts I and II use AI as Collaborator (**AIC**) as a coursework category. There it names a working role within an assignment. It makes no claim about authorship or responsibility, and those are the subject of this section.

This distinction matters when journals ask whether AI tools should be listed as authors. The professional consensus across publishing bodies is that they should not, because authorship implies responsibility: the standing to answer criticism, disclose uncertainty, authorize correction, and be held accountable for error. Present AI systems cannot undertake those obligations, which is why responsibility stays with the human authors. This reflects what today's systems and today's scholarly institutions can support, not a permanent law of the subject. If systems ever come to hold commitments and answer for them the way a responsible author does, that boundary will need to be redrawn. The paragraphs that follow describe the standard as it stands now.

Transparency proportional to impact. The level of AI disclosure should match the level of AI involvement. Minor uses, such as grammar checking and formatting, warrant a brief acknowledgment. Moderate uses, such as drafting text or assisting with code, warrant a clearer description in the acknowledgments. Substantial uses, such as significant drafting, analysis, or exploration, warrant a detailed description in both the acknowledgments and the methods section.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Central uses, where the system supplied the conjecture, the counterexample, the strategy, or the proof, warrant specific attribution of that contribution. A generic label such as AI-assisted understates the system's role. Understating that role is also a failure of disclosure.

Confidentiality, and the researcher as reviewer. What a researcher puts *into* an AI tool matters as much as what comes out of it. Uploading unpublished work, whether one's own or a colleague's, sensitive or proprietary data, or another person's personal data risks exposing it, because many tools may retain inputs or use them to train future models. Researchers should check a tool's data-retention terms before use and keep anything not already public in institutional or closed environments. The same concern explains why a researcher should not run a manuscript or proposal they are reviewing through an external tool, since doing so exposes a colleague's confidential, unpublished work. Major funders have made this a rule: the National Institutes of Health prohibits generative AI in its peer review process [95], and the European Research Area guidelines allow AI in peer review and proposal evaluation only for supporting tasks the process permits, never to judge merit, both to protect confidentiality and to avoid the unfair treatment the tools' limitations can produce [36].

Responsibility to the shared record. Peer review and the published literature run on a reserve of trust. A reviewer can usually treat a manuscript's references and claims as good-faith starting points rather than verifying each one before the review can begin. The literature is usable because most of what enters it was checked by someone. AI strains that trust, and not only through the confidentiality concern above. The volume of AI-assisted writing entering the literature has risen sharply [65], and the same pressure has reached peer review: reviewers increasingly lean on AI, and in one 2025 episode authors were found to have hidden instructions in their manuscripts, in white text or tiny fonts, directing any AI reviewer to return a positive report [80]. The author's responsibility runs to this shared record, not only to the manuscript in hand. Submitting AI output one has not verified, or attempting to game an AI-assisted reviewer, converts a private convenience into a public cost that every colleague pays in lost trust and added work.

A.2 ▪ The verification cornerstone

The author must verify all AI-assisted content independently. Only then is it work the author can defend before peer reviewers and critics in the field. Without verification, AI-assisted text is at best a draft and at worst a hazard.

Recent developments make this cornerstone more pressing, not less. AI systems have begun to produce genuine research-level results: in 2026 OpenAI reported that an internal model had generated a counterexample disproving the prevailing conjecture for the Erdős unit-distance problem, a question in discrete geometry open for roughly eighty years [99]. What matters here is why it was accepted. Outside mathematicians checked the proof, and one, Thomas Bloom, judged it by what it taught the field. The same standard applies when AI produces results in large numbers. In one 2026 evaluation, an autonomous system's internal verifier flagged 212 of 700 attempted Erdős problems as potentially correct; expert review confirmed 13 as meaningful, most of which rediscovered results already in the literature [39]. Checking that a result is correct is not the same as judging whether it matters mathematically. Tao has argued that the amount of AI automation a field can absorb before its output degrades into slop is roughly proportional to how stringently that output is verified [124], and has warned that AI-written proofs can be plausible-looking nonsense [122]. The same holds for the individual researcher: one who can verify what AI produces can put it to serious use, while one who cannot is not assisted by it but exposed by it. That verification capacity, built through training and practice, is what separates

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

them.

The sharpest case is a finder from outside the field. In 2026 an amateur without advanced mathematical training gave a public model an open Erdős problem on primitive sets [111]. The answer held a method the field had apparently overlooked. Experts rewrote it as a clear proof and extended it to related conjectures [2]. Such a finder holds a right answer they cannot explain: the student's problem at research scale. A growing number post their results on arXiv, saying they are not experts and inviting others to check the work. The candor is right, but it does not shift responsibility. arXiv holds each author responsible for everything in a paper, however it was produced [7]. Such a posting is a claim, not yet a result. Mathematicians rely on a result only after it has been verified, explained, and accepted [123].

A formal proof helps, but someone qualified must still confirm that it proves the intended statement. A finder can do four things. Search the literature first, since many early AI solutions proved to be known results. Find a qualified partner who will check the work and answer for it. State what was checked, how, and by whom. Disclose the tools [76]. Credit follows contribution and responsibility, and the contribution can be the question. Choosing the right problem, or framing it well, is often the hardest part of solving it. A finder who did that can be an author without writing a line of the proof, if they approve the paper and can answer for what was asked, how, and why. The primitive-sets paper lists the amateur, and the undergraduate who took the result to experts, as co-authors. A finder who will not take on that responsibility belongs in the acknowledgments. The AI is disclosed, not listed as an author.

Verification works better when the AI's output is calibrated to the author's actual technical level and goals. Researchers who configure their AI accounts with persistent preferences specifying their field, current projects, and uncertainty-flagging requirements receive output that is easier to verify. The AI has been told what kind of confidence claims to make and how to mark unfamiliar territory. The student-facing version of this configuration practice is described in section 1.2 of Part I (with the full configuration template at section 7.2); the principles transfer directly to research use, with the technical-work accuracy guard (showing work, flagging claims to verify and naming how to check them) becoming particularly valuable for research-grade writing.

A practical verification workflow has five steps. On collaborative work, the author who owns the relevant part carries them out; "the author" below means that person, not necessarily every author.

1. AI assists with draft content or analysis.
2. The owning author reviews the output critically: does this make sense?
3. The owning author verifies the output technically: is this correct?
4. The owning author confirms meaning: does this say what I intend?
5. Once verified, the content represents the authors' verified work.

Each step is necessary. Reviewing for plausibility (step 2) is not the same as verifying technical correctness (step 3). A passage that reads smoothly and reaches a familiar conclusion can still contain errors that only careful technical inspection will surface. A passage that is technically correct can still misrepresent the author's intended meaning if the author has not examined it for fidelity to the underlying scientific argument.

These tools are also stochastic: the same prompt can return different output on different runs, so anything a researcher intends to report should be checked for reproducibility rather than trusted from a single run.

One further point belongs here, because this cornerstone rests on a condition that will not hold forever: that a human can, in principle, verify what AI produces. Today's journal policies are built on that assumption. They keep a person responsible for every result and offer no category for a

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

result no human can check. The assumption is already under strain. AI systems have begun to produce proofs and constructions at the edge of what any individual can follow, and results may soon arrive that are machine-verified yet too deep for a person to reconstruct. When such results become common, publishers will need a way to accept them without pretending that a human did the checking. The responsibility principles in this appendix will then need restating. In that setting, only other machines will be able to check some essential parts of the work. That setting is not yet here. This guidance is written for the one that is, and readers should expect the ground to move.

A.3 ▪ Disclosure guidelines

Where to disclose.

- Acknowledgments (the primary location for AI disclosure).
- Cover letter, when AI use was substantial enough to merit editorial attention.
- Methods section, when AI assisted with analysis or modeling.
- Data availability or supplementary materials, when AI assisted with code that affects reproducibility.

Sample disclosure language. *Minimal* (grammar checking, prose refinement only):

> The authors used *[AI tool]* for grammar checking and prose refinement. All scientific content was developed independently by the authors.

Standard (drafting and revising portions of exposition):

> During the preparation of this article, the authors used *[AI tool and version]* for drafting and revising portions of the exposition. All scientific results and interpretations were independently verified and are the sole responsibility of the authors.

Comprehensive (substantial AI involvement in drafting, code prototyping, or analysis exploration):

> This manuscript was prepared with AI assistance using *[tool and version]*. Initial drafts of *[sections]* were generated with AI assistance and substantially revised by the authors. Code was prototyped with AI assistance and independently tested. All scientific results were independently verified. *[Name any result, method, or proof that originated with AI.]* The authors assume responsibility for all content.

The right disclosure level is the one that accurately represents what AI contributed to the work. Under-disclosure is unethical; over-disclosure can imply more AI involvement than actually occurred and may misrepresent the author's contribution in the other direction.

Capstones and theses. A semester project is a document too, and the three levels above apply to it. A capstone that used AI only to tidy prose sits at the minimal level. One that drafted sections or prototyped code with AI sits at the standard or comprehensive level, and the student says which. What changes, for a capstone, is the reader. An editor reads a disclosure to decide whether the work meets a journal's standard. A supervisor reads it to decide whether the student has learned what the project was for, and a defense committee reads it to decide what to ask. The disclosure is also a study document: it tells the student where the questions will come from.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

For that reason a capstone disclosure uses the student block in section 18.3.3 rather than the paper templates above. Its six fields already ask what the tools did, what the student did, what was accepted or set aside, how claims were checked, and what the student can now explain without AI. That last field is the defense, written down in advance. A student who fills it honestly has a list of what to prepare, and a committee that reads it has a list of what to ask. For a capstone or thesis the block gains a seventh field:

> **Responsibility:** State that you assume responsibility for all content.

This is the commitment the paper templates close with, in the student's own words. The process log kept during the project supplies the material for every field. A student who kept none will find the block hard to fill, and the supervisor should know that too.

What counts as substantial, and sharing the record. Draw the line between minor and substantial use explicitly. Basic editorial support is not substantial use. Using AI as a method is substantial, for example to run or interpret an analysis, to conduct a literature review, or to formulate hypotheses. Document it as you would any other method [36]. Where it is feasible and confidentiality allows, authors can make their prompts and the resulting outputs available. Sharing them follows open-science and FAIR (findable, accessible, interoperable, reusable) practice. It lets others judge how AI shaped the work and supports reproducibility [48].

Fairness considerations in detector-based screening. Some journals use AI-text detectors during submission screening. These tools have documented limitations that disproportionately affect non-native-English writers. In an analysis of papers from the 2023 International Conference on Learning Representations, those whose first authors were from non-English-native countries showed systematically lower text perplexity than papers by English-native first authors [79]. The same study measured false-positive rates on a separate set of texts, TOEFL essays rather than the conference papers. Part III (section 16.5) gives those rates and explains how perplexity produces them. Authors writing in English as a second language may face higher false-positive rates from journal screening tools and should be prepared to provide process evidence (drafts, revision history, or prompt logs) if questioned. The detailed-disclosure standards in this appendix support exactly that kind of evidence-based response. The broader implication is that the research-publishing community should treat detector output as one signal among several rather than as a basis for editorial decisions. Authors who can document their writing process protect themselves against detector-based misclassification.

A.4 ▪ Common pitfalls

Citation hallucination. AI tools sometimes suggest citations that do not exist or that exist but do not support the claim being made. Empirical work documents substantial fabrication rates in AI-generated bibliographies, often in the range of tens of percent depending on the model version, the field, and the kind of work cited, with errors common even in citations that point to real papers [131]. The risk is high enough that researchers should treat every AI-suggested citation as unverified until checked. The remedy is straightforward: verify that every cited source exists, read every cited paper, and never cite an unexamined source. This is not new advice for scholarly work; AI simply makes it more important by increasing the volume of plausible-looking but incorrect citations a researcher may encounter. The failure is not limited to invented references. A citation can be perfectly real and correctly formatted and still be paired with an AI-written summary that misstates what the paper says, so the standard is reading the source, not confirming that it exists. A citation is a claim, not a decoration. Verify it.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Accepting unverified analysis. AI presents analysis confidently. The temptation to accept it without verification is real, especially when the analysis appears to confirm a hypothesis the researcher already favors. This pattern is documented in the educational research literature on AI use in learning, where it has been characterized as metacognitive laziness: individuals who use generative AI to complete cognitive tasks tend to disengage from monitoring whether the task has actually been completed correctly [38]. The studies focus on student learning, and the same pull is easy to feel in research writing. Confident output tempts the reader to accept it, even when the norm is to verify. Part of that pull is the tool's own tendency toward sycophancy, the habit of shaping answers to match what the user seems to expect, so an AI will often return analysis that flatters a favored hypothesis. Knowing the output came from a machine does not blunt the effect. Participants received identical advice, labeled as coming either from an AI or from a person. Those told it came from an AI trusted it less, but were moved by it just as much [24]. The remedy is to work through every step independently, to test code with known inputs, and to refuse to include any analysis the researcher cannot follow on their own.

Meaning drift in prose. AI revision can subtly change the meaning of scientific claims. A sentence that originally said a result "suggests" a conclusion may be revised to say it "shows" or "proves" that conclusion, an inflation of certainty that the underlying evidence may not support. The remedy is to review every paragraph after AI revision, check that hedging language matches the author's actual confidence, and rewrite any claim the author does not recognize as their own.

Invisible code errors. AI-generated code that runs successfully can still produce incorrect results. Compilation success and execution success are not correctness. The remedy is to test with known inputs whose correct outputs the researcher can verify independently, to check boundary conditions, and to compare results against analytical solutions or established benchmarks where they exist.

One habit behind all four. These pitfalls look different, but they share a structure. In each, smooth and confident text hides a gap between what the text claims and what the author has actually checked. They share a single remedy as well. Citation hallucination, unverified analysis, meaning drift, and silent code errors are all closed by the same act, the author opening the source, redoing the step, and running the test before the claim leaves their hands. The discipline takes no special tools and no unusual talent, only the willingness to treat verification as part of the cost of using the tool, which is the cornerstone this appendix began with.

A.5 ▪ Pre-submission checklist

The following checklist, completed before manuscript submission, encodes the standards in this appendix.

- ☐ I understand every equation and analysis I answer for, and I can name the author who answers for the rest.
- ☐ If a question is raised about any part of this paper, I will see that it is resolved.
- ☐ I have verified all AI-assisted content independently.
- ☐ I have read all cited papers.
- ☐ I have tested all code and can reproduce key results.
- ☐ My disclosure accurately reflects AI use in this work.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

☐ I have checked the target journal's AI policy and complied with its specific requirements.

A manuscript that passes this checklist is not necessarily ready for submission, but a manuscript that fails any item is not ready. The failures should be addressed before any further submission preparation.

A.6 ▪ Verification standards by AI contribution type

The following table summarizes what verification looks like for each common type of AI contribution to a research manuscript.

AI contribution	Verification standard	What the author must do
Prose drafting	Read critically	Ensure the text says what the author means; rewrite anything that drifts in meaning or certainty.
Analytical approach	Work through independently	Verify every step of the analysis on the author's own; do not include analysis the author cannot reconstruct.
Code generation	Test extensively	Validate correctness through known-input testing, boundary checks, and benchmark comparisons; do not assume execution implies correctness.
Citation suggestions	Read the sources	Confirm each cited source exists and supports the claim; treat unverified citations as unsuitable for inclusion.
Numerical results	Reproduce independently	Check against known values, analytical solutions, or independent reimplementation.

A.7 ▪ Summary

AI-assisted scientific writing is ethically sound when four conditions hold. The researcher's intellectual contribution is genuine: the researcher chooses the question, directs the work, and answers for the result. Verification is thorough: every AI contribution is independently checked. Disclosure is proportional: transparency matches AI involvement. Responsibility is the authors': the authors can defend everything in the manuscript without reference to AI.

These four conditions are easier to satisfy when the researcher has already developed strong verification habits in graduate training. Researchers whose graduate training predates the AI era can develop the same habits through deliberate practice and through the verification workflow described above.

The standards in this appendix are general. Specific journals, funding agencies, and professional societies may impose additional requirements. Researchers should treat this appendix as a baseline rather than a ceiling.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Appendix B

Math-Specific Quick Reference

In mathematics, the question is rarely whether the answer is right.
It is whether the student understands why.

This appendix is a one-stop reference for using the framework in mathematics courses. It consolidates the practical patterns scattered across Parts I and II: student-side AI study habits for proof-based and computational mathematics, instructor-side category mixes and assessments for typical math course types, and a few fast design rules for instructors under time pressure. The student-side material is in second person; the instructor-side material is in third-person summary.

B.1 ▪ Student use in mathematics

The student-side principle is simple. AI should support mathematical thinking, not replace it. In the language of the book, students should preserve Core Competence, use AI for AI-assisted Practice, and show Trustworthiness through verification, disclosure, and explainability.

Proof-based mathematics. Proof-based courses include discrete mathematics, transition-to-proof, linear algebra with proofs, abstract algebra, real analysis, topology, and most graduate theory courses. In these courses, the main danger is accepting a polished proof that the student cannot reconstruct, defend, or adapt under questioning.

A compact proof-based workflow is:

1. State the theorem and rewrite the relevant definitions from memory.
2. Sketch a plan before consulting AI.
3. When the goal is learning to write proofs, use AI only for hints, audits, or comparison of approaches, not for full proofs.
4. Rebuild the proof independently in your own words, ideally from memory or at the board.
5. Check whether the argument survives a small variation of the statement.

Recommended habits as you work through the spiral:

- Try the problem first, even briefly.
- Ask AI for a hint, a simpler analogous problem, or an audit of a proof attempt rather than a complete proof.
- Verify every major step by checking where each hypothesis is used.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

- Use AI to generate examples, non-examples, and borderline cases for definitions, especially in analysis and algebra, where understanding often depends on seeing what fails when a hypothesis is dropped.

Computational and applied mathematics. Computational and applied courses include calculus, differential equations, numerical analysis, optimization, modeling, scientific computing, statistics, and data-oriented math courses. In these courses, the main danger is accepting output that looks plausible because the algebra works or the code runs, even when the method, assumptions, or interpretation are wrong.

A compact computational workflow is:

1. Write down the model, formula, or plan first.
2. Use AI for hints, diagnostics, or method comparison.
3. Solve or code independently.
4. Run verification checks on cases where the answer is known or constrained, using the verification kit in section 4.4.1.
5. Be prepared to explain the method, not just the output.

Recommended habits:

- Identify variables, assumptions, equations, or algorithmic goals before asking AI for help.
- Use AI to suggest solution strategies, debug code by diagnostic questions, or generate test cases.
- Verify quantitative work with the math verification kit: special case, limiting case, units, known formula, independent route.
- Test edge cases and reproducibility for code.
- Ask AI to explain why a method is appropriate, what it assumes, and when it can fail.

Strong AI prompt patterns for mathematics. The strongest AI prompts in mathematics preserve the student's active role. Table B.1 lists patterns that work across course types.

B.2 ▪ Instructor use in mathematics

For instructors, the main design problem is to separate what students must demonstrate independently from what they may do responsibly with AI assistance. The **NAI**, **AIT**, **AIC**, and **AIS** assignment categories from section 9.2 and the Oral-Defense, Evidence-of-Thinking, and Hybrid assessment models already provide that structure.

Typical course patterns. Table B.2 gives practical default mixes for common mathematics course types. These are starting points, not rigid prescriptions.

Assessments that work especially well in mathematics. The strongest mathematics assessments make reasoning visible rather than grading only final polish. Across course types, the following tasks are especially effective:

- *First-attempt plans*: a 2–3 line proof idea, setup, or strategy written before AI use.
- *Homework-to-quiz checks*: AI-allowed homework paired with a short no-AI in-class question on the same underlying idea.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Table B.1. ***Strong AI prompt patterns for mathematical work.*** *The pattern is consistent across goals: ask AI to help you do the thinking, not to replace it.*

Goal	Strong prompt pattern
Start a proof	Give me one hint about the main idea, but do not prove it.
Check a proof	Audit my proof attempt and identify the first step that needs more justification.
Learn a definition	Give me an example, a non-example, and a borderline case of this definition.
Debug code	Ask me diagnostic questions until I find the bug.
Prepare for oral defense	Ask me to explain each step of this argument until you find what I do not understand.
Verify a method	What assumptions does this method use, and how would I test whether they hold here?

- *Audit-and-repair tasks*: students critique and repair flawed AI-generated proofs, derivations, or code.
- *Verification notes*: short statements showing how a solution, derivation, or computation was checked.
- *Oral defenses*: brief conversations in which students explain a proof, method, or design choice in their own words.
- *Transfer questions*: nearby variants of a problem that reveal whether students learned the structure rather than memorized a polished answer.

Fast design rules for mathematics instructors. When redesign time is limited, three moves usually provide the largest benefit. A close variant appears in the implementation workflow in section 12.2, where the third move asks only for disclosure.

1. Label each assignment clearly as **NAI**, **AIT**, **AIC**, or **AIS**.
2. Add at least one meaningful independent-competence check to each unit.
3. Require one small process record when AI is allowed: a plan, one representative AI exchange, and a verification note are often enough.

For small proof-based classes, the Oral-Defense Model is often the most robust design because it allows substantial AI use while preserving standards of proof and explanation. For large service courses, the Evidence-of-Thinking Model is usually more feasible: homework-to-quiz checks, random oral audits, brief process logs, and AI-critique items scale better than full oral examinations.

In mathematics, AI is most helpful when it acts as a tutor, critic, or diagnostic partner rather than a substitute author. Students should use it to get unstuck, test understanding, generate examples, audit arguments, and debug computations. Instructors should design assessments that preserve independent competence while making verification, process, and judgment visible.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Table B.2. ***Default AI-policy mixes by mathematics course type.*** *The mixes are starting points; individual courses should adjust to local conditions.*

Course type	Typical category mix	Strong assessments
Introductory procedural (Calculus I, College Algebra)	Mostly **NAI** and **AIT**, with selective **AIS** for critique tasks.	Short no-AI quizzes, homework-to-quiz transfer checks, brief explanation of one step, AI-generated error critique, formula or concept checks.
Intermediate proof transition (Discrete Math, Intro to Proof)	Balanced **NAI** and **AIT** with selective **AIC** and **AIS**.	In-class proof sketches, definition recall, proof audit-and-repair tasks, short oral explanations, annotated correction of flawed AI proofs.
Upper-division proof (Real Analysis, Abstract Algebra, Topology)	Mixed **NAI**, **AIC**, and **AIS**; **AIT** remains useful for study support.	Board proofs, oral defense of homework, Plan-Use-Verify submissions, proof comparison tasks, hypothesis-stress tests, AI-proof audit tasks.
Computational or numerical (Numerical Analysis, Scientific Computing)	Balanced **AIC** with meaningful **NAI** and **AIS** components.	Code test reports, reproducible notebooks, boundary-case checks, short no-AI conceptual quizzes, model comparison memos, critique of AI-generated code.
Applied modeling (ODE/PDE modeling, mathematical biology, operations research)	**AIC** and **AIS** with targeted **NAI** checks on core concepts.	Assumption-to-model notes, verification of outputs against benchmarks, sensitivity checks, short oral or video defense, process logs, AI-assisted model critique.
Graduate proof-intensive seminar or core course	Substantial **AIC** and **AIS** with **NAI** oral or written competence checks.	Blackboard examinations, oral defense, proof repair, comparative AI/human solution analysis, disclosure blocks, verification notes, mini research projects.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Appendix C

Disclosure of AI Assistance in Preparing This Book

This book was written the way it recommends. For many months I worked with AI in the mode Part I calls AI as Collaborator (**AIC**): AI could assist with any part of the work, and I verified the content and took responsibility for it.

I owe the reader a brief note on what that meant in practice.

There were nights when a paragraph I had drafted felt clumsier than the one the AI returned. The polished version made me sound a little more confident than I felt about that paragraph's argument, a little more current than I am with the AI literature, and a little more sure-footed than the underlying thinking yet was. The temptation to let the polished version stand was real, and not only once.

Resisting the temptation was the work. Where I have used AI assistance, I have tried to use it in the way I recommend students and colleagues use it: as a way to clarify thinking that was already mine, not as a substitute for thinking I had not yet done.

Day to day, the work was a long conversation. Claude (Anthropic, 2025–2026) was the primary tool. With it I worked out the book's direction and topics, analyzed the literature, weighed which of my colleagues' suggestions to take up, and revised the narrative. Some of Claude's suggestions were better than my first ideas, and I took them. Others I rejected. Claude also drafted and revised prose from my arguments, ran structural checks, and helped produce the figures and dialogues. It checked every reference in a fresh conversation that had not seen the drafting. I also asked ChatGPT (OpenAI, 2025–2026) for harsh reviews. It found weaknesses I had missed and strengths worth keeping, and the book is much better for those reviews. Every sentence, figure, and table went through this cycle of discussion, critique, and revision.

The decisions are mine: which arguments to make, which suggestions to keep, and what the book recommends. An idea that began as an AI's suggestion stayed only once I could defend it. I checked claims, citations, and examples against their sources, and errors that remain are mine. I thank Anthropic for the Claude models; the relationship is that of a tool user, not a sponsorship.

Readers can copy the pattern. Talk the work through before drafting. Ask a second model for the harshest review it can give. Check references in a fresh conversation. Keep the final call, and say who did what. Appendix A gives the standards.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Glossary

This book is written to be entered at any point. Readers who start at the Instructor Guide or the Departmental and Institutional Guide will meet terms that are introduced in the Student Guide. Most terms defined here are used in more than one Part.

AI. General-purpose generative tools that produce text, code, proofs, explanations, or analyses in response to an ordinary request, and whose output can vary between runs and therefore has to be verified rather than trusted. The term is defined by how a tool behaves rather than by product name, so that policy does not need rewriting each time the tools change. Bounded, deterministic aids such as calculators, computer algebra systems, spelling and grammar checkers, and reference software fall outside the term unless a course states otherwise.

AI as Collaborator (AIC). An assignment category in which AI may co-produce what you hand in, within stated limits. What you submit is partly AI's work, and you say so.

AI as Object of Study (AIS). An assignment category in which the AI system is itself what you are studying. It sits off the permission scale used by the other three categories, because the question is not how much help you may take.

AI as Tutor (AIT). An assignment category in which AI may be used for study support but its wording, code, and worked solutions stay out of what you submit. Once they enter the submission, the work has moved to AI as Collaborator.

AI-assisted Practice. The second dimension of the CAT framework. What responsible use of AI tools looks like in a given course, and what you should be able to do with them.

AI-assisted work. Work shaped in some way by an AI tool, including brainstorming, drafting, revision, coding, explanation, summarization, critique, or feedback.

AI-generated output. Text, code, images, explanations, summaries, solutions, data analyses, or other content produced by an AI system.

Assessment validity. The degree to which an assessment measures the competence it claims to measure. Distinct from reliability (consistency of measurement) and fairness (procedural justice in the assessment process).

Bounded-AI work. Work completed with limited, specified AI assistance, such as a single approved tool, a restricted prompt type, or a fixed time window.

CAT framework. The framework used throughout this book, named for its three dimensions: Core Competence, AI-assisted Practice, and Trustworthiness. It separates what you must be able to do yourself from what you may do with tools and from how you account for the difference. See section 3.1.

Calibration error. The gap between what a student believes they understand and what they can actually reproduce, apply, or defend under questioning. AI can widen the real gap while

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

making it harder to see.

Competence costume. AI-polished material that signals more current and confident engagement than the underlying work supports.

Core Competence. The first dimension of the CAT framework. What you must be able to do without AI, or with only tightly limited help. Every course has a core, and it differs by course.

Disclosure. A factual statement of which AI tools were used, what they contributed, and how the student verified the result.

Distributed practice. Also called spacing. Reviewing material at intervals, with forgetting and re-retrieval in between, holds it more durably than the same review in one block. The interval is the work.

Evidence of thinking. Process records that show how a student reasoned, revised, verified, explained, or chose. Large classes use these where one-to-one oral examination will not scale.

Hollow understanding. One of the three failure modes of a trilemma. The work looks finished and the account of it holds up, but the learning did not happen. The instructor and institutional forms carry the same name.

Institution's Trilemma. The three-way pull on an institution among remaining financially viable, defending what its credential certifies, and serving its educational mission. Any two can be held, but only at the cost of the third.

Instructor's Trilemma. The three-way pull on an instructor among a sustainable workload, judgments that can be defended, and student learning. Any two can be held, but only at the cost of the third.

Integrity penalty. One of the three failure modes of a trilemma. The student is fully honest about AI use and about the limits of their understanding, and pays a competitive price for the transparency. The instructor and institutional forms carry the same name.

Learning Spiral. The main habit this guide asks you to practice. A clear question sits at the center, and five steps run around it: try to solve, ask for help, work independently, verify, and reflect. See section 3.2.

Mentor prompt. A standing prompt that tells an AI how to guide your learning across many tasks: how to pitch its help to where you are, and how to notice when that needs to change.

Metacognitive laziness. The measured drop in self-monitoring, especially evaluation and orientation, when generative AI is used on revision tasks. It is one mechanism by which polished work can hide reduced learning.

No AI (NAI). An assignment category in which AI use is not allowed. It is not a claim that AI is bad. It marks work whose purpose is to measure what you can do unaided.

No-AI work. Work completed without AI assistance, except for accommodations or tools explicitly approved by the instructor.

Oral defense. A live conversation or examination in which a student explains submitted work and answers questions about it.

Plan-Use-Verify. A design pattern for AI-assisted assignments. The student states a plan before using the tool and verifies the result afterward, which makes self-monitoring explicit at both ends. It is one pass of the Learning Spiral.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Practice testing. Also called retrieval practice. Trying to recall, produce, or reconstruct something builds memory more strongly than re-reading it. The retrieval is itself a learning event.

Quiet overclaim. One of the three failure modes of a trilemma. The learning happens and the grade is earned, but the account of how AI helped is shaped to make its part look smaller than it was. The instructor and institutional forms carry the same name.

Random audit. A routine assessment check in which a student is asked to explain or verify submitted work. It is not, by itself, an accusation.

Spot check. An audit given to some students or submissions rather than to all, chosen at random or by a stated rule.

Student's Trilemma. The three-way pull on a student among earning the grade, being honest about how the work happened, and showing genuine understanding. A dilemma forces a choice between two things; this one has three. Any two can be held, but only at the cost of the third.

Study Partner Protocol. The paste-in prompt in the companion repository that sets an AI up as a tutor: you try first, it gives hints rather than solutions, you explain the idea back, and it stays within the course (`companion/spp/`).

Translation test. A diagnostic. Take a piece of work and re-present it in a different form. The broad version asks whether you can say the same thing three ways: visually, mathematically, and in words.

Trustworthiness. The third dimension of the CAT framework. Your ability to verify, disclose, critique, and take ownership of AI-assisted work. It scores what you did, not what kind of person you are.

Verification. Checking AI-assisted content against reliable evidence, sources, data, tests, methods, or reasoning. It is the habit that turns AI assistance into responsible academic work.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Bibliography

[1] Janice Ahn, Rishu Verma, Renze Lou, Di Liu, Rui Zhang, and Wenpeng Yin. Large language models for mathematical reasoning: Progresses and challenges. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop*, pages 225–237, St. Julian's, Malta, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.eacl-srw.17.

[2] Boris Alexeev, Kevin Barreto, Yanyang Li, Jared Duker Lichtman, Liam Price, Jibran Iqbal Shah, Quanyu Tang, and Terence Tao. Primitive sets and von Mangoldt chains: Erdős problem #1196 and beyond, 2026. Preprint, arXiv:2605.00301, submitted 1 May 2026.

[3] Susan A. Ambrose, Michael W. Bridges, Michele DiPietro, Marsha C. Lovett, and Marie K. Norman. *How Learning Works: Seven Research-Based Principles for Smart Teaching*. Jossey-Bass, San Francisco, 2010. ISBN 9780470484104.

[4] American Association of University Professors. Artificial intelligence and academic professions. Technical report, American Association of University Professors, Washington, DC, 2025. URL `https://www.aaup.org/sites/default/files/2025-07/Artificial-Intelligence-and-Academic-Professions-Bulletin.pdf`. Report of the ad hoc Committee on Artificial Intelligence and Academic Professions.

[5] American Association of University Professors. Background facts on contingent faculty positions. AAUP, 2025. URL `https://www.aaup.org/background-facts-contingent-faculty-positions`. AAUP figures on contingent appointments: nearly 70% of instructional staff appointments in U.S. higher education are off the tenure track.

[6] Chenyang An, Qihao Ye, Minghao Pan, and Jiayun Zhang. QED: An open-source multi-agent system for generating mathematical proofs on open problems, 2026. Preprint, arXiv:2604.24021v4, submitted 27 April 2026, revised 26 June 2026.

[7] arXiv. Policy for authors' use of generative AI language tools. arXiv moderation page, January 2023. URL `https://info.arxiv.org/help/moderation/index.html`. Restated by T. G. Dietterich, chair of the arXiv Computer Science section, 14 May 2026.

[8] Association of College and Research Libraries. Framework for information literacy for higher education. American Library Association, Chicago, IL, 2016. URL `https://www.ala.org/acrl/standards/ilframework`. Adopted by the ACRL Board of Directors, January 11, 2016; revisions approved September 2, 2026.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[9] Fritz D. Bacalso, Francis E. Villamor, Cyre P. Paclipan, Clifford John B. Orquia, Irvin Paul Rendon, Antonietta T. Albiso, and Celbert M. Himang. Institutional approaches to generative AI management in higher education: A systematic review. *Frontiers in Education*, 11: 1814426, 2026. doi: 10.3389/feduc.2026.1814426.

[10] J. Elin Bahner, Anke-Dorothea Hüper, and Dietrich Manzey. Misuse of automated decision aids: Complacency, automation bias and the impact of training experience. *International Journal of Human-Computer Studies*, 66(9):688–699, 2008. doi: 10.1016/j.ijhcs.2008.06.001.

[11] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

[12] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*, pages 610–623, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 978-1-4503-8309-7. doi: 10.1145/3442188.3445922.

[13] Tricia Bertram Gallant. *Academic Integrity in the Twenty-First Century: A Teaching and Learning Imperative*. ASHE Higher Education Report 33(5). Jossey-Bass, San Francisco, 2008. ISBN 9780470373668.

[14] Tricia Bertram Gallant. Academic integrity as a teaching and learning issue: From theory to practice. *Theory Into Practice*, 56(2):88–94, 2017. doi: 10.1080/00405841.2017.1308173.

[15] Tricia Bertram Gallant and David A. Rettinger. *The Opposite of Cheating: Teaching for Integrity in the Age of AI*, volume 4 of *Teaching, Engaging, and Thriving in Higher Ed*. University of Oklahoma Press, Norman, 2025. ISBN 9780806194967.

[16] Elizabeth Ligon Bjork and Robert A. Bjork. Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In Morton Ann Gernsbacher, Richard W. Pew, Leaetta M. Hough, and James R. Pomerantz, editors, *Psychology and the Real World: Essays Illustrating Fundamental Contributions to Society*, pages 56–64. Worth Publishers, New York, 2011. ISBN 9781429230438.

[17] Paul Black and Dylan Wiliam. Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice*, 5(1):7–74, 1998. doi: 10.1080/0969595980050102.

[18] Maura Borrego and Charles Henderson. Increasing the use of evidence-based teaching in STEM higher education: A comparison of eight change strategies. *Journal of Engineering Education*, 103(2):220–252, 2014. doi: 10.1002/jee.20040.

[19] José Antonio Bowen and C. Edward Watson. *Teaching with AI: A Practical Guide to a New Era of Human Learning*. Johns Hopkins University Press, Baltimore, 2024. ISBN 9781421449227.

[20] Derek Bruff. *Intentional Tech: Principles to Guide the Use of Educational Technology in College Teaching*. Teaching and Learning in Higher Education. West Virginia University Press, Morgantown, 2019. ISBN 9781949199161.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[21] Elizabeth A. Canning, Judith M. Harackiewicz, Stacy J. Priniski, Cameron A. Hecht, Yoi Tibbetts, and Janet S. Hyde. Improving performance and retention in introductory biology with a utility-value intervention. *Journal of Educational Psychology*, 110(6):834–849, 2018. doi: 10.1037/edu0000244.

[22] Angxuan Chen, Mengtong Xiang, Junyi Zhou, Jiyou Jia, Junjie Shang, Xinyu Li, Dragan Gašević, and Yizhou Fan. Unpacking help-seeking process through multimodal learning analytics: A comparative study of ChatGPT vs human expert. *Computers & Education*, 226:105198, 2025. doi: 10.1016/j.compedu.2024.105198.

[23] Lingjiao Chen, Matei Zaharia, and James Zou. How is ChatGPT's behavior changing over time? *Harvard Data Science Review*, 6(2), 2024. doi: 10.1162/99608f92.5317da47.

[24] Myra Cheng, Cinoo Lee, Pranav Khadpe, Sunny Yu, Dyllan Han, and Dan Jurafsky. Sycophantic AI decreases prosocial intentions and promotes dependence. *Science*, 391 (6792):eaec8352, March 2026. doi: 10.1126/science.aec8352. Cover story. Corrections issued 10 April 2026 (summary figure; one duplicate reference removed and references renumbered) and 27 May 2026 (two swapped percentages in the text corrected; reference list updated).

[25] Igor Chirikov, Ivan Smirnov, and René F. Kizilcec. Generative AI use and misuse call for assessment reform in higher education. *Science*, 392(6800):818–820, 2026. doi: 10.1126/science.aec5115. URL https://www.science.org/doi/10.1126/science.aec5115.

[26] Thomas Corbin, Phillip Dawson, and Danny Liu. Talk is cheap: Why structural assessment changes are needed for a time of GenAI. *Assessment & Evaluation in Higher Education*, 50(7):1087–1097, 2025. doi: 10.1080/02602938.2025.2503964. URL https://doi.org/10.1080/02602938.2025.2503964.

[27] Debby R. E. Cotton, Peter A. Cotton, and J. Reuben Shipway. Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*, 61(2):228–239, 2024. doi: 10.1080/14703297.2023.2190148.

[28] Louis Deslauriers, Logan S. McCarty, Kelly Miller, Kristina Callaghan, and Greg Kestin. Measuring actual learning versus feeling of learning in response to being actively engaged in the classroom. *Proceedings of the National Academy of Sciences*, 116(39):19251–19257, 2019. doi: 10.1073/pnas.1821936116.

[29] Digital Education Council. Global AI faculty survey 2025. Technical report, Digital Education Council, 2025. URL https://www.digitaleducationcouncil.com/post/digital-education-council-global-ai-faculty-survey.

[30] Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28):eadn5290, 2024. doi: 10.1126/sciadv.adn5290.

[31] John Dunlosky and Janet Metcalfe. *Metacognition*. SAGE Publications, Thousand Oaks, CA, 2009. ISBN 9781412939720.

[32] John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham. Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, 14(1):4–58, 2013. doi: 10.1177/1529100612453266.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[33] EDUCAUSE. 2025 EDUCAUSE AI landscape study: Into the digital AI divide. Technical report, EDUCAUSE, February 2025. URL https://library.educause.edu/resources/2025/2/2025-educause-ai-landscape-study.

[34] European Commission. AI Act. Shaping Europe's Digital Future, 2026. URL https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai. Page last updated 19 October 2026. Accessed 2026-10-30.

[35] European Commission, Directorate-General for Education, Youth, Sport and Culture. Ethical guidelines on the use of artificial intelligence (AI) and data in teaching and learning for educators. Technical report, Publications Office of the European Union, Luxembourg, 2022. URL https://op.europa.eu/en/publication-detail/-/publication/d81a0d54-5348-11ed-92ed-01aa75ed71a1/language-en.

[36] European Commission, Directorate-General for Research and Innovation. Living guidelines on the responsible use of generative AI in research. ERA Forum Stakeholders' document, fourth version, European Research Area Forum, October 2026. URL https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/era-living-guidelines-responsible-use-generative-ai-research-updated-2026-10-16_en. Published 15 October 2026.

[37] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), as amended by Regulation (EU) 2026/1744. Official Journal of the European Union, L 2024/1689; consolidated version as at 27 July 2026, AI Act Service Desk, 2024. URL https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3. Annex III point 3(b); Article 26. Accessed 2026-10-30.

[38] Yizhou Fan, Luzhen Tang, Huixiao Le, Kejie Shen, Shufang Tan, Yueying Zhao, Yuan Shen, Xinyu Li, and Dragan Gašević. Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. *British Journal of Educational Technology*, 56(2):489–530, 2025. doi: 10.1111/bjet.13544.

[39] Tony Feng, Trieu Trinh, Garrett Bingham, Jiwon Kang, Shengtong Zhang, Sang-hyun Kim, Kevin Barreto, Carl Schildkraut, Junehyuk Jung, Jaehyeon Seo, Carlo Pagano, Yuri Chervonyi, Dawsen Hwang, Kaiying Hou, Sergei Gukov, Cheng-Chiang Tsai, Hyunwoo Choi, Youngbeom Jin, Wei-Yuan Li, Hao-An Wu, Ruey-An Shiu, Yu-Sheng Shih, Quoc V. Le, and Thang Luong. Semi-autonomous mathematics discovery with Gemini: A case study on the Erdős problems, 2026. Preprint, arXiv:2601.22401v3, submitted 29 January 2026, revised 5 February 2026.

[40] Colleen Flaherty. Survey: College students' views on AI. Inside Higher Ed Student Voice survey, August 2025. URL https://www.insidehighered.com/news/students/academics/2025/08/29/survey-college-students-views-ai. Survey of 1,047 students from 166 two- and four-year institutions, public and private nonprofit, conducted in July 2025 in partnership with Generation Lab. Margin of error plus or minus three percentage points.

[41] Colleen Flaherty. Presidents puzzled on rebuilding public trust in higher ed. Inside Higher Ed, April 2026. URL https://www.insidehighered.com/

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

news/governance/executive-leadership/2026/04/09/presidents-puzzled-rebuilding-public-trust-higher. Reports Inside Higher Ed's 2026 Survey of College and University Presidents, conducted with Hanover Research: N=430 presidents of public and private nonprofit two- and four-year institutions; margin of error 4 percentage points.

[42] Josh Freeman. Student generative AI survey 2025. HEPI / Kortext, Policy Note 61, February 2025. URL https://www.hepi.ac.uk/reports/student-generative-ai-survey-2025/. UK survey reporting that 92% of students use AI in some form in 2025, up from 66% in 2024. Reported widely including by Fortune (July 2025).

[43] Scott Freeman, Sarah L. Eddy, Miles McDonough, Michelle K. Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, 111(23):8410–8415, 2014. doi: 10.1073/pnas.1319030111.

[44] Leon Furze, Mike Perkins, Jasper Roe, and Jason MacVaugh. The AI Assessment Scale (AIAS) in action: A pilot implementation of GenAI-supported assessment. *Australasian Journal of Educational Technology*, 40(4):38–55, 2024. doi: 10.14742/ajet.9434. URL https://ajet.org.au/index.php/AJET/article/view/9434.

[45] Peter M. Gollwitzer. Implementation intentions: Strong effects of simple plans. *American Psychologist*, 54(7):493–503, 1999. doi: 10.1037/0003-066X.54.7.493.

[46] Chahna Gonsalves. Addressing student non-compliance in AI use declarations: Implications for academic integrity and assessment in higher education. *Assessment & Evaluation in Higher Education*, 50(4):592–606, 2025. doi: 10.1080/02602938.2024.2415654. URL https://www.tandfonline.com/doi/full/10.1080/02602938.2024.2415654. Single-institution study at King's Business School: up to 74% of students, varying by module, did not complete a mandatory AI-use declaration; reasons included fear of academic consequences and unclear guidance on what counted as AI use.

[47] John Hattie and Helen Timperley. The power of feedback. *Review of Educational Research*, 77(1):81–112, 2007. doi: 10.3102/003465430298487.

[48] Mohamed Helmy, Lingling Jin, Amr Alhossary, Tamer Mansour, Diogo Pellagrina, and Kumar Selvarajoo. Ten simple rules for optimal and careful use of generative AI in science. *PLOS Computational Biology*, 21(10):e1013588, 2025. doi: 10.1371/journal.pcbi.1013588.

[49] Charles Henderson, Andrea Beach, and Noah Finkelstein. Facilitating change in undergraduate STEM instructional practices: An analytic review of the literature. *Journal of Research in Science Teaching*, 48(8):952–984, 2011. doi: 10.1002/tea.20439.

[50] Kashmir Hill. The professors are using ChatGPT, and some students aren't happy about it. *The New York Times*, May 2025. URL https://www.nytimes.com/2025/05/14/technology/chatgpt-college-professors.html.

[51] Irene Hou, Owen Man, Kate Hamilton, Srishty Muthusekaran, Jeffin Johnykutty, Leili Zadeh, and Stephen MacNeil. "All roads lead to ChatGPT": How generative AI is eroding social interactions and student learning communities. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1 (ITiCSE*

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

2025), pages 79–85, New York, NY, USA, June 2025. Association for Computing Machinery. ISBN 979-8-4007-1567-9. doi: 10.1145/3724363.3729024. Preprint available as arXiv:2504.09779.

[52] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)*, Vienna, Austria, 2024. URL https://openreview.net/forum?id=IkmD3fKBPQ. arXiv:2310.01798.

[53] Chris S. Hulleman and Judith M. Harackiewicz. Promoting interest and performance in high school science classes. *Science*, 326(5958):1410–1412, 2009. doi: 10.1126/science.1177067.

[54] James M. Hyman. Learning to create with AI: Disciplines for deep learning, critical inquiry, and creative thought. Companion paper to *Learning with AI*. Available at the book's GitHub repository, 2026. URL https://github.com/machyman/hyman2026learning/tree/main/companion-papers/hyman2026creative.

[55] James M. Hyman. AI and the next layer of human work: A position paper for students, faculty, and administrators. Companion paper to *Learning with AI*. Available at the book's GitHub repository, 2026. URL https://github.com/machyman/hyman2026learning/tree/main/companion-papers/hyman2026next.

[56] James M. Hyman. Teaching AI responsibly to principled skeptics: A framework for students, faculty, and administrators. Companion paper to *Learning with AI*. Available at the book's GitHub repository, 2026. URL https://github.com/machyman/hyman2026learning/tree/main/companion-papers/hyman2026teaching.

[57] International Committee of Medical Journal Editors. Recommendations for the conduct, reporting, editing, and publication of scholarly work in medical journals. https://www.icmje.org/recommendations/browse/, January 2026. January 2026 update.

[58] Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011. ISBN 9780374275631.

[59] Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. Preprint, arXiv:2509.04664.

[60] Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günnemann, Eyke Hüllermeier, Stephan Krusche, Gitta Kutyniok, Tilman Michaeli, Claudia Nerdel, Jürgen Pfeffer, Oleksandra Poquet, Michael Sailer, Albrecht Schmidt, Tina Seidel, Matthias Stadler, Jochen Weller, Jochen Kuhn, and Gjergji Kasneci. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103:102274, 2023. doi: 10.1016/j.lindif.2023.102274.

[61] Greg Kestin, Kelly Miller, Anna Klales, Timothy Milbourne, and Gregorio Ponti. AI tutoring outperforms in-class active learning: An RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports*, 15(1):17458, 2025. doi: 10.1038/s41598-025-97652-6.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[62] Adrianna Kezar. *How Colleges Change: Understanding, Leading, and Enacting Change*. Routledge, New York, 2013. ISBN 9780415532068.

[63] Tyler Kingkade. To avoid accusations of AI cheating, college students are turning to AI. NBC News, January 2026. URL https://www.nbcnews.com/tech/internet/college-students-ai-cheating-detectors-humanizers-rcna253878.

[64] Miles Klee. She was falsely accused of cheating with AI — and she won't be the last. *Rolling Stone*, June 2023. URL https://www.rollingstone.com/culture/culture-features/student-accused-ai-cheating-turnitin-1234747351/.

[65] Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, and Jan Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27):eadt3813, 2025. doi: 10.1126/sciadv.adt3813. URL https://www.science.org/doi/10.1126/sciadv.adt3813.

[66] Nate Kornell, Matthew Jensen Hays, and Robert A. Bjork. Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35(4):989–998, 2009. doi: 10.1037/a0015729.

[67] Gerd Kortemeyer. Toward AI grading of student problem solutions in introductory physics: A feasibility study. *Physical Review Physics Education Research*, 19(2):020163, 2023. doi: 10.1103/PhysRevPhysEducRes.19.020163.

[68] Gerd Kortemeyer and Julian Nöhl. Assessing confidence in AI-assisted grading of physics exams through psychometrics: An exploratory study. *Physical Review Physics Education Research*, 21(1):010136, 2025. doi: 10.1103/PhysRevPhysEducRes.21.010136.

[69] Gerd Kortemeyer, Julian Nöhl, and Daria Onishchuk. Grading assistance for a handwritten thermodynamics exam using artificial intelligence: An exploratory study. *Physical Review Physics Education Research*, 20(2):020144, 2024. doi: 10.1103/PhysRevPhysEducRes.20.020144.

[70] Tomaž Kosar, Dragana Ostojić, Yu David Liu, and Marjan Mernik. Computer science education in ChatGPT era: Experiences from an experiment in a programming course for novice programmers. *Mathematics*, 12(5):629, 2024. doi: 10.3390/math12050629.

[71] Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes. Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task, 2025. URL https://arxiv.org/abs/2506.08872.

[72] David Krakauer. Will A.I. harm us? Better to ask how we'll reckon with our hybrid nature. *Nautilus*, September 2016. URL https://nautil.us/will-ai-harm-us-better-to-ask-how-well-reckon-with-our-hybrid-nature-236098/.

[73] Tzu-Sheng Kuo, Quan Ze Chen, Amy X. Zhang, Jane Hsieh, Haiyi Zhu, and Kenneth Holstein. PolicyCraft: Supporting collaborative and participatory policy design through case-grounded deliberation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*, pages 805:1–805:24, New York, NY, USA, 2025. Association for Computing Machinery. doi: 10.1145/3706598.3713865.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[74] Jason LaFrance. Governing generative artificial intelligence: Institutional policies and guidelines at America's flagship universities. *Educational Policy*, 2026. doi: 10.1177/08959048261450995. Online first.

[75] Learning Innovation Practice. The AI assessment scale: Resources and information. https://aiassessmentscale.com/, 2026. Accessed 2026-08-26.

[76] Leiden Declaration Working Group. Leiden declaration on artificial intelligence and mathematics. Published 2 June 2026; endorsed by the International Mathematical Union, June 2026. URL https://leidendeclaration.ai/.

[77] Arthur Levine and Scott Van Pelt. *From Upheaval to Action: What Works in Changing Higher Ed*. Johns Hopkins University Press, Baltimore, March 2026. ISBN 978-1421454030. doi: 10.56021/9781421454030.

[78] Hongli Li, Che Han Chen, Kevin Fan, Chiho Young-Johnson, Soyoung Lim, and Yali Feng. Agreement between large language models and human raters in essay scoring: A research synthesis, 2025. URL https://arxiv.org/abs/2512.14561. Preprint, arXiv:2512.14561v2, submitted 16 December 2025, revised 25 May 2026.

[79] Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou. GPT detectors are biased against non-native English writers. *Patterns*, 4(7):100779, 2023. doi: 10.1016/j.patter.2023.100779.

[80] Zhicheng Lin. Hidden prompts in manuscripts exploit AI-assisted peer review. *Communications of the ACM*, 69(7):53–56, June 2026. doi: 10.1145/3779116. URL https://cacm.acm.org/opinion/hidden-prompts-in-manuscripts-exploit-ai-assisted-peer-review/.

[81] Grace Liu, Brian Christian, Tsvetomira Dumbalska, Michiel A. Bakker, and Rachit Dubey. AI assistance reduces persistence and hurts independent performance. In *Proceedings of the Third Conference on Language Modeling (COLM 2026)*, San Francisco, October 2026. URL https://arxiv.org/abs/2604.04721. OpenReview u9VvZd6CiY; arXiv:2604.04721v6, revised 13 October 2026.

[82] Xiaohong Liu, Baoxin Guo, Wei He, and Xiaoyong Hu. Effects of generative artificial intelligence on K-12 and higher education students' learning outcomes: A meta-analysis. *Journal of Educational Computing Research*, 63(5):1249–1291, 2025. doi: 10.1177/07356331251329185.

[83] Nan Ma and Zhiyong Zhong. A meta-analysis of the impact of generative artificial intelligence on learning outcomes. *Journal of Computer Assisted Learning*, 41(5):e70117, 2025. doi: 10.1111/jcal.70117.

[84] Kim Manturuk. The best defense against AI cheating (opinion). Inside Higher Ed, April 2026. URL https://www.insidehighered.com/opinion/career-advice/teaching/2026/04/13/best-defense-against-ai-cheating-opinion. IHE opinion essay describing faculty postures of surveillance vs. supplication in response to AI cheating.

[85] Eric Mazur. *Peer Instruction: A User's Manual*. Prentice Hall Series in Educational Innovation. Prentice Hall, Upper Saddle River, NJ, 1997. ISBN 9780135654415.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[86] Donald L. McCabe and Linda Klebe Treviño. Academic dishonesty: Honor codes and other contextual influences. *The Journal of Higher Education*, 64(5):522–538, 1993. doi: 10.1080/00221546.1993.11778446.

[87] Beth McMurtrie. Professors ask: Are we just grading robots? *The Chronicle of Higher Education*, June 2024. URL https://www.chronicle.com/article/professors-ask-are-we-just-grading-robots. Print version appeared in the July 5, 2024 issue.

[88] Fengchun Miao and Mutlu Cukurova. AI competency framework for teachers. Technical report, UNESCO, Paris, 2024. URL https://www.unesco.org/en/articles/ai-competency-framework-teachers.

[89] Fengchun Miao and Wayne Holmes. Guidance for generative AI in education and research. Technical report, UNESCO, Paris, 2023. URL https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research.

[90] Fengchun Miao, Kelly Shiohira, and Natalie Lao. AI competency framework for students. Technical report, UNESCO, Paris, 2024. URL https://www.unesco.org/en/articles/ai-competency-framework-students.

[91] Modern Language Association and Conference on College Composition and Communication. MLA-CCCC joint task force on writing and AI working paper: Overview of the issues, statement of principles, and recommendations. Technical report, MLA-CCCC Joint Task Force on Writing and AI, 2023. URL https://hcommons.org/app/uploads/sites/1003160/2023/07/MLA-CCCC-Joint-Task-Force-on-Writing-and-AI-Working-Paper-1.pdf.

[92] Ethan Mollick. *Co-Intelligence: Living and Working with AI*. Portfolio/Penguin, New York, 2024. ISBN 9780593716717.

[93] Ethan R. Mollick and Lilach Mollick. Assigning AI: Seven approaches for students, with prompts. Working paper, The Wharton School, University of Pennsylvania, 2023. URL https://arxiv.org/abs/2306.10052. arXiv:2306.10052.

[94] National Academies of Sciences, Engineering, and Medicine. *How People Learn II: Learners, Contexts, and Cultures*. National Academies Press, Washington, DC, 2018. ISBN 9780309459648. doi: 10.17226/24783.

[95] National Institutes of Health. The use of generative artificial intelligence technologies is prohibited for the NIH peer review process. NIH Guide Notice NOT-OD-23-149, 2023. URL https://grants.nih.gov/grants/guide/notice-files/NOT-OD-23-149.html.

[96] National Institutes of Health. Supporting fairness and originality in NIH research applications. Notice NOT-OD-25-132, July 2025. URL https://grants.nih.gov/grants/guide/notice-files/NOT-OD-25-132.html. Released 17 July 2025; effective for applications submitted to the 25 September 2025 receipt date and later.

[97] Donald A. Norman. Cognitive artifacts. In John M. Carroll, editor, *Designing Interaction: Psychology at the Human-Computer Interface*, pages 17–38. Cambridge University Press, Cambridge, 1991.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[98] OECD. *OECD Digital Education Outlook 2026: Exploring Effective Uses of Generative AI in Education*. OECD Publishing, Paris, 2026. ISBN 978-92-64-74128-7. doi: 10.1787/062a7394-en.

[99] OpenAI. An OpenAI model has disproved a central conjecture in discrete geometry. OpenAI, 2026. URL https://openai.com/index/model-disproves-discrete-geometry-conjecture/. Announced 20 May 2026.

[100] Meghan O'Rourke. I teach creative writing. This is what A.I. is doing to students. *The New York Times*, July 2025. URL https://www.nytimes.com/2025/07/18/opinion/ai-chatgpt-school.html. Guest essay, Opinion section. The author is executive editor of *The Yale Review* and a professor of creative writing at Yale University.

[101] Mike Perkins, Leon Furze, Jasper Roe, and Jason MacVaugh. The Artificial Intelligence Assessment Scale (AIAS): A framework for ethical integration of generative AI in educational assessment. *Journal of University Teaching and Learning Practice*, 21(6):49–66, 2024. doi: 10.53761/q3azde36. URL https://open-publishing.org/journals/index.php/jutlp/article/view/810.

[102] Mike Perkins, Jasper Roe, and Leon Furze. Reimagining the Artificial Intelligence Assessment Scale (AIAS): A refined framework for educational assessment. *Journal of University Teaching and Learning Practice*, 22(7), 2025. doi: 10.53761/rrm4y757. URL https://open-publishing.org/journals/index.php/jutlp/article/view/1707.

[103] George Pólya. *How to Solve It: A New Aspect of Mathematical Method*. Princeton University Press, Princeton, NJ, 1945.

[104] Geraint Rees and James Wilsdon. Could agentic AI topple grant-funding systems? *Nature*, 652(8112):1119–1121, April 2026. doi: 10.1038/d41586-026-01297-y. URL https://www.nature.com/articles/d41586-026-01297-y. Lead author affiliation: Geraint Rees, UCL Vice-Provost of Research, Innovation & Global Engagement.

[105] Stephanie Reeves. AI as the new digital divide for first-generation students. WGU Labs blog (EdTech Survey), August 2023. URL https://www.wgulabs.org/posts/ai-as-the-new-digital-divide-for-first-generation-students. WGU Labs EdTech Survey reporting that 48% of continuing-generation students were aware of ChatGPT compared with 34% of first-generation students; 11% of continuing-generation students had used ChatGPT for coursework compared with 7% of first-generation students. Survey covered nine diverse colleges and universities.

[106] Henry L. Roediger, III and Jeffrey D. Karpicke. Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3):249–255, 2006. doi: 10.1111/j.1467-9280.2006.01693.x.

[107] Everett M. Rogers. *Diffusion of Innovations*. Free Press, New York, 5 edition, 2003. ISBN 9780743222099.

[108] Doug Rohrer and Kelli M. Taylor. The shuffling of mathematics problems improves learning. *Instructional Science*, 35(6):481–498, 2007. doi: 10.1007/s11251-007-9015-8.

[109] Anita Samuel. Faculty + AI: A retrospective of 2025. Tech-Savvy Teaching (Substack), December 2025. URL https://anitasamuel.substack.com/p/faculty-ai-a-retrospective-of-2025.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[110] Andreas Schleicher. Quality teaching in the age of AI: Shining glory of humanity in teaching profession. *ECNU Review of Education*, 9(2), 2026. doi: 10.1177/20965311261421964.

[111] Scientific American. Amateur armed with ChatGPT 'vibe-maths' a 60-year-old problem, April 2026. URL https://www.scientificamerican.com/article/amateur-armed-with-chatgpt-vibe-maths-a-60-year-old-problem/. Editor's note dated 28 April 2026.

[112] Judy Hanwen Shen and Alex Tamkin. How AI impacts skill formation, 2026. Preprint, arXiv:2601.20245.

[113] Elise Silva. University students feel 'anxious, confused and distrustful' about AI in the classroom and among their peers. The Conversation, July 2025. URL https://theconversation.com/university-students-feel-anxious-confused-and-distrustful-about-ai-in-the-classroom-and-among-their-peers-258665. Reports findings from focus groups with 95 University of Pittsburgh students (Spring 2025). Silva is a member of the study team and writes in the first person; the full study is Vee et al. [128], on which she is ninth author and Vee is first. Other members of the team, acknowledged in her article: Annette Vee, Patrick Manning, Jessica FitzPatrick, Jessica Ghilani, Catherine Kula, Patty Wharton-Michael, Jialei Jiang, Sean DiLeonardi, Birney Young, Mark DiMauro, Jeff Aziz, Gayle Rogers.

[114] Linda J. Skitka, Kathleen L. Mosier, and Mark Burdick. Does automation bias decision-making? *International Journal of Human-Computer Studies*, 51(5):991–1006, 1999. doi: 10.1006/ijhc.1999.0252.

[115] Linda J. Skitka, Kathleen L. Mosier, and Mark D. Burdick. Accountability and automation bias. *International Journal of Human-Computer Studies*, 52(4):701–717, 2000. doi: 10.1006/ijhc.1999.0349.

[116] Nicholas C. Soderstrom and Robert A. Bjork. Learning versus performance: An integrative review. *Perspectives on Psychological Science*, 10(2):176–199, 2015. doi: 10.1177/1745691615569000.

[117] Jane Southworth, Kati Migliaccio, Joe Glover, Ja'Net Glover, David Reed, Christopher McCarty, Joel Brendemuhl, and Aaron Thomas. Developing a model for AI across the curriculum: Transforming the higher education landscape via innovation in AI literacy. *Computers and Education: Artificial Intelligence*, 4:100127, 2023. doi: 10.1016/j.caeai.2023.100127.

[118] Miloš Stanković, Ella Hirche, Sarah Kollatzsch, and Julia Nadine Doetsch. Comment on: Your Brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing tasks, 2026. URL https://arxiv.org/abs/2601.00856.

[119] Rose Stephenson and Charlotte Armstrong. Student generative AI survey 2026. HEPI / Kortext, Report 199, 2026. URL https://www.hepi.ac.uk/reports/student-generative-ai-survey-2026/. UK survey of 1,054 full-time undergraduates (Savanta, December 2025): 95% use AI in some form, 94% for assessed work, up from 92% (2025) and 66% (2024). Full report PDF: https://www.hepi.ac.uk/wp-content/uploads/2026/03/HEPI-Report-199-Gen-AI-Survey-2026.pdf.

[120] Lihui Sun and Liang Zhou. Does generative artificial intelligence improve the academic achievement of college students? A meta-analysis. *Journal of Educational Computing Research*, 62(7):1676–1713, 2024. doi: 10.1177/07356331241277937.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[121] John Sweller, Paul Ayres, and Slava Kalyuga. *Cognitive Load Theory*, volume 1 of *Explorations in the Learning Sciences, Instructional Systems and Performance Technologies*. Springer, New York, 2011. ISBN 9781441981257. doi: 10.1007/978-1-4419-8126-4.

[122] Terence Tao. Machine-assisted proof. *Notices of the American Mathematical Society*, 72 (1):6–13, 2025. doi: 10.1090/noti3041. URL https://www.ams.org/notices/202501/rnoti-p6.pdf.

[123] Terence Tao. Mathematics in the age of AI, 2026. Preprint, arXiv:2608.16753, dated 18 August 2026; written version of a public lecture at the International Congress of Mathematicians, July 2026.

[124] Terence Tao and Mark Chen. Fireside chat with James Donovan. *Accelerating Math and Theoretical Physics with AI*, OpenAI Forum and Institute for Pure and Applied Mathematics, University of California, Los Angeles; recordings on the OpenAI Forum and IPAM's YouTube channel, March 2026. URL https://www.ipam.ucla.edu/programs/special-events-and-conferences/accelerating-math-and-theoretical-physics-with-ai/. Recorded 4 March 2026.

[125] United States Court of Appeals for the Sixth Circuit. Whiting v. City of Athens, 2026. URL https://www.opn.ca6.uscourts.gov/opinions.pdf/26a0080p-06.pdf. Nos. 24-5918/5919, 25-5424, 2026 WL 710568 (6th Cir. 2026). The court imposed sanctions under Rule 38 and its inherent authority on attorneys whose appellate briefs contained over two dozen fake citations and misrepresentations of fact.

[126] U.S. Department of Education, Office of Educational Technology. Artificial intelligence and the future of teaching and learning: Insights and recommendations. Technical report, U.S. Department of Education, Washington, DC, 2023. URL https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf.

[127] U.S. Department of Education, Student Privacy Policy Office. Who is a "school official" under FERPA? Protecting Student Privacy (studentprivacy.ed.gov), frequently asked questions, 2026. URL https://studentprivacy.ed.gov/faq/who-school-official-under-ferpa. Accessed 2026-09-20.

[128] Annette Vee, Patrick Manning, Jessica FitzPatrick, Jessica Ghilani, Catherine Kula, Patty Wharton-Michael, Jialei Jiang, Sean DiLeonardi, Elise Silva, Birney Young, Mark DiMauro, Jeff Aziz, and Gayle Rogers. GenAI conversations: Student use of generative AI at Pitt. Report, University of Pittsburgh, June 2026. URL https://www.cyber.pitt.edu/sites/default/files/GenAI%20Conversations%20Report%20v.%202.pdf. Version 2, 17 June 2026; supersedes Version 1 of 16 May 2025.

[129] Annette Vee, Marc Watkins, and Derek Bruff. *The Norton Guide to AI-Aware Teaching*. W. W. Norton, New York, 2026. ISBN 9781324127826. URL https://wwnorton.com/books/9781324127673. Ebook (ISBN 9781324127826), 1 July 2026; paperback (ISBN 9781324127673), 24 September 2026. Cited from the ebook.

[130] James D. Walsh. Everyone is cheating their way through college. *New York Magazine / Intelligencer*, May 2025. URL https://nymag.com/intelligencer/article/openai-chatgpt-ai-cheating-education-college-students-school.html.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

[131] William H. Walters and Esther Isabelle Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi: 10.1038/s41598-023-41032-5.

[132] Rose E. Wang, Ana T. Ribeiro, Carly D. Robinson, Susanna Loeb, and Dorottya Demszky. Tutor CoPilot: A human-AI approach for scaling real-time expertise. EdWorkingPaper 24-1054, Annenberg Institute at Brown University, November 2025. Version 2, November 2025.

[133] Joan Monahan Watson, José Antonio Bowen, and C. Edward Watson. *Learning with AI: The K–12 Teacher's Guide to a New Era of Human Learning*. Johns Hopkins University Press, Baltimore, 2024.

[134] Max Weinreich. The crisis of AI-generated mathematics, August 2026. URL `https://arxiv.org/abs/2608.02859`. arXiv preprint, math.HO, MSC 00A30.

[135] Grant Wiggins and Jay McTighe. *Understanding by Design*. Association for Supervision and Curriculum Development (ASCD), Alexandria, VA, 2 edition, 2005. ISBN 9781416600350.

[136] David M. Williamson, Xiaoming Xi, and F. Jay Breyer. A framework for evaluation and use of automated scoring. *Educational Measurement: Issues and Practice*, 31(1):2–13, 2012. doi: 10.1111/j.1745-3992.2011.00223.x.

[137] Sarah Shi Hui Wong and Sophia Xuefei Qiu. Think first, ChatGPT later: Guiding human–AI collaboration for learning gains in independent human creativity. *Educational Psychology Review*, 38(1):45, 2026. doi: 10.1007/s10648-026-10118-7.

[138] Kaixun Yang, Mladen Raković, Dragan Gašević, and Guanliang Chen. Does the prompt-based large language model recognize students' demographics and introduce bias in essay scoring? In Alexandra I. Cristea, Erin Walker, Yu Lu, Olga C. Santos, and Seiji Isotani, editors, *Artificial Intelligence in Education: 26th International Conference, AIED 2025, Palermo, Italy, July 22–26, 2025, Proceedings, Part II*, pages 75–89, Cham, 2025. Springer. doi: 10.1007/978-3-031-98417-4_6. Preprint available as arXiv:2504.21330.

[139] Yan Zhao, Yuhe Yue, Zhonghua Sun, Qiang Jiang, and Gangsheng Li. Does generative artificial intelligence improve students' higher-order thinking? A meta-analysis based on 29 experiments and quasi-experiments. *Journal of Intelligence*, 13(12):160, 2025. doi: 10.3390/jintelligence13120160.

[140] Michael J. Zyphur. The institutional AI readiness pack: Self-assessment and implementation tools for responsible AI in academic research. Instats policy series, Instats, 2026. URL `https://github.com/mzyphur/ai-readiness`. CC BY 4.0.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

Index

academic integrity, 118
  first response, 135
  procedure, 160
accessibility
  assistive routes to a text, 49
accreditation, 153
accreditors, 175
adjunct faculty, 82
  verification labor, 83
adoption summary, one-page, 182
AI
  dangerous uses of, 44
  good uses of, 37
  when not to use, 59
AI account configuration, 8
AI as Object of Study, *see* assignment categories
AI Assessment Scale, 90
AI detectors, 118
  bias against non-native speakers, 118
  constructive use of, 121
  false positives, 118
  perplexity-based detection, 118
AI literacy, 167
AI policy
  posting for students' AI tools, 98
AI-assisted grading, 114, 159
AI-assisted work, **26**
AI-first trap, **8**, 44
  cost to intuition, 47
assessment
  transfer task, 112
assessment models, 155
assessment validity, 99
assignment categories, 33, 89
  AI as Collaborator (AIC), 33
  AI as Object of Study (AIS), 33
  AI as Tutor (AIT), 33
  AI Assessment Scale crosswalk, 90
  No AI (NAI), 33
assignment design, 92
  recasting single-answer problems, 95
authorship
  non-expert finders, 194
automation bias, 103

Bastani field experiment, xvii, **61**
budget incentives, 151
buy-in, faculty, 171
buy-in, student, 92
  and safety to be wrong, 92
  evidence for, 93

calibration error, **20**
calibration problem, instructor-side, 78
capacity allocation, 6
CAT framework, 25, 87, 153
  AI-assisted Practice, **25**, **87**
  Core Competence, **25**, **87**
  Trustworthiness, **25**, **87**
ChatGPT, 97
cheating, *see* academic integrity
checklist, student, 72
Cheng sycophancy study, 11, 38
class notes
  as the tutor's map, 14
class size, 85
  large classes, 53
  small classes, 53
Claude, 97
code, verifying, 45
cognitive offloading, 62
collective diversity, 146
compact, student, *see* Student Compact
companion files
  adapting for a course, 98

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

companion repository, 65
compensation, *see* stipends
competence costume, **151**
computing and programming, 58
conscientious objector, 60
contingent faculty, 82
  and faculty development, 171
  differentiated support, 172
  structural failure mode, 83
course level, 85
course materials
  posting for AI use, 98
course redesign
  one term at a time, 129
  where to start, 129
course scope, 13
  disciplinary framing, 97
  instructor scope note, 97
  syllabus and table of contents, 13
course-level AI-use categories, 154
creative work, 126
  assignment design, 98
  capability versus output, 32

data governance, 158
deans, 174
Define Personal Preferences, 9
department chairs, 143
  argument to deans, 150
departmental adoption kit, 180
departmental practice, shaping policy, 174
departmental responsibilities, 166
desirable difficulties, **20**
detectors, *see* AI detectors
disability accommodation, 60
  campus office, 60
disciplinary societies, 175
discipline overlays, 85, 123
disclosure, 50
  standard, 157
  templates, 50
distributed practice, **19**

employment status, 82
enrollment pressure, 151
equity and access, 122, 159
  accessibility, 123
  in oral assessment, 105
  multilingual students, 123
escalation of AI requests, 29
established courses
  redesign cost, 130
EU AI Act, 159
evaluation metrics, 185
evidence base and limits, 61
  uneven effects, 131
evidence of thinking, **53**
evidence status, 185
Evidence-of-Thinking Model, 53, 106
expectations
  student, 62

faculty buy-in, *see* buy-in, faculty
faculty development
  attendance, 171
  barriers, 171
  change management, 170
  faculty paid by the course, 171
  training, 168
faculty use of AI, 162
feedback and grading with AI, 113
FERPA, 159
Feynman, Richard, 44
first move, independent, 98

generated citation, 44
generation effect, **20**
governance, 165
grade weighting, 103
  splitting a component, 103
grading consistency, 129
grading-agreement check, **114**
group assignments, 60, 112
guiding principles, 152

hallucination, 44
help ladder
  in the course tutor setup, 90
help, asking for
  escalation, 29
  going backward, 31
hollow understanding
  institutional, **149**
  instructor, **81**
  student, 16
humanities, 56
  interpretive work, 56
Hybrid Model, 108

illusion of competence, 20
implementation plan, 168

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

cost, 170
implementation workflow, 129
institutional commitments
student view, 62
institutional problem, 151
instructor orientation to AI, 77
integrity penalty
institutional, **149**
instructor, **82**
student, 16
who can afford it, 83
inter-rater reliability, *see* TA calibration
interleaving, **20**
interpretive disciplines, 56
intuition
for correct answers, 47

judgment
generative, 48

Kalai hallucination analysis, 31
Kestin AI-tutor study, **61**, 93

learning how to learn, 8
learning outcomes
institutional, 146
instructor, 78
student, 14
learning science, 19
Learning Spiral, 27
ask for help, 29
frame the question, 28
reflect and disclose, 32
try to solve, 29
verify, 31
work independently, 31
writing a problem to start it, 95
learning-with-AI skill, 11
lecturers, 83
Liang detector study, 160
library, 165
Liu persistence trial, 62
looking forward, 183

mathematics
asking better questions, 54
quick reference, 199
mentor prompt, 11
required, 95
metacognitive laziness, 20, **80**
minimum independent-competence requirement, 157
model institutional policy statement, 189

narrowing of student output, 146
neutrality, asking AI for, 38
Northeastern case, 119

O'Rourke capacity account, 6
office hours, 45
as an early-warning signal, 78
decline in attendance, 78
OpenStax, 13
oral defense, **53**
oral examination, 104
Oral-Defense Model, 53, 104

part-time faculty, *see* adjunct faculty
persistent settings, 8
pitfalls, 70
instructor, 138
Plan-Use-Verify, **93**
policy self-audit, 180
Pólya, George, 95
practice testing, **19**, 37
presumption of AI use, 162
privacy and data protection, 52, 122, 158
problems, writing and recasting, 95
process evidence, 106
process log, **107**
productive struggle, 19, 92
professional programs, 56

quiet overclaim
institutional, **149**
instructor, **82**
student, 16

random audit, 108
RCM, *see* Responsibility Center Management
refusing AI, 60
release time, 172
college-controlled, 174
relevance, writing about, 93
replacing people with AI, 45
research
choosing the next move, 71
Responsibility Center Management, 151
restorative practice, **161**
retrieval practice, *see* practice testing

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.

risk register, 187
rubrics, 127
  AIC worked example, 128

safeguards
  against AI substitution, 138
sampling
  live audits, 110
scientific writing
  pre-submission checklist, 197
  results found outside one's field, 194
  verification cornerstone, 193
scope of the instructor guide, 77
scope, course, *see* course scope
second reader, 104
secure assessment, 162
seeded-error check, 108
self-study protocol, 187
sharing the conversation link, 51
Shen and Tamkin skill-formation study, 62
social dilemma, 146
social sciences, 56
spot checks
  choosing whom, 110
  not a penalty, 84
STEM, 123
stipends, 170
Student Compact, 22
Study Partner Protocol, 34
  course version, 90
study routine, 65
  in a group, 65
support units, 165
suspected AI misuse, *see* academic integrity
syllabus
  grade formula, 103
  required elements, 162
  sample language, 133

TA calibration, 108, 129
  for oral defense, 104
teaching centers, 166
tenure-track faculty, 83
thread review
  as teaching feedback, 96
tool approval, 158
tool-use illusion, 21
tool-use skill, 11
traffic-light metaphor, 33
training plan, 167
transfer, 19
transfer of learning, 8
transfer task, 112
translation test, **23**
Trilemma
  Institution's, **147**
  Instructor's, **80**
  Student's, **15**
trilemma, *see* Trilemma
tutor, AI
  keeping inside the course, 13
two-lane assessment framework, 162

University of Sydney assessment framework, 162

Verification, **31**
verification, 45
  in graduate research, 50
  in professional fields, 49
  in STEM, 48
  in the humanities, 49
  in the social sciences, 49
  in writing, 49
  measuring, 108
  of citations, 44
  order of, 47
verification discipline, 128

Wendy, 15, 63
who benefits from AI, 131
workload, 83
  disparities across course sizes, 172
writing courses, 125

Author's pre-edit draft, v6.1.1, November 2026, posted with SIAM's permission. For the final edition, see SIAM Books.